

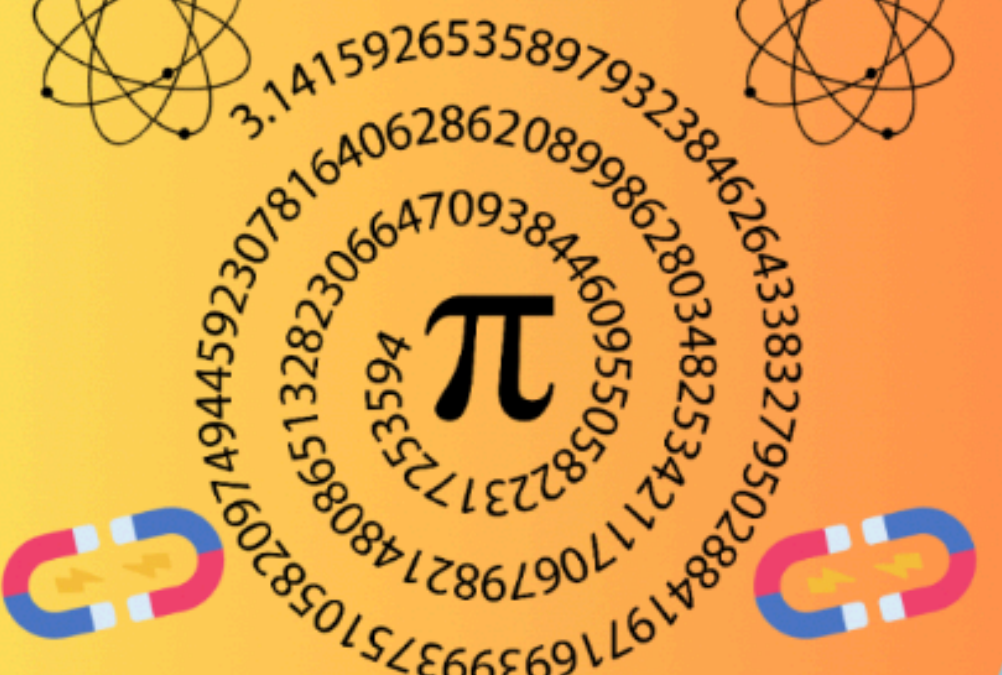
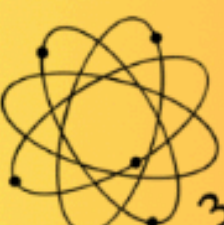


ACE THE AMC 10/12

October 2023

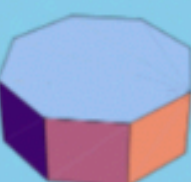
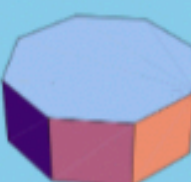
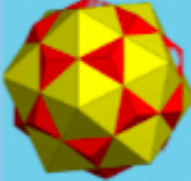


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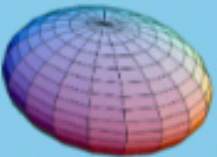
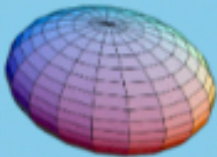


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RITVIK RUSTAGI



tmasacademy.com

Version 1



Ritvik Rustagi

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1 Information

§1.1 About TMAS Academy

This book is brought to you by me (Ritvik Rustagi). TMAS Academy, previously known as Explore Math, was started by me in 2020. TMAS stands for The Math and Science. You can find more info about this program on my website linked below.

Website: <https://www.tmasacademy.com/>

§1.2 Opportunities For You To Contribute To TMAS Academy

Contributing to TMAS Academy is simple.

You can **join the team** by going to our website: <https://www.tmasacademy.com/opportunitieswithtmas>

Donations: If you want to assist me in my monthly payments to run this program which includes website costs, Overleaf costs (the platform used to write such long booklets), and filming/editing costs, then please consider donating! For those that are willing to contribute, I have listed a few ways below. **Don't forget to write a message so I know who you are which will allow me to send you a thank you note.**

- You can donate through PayPal to the email: ritvikrustagi7@gmail.com
- You can donate on <https://www.buymeacoffee.com/tmasacademy>
- If you want to donate and the above 2 methods don't work for you, then you can send an email to weexploremath@gmail.com

You can also contribute by **subscribing** to the Youtube channel: <https://www.youtube.com/@tmasacademy>

Also, don't forget to join the Discord server to connect with other students and the owner: <https://discord.gg/xFRN3TWd>!

You can also follow all of our socials such as the LinkedIn page and the Instagram account that is run by the media team, please join the mailing list to learn about all updates and our upcoming books and videos. All of that can be found at the bottom of the site: <https://www.tmasacademy.com/>

§1.3 About The Author: Ritvik Rustagi

My name is Ritvik Rustagi, and I am a student at Prospect High School. Some information about me is that I enjoy doing math, physics, and programming. Although I did some Math Olympiad contests starting 6th grade, I started my serious journey of contest math in 8th grade. That is when I truly utilized online school to absorb these concepts that show up in math competitions.

Some of my qualifications include qualifying for USAJMO (United States of America Junior Mathematical Olympiad) 2 times, qualifying for USAPHO (United States of America Physics Olympiad), achieving a gold medal in the finals round for MathCon in Chicago, and earning DHR (Distinguished Honor Roll) 2 times for the AMC 10.

During Covid, I discovered my passion for teaching math competition topics through my Youtube channel. It also allowed me to absorb these complicated topics more efficiently since teaching can help one improve their own skills.

This book has been written to help any student aiming to qualify for the AIME. Due to the increasing difficulty of this exam, a good guide is necessary with a rich problem set for students to practice with. This is what the book aims to do. Hundreds of problems are contained within this book with well written solutions. This will allow even the most inexperienced students to have a productive session of math while comprehending the problems.



§1.4 Benefits of Preparing for the AMC 10 and AMC 12

Preparing for math competitions is a great way to expand your math knowledge from school. These competitions go a step further to deepen your knowledge of topics you learned in school. On top of that, you will learn many concepts that schools don't cover. It's a great learning experience and can give you the opportunity to enrich your journey in math. It also improves your problem solving skills which can serve as a life skill in many situations.

§1.5 What if there is an error in the book?

There are possibilities for minor errors such as typos or a mistake in latex for some of the solutions to the problems. If that's the case, then please click on this link (<https://forms.gle/3mxZb4izUuBZLkmz5>) to report the mistake.

§1.6 Any other questions or concerns?

If you have any other questions or concerns, then please feel free to reach out to weexploremath@gmail.com

§1.7 Credits

I would like to thank **The Art of Problem Solving** for some of their diagrams and problems that were used in this book.



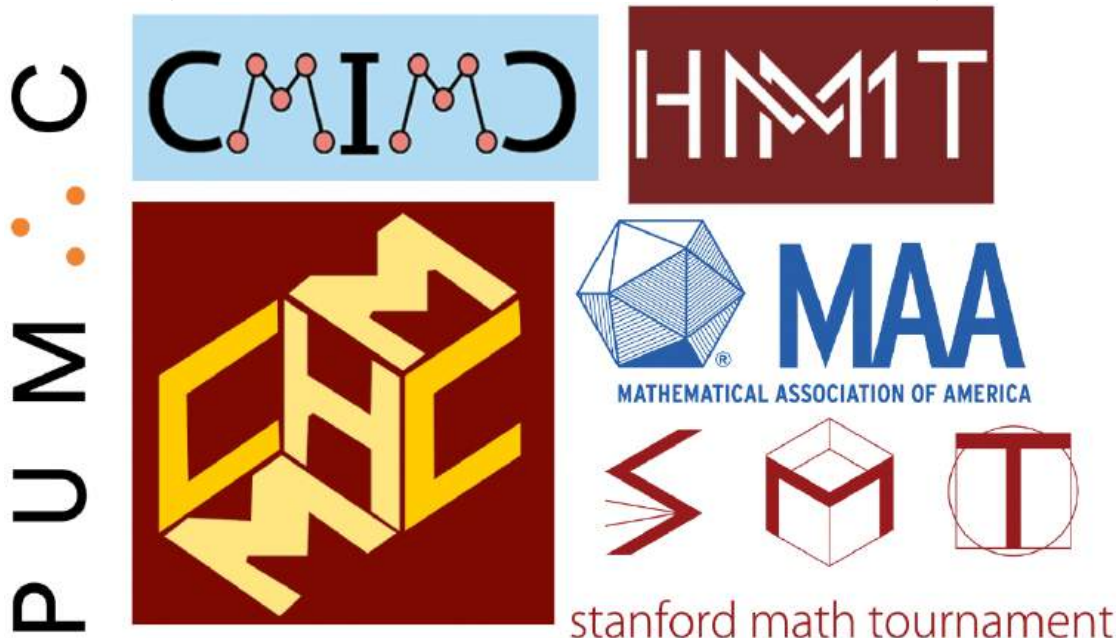
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Art of Problem Solving

I would also like to thank **Evan Chen**, a PhD student at MIT (Massachusetts Institute of Technology), for his latex template which made it easy to format this book. I would also like to thank him for answering all of my questions regarding the format of this book.

I would also like to thank these following organizations for their rich source of problems which gave me a large question bank to explore for this book

- **MAA** (The Mathematical Association of America)
- **SMT** (Stanford Math Tournament)
- **HMMT** (Harvard MIT Mathematics Tournament)
- **PUMaC** (Princeton University Mathematics Competition)
- **CHMMC** (Caltech Harvey Mudd Math Competition)
- **CMIMC** (Carnegie Mellon Informatics and Mathematics Competition)



I would also like to thank my parents for encouraging me to do contest math and for everything else that they have done.

§1.8 AMC 10/12 Information That You Should Know

- 25 questions to complete in 75 minutes
- 3 minutes per question on average
- 6 points for all correct answers, 1.5 points for unanswered questions, and 0 points for wrong answers
- Fortunately, anyone can take the AMC 10/12 as long as they're in tenth/twelfth grade or below depending on the contest
- There is an AMC 10/12 A and B and the information above applies to both A and B
- The AMC 10/12 is the first exam to decide IMO representatives for America
- Don't bring graph paper to the contest day! It's not allowed.
- People who score well on the AMC 10/12 take the AIME
- Those who do well on both and get a certain score can get into USA(J)MO and continue their journey into MOP (Math Olympiad Summer Program) and even IMO

§1.9 Strategies for a Higher Score

- DON'T Panic at ALL
- Leave a problem blank if you can't understand it
- If you can eliminate some answer choices, then always do so.
- Avoid guessing because you can get 1.5 points for every unanswered question!
- Try to remove answer choices by finding the range of the answers. Then, you can remove the choices that are too high or low.
- In Geometry Problems
 - Draw a diagram for all the problems
 - Estimate the dimensions if you don't have enough time
 - If something looks the same to you (such as 2 side lengths), then it probably is the same (**Note:** Only use this tip if you don't have enough time)

§1.10 Expected Paths for AIME and USA(J)MO

For those that don't know, AIME stands for the **American Invitational Mathematical Examination**. This exam has 15 difficult questions to solve in 3 hours. To make this exam, you must pass the cutoffs for either the AMC 10 or AMC 12 exam.

USAJMO stands for the **United States of America Junior Mathematical Olympiad** while USAMO stands for the **United States of America Mathematical Olympiad**. To make USAJMO, you must do well on specifically the AMC 10 and AIME. To make USAMO, you must do well on the AMC 12 and AIME.

For a guaranteed AIME qualification from the AMC 10, you will need above 110 out of 150. However, these days the cutoffs are going down due to the drastically increasing difficulty of the test. The cutoffs lately have been barely above 100 (sometimes even less than 100).

For the AMC 12, the required cutoff to make AIME is lower than what's necessary for AMC 10. A score that's above 100 is almost certain to make it. Even a score in the 90s can be enough; however, you should still aim higher just in case the cutoffs are high.

Before talking about USA(J)MO, you should know what a score index is for this exam. Your index is your AMC Score + $10 \times$ Your AIME Score.

For example, if your AMC score is 120 while your AIME score is 10, then your index is $120 + 10 \cdot 10$ which is 220.

USAMO cutoffs are significantly higher than USAJMO cutoffs. You should aim for a 240 index if you want to maximize your chances of qualifying for the USAMO.

For USAJMO, you should aim for a 215 index. An index of 215 should be significantly above the cutoff, as it has been in recent years. The increasing difficulty of both the AMC 10 and AIME has caused USAJMO cutoffs to decrease to slightly above 200 to even less than 200.

2 Number Theory

§2.1 The Basics: Prime and Composite Numbers

Definition 2.1.1

A prime number isn't divisible by any number other than 1 and itself. A composite number is any integer that isn't a prime number. It has more than 2 divisors.

It's important to learn your divisibility rules.

Rule for 2: The units digit must be even.

Rule for 3: The sum of the digits must be divisible by 3.

Rule for 4: The last 2 digits must be divisible by 4.

Rule for 5: The last digit must be 0 or 5.

Rule for 8: The last 3 digits must be divisible by 8.

Rule for 9: The sum of the digits must be divisible by 9.

Rule for 11: The difference between the alternating sums of the digits must be divisible by 11

What if you want to know whether or not a number is divisible by 6 or 10 or 12? Then, we simply write the prime factorization of those numbers and use the divisibility rules for each of the prime factors separately. For example, 6 is $2 \cdot 3$, so we simply use both the rules for divisibility by 2 and 3. For 10 we use the rules for 2 and 5 while for 12 we use the rules for 3 and 4.

Problem 2.1.2 — The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?

(A) 2 (B) 3 (C) 5 (D) 7 (E) 11

Source: 2020 AMC 10

Solution: Since the sum of all of the angles in any triangle is 180° , we know that a and b must sum to 90° since one of our angles is already 90° . Since we want to minimize b , we'll start testing our values for b starting from 1° and move up. Note: We only need to test the **odd** numbers since an even number can never be prime other than 2° .

Trials for $\angle B$ ($\angle A = 90 - \angle B$)

1: Not a prime number

2: $A = 90 - 2 = 88$ (even)

3: $A = 90 - 3 = 87$ (divisible by 3)

5: $A = 90 - 5 = 85$ (divisible by 5)

7: $A = 90 - 7 = 83$ (prime)

Thus, the answer is **7 (D)**

Problem 2.1.3 — Let p, q , and r be prime numbers such that $2pqr + p + q + r = 2020$. Find $pq + qr + pr$.

Source: 2020 Purple Comet

Solution: In this problem, let's assume that all of our prime numbers (p, q , and r are odd since that's always the case unless the prime number is 2.

We can see that $2pqr$ is always even. Also, if p, q, r are odd, then the sum of $2pqr + p + q + r$ will be odd. However, that's not possible because 2020 is even.

Thus, one of the prime numbers must be 2. Let's assume that p is 2. Then our equation becomes $4qr + q + r = 2018$.

From Simon's Favorite Factoring Trick (this will be covered in the Algebra chapter later), we can factor the expression as

$$(2q + \frac{1}{2})(2r + \frac{1}{2}) = 2018 + \frac{1}{4}$$

We can multiply both sides by 4 to get

$$(4q + 1)(4r + 1) = 8073$$

We prime factorize: $8073 = 3^3 \cdot 13 \cdot 23$

Now we must find a pair of two numbers that multiplies to the prime factorization above. We will equate each of the numbers to $4q + 1$ and $4r + 1$ to see if q and r are prime numbers.

We can test the pairs one at a time, and the one that works is $3^2 \cdot 13$ and $3 \cdot 23$. If we equate $3^2 \cdot 13$ (117) to $4q + 1$ and $3 \cdot 23$ (69) to $4r + 1$, then we get that $q = 29$ and $r = 17$.

Since we know that $p = 2, q = 29, r = 17$, we can plug this into $pq + qr + pr$ to get that the answer is **585**.

§2.2 Multiples, Divisors, Prime Factorization

Definition 2.2.1

The LCM (Least Common Multiple) of a set of numbers is the smallest integer that is a multiple of the entire set. Often, you can find that LCM by prime factorizing all of the numbers of that set.

Theorem 2.2.2

Suppose the prime factorization of a and b are:

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdots p_n^{a_n} \text{ and } b = p_1^{b_1} \cdot p_2^{b_2} \cdots p_n^{b_n},$$

$$\text{then the } \text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

The theorem above is used to find the LCM of two numbers. If you prime factorize any set of numbers, then you take the largest exponent for ALL of the prime numbers. If you don't understand the theorem above, then do not panic since examples will be presented soon.

Definition 2.2.3

The GCD (Greatest Common Denominator) of a set of numbers is the largest integer that can divide the entire set. Often, you can find that GCD by prime factorizing all of the numbers of that set.

Theorem 2.2.4

Suppose the prime factorization of a and b are:

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdots p_n^{a_n} \text{ and } b = p_1^{b_1} \cdot p_2^{b_2} \cdots p_n^{b_n},$$

$$\text{then the } \text{gcd}(a, b) = p_1^{\min(a_1, b_1)} \cdot p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

The image above is the best viewpoint to learn GCD from. If you prime factorize any set of numbers, then you take the SMALLEST exponent for ALL of the prime numbers. For example, if the two numbers that I am finding the GCD for are divisible by 2^6 and 2^8 respectively, then 2^6 will be used to find the GCD since 6 is smaller than 8. A simple example will be presented below.

Example 2.2.5

What is the GCD of 15 and 24?

Solution: Prime factorizing gives that 15 is $3 \cdot 5$ and 24 is $2^3 \cdot 3$.

We will now account for all of the unique prime divisors for both of those numbers even if it only divides one of the numbers. Our unique prime divisors are 2, 3, and 5. Since the lowest exponent of the divisor 2 is 0 (since it doesn't divide 15), our GCD will include 2^0 . Similarly, the lowest exponent is simply 1 for the prime factor 3. For the prime factor 5, even though it divides 15, the exponent of 5 in the prime factorization of 24 is 0 (which is smaller than 1).

Thus, our GCD is $2^0 \cdot 3^1 \cdot 5^0 = 3$.

Key Takeaway: Remember that even if a prime factor only divides one of the numbers from that set, the minimum exponent will still be 0 because it doesn't divide all.

Similar for the LCM, you take the maximum of the exponents for all of the unique prime divisors.

Example 2.2.6

What is the LCM of 18, 22, and 24.

Solution: We will write out the prime factorization of all of the numbers.

$$18 = 2^1 \cdot 3^2$$

$$22 = 2^1 \cdot 11$$

$$24 = 2^3 \cdot 3$$

Since the maximum exponents for 2, 3, and 11 are 3, 2, and 1 respectively, the LCM is $2^3 \cdot 3^2 \cdot 11^1 = 792$

Note: Just remember that the LCM is the maximum of all the exponents of each unique divisor while the GCD is the minimum of all exponents of each unique divisor.

Problem 2.2.7 — The least common multiple of a positive integer n and 18 is 180, and the greatest common divisor of n and 45 is 15. What is the sum of the digits of n ?

(A) 3 (B) 6 (C) 8 (D) 9 (E) 12

Source: 2022 AMC

Solution: Since the LCM involves prime factorizing the set of numbers and taking the largest exponent of all unique prime divisors (the union of all the prime divisors), we will do that for n and 18. We will similarly do that for 45.

$$18 = 2 \cdot 3^2$$

$$180 = 2^2 \cdot 3^2 \cdot 5$$

$$45 = 3^2 \cdot 5$$

$$15 = 3 \cdot 5$$

Using our first 2 prime factorizations, we know that the 2^2 must divide n since 2^2 divides the LCM (180) but it doesn't divide 18 (We want the maximum exponent for the prime factor to divide the LCM). We can't use the prime divisor of 3 to make any conclusions since 3^2 divides both the LCM and 18. This means that the exponent of 3 must be between 0 and 2 for n because the maximum condition for the prime divisor has already been satisfied.

Now we will work with 45 and 15. Clearly 5 must divide n since it divides the GCD of 45 and n which is 15. We also know that 3 must divide n to make the GCD 15 (if n has no factor of 3, then the GCD won't be divisible by 3).

Thus, n is $2^2 \cdot 3 \cdot 5$ which evaluates to 60. Our answer is $6 + 0$ which is **6 (B)**.

Problem 2.2.8 — How many positive integers n are there such that n is a multiple of 5, and the least common multiple of $5!$ and n equals 5 times the greatest common divisor of $10!$ and n ?

(A) 12 (B) 24 (C) 36 (D) 48 (E) 72

Source: 2021 AMC

Solution: The given equation is $\text{lcm}(5!, n) = 5 \cdot \text{gcd}(10!, n)$

We can prime factorize $5!$ to get $2^3 \cdot 3 \cdot 5$.

We can prime factorize $10!$ to get $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

Note: The numbers above were prime factorized using legendre's theorem which you will be learning soon. Don't worry about that step right now.

The LCM involves taking the largest exponent of each unique prime divisor while the gcd is the smallest exponent of each unique prime divisor. We will focus on each unique prime factor.

Starting with 2: Since the LCM involves the largest exponent of any unique prime divisor, we know that the LCM must be divisible by at least 2^3 because of $5!$.

Looking at 2 in the GCD, the exponent of 2 must be less than or equal to 8. Thus, n can take all values from 3 to 8.

For 3, the LCM will have at least an exponent of 1 for 3. The GCD relation tells us that the maximum exponent for 3 is 4. Thus, the prime factor 3 of n can have any exponent from 1 to 4.

For 5, we have to use the given condition that 5 divides n . Let's assume that $n = 5^a \cdot x$. We can now write out an equation for the exponent of 5: $\max(1, a) = 1 + \min(2, a)$. Testing values for a gives that the only one that works is 3 which means that n must be divisible by 5^3 .

For 7, we can write out an equation for the exponents. Let's assume that 7^b divides n . Then, we get $\max(0, b) = \min(1, b)$. Testing values shows that both 0 and 1 work. This means that the possible exponents for the prime factor 7 are 0 and 1.

Since we have 6 possible exponents for 2, 4 for 3, 1 for 5, and 2 for 7, our answer is all of those numbers multiplied ($6 \cdot 4 \cdot 1 \cdot 2$) which gives **48 (D)**.

Problem 2.2.9 — Find the number of 7-tuples of positive integers (a, b, c, d, e, f, g) that satisfy the following system of equations:

$$abc = 70$$

$$cde = 71$$

$$efg = 72.$$

Source: 2019 AIME

Solution: Since c is common in the first and second equation, and e is common in the

second and third equation, we know that c must divide 71 and 70 while e must divide 71 and 72.

Working with c , there clearly is no divisor that divides both 71 and 70 other than 1. Thus, $c = 1$. Similarly, there is no divisor that divides both 71 and 72 other than 1, so $e = 1$.

Our 3 equations now become

$$ab = 70$$

$$d = 71$$

$$fg = 72$$

Since we already know our values c, d, e which are 1, 71, 1 respectively, we just have to work with the equations $ab = 70$ and $fg = 72$ to find the number of possible values for these 4 variables.

Since $70 = 2 \cdot 5 \cdot 7$, there are 8 factors in this number (if you don't understand why there are 8 factors in the number, then don't panic. The theorem for finding that out will be explained later in the book). We can simply pair up the factors that multiply to 70 such as for (a, b) some possible values are $(1, 70)$, $(70, 1)$, $(2, 35)$, and $(35, 2)$. Clearly there are 8 ways to choose our values of a and b .

Doing this for $fg = 72$ first gives that $72 = 2^3 \cdot 3^2$. There are $(3 + 1)(2 + 1) = 12$ factors which means that it is the number of ways to choose values for f and g .

Our total number of ways to choose the variables is $8 \cdot 12$ which is **96**

Problem 2.2.10 — Let a_1, a_2, a_3, a_4, a_5 be positive integers such that a_1, a_2, a_3 and a_3, a_4, a_5 are both geometric sequences and a_1, a_3, a_5 is an arithmetic sequence. If $a_3 = 1575$, find all possible values of $|a_4 a_2|$.

Source: 2017 CMIMC

Solution: In this problem, we will assume that for the geometric sequence a_1, a_2, a_3 , the ratio is r . We work backwards to write the terms a_1 and a_2 in terms of a_3 (a_3 is given as 1575) and r .

$$a_1 = \frac{1575}{r^2}$$

$$a_2 = \frac{1575}{r}$$

Let's assume the ratio is s for the geometric sequence a_3, a_4, a_5 which gives

$$a_4 = 1575s$$

$$a_5 = 1575s^2$$

Since a_1, a_3, a_5 forms an arithmetic sequence, we know that $a_5 - a_3 = a_3 - a_1$ (since the difference between consecutive elements must be the same) which becomes $a_1 + a_5 = 2a_3$. We substitute our values to get

$$\frac{1575}{r^2} + 1575s^2 = 2 \cdot 1575$$

Dividing the equation by 1575 gives

$$\frac{1}{r^2} + s^2 = 2$$

Since we know that both r and r^2 divide 1575 because all the terms are an integer, we can prime factorize 1575 to get $3^2 \cdot 5^2 \cdot 7$.

From here, the possible values of r that makes r and r^2 divide 1575 are 1, 3, 5, and 15. We will test all values of r , and plug them into $\frac{1}{r^2} + s^2 = 2$. Then, we'll find the value of s using that equation and see if $1575s$ and $1575s^2$ become integers.

Testing all our values of r gives that the only values that will work are 1 and 5 which respectively gives 0 and $\frac{7}{5}$ for s .

Now we will compute the value of $a_4 - a_2$, and it is $1575s - \frac{1575}{r}$. We plug in our found values of r, s to give that the possible values are 0 **and** 1890.

Problem 2.2.11 — A positive integer n has 4 positive divisors such that the sum of its divisors is $\sigma(n) = 2112$. Given that the number of positive integers less than and relative prime to n is $\phi(n) = 1932$, find the sum of the proper divisors of n .
Source: 2021 SMT

Solution: Since we know that n has 4 divisors, the prime factorization must be in the form $p \cdot q$ or p^3 .

Case 1: $n = p \cdot q$

The sum of the divisors in this case is $(1 + p)(1 + q) = 2112$

The number of numbers relatively prime to n can be found using Euler's Totient Function, and we get $p \cdot q \cdot \frac{p-1}{p} \cdot \frac{q-1}{q}$

We now simplify it to $(p-1)(q-1) = 1932$

We work with the equation $(1 + p)(1 + q) = 2112$ and prime factorize 2112 as $2^6 \cdot 3 \cdot 11$.

We will now make pairs for $1 + p$ and $1 + q$ that multiply to 2112, and we get one pair to work as 48 and 44. This gives that one prime number will be 47 while the other will be 43.

We test this by plugging it into $(p-1)(q-1) = 1932$, and we see that it works. Thus, the sum of the proper divisors (all divisors except the number itself) is $1 + 43 + 47 = \mathbf{91}$.

You can test out the second case ($n = p^3$) for the problem above. You'll simply find out there's no possible value of n for it.

§2.3 Number of Factors and Sum of all Factors

To find the number of factors, we must prime factorize that number. Then, we add 1 to all of the exponents and multiply all of those numbers with each other.

For example, for the number 18, the prime factorization is $3^2 \cdot 2^1$. Now we simply add 1 to all of the exponents of the prime divisors. The exponents are 2 and 1. When we add 1

to all, we get 3 and 2. We multiply those two numbers to get 6 factors.

We'll use another example of 180. Since 180 is $2^2 \cdot 3^2 \cdot 5^1$, the exponents of all 3 unique prime divisors are 2, 2, and 1. We add 1 to each to get 3, 3, 2. Multiplying the 3 numbers gives us 18 factors.

Theorem 2.3.1

The sum of the divisors is represented as $\sigma_1(n)$. $\sigma_1(n) = (1 + p_1 + p_1^2 + \cdots + p_1^{e_1})(1 + p_2 + p_2^2 + \cdots + p_2^{e_2}) \cdots (1 + p_k + p_k^2 + \cdots + p_k^{e_k})$.

The above formula tells us how to calculate the sum of all factors of any number. We will work with each individual prime divisor at a time. We will add 1 and all powers of that specific prime divisor till we hit the maximum exponent for that divisor that divides n . Then, we just multiply those individual sums.

Example 2.3.2

What is the sum of the factors of 140?

Solution: To find the sum of the factors for any number, we must first prime factorize it.

$$140 = 2^2 \cdot 5^1 \cdot 7^1$$

We compute $(1 + 2^1 + 2^2)(1 + 5^1)(1 + 7^1)$ which results in the answer being 336.

What if you want to find how many powers of a specific prime number divide a factorial ($n!$)? What if you want to prime factorize a factorial number?

Theorem 2.3.3

Legendre's Theorem allows us to find the largest exponent of any prime number that divides a factorial.

$$e_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$$

The formula above works when p is a prime number and $e_p(n!)$ is the largest exponent of p found when you prime factorize $n!$. Read the example below to get a feel for how to use this formula if you're still struggling.

We can find the highest power of a prime number that divides n factorial by just using n and dividing it by p once. We also have to round down each time (floor function). Then, we divide n by p^2 and round down. We continue to increase the exponent of p by 1 until we round down and get 0 which is when we can stop and add up all the values we found.

Example 2.3.4

Find the greatest power of 5 that divides $156!$.

Solution: $e_p(156!) = \sum_{i=1}^{\infty} \left\lfloor \frac{156}{5^i} \right\rfloor$

All we have to do is calculate the summation on the right now. $\left\lfloor \frac{156}{5} \right\rfloor + \left\lfloor \frac{156}{25} \right\rfloor + \left\lfloor \frac{156}{125} \right\rfloor + \left\lfloor \frac{156}{625} \right\rfloor$

Note: Floor function just means we have to round down.

Our sum simply is $31 + 6 + 1 + 0$ which equals to 38. Thus, 5^{38} can divide $156!$

What if we want to find the greatest power of a non-prime number that can divide a factorial?

To find the greatest power of a non-prime number that can divide a factorial, we simply prime factorize that non-prime number. Then, we find the largest exponent for each individual prime factor that can divide the factorial. After that, we will use the minimum exponent out of those. However, there's a catch to this. Non prime numbers can also include 12 which is divisible by more than 1 power of 2. That is why you can't blindly apply this technique. There will be an example that will involve non prime numbers with at least one prime factor that has an exponent larger than 1 in the problems section.

Example 2.3.5

Find the largest exponent of 14 that divides $180!$.

Solution: We first prime factorize 14 to simply get $2 \cdot 7$. Now, we individually find the largest exponent for 2 and 7 separately that can divide $180!$. We will use the smallest one since that's the "limiting factor" (for people that have done chemistry and know limiting reactants should make sense of this with ease).

$$e_2(180!) = \left\lfloor \frac{180}{2} \right\rfloor + \left\lfloor \frac{180}{4} \right\rfloor + \left\lfloor \frac{180}{8} \right\rfloor + \left\lfloor \frac{180}{16} \right\rfloor + \left\lfloor \frac{180}{32} \right\rfloor + \left\lfloor \frac{180}{64} \right\rfloor + \left\lfloor \frac{180}{128} \right\rfloor + \left\lfloor \frac{180}{256} \right\rfloor.$$

This sum equals to $90 + 45 + 22 + 11 + 5 + 2 + 1 + 0 = 176$

$$e_7(180!) = \left\lfloor \frac{180}{7} \right\rfloor + \left\lfloor \frac{180}{49} \right\rfloor + \left\lfloor \frac{180}{343} \right\rfloor.$$

This sum equals to $25 + 3 + 0 = 28$.

Thus, the largest exponent of 2 that can divide $180!$ is 176 while the largest for 7 is 28. The exponent for 2 and 7 must be the same for us since we're trying to find the largest exponent of 14. If we assume that largest exponent is n then $14^n = 2^n \cdot 7^n$. Thus, to maximize our value of n we must choose 28 since any higher value won't divide $180!$ since 7 restricts it to be below 28.

Final Answer: 28

Theorem 2.3.6

We also know that the **product of the divisors** of any integer n is

$$n^{\frac{d(n)}{2}}$$

$d(n)$ represents the number of divisors of the number n .

Problem 2.3.7 — Compute the sum of all positive integers whose positive divisors sum to 186.

Source: 2020 PUMAC Number Theory

Solution: Let's assume that the prime factorization of the number as $p_1^{e_1} \cdot p_2^{e_2} \cdot p_3^{e_3} \dots$

The sum of the divisors are $(1 + p_1 + p_1^2 \dots + p_1^{e_1})(1 + p_2 + p_2^2 \dots + p_2^{e_2})(1 + p_3 + p_3^2 \dots + p_3^{e_3}) \dots$

$$186 = 2 \cdot 3 \cdot 31$$

Now we need to write out pairs that multiply to 186 to find the prime factorization.

2 won't work because $(1 + p_1 + p_1^2 \dots + p_1^{e_1})$ can't equal 2 since it means p_1 will have to be 1 which isn't a prime number.

Thus, the pairs that we need to test are (6, 31) and (3, 62).

$$6 = 1 + 5$$

$$31 = 1 + 2 + 4 + 8 + 16$$

For this combination, the product is $5 \cdot 2^4$ which is 80.

$$3 = 1 + 2$$

$$62 = 1 + 61$$

In this case, our number that's sum of divisors is 186 is $2 \cdot 61$ which is 122.

Thus, we add up $80 + 122$ to get **202** as our final answer.

Problem 2.3.8 — For some positive integer n , the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. How many positive integer divisors does the number $81n^4$ have?

(A) 110 (B) 191 (C) 261 (D) 325 (E) 425

Source: 2016 AMC

Solution: $110n^3$ equals to $2 \cdot 5 \cdot 11 \cdot n^3$.

We know that this number has 110 divisors. We also know that there are a minimum of 3 unique divisors (since we already know that 2, 5, and 11 are divisors).

Let's prime factorize 110 to get $2 \cdot 5 \cdot 11$. We know from our formula that to find the number of divisors that we simply add 1 to each exponent of the unique prime divisors.

Thus, now we work REVERSE and our exponents are 1, 4, and 10 (after subtracting 1 from 2, 5, and 11). Thus, we can assume that n^3 is $5^3 \cdot 11^9$. This assumption stems from the fact that $110n^3$ prime factorized should have exponents of 1, 4, 10 for it's unique prime factors.

We take the cube root for n^3 to get that n equals to $5 \cdot 11^3$. Now we find the number of divisors of $81n^4$. We know this equals to $3^4 \cdot 5^4 \cdot 11^{12}$. Now we just add 1 to each exponent to get 5, 5, and 13. We multiply the 3 numbers to get **325 (D)**.

Problem 2.3.9 — The following number is the product of the divisors of n .

$$2^6 \cdot 3^3$$

What is n ?

Source: 2015 CHMMC

Solution: In this problem, we can apply our formula that the product of the divisors of any number n is $n^{\frac{d(n)}{2}}$ where $d(n)$ represents the number of divisors. We can equate this to our value of $2^6 \cdot 3^3$.

$$n^{\frac{d(n)}{2}} = 2^6 \cdot 3^3$$

Squaring both sides gives

$$n^{d(n)} = 2^{12} \cdot 3^6$$

Clearly, the only prime factors of n must be 2 and 3 (if it's something else, then it won't equate to the right side as it only has prime factors 2 and 3).

Let's assume $n = 2^a \cdot 3^b$. $d(n)$ for this is simply $(a+1)(b+1)$

$$(2^a \cdot 3^b)^{(a+1)(b+1)} = 2^{12} \cdot 3^6$$

We can multiply the exponent of 2 which is currently a to $(a+1)(b+1)$ and similarly do the same for the exponent b of 3.

$$\text{This gives } 2^{a(a+1)(b+1)} = 2^{12}$$

$$\text{This gives } 3^{b(a+1)(b+1)} = 3^6$$

We can equate the exponents for both equations we wrote above to get

$$a(a+1)(b+1) = 12$$

$$b(a+1)(b+1) = 6$$

Dividing both of the equations above gives $\frac{a}{b} = 2$ which means that $a = 2b$

We can plug this into $b(a+1)(b+1) = 6$ to get that $b(2b+1)(b+1) = 6$ and b obviously has to be 1. This means that a is 2.

Since we know our values of a and b , we know that n is $2^2 \cdot 3^1$ which is **12**

Problem 2.3.10 — How many positive integer divisors of 2004^{2004} are divisible by exactly 2004 positive integers?

Source: 2004 AIME

Solution: We first write out the prime factorization of both 2004^{2004} and 2004. Both respectively are $2^{4008} \cdot 3^{2004} \cdot 167^{2004}$ and $2^2 \cdot 3 \cdot 167$.

Out of all of the divisors of 2004^{2004} , we can have a maximum of 3 unique prime factors (2, 3, 167) as seen in our prime factorization.

Let's assume our divisor of 2004^{2004} is $2^x \cdot 3^y \cdot 167^z$. Then, the number of divisors that it has is $(x+1)(y+1)(z+1)$. We equate this to 2004.

$$(x+1)(y+1)(z+1) = 2^2 \cdot 3 \cdot 167$$

Now we can distribute each prime factor to the numbers $x+1$, $y+1$, and $z+1$.

We can have two cases: one in which both factors of 2 divide one of the numbers and for the other 2 will divide two separate numbers.

If all factors of 2 divide only one number, then there are 3 cases for that (either $x+1$, $y+1$, or $z+1$). If one factor of 2 divides 2 of the numbers, then there are $\binom{3}{2}$ cases for that which is 3. The total possible number of cases are $3+3$ which is 6 (for the prime factor of 2

Now we split the factor of 3. There are simply 3 places for it to go: either $x+1$, $y+1$, or $z+1$. The same thing applies to 167.

Our answer is $6 \cdot 3 \cdot 3$ which is **54**.

Problem 2.3.11 — Let $n = 2^{31}3^{19}$. How many positive integer divisors of n^2 are less than n but do not divide n ?

Solution: 1995 AIME

Solution: n^2 is simply $2^{62} \cdot 3^{38}$.

We notice that each factor pair of n^2 will have one factor less than n and one more than n . To see this, you can observe some examples and experiment with a few factor pairs that multiply to n^2 .

However, there will be one divisor that won't have a unique pair which is n since it is paired to itself to get n^2 .

Thus, to find the number of divisors that are less than n , we first find the number of factors in n^2 : $(62+1)(38+1)$. Then, we subtract by 1 to get rid of n from the factors. After that, we divide by 2 (since only half will be less than n) to get 1228. However, some of those factors divide n (which is something we don't want due to the given conditions). Thus, we now need to subtract the number of factors that are less than n and divide n .

Clearly, all factors of n are less than n except for n itself. Thus, we find the number of divisors of n and subtract that by 1. Then, we'll subtract that value from 1228. The number of factors in n is $(31+1)(19+1)$ which is 640. Subtracting 1 from 640 gives us 639. Then, we subtract that from 1228 to get $1228-639 = \mathbf{589}$

§2.4 Factorials and Palindromes

Factorials and palindromes both show up often on the AMCs and AIME. It's important to know problem solving strategies for these types of problems. Often, factorial problems involve Legendre's theorem that we saw in this book already. The best way to solve factorial/palindrome problems is to simply practice as there isn't much theory to go with it.

Definition 2.4.1

A palindrome is a number that reads the same if you read it left to right or from right to left. For example, 121, 1441, 3883, 39493 are all palindromes as they are the same numbers whether you read it from left to right or right to left.

Often, to solve problems related to palindromes, you can write the palindrome out in its expanded form. If you're not given the palindrome, then you can use variables to represent the digits. Then, you can pair up the digits that are the same to write an expression.

For example, if you want to represent a palindrome with 3 digits, you can write it as aba where a and b are integers. aba is equivalent to $a \cdot 100 + b \cdot 10 + a \cdot 1$ which is equivalent to $101a + 10b$.

Problem 2.4.2 — A palindrome between 1000 and 10,000 is chosen at random. What is the probability that it is divisible by 7?

- (A) $\frac{1}{10}$ (B) $\frac{1}{9}$ (C) $\frac{1}{7}$ (D) $\frac{1}{6}$ (E) $\frac{1}{5}$

Source: 2010 AMC

Solution: Our palindrome clearly has 4 digits. That means we can represent it as $abba$. We can rewrite this as $a \cdot 10^3 + b \cdot 10^2 + b \cdot 10^1 + a \cdot 10^0$. This is equivalent to $1001a + 110b$.

From here, we will now investigate with the numbers a and b to find the probability that the number is divisible by 7. We notice that $1001a$ is always divisible by 7 because 1001 is. Thus, we only need to work with $110b$ which is equivalent to $5b$ modulo 7. $5b$ will be divisible by 7 when b is 0 or 7. We have 10 possible digits for b from 0 to 9. There are 2 possibilities that work which means the probability is $\frac{2}{10} = \frac{1}{5}$ (E).

Problem 2.4.3 — Many states use a sequence of three letters followed by a sequence of three digits as their standard license-plate pattern. Given that each three-letter three-digit arrangement is equally likely, the probability that such a license plate will contain at least one palindrome (a three-letter arrangement or a three-digit arrangement that reads the same left-to-right as it does right-to-left) is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Source: 2002 AIME

Solution: We can separately solve for the chance of having palindromic digits or letters. For a combination to be palindromic for digits, we have 10^2 palindromes. We know this is true because a 3 digit palindrome is equivalent to aba where a and b are digits. In this case, whatever we choose for the first digit forces the third digit to be that since it's a palindrome. The number of possible numbers is simply 10^3 . Thus, the

probability is $\frac{10^2}{10^3}$ which reduces to $\frac{1}{10}$.

When we solve for the palindromic letters, then there are 26^2 palindromes and 26^3 possible strings. The probability for this case is $\frac{26^2}{26^3}$ which equals to $\frac{1}{26}$.

Before we simply add both of these probabilities, by the principle of inclusion exclusion theorem (PIE), we must subtract the chance of both occurring at once. For those that don't know this theorem just remember that it will be covered eventually so don't panic.

The chance for both occurring at once is $\frac{1}{26} \cdot \frac{1}{19}$ which is $\frac{1}{260}$. Now we compute: $\frac{1}{26} + \frac{1}{10} - \frac{1}{260}$ to get $\frac{7}{52}$. We add the numerator and denominator to get $7 + 52 = \mathbf{59}$.

Problem 2.4.4 — A palindrome is a nonnegative integer number that reads the same forwards and backwards when written in base 10 with no leading zeros. A 6-digit palindrome n is chosen uniformly at random. What is the probability that $\frac{n}{11}$ is also a palindrome?

(A) $\frac{8}{25}$ (B) $\frac{33}{100}$ (C) $\frac{7}{20}$ (D) $\frac{9}{25}$ (E) $\frac{11}{30}$

Source: 2013 AMC

Solution: To solve this problem, it's best to work backwards. We will only get a 6 digit palindrome if we multiply a 4 or 5 digit palindrome by 11.

Case 1: We get a 4 digit palindrome after dividing the 6-digit palindrome by 11. This number will be in the form of ABBA. A and B represent integers; B can be anything from 0 to 9 while A can be anything from 1 to 9. When we multiply this by 11, we get the result shown below.

$$\begin{array}{r} \text{ABBA} \\ \times 11 \\ \hline \text{ABBA} \\ + \text{ABBA} \\ \hline \end{array}$$

The digits from left to right are $A, A + B, 2B, A + B, A$ if there are no carryovers.

However, we want to get a 6 digit palindrome when we multiply by 11. Thus, that is only possible when A is 9 and we carry a 1 from summing $A + B$. This means the first digit will be 1, and the last digit which is A will simply be 9. However, since the first and last digit are not equal, this case is not possible (a palindrome's first and last digits must be equal).

Case 2 (We get a 5 digit palindrome after dividing the 6 digit palindrome by 11)
 A 5 digit palindrome will be in the form ABCBA. Multiply this by 11 to get what is shown below

$$\begin{array}{r}
 \text{ABCBA} \\
 \times 11 \\
 \hline
 \text{ABCBA} \\
 + \text{ABCBA} \\
 \hline
 \end{array}$$

If there are no carryovers, then the digits from right to left are $A, A + B, B + C, B + C, A + B, A$. We can see immediately that this is a 6 digit number. The most important thing to note is that we don't want any carryovers or else it won't be a palindrome anymore. You can plug in some numbers to notice this.

We will do some casework now. The digit A must be at least 1.

Value of A	Value of B	Range of Values for C	Total Number of Cases
1	0	0-9	10
	1	0-8	9
	2	0-7	8
	3	0-6	7
	4	0-5	6
	5	0-4	5
	6	0-3	4
	7	0-2	3
	8	0-1	2

The image above shows an example of casework for the digit A being 1. The number of possible cases for it is simply the sum of all integers from 2 to 10 inclusive which is 54.

We can continue this pattern all the way to $A = 9$ and see that the amount of possibilities are $54 + 52 + 49 + 45 + 40 + 34 + 27 + 19 + 10 = 330$. The total number of cases are simply the amount of 5 digit palindromes representing ABCBA. There are 9 choices for digit A (1 – 9) and 10 for each of B and C (0 – 9). The total is $9 \cdot 10 \cdot 10$ which is 900. The final answer is $\frac{330}{900}$ which reduces to $\frac{11}{30}$.

Problem 2.4.5 — Find the probability that a randomly selected divisor of $20!$ is a multiple of 2000.

Source: 2020 SMT

Solution: We will use Legendre's theorem to first prime factorize $20!$. We do this for the prime numbers that are less than 20 which are 2, 3, 5, 7, 11, 13, 17, and 19.

$$20! = 2^{18} \cdot 3^8 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19$$

Now we also prime factorize 2000 which gives $2^4 \cdot 5^3$

Any multiple of 2000 must have an exponent of at least 4 for 2 and at least 3 for 5.

We must find the number of divisors in $20!$ that satisfy this. This means that the exponent of 2 can be anything from 4 to 18 (15 choices out of 19), and the exponent of 5 can be anything from 3 to 4 (2 choices out of 5).

We can write this probability as $\frac{15}{19} \cdot \frac{2}{5}$ to get $\frac{6}{19}$

Problem 2.4.6 — Given that

$$\frac{((3!)!)!}{3!} = k \cdot n!$$

where k and n are positive integers and n is as large as possible, find $k + n$.

Source: 2003 AIME

Solution: For this problem, we first compute the left-hand side. $3!$ is 6 and $6!$ is 720 and now we're left with $\frac{720!}{3!}$.

Since we want to maximize n , we need to find the largest factorial that can divide the expression $\frac{720!}{3!}$. We can simply reduce $\frac{720!}{3!}$ to $\frac{720 \times 719!}{3!}$ which gives us $120 \cdot 719!$. This is the format of the answer we want, and we can't have a higher number than 719 for n . Thus, the answer is $719 + 120$ which is **839**.

Problem 2.4.7 — How many positive integers less than or equal to 240 can be expressed as a sum of distinct factorials? Consider $0!$ and $1!$ to be distinct.

Source: 2020 HMMT

Solution: The factorials that are less than 240 are $0!$, $1!$, $2!$, $3!$, $4!$, and $5!$. If we try to add up $0!$, $1!$ or $2!$, then we'll notice that some of the sums repeat such as $0! + 1!$ being equivalent to $2!$. However, no such problem occurs with $3!$, $4!$ or $5!$.

Thus, any of the 3 factorials can be present or not in our sum which means there are 2 cases for each (present or not). $2 \cdot 2 \cdot 2$ is simply 8.

Now we will deal with $0!$, $1!$, and $2!$. The 3 numbers are equivalent to 1, 1, and 2. We can either exclude all which gives us 0 or add combinations of them to get all numbers from 1 to 4 inclusive. Thus, there are 5 possible values. We multiply this to 8 to get 40.

However, we must subtract the case in which $3!$, $4!$, and $5!$ are not present since it

allows us to get the number 0 which isn't positive. Thus, the answer is $40 - 1$ which is **39**.

Problem 2.4.8 — Let N be the number of consecutive 0's at the right end of the decimal representation of the product $1!2!3!4!\cdots 99!100!$. Find the remainder when N is divided by 1000.

Source: 2006 AIME

Solution: To find the number of consecutive 0s, we need to know how many times 10 divides the number. From Legendre's theorem, we remember that we can simply split an integer we want to find the highest power for into its prime factors. In this case, we split 10 to 2 and 5. Clearly there will be much more 2s than 5s. Thus, we just solve for the number of 5s. We will go through and divide everything by 5 and round down.

$$e_5(n!) = \sum_{n=1}^{100} \left\lfloor \frac{n}{5} \right\rfloor$$

We can solve the above expression by noticing a pattern. The first 4 numbers from 1 to 4 simply give us 0. However, when $n = 5$ to 9 we get 1. When n is a value between 10 and 14 inclusive, we get 2 and so on.

There are 5 repeats for the same values from 1 through 19. For the value $n=100$, we get 20 in the summation. Thus, we simply sum 1 through 19 using the sum of the first n terms to get $\frac{19 \cdot 20}{2}$ which is 190. Multiplying by 5 gives 950. Now, we add the 20 from $n = 100$ to get **970**.

However, there are more factors of 5. Whenever 25 divides any of the factorials, we get another factor 5. Thus, we will repeat this process but with 25 instead of 5.

$$e_{25}(n!) = \sum_{n=1}^{100} \left\lfloor \frac{n}{25} \right\rfloor$$

We can find a pattern again. Whenever n is between 0 and 24 inclusive, we get 0. When n is between 25 and 49, we get 1 factor of 25 for each. When n is between 50 and 74 we get 2 factor of 25 for each. When n is between 75 and 99 we get 3 factor of 25 for each. When n is 100, we get 4 factors of 25. We compute this as $(24 \cdot 0) + (25 \cdot 1) + (25 \cdot 2) + (25 \cdot 3) + (1 \cdot 4)$ which is 154. We add 970 and 154 to get 1124 0s in this number. Dividing this by 1000 gives us **124** as the final answer.

§2.5 Numbers in Different Bases

Problem 2.5.1 — Let n be a positive integer and d be a digit such that the value of the numeral $\underline{32d}$ in base n equals 263, and the value of the numeral $\underline{324}$ in base n equals the value of the numeral $\underline{11d1}$ in base six. What is $n + d$?

- (A) 10 (B) 11 (C) 13 (D) 15 (E) 16

Source: 2021 AMC

Solution: To solve this problem, we can rewrite the numbers in different bases as an algebraic expression. We convert it to base 10. $\underline{32d}_n$ in base 10 is $3n^2 + 2n + d$ and this

equals to 263. We now convert 324_n to base 10 and equate it to $11d1_6$ in base 10 which is $253 + 6d$. Our two equations are listed below.

1. $3n^2 + 2n + d = 263$
2. $3n^2 + 2n + 4 = 253 + 6d$

From here, we can rewrite the first equation as $3n^2 + 2n = 263 - d$. Then, we can substitute that into the second one to get

$$253 + 6d + 4 = 263 - d$$

We solve for d to get that $d = 2$. Then, we can plug that into the first equation and solve the quadratic to get $n = 9$. Thus, the answer is simply $2 + 9$ which is **11 (B)**.

Our next problem again shows an example of how many of these problems are simply about converting the numbers in different bases to base 10.

Problem 2.5.2 — The number N_b is the number such that when written in base b , it is 123. What is the smallest b such that N_b is a cube of a positive integer?

Source: 2019 SMT

Solution: In this problem, we will convert 123_b to base 10.

We get $1 \cdot b^2 + 2 \cdot b + 3 = b^2 + 2b + 3$.

The base must be at least 4 because the largest digit in that number is 3.

We plug in 4 for b to test it which gives 27. Since 27 is the cube of 3, our base b is 4.

Problem 2.5.3 — A rational number written in base eight is $\underline{ab}.\underline{cd}$, where all digits are nonzero. The same number in base twelve is $\underline{bb}.\underline{ba}$. Find the base-ten number $\underline{abc}$.

Source: 2017 AIME

Solution: We notice that for the two numbers to be equal, the integer parts must be equal to each other while the fractional parts for both numbers are also equal to each other. ab_8 must be equal to bb_{12} . Similarly, the fractional parts in the different bases must also equate to each other. Working with the first and second digit to the left of the decimal point, we get

$$8^1 \cdot a + 8^0 \cdot b = 12^1 \cdot b + 12^0 \cdot b$$

This is equivalent to $8a + b = 12b + b$ which simplifies to $2a = 3b$. From here, we will do some casework to find the possible values of a and b . a and b both must be strictly less than 8 because all the digits of a number in base b must be less than b (ab is a number in base 8 given in the problem statement). Thus, the possible sets for (a, b) are $(3, 2)$ and $(6, 4)$. We will work with both cases.

Case 1: $(a, b) = (3, 2)$

We just need to check if the fractional parts equate since we already found the solutions that make the non fractional parts equate. The fractional part for $\underline{ab}.\underline{cd}$ from base 8 to 10 is $\frac{c}{8} + \frac{d}{64}$.

We equate that to the fractional part of base 12 converted to get the final equation of

$\frac{c}{8} + \frac{d}{64} = \frac{b}{12} + \frac{a}{144}$. We now plug in 2 for b and 3 for a . Simplifying gives that $8c + d = 12$.

From here the only possible solution set for $(c, d) = (1, 4)$. Thus, our value of abc is 321.

Although clearly there is only one possible answer for this problem which we found as 321, let's see if there is something possible for the other solution set.

Case 2: $(a, b) = (6, 4)$

We just need to check the fractional parts. We can convert both fractional parts to get the equation

$\frac{c}{8} + \frac{d}{64} = \frac{b}{12} + \frac{a}{144}$. Plugging in 6 for a and 4 for b and simplifying gives us $c + 8d = 24$. However, there are no solutions for this since c and d must be nonzero. This means d can either be 1 or 2. The value of c will be 16 and 8 respectively for both cases, and that isn't possible since c is part of a base 8 number which means the digits must be less than or equal to 7.

Thus, our final answer is **321**.

Problem 2.5.4 — Hexadecimal (base-16) numbers are written using numeric digits 0 through 9 as well as the letters A through F to represent 10 through 15. Among the first 1000 positive integers, there are n whose hexadecimal representation contains only numeric digits. What is the sum of the digits of n ?

(A) 17 (B) 18 (C) 19 (D) 20 (E) 21

Source: 2015 AMC

Solution: We will first convert 1000 to base 16. We get $3 \cdot 16^2 + 14 \cdot 16^1 + 8$.

14 in base 16 is equivalent to E. Since we want to find the number of integers who only have numeric digits in hexadecimal base (meaning no 10 through 15 for the digit), we have 4 possible values for the first digit from 0 – 3. Similarly, there are 10 for the remaining 2 digits since they can take anything from 0 through 9 and still be less than 1000. Thus, the number of possibilities are $4 \cdot 10 \cdot 10$ which is 400. However, we only want positive integers so we need to subtract one case when all the digits are 0. Thus, the final answer is **399** and the sum of the digits are **21 (E)**.

Note: If you're asked to add/subtract/divide/multiply two numbers in different bases, you can simply convert all the numbers to base 10.

Problem 2.5.5 — A positive integer N has base-eleven representation $\underline{abc}$ and base-eight representation $\underline{1bca}$, where a, b , and c represent (not necessarily distinct) digits. Find the least such N expressed in base ten.

Source: 2020 AIME

Solution: We can convert the two integers that are in base 11 and base 8 to base 10. Then, we can equate both of them since they both equal to N .

abc from base 11 to 10 is simply $11^2 \cdot a + 11 \cdot b + c = 121a + 11b + c$

$1bca$ from base 8 to 10 is $8^3 \cdot 1 + 8^2 \cdot b + 8 \cdot c + a = 512 + 64b + 8c + a$.

Now we know that $121a + 11b + c = 512 + 64b + 8c + a$

Rearranging the terms gives $120a = 512 + 53b + 7c$

Now we can use the idea that for any number in base b , all of its digits must be between 0 and $b - 1$ inclusive. Since a, b, c are part of two numbers that are in base 8 and 11, the 3 variables must be between 0 and 7 inclusive.

Since we want the left side (which is $120a$) to equate to the right side, the value of a cannot be 4 (because 512 is present on the right side). It must be at least 5. Let's assume that it is 5. After that, we get $600 = 512 + 53b + 7c$ which simplifies to $88 = 53b + 7c$

From here, we notice that if $b = 1$, then $c = 5$. Since $a = 4, b = 1, c = 5$ and all of those digits are ≤ 7 , we have found our solution set.

We can plug in these numbers into either the base 8 or base 11 number to find the value of N to be **621**.

Problem 2.5.6 — Find the sum (in base 10) of the three greatest numbers less than 1000 in base 10 that are palindromes in both base 10 and base 5.

Source: 2020 PUMAC

Solution: In this problem, we will first convert 1000 to base 5 to get 13000_5 .

We know that the maximum number of digits in base 5 for our numbers must be 5. We can do casework on the number of digits in base 5.

Case 1: Our base 5 number has 5 digits

It's obvious that the first digit must be 1 or else it will be more than 13000_5

Our number is $1aba1 = 626 + 130a + 25b$

Digit a must be 0, 1, or 2 because if it is 3, then it will be greater than 13000_5 .

Subcase 1.1: a is 2. This gives $626 + 260 + 25b$ which is $886 + 25b$.

There will be no numbers that will be a palindrome in base 10 in this case.

Subcase 1.2: a is 1. This gives $626 + 130 + 25b$ which is $756 + 25b$, and there will be nothing that works in this case.

Subcase 1.3 : a is 0. This gives $626 + 25b$. In this case however, b can be 1 or 0, and we'll get 626 and 656 as our possible values.

Case 2: Our base 5 number has 4 digits

Our number is $abba$. Expanding it in base 5 gives $5^3 \cdot a + 5^2 \cdot b + 5^1 \cdot b + 5^0 \cdot a$ which gives $126a + 30b$.

Since we know that a and b can't be greater than 4, let's assume a is 4 to maximize that number. However, we realize we form no palindromes in that case. Similarly for $a = 3$, there will be no possible palindromes. However, for $a = 2$, we get $252 + 30b$. 252 is a palindrome, but if we substitute 1 for b , then we get a larger palindrome which is 282.

Our 3 palindromes are $282 + 626 + 656$ which is **1584**.

§2.6 Working with unit digits

Many AMC and AIME problems have problems in which you have to observe the unit digits. Often, the unit digit repeats in cycles. The only way to learn how to do these problems is to practice. There isn't much theory to it.

Problem 2.6.1 — What is the units digit of 13^{2003} ?

(A) 1 (B) 3 (C) 7 (D) 8 (E) 9

Source: 2003 AMC 10

Solution: For this problem, we only need to find a pattern to see what the unit digit will be when the exponent of 13 is 2003.

Units Digits of a Few Powers of 13

$$13^1 = 3$$

$$13^2 = 9$$

$$13^3 = 7$$

$$13^4 = 1$$

$$13^5 = 3$$

$$13^6 = 9$$

$$13^7 = 7$$

$$13^8 = 1$$

Clearly, the units digit cycles with a period of 4. It repeats the pattern of 3, 9, 7, 1. We notice that whenever the exponent's remainder divided by 4 is 1, 2, 3, 0, the units digit respectively is 3, 9, 7, 1. Since 2003 has a remainder of 3 when divided by 4, that ultimately means the unit digit is **7 (C)**.

Problem 2.6.2 — For a positive integer n , let d_n be the units digit of $1 + 2 + \cdots + n$.

Find the remainder when

$$\sum_{n=1}^{2017} d_n$$

is divided by 1000.

Source: 2017 AIME

Solution: This is another problem in which we need to find a pattern. We are supposed to find the units digit for 2017 summations. We will try to find a pattern by finding the units digit of a values of d_n .

$$d_n = \sum_{i=1}^n i$$

$$d_1 = \sum_{i=1}^1 i = 1$$

$$d_2 = \sum_{i=1}^2 i = 3$$

$$d_3 = \sum_{i=1}^3 i = 6$$

$$d_4 = \sum_{i=1}^4 i = 0$$

$$d_5 = \sum_{i=1}^5 i = 5$$

$$d_6 = \sum_{i=1}^6 i = 1$$

$$d_7 = \sum_{i=1}^7 i = 8$$

$$d_8 = \sum_{i=1}^8 i = 6$$

If we continue this pattern then we get a pattern with a period of 20. The digits alternate in the pattern

$$1, 3, 6, 0, 5, 1, 8, 6, 5, 5, 6, 8, 1, 5, 0, 6, 3, 1, 0, 0,$$

.

This cycle sums to 70. We find how many times this cycle fully repeats itself.

$$\left\lfloor \frac{2017}{20} \right\rfloor = 100.$$

However, in the last 17 times there won't be a full cycle so we will individually add those terms to find the total sum. $100 \cdot 70 = 7000 + 69$. The total sum is 7069, and after dividing this by 1000, we get **69**.

§2.7 Modular Arithmetic Introduction

People sometimes get confused by the word mod, but it's time to clear it up. When you say 8 is divisible by 4, you can simply say $8 \equiv 0 \pmod{4}$. All this means is that when 8 is divisible by 4, the remainder is 0. For example, 4, 8, 12, 16, 20, 24 are all 0 in mod 4. This means that when you divide all of those numbers by 4, you always get a remainder of 0.

$\equiv$ is the sign used in modular arithmetic. It just means that two numbers are congruent (meaning the remainder is the same when divided by a specific number).

Modular Arithmetic Properties

Theorem 4.6. If a, b, c, d , and m are integers such that $m > 0$, $a \equiv b \pmod{m}$, and $c \equiv d \pmod{m}$, then

- (i) $a + c \equiv b + d \pmod{m}$,
- (ii) $a - c \equiv b - d \pmod{m}$,
- (iii) $ac \equiv bd \pmod{m}$.

Now we will solve some problems before moving onto more complex topics like Fermat's Little Theorem and Euler's Totient Theorem.

Problem 2.7.1 — The base-nine representation of the number N is $27,006,000,052_{\text{nine}}$. What is the remainder when N is divided by 5?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Source: 2021 AMC 12

Solution: We need to solve this problem by converting this base 9 number to base 10 using the methods discussed before. We will rewrite this number below.

$$2 \cdot 9^{10} + 7 \cdot 9^9 + 6 \cdot 9^6 + 5 \cdot 9^1 + 2 \cdot 9^0$$

Now, we will take this number (mod) 5 meaning we want to find the remainder when we divide it by 5. Since $9 \bmod 5$ is simply -1 , we can easily find the value of our expression in (mod) 5. Plugging in -1 in the place of 9 and solving it gives us $2 - 7 + 6 - 5 + 2$ which evaluates to -2 . Since -2 isn't directly a remainder, we simply add 5 to this to get **3(D)**.

Problem 2.7.2 — Find the remainder when $9 \times 99 \times 999 \times \cdots \times \underbrace{99 \cdots 9}_{999 \text{ 9's}}$ is divided by 1000.

Source: 2010 AIME

Solution: In this problem, we will try to reduce the numbers mod 1000. We will leave 9 and 99 the way they are. However, we notice that whenever we divide 999, or 9999, or any more numbers like that, then the remainder mod 1000 is simply -1 .

We are simply left to evaluate $9 \cdot 99 \cdot -1^{997}$. This evaluates -891 . We simply add 1000 to this to get **109**.

Problem 2.7.3 — Positive integers a , b , and c are randomly and independently selected with replacement from the set $\{1, 2, 3, \dots, 2010\}$. What is the probability that $abc + ab + a$ is divisible by 3?

- (A) $\frac{1}{3}$ (B) $\frac{29}{81}$ (C) $\frac{31}{81}$ (D) $\frac{11}{27}$ (E) $\frac{13}{27}$

Source: 2010 AMC

Solution: $abc + ab + a$ is equivalent to $a(bc + b + 1)$. Either a can be divisible by 3 or $bc + b + 1$ or both.

If a is divisible by 3, then b and c can be anything. There is a $\frac{1}{3}$ chance of this happening (since $\frac{1}{3}$ is the ratio of the amount of numbers divisible by 3 in the set from 1 to 2010).

Now, we will take the case when a isn't divisible by 3. In this case, b and c can't take up all the values so we have to consider when $bc + b + 1$ is divisible by 3. Instead of checking each big number, we will simply just check 3 digits for each which are 0, 1, 2 as those are the only possible remainders modulo 3.

If b is $0 \bmod 3$, then we can plug that into $bc + b + 1$ to see what the remainder

will be. $0 \cdot c + 0 + 1$ is always $1 \pmod 3$. Thus, that isn't a possible case.

Now we will check when b is $1 \pmod 3$. We plug that into $bc + b + 1$ to get $c + 2$. This will be divisible by 3 when c is $1 \pmod 3$. Thus, our expression is divisible by 3 when b is $1 \pmod 3$ and c is $1 \pmod 3$.

Now we will check when b is $2 \pmod 3$. We plug in 2 in replacement of b in our expression to get $2c + 3$ which reduces to $2c \pmod 3$. This is only divisible by 3 when c is $0 \pmod 3$.

Since a isn't divisible by 3 in the subcase, there is a $\frac{2}{3}$ chance of that happening. For each subcase that works, there is a $\frac{1}{3}$ chance of getting the condition for b and the same probability applies for c . Multiply the probabilities for satisfying the conditions for a , b , and c gives $\frac{2}{27}$. We need to multiply this number by 2 since 2 subcases work. We now get $\frac{4}{27}$.

Now we add up $\frac{1}{3}$ and $\frac{4}{27}$ to get $\frac{13}{27}$ (E).

§2.8 Fermat's Little Theorem and Euler's Totient Function

This is one of the most important and most common topic that shows up on the AMC and AIME.

The simple example below will be teaching you **Euler's Totient Function**.

Example 2.8.1

How many numbers are relatively prime to 48?

Solution: Relatively prime means that the GCD of 48 and the numbers less than that are just 1. This can be done by using this function. If you don't know this function, you're probably gonna count by listing out 1, 5, 7, and so on. This will take a long time. However, this function just involves us to find the prime factorization of 48. The prime factorization of 48.

We know that the prime factorization is just $2^4 \cdot 3$. This theorem states that we take the reciprocal of each of the prime numbers that divide a certain number. After that, we subtract that reciprocal from 1.

If we take those exact steps, we'll get $1 - \frac{1}{2} = \frac{1}{2}$ and $1 - \frac{1}{3} = \frac{2}{3}$. Now we have two fractions that are $\frac{1}{2}$ and $\frac{2}{3}$. We find the number of relatively prime numbers to 48 by multiplying those 2 fractions to 48. After we multiply 48 with $\frac{1}{2}$ and $\frac{2}{3}$, we get 16 which is indeed our answer.

There is a picture of the theorem below, but it's fine if you don't understand the symbols since we made this hand typed explanation for that reason. The symbol for representing the value you get for a number by using the euler's totient function is ϕ . For example, ϕ of 48 is 16.

I have attached the formula with the original terminology below, but even if you don't understand it, don't worry because we just explained it to you.

$$\begin{aligned}
\varphi(n) &= \varphi(p_1^{k_1}) \varphi(p_2^{k_2}) \cdots \varphi(p_r^{k_r}) \\
&= p_1^{k_1-1} (p_1 - 1) p_2^{k_2-1} (p_2 - 1) \cdots p_r^{k_r-1} (p_r - 1) \\
&= p_1^{k_1} \left(1 - \frac{1}{p_1}\right) p_2^{k_2} \left(1 - \frac{1}{p_2}\right) \cdots p_r^{k_r} \left(1 - \frac{1}{p_r}\right) \\
&= p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r} \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_r}\right) \\
&= n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_r}\right).
\end{aligned}$$

We can use the Euler's Totient Function in key modular arithmetic problems.

Theorem 2.8.2

Euler's Totient Function: If a is an integer that is relatively prime to n , then $a^{\phi(n)} \equiv 1 \pmod{n}$ is true

The Fermat's Little Theorem is similar and is widely used in number theory problems.

Theorem 2.8.3

The Fermat's Little Theorem: If there is an integer a , and p is a prime number, and the number a isn't divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$.

The best way to solidify your understanding with both of these theorems is to practice a few problems.

Problem 2.8.4 — Find the largest positive integer n such that $n\phi(n)$ is a perfect square. ($\phi(n)$ is the number of integers k , $1 \leq k \leq n$ that are relatively prime to n)

Source: 2010 PUMAC

Solution: In this problem, we will rewrite the expression $n\phi(n)$. We will rewrite n as a product of its prime factors. Pretend there's a prime factor p_1 with an exponent of e_1 . $\phi(n)$ for this prime factor will simply be $p_1^{e_1} \cdot (1 - \frac{1}{p_1})$.

The expression multiplied above gives $(p_1^{e_1-1} \cdot (p_1 - 1))$. We multiply this to the value $p_1^{e_1}$ to get $p_1^{2e_1-1} \cdot (p_1 - 1)$.

$\prod_{i=1}^x p_i^{2e_i-1} \cdot (p_i - 1)$ is the value of the expression $n\phi(n)$. $p_1, p_2, p_3, \dots, p_x$ represent the x prime factors of n .

From the expression above, we notice that our largest prime factor will always have an odd exponent. The reason is that $2e_1 - 1$ is always going to be an odd number. Thus, that means our expression can never be a perfect square.

However, it's obvious that $n = 1$ works as an answer.

Problem 2.8.5 — An integer N is selected at random in the range $1 \leq N \leq 2020$. What is the probability that the remainder when N^{16} is divided by 5 is 1?

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$ (E) 1

Source: 2017 AMC 10

Solution: Using the Fermat's Little Theorem, we know that $N^4 \pmod{5}$ is always 1 whenever N is relatively prime to 5 (meaning $N \pmod{5}$ cannot be 0). If we raise N^4 to the fourth power then we simply get $N^{16} \equiv 1 \pmod{5}$. This is what we want, and it always works whenever N isn't $0 \pmod{5}$. That means that N must be 1, 2, 3, or 4 mod 5 giving a probability of $\frac{4}{5}$.

§2.9 Chinese Remainder Theorem

The Chinese Remainder Theorem is useful for solving multiple linear congruence equations and combining the divisors.

Chinese Remainder Theorem

- The solution to the following equations:

$$x = a_1 \pmod{n_1}$$

$$x = a_2 \pmod{n_2}$$

$$x = a_k \pmod{n_k}$$

where $n_1, n_2, \dots, n_k$ are relatively prime is found as follows:

$$N = n_1 n_2 \dots n_k$$

$$N_i = N/n_i$$

Find s_i such that $r_i n_i + s_i N_i = 1$

Let $e_i = s_i N_i$, then

$$x = \sum_i^k a_i e_i \pmod{N}$$

The picture above can be very confusing to understand. It's just full of variables, but it's time to clear it up. The chinese remainder theorem solves a set of linear congruences. The example below will clear it up.

$$x \equiv 1 \pmod{5}$$

$$x \equiv 2 \pmod{6}$$

$$x \equiv 3 \pmod{7}$$

First, make a chart. If you have 3 linear congruences, then make a 3 by 3 table. You need to have 3 columns always, but the amount of rows is the same thing as the amount of expressions you have. Designate the first row to the first expression, the second to the second one, and the third to the third one. In the middle column, multiply all of the divisors except the one that is used in the expression. For example, in the first one, I will only multiply 6 and 7, but exclude 5. Do the same for all 3 steps. In the left most column, write down all the remainders that you want. Remember that they all have to be in order. Order is what matters the most. Now our next step is to literally just use the linear statements we have at the top, and multiply the variable with the values in

our middle column in order.

$$42x \equiv 1 \pmod{5}$$

$$35x \equiv 2 \pmod{6}$$

$$30x \equiv 3 \pmod{7}$$

Now, we know that $42x$ is the same thing as $2x$ in mod 5 because 40 is a multiple of 5. We do this to all the expressions.

$$2x \equiv 1 \pmod{5}$$

$$5x \equiv 2 \pmod{6}$$

$$2x \equiv 3 \pmod{7}$$

Now we will individually solve each of the x terms and put it in the rightmost column. We want to see what value of x makes $2x \equiv 1 \pmod{5}$. This can be done by simple trial and error. After we do that, we find that the value of x is just 3. We do the same for all the equations to get 3, 5, and 4.

$$\begin{array}{rcl} 1 & 6 \times 7 = 42 & 3 \\ 2 & 5 \times 7 = 35 & 5 \\ 3 & 5 \times 6 = 30 & 4 \end{array}$$

Our last step is to multiply the terms in each row. After we do that, we get $1 \times 42 \times 3$, $2 \times 35 \times 5$, and $3 \times 30 \times 4$.

Then, you add all of those terms to each other to get your answer. After we add, we get $126 + 350 + 360 = 836$.

Definition 2.9.1

When using the Chinese Remainder Theorem, split the large divisor into smaller ones, but make sure that the smaller ones are relatively prime to each other. For example, if I want to check if a number is divisible by 72, I can work with 8 and 9 separately.

The Chinese Remainder Theorem shows up often on the AIME, but it's not that common on the AMC. The most important idea that you need to know has already been listed above. It's about splitting the divisor into smaller numbers. In many cases, many numbers that you will test will be modulo 10, 100, and 1000. You can split them as 2 and 5, 4 and 25, and 8 and 125 respectively.

Problem 2.9.2 — Let $N = 123456789101112 \dots 4344$ be the 79-digit number that is formed by writing the integers from 1 to 44 in order, one after the other. What is the remainder when N is divided by 45?

(A) 1 (B) 4 (C) 9 (D) 18 (E) 44

Source: 2017 AMC

Solution: In this problem, we can use the Chinese Remainder Theorem. We can split 45 to two different divisors which are 5 and 9 and work separately with both. If N is divisible by 45 then it must also be divisible by 5 and 9.

Directly trying to divide the 79 digit number by 9 is a tough job. However, we can use divisibility rules. We know that the sum of the digits of any number modulo 9 is equal to the number modulo 9.

Sum of Digits of $N \equiv N \pmod{9}$

The sum of the digits is simply the sum of the numbers from 1 to 44 which is $\frac{44 \cdot 45}{2}$. This is 990 which is $0 \pmod{9}$ which means that N must also be.

$$N \equiv 0 \pmod{9}$$

To see the remainder when dividing by 5, we simply have to look at the units digit only and take that mod 5. In this case, the remainder when we divide the last digit by 5 is simply 4.

$$N \equiv 4 \pmod{5}$$

$$N \equiv 0 \pmod{9}$$

Now using the Chinese Remainder Theorem, we get the remainder as **9 (C)**.

Problem 2.9.3 — It is known that, for all positive integers k , $1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$. Find the smallest positive integer k such that $1^2 + 2^2 + 3^2 + \dots + k^2$ is a multiple of 200.

Source: 2002 AIME

Solution: In this problem, we know that the sum of the squares must be divisible by 200. However, we can use the formula for the sum of the squares and take that mod 200. Before we directly use the formula, we can take the 6 from the denominator out and take the numerator mod 1200.

$$k(k+1)(2k+1) \equiv 0 \pmod{1200}$$

Now because of the Chinese Remainder Theorem, we can split 1200 into 3 and 16 and 25 and write separate linear congruency statements for those 3 divisors.

For the case of dividing by 3, we will test equating k to 0 and 1 and 2 mod 3 separately and seeing if the product is $0 \pmod{3}$. When we do it for all 3 cases, we notice it is always $0 \pmod{3}$ which means there are no restrictions for this case. We can move on.

$$k(k+1)(2k+1) \equiv 0 \pmod{16}$$

For the case of dividing by 16, k must either be 0 or 15 mod 16 for the whole product to be divisible by 16.

$$k(k+1)(2k+1) \equiv 0 \pmod{25}$$

For the case of dividing by 25, k must either be 0 or 12 or 24 mod 25 for the whole product to be divisible by 25.

Applying the Chinese Remainder Theorem for each pair of possibilities for the modulo 16 case and the modulo 25 case, we get the answer to be **112**.

Problem 2.9.4 — Let $2^{1110} \equiv n \pmod{1111}$ with $0 \leq n \leq 1111$. Compute n

Source: 2019 BMT

Solution: In this problem, we know that 1111 is $11 \cdot 101$. We will separate the modular

equation and solve it separately for modulo 11 and 101.

$$2^{1110} \equiv ? \pmod{11}$$

$$2^{1110} \equiv ? \pmod{101}$$

In the equation with mod 11, we know that 2^{10} is 1 mod 11 because of Fermat's Little Theorem. At the same time, 2^{1110} is just $2^{10 \cdot 111}$ so it must also be 1 mod 11.

$$2^{1110} \equiv 1 \pmod{11}$$

For modulo 101: from Fermat's little theorem, we know that $2^{100} \equiv 1 \pmod{101}$.

We can raise that to the power of 11 to get $2^{1100} \equiv 1 \pmod{101}$. Thus, we are simply left with 2^{10} and we know that $2^{10} \pmod{101}$ is simply 14. We now have solved our two individual equations.

$$2^{1110} \equiv 1 \pmod{11}$$

$$2^{1110} \equiv 14 \pmod{101}$$

Solving them gives $2^{1110} \equiv 1024 \pmod{1111}$. Our final answer is just **1024**.

Problem 2.9.5 — The positive integers N and N^2 both end in the same sequence of four digits $abcd$ when written in base 10, where digit a is not zero. Find the three-digit number abc .

Source: 2014 AIME

Solution: We will write down a few modular equations and work from there.

$$N^2 \equiv abcd \pmod{10000}$$

$$N \equiv abcd \pmod{10000}$$

We can now subtract the two statements from each other since both are being divided by the same number which is 10000 in this case. If you still don't understand why I was able to subtract both statements, then look back to the very basics of modular arithmetic.

$$N^2 - N \equiv 0 \pmod{10000}$$

$$N(N-1) \equiv 0 \pmod{10000}$$

We now can distribute 10000 and write it out as 16 and 625. (This comes from our techniques that we learned for CRT)

$$N(N-1) \equiv 0 \pmod{16}$$

$$N(N-1) \equiv 0 \pmod{625}$$

Now we notice one thing. Only one of N and $N - 1$ can be divisible by 2 because one must be even while the other must be odd since they are consecutive numbers. Two consecutive numbers can't both be divisible by 2. Similarly, only one can be divisible by 5. Thus, we have two cases.

Either N is divisible by 16 while $N - 1$ is divisible by 625 or N is divisible by 125 and $N - 1$ is divisible by 16.

We will try one case first to see if we get an answer.

$$N \equiv 0 \pmod{16}$$

$$N-1 \equiv 0 \pmod{625}$$

Using CRT for the system above, we get the answer to be $9376 \pmod{10000}$. 9376 represents abcd, and we only want the first 3 digits. Thus, our final answer is **937**.

§2.10 Cumulative Problems

Now it's time to work on more problems because this is the only way to solidify your understanding.

Problem 2.10.1 — In base 10, the number 2013 ends in the digit 3. In base 9, on the other hand, the same number is written as $(2676)_9$ and ends in the digit 6. For how many positive integers b does the base- b -representation of 2013 end in the digit 3?

(A) 6 (B) 9 (C) 13 (D) 16 (E) 18

Source: 2013 AMC 10

Solution: The units digit in base b is always the number given modulo b . Similarly, the last 2 digits is the number given modulo b^2 . In this case, we just want the units digit so we take 2013 modulo b .

$$2013 \equiv 3 \pmod{b}$$

$$2010 \equiv 0 \pmod{b}$$

From our equation, we know that b divides 2010. We will find the number of factors in 2010 by prime factorizing first.

$$2010 = 2 \times 3 \times 5 \times 67$$

The number of factors are found by adding 1 to all the exponents and multiplying those numbers.

Number Of Factors is $(1+1) \cdot (1+1) \cdot (1+1) \cdot (1+1)$ which is 16

However, this isn't the answer. We must subtract a few cases. A number in base b can't have any digits greater than or equal to b . That means our base must be 4 or greater since 2013 has 3 as the largest digit. However, 1, 2, and 3 are all factors of 2010 that were included in the number 16 we calculated. Thus, we must subtract 3 to get 13 which is (C).

Problem 2.10.2 — What is the remainder when $12^{2011} + 11^{2012}$ is divided by seven?

Source: 2012 UNCO

Solution: In this problem, we will use the Fermat's little theorem. Since we want to take the sum in modulo 7 and find the remainder, we already know that 7 is prime. Since 7 is relatively prime to both 12 and 11, we can use Fermat's little theorem separately to them.

$$12^{2011} \equiv ? \pmod{7}$$

$$11^{2012} \equiv ? \pmod{7}$$

We know that any number relatively prime to 7 raised to the power of 6 will have a remainder of 1 when divided by 7.

$$12^6 \equiv 1 \pmod{7}$$

$$11^6 \equiv 1 \pmod{7}$$

We will transform the equations we got using Fermat's little theorem to find the remainder for the large exponents.

$$(12^6)^{335} \cdot 12 \equiv 1^{335} \cdot 12 \pmod{7}$$

$$(11^6)^{335} \cdot 11^2 \equiv 1^{335} \cdot 11^2 \pmod{7}$$

Simplifying the value for both equations gives us 5 and 2 modulo 7. Adding both of those and taking it mod 7 gives us a final answer of **0**.

Problem 2.10.3 — The smallest three positive proper divisors of an integer n are $d_1 < d_2 < d_3$ so that $d_1 + d_2 + d_3 = 57$. Find the sum of the possible values of d_2 .

Source: 2021 PUMAC

Solution: Since d_1, d_2, d_3 are the smallest divisors of n , then the smallest one is obviously 1. Thus, d_1 is 1.

Plugging it in gives $1 + d_2 + d_3 = 57$.

Subtracting both sides by 1 gives $d_2 + d_3 = 56$

Let's assume that the smallest divisors are 2 prime numbers (prime numbers will obviously be the smallest since we can't have something like 15 be the smallest when there are prime numbers that divide 15 that are smaller than 15 such as 3 and 5)

$d_2 = p$ and $d_3 = q$ (p and q are prime numbers)

$p + q = 56$ (we need to find the prime number pairs that sum to 56)

(p, q) can be $(3, 53), (13, 43), (19, 37)$

We can have another case in which $d_2 = p$ and $d_3 = p^2$. Plugging it in gives

$$p + p^2 = 56$$

Testing prime numbers gives that 7 works for p .

Our possible values of d_2 are 3, 13, 19, and 7. The sum of these values are **42**.

Problem 2.10.4 — Determine all positive integers a such that $a < 100$ and $a^3 + 23$ is divisible by 24.

Source: 2018 UNM-PNM

Solution: This is another problem in which we use our modular arithmetic techniques.

$$a^3 + 23 \equiv 0 \pmod{24}$$

We can subtract 24 from the left side to get

$$a^3 - 1 \equiv 0 \pmod{24}$$

We can bring 1 over to the right side to get

$$a^3 \equiv 1 \pmod{24}$$

From here, we can use the Chinese remainder theorem techniques to separate the

divisor 24 into smaller terms that are relatively prime to each other. We will use 8 and 3.

$$a^3 \equiv 1 \pmod{3}$$

$$a^3 \equiv 1 \pmod{8}$$

Now we will first test all the possible remainders separately for 3 and 8 since both are small numbers. We will cube them and see which ones have a remainder of 1.

a	a^3	$a^3 \bmod 3$
0	0	0
1	1	1
2	8	2

Clearly, a must be 1 (mod 3).

a	a^3	$a^3 \pmod{8}$
0	0	0
1	1	1
2	8	0
3	27	3
4	64	0
5	125	5
6	216	0
7	343	7

We see that a must be 1 mod 8 since that's the only case that makes a^3 have a remainder of 1 when divided by 8.

$$a \equiv 1 \pmod{3}$$

$$a \equiv 1 \pmod{8}$$

Both the equations solve to $a \equiv 1 \pmod{24}$. Our solutions less than 100 simply are 1, 25, 49, 74, and 99.

Problem 2.10.5 — A number m is randomly selected from the set $\{11, 13, 15, 17, 19\}$, and a number n is randomly selected from $\{1999, 2000, 2001, \dots, 2018\}$. What is the probability that m^n has a units digit of 1?

(A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{3}{10}$ (D) $\frac{7}{20}$ (E) $\frac{2}{5}$

Source: 2018 AMC 10

Solution: In this problem, we will be dealing with units digits only. Thus, from all the numbers m , we just need to look at their units digit since this is a mod 10 problem. We can take all of the bases mod 10 to change the set from $\{11, 13, 15, 17, 19\}$ to $\{1, 3, 5, 7, 9\}$. Now with this new set, we can work separately with each digit starting with 1.

1 raised to any exponent will always have a units digit of 1. Thus, there are 20 cases here since all exponents from 1999 to 2018 will work.

Now we will work with 3. We will look for the period for the units digit when raising 3. $3^1=3$

$$3^2=9$$

$$3^3=27 \equiv 7 \pmod{10}$$

$$3^4=81 \equiv 1 \pmod{10}$$

$$3^5=243 \equiv 3 \pmod{10}$$

$$3^6=729 \equiv 9 \pmod{10}$$

We see that the units digits simply repeat with a period of 4. Thus, this case will work with 5 exponents.

We will now work with 5. 5 raised to any power will always end with 5, so there are 0 cases for this.

We will now work with 7.

$$7^1=7$$

$$7^2=49 \equiv 9 \pmod{10}$$

$$7^3=343 \equiv 3 \pmod{10}$$

$$7^4=2401 \equiv 1 \pmod{10}$$

$$7^5=16807 \equiv 7 \pmod{10}$$

$$7^6=117649 \equiv 9 \pmod{10}$$

This case also repeats with a period of 4. Thus, this case will work with 5 exponents.

We will now work with 9. We do the same thing that we did for 3 and 7 to get 5 again.

The total number of cases that work are $20+5+0+5+5$ which is 40. The total number of cases in general are 5×20 which is 100. $40 / 100$ simplifies to $\frac{2}{5}$ (E).

Problem 2.10.6 — What is the sum of all positive integers n such that $\text{lcm}(2n, n^2) = 14n - 24$?

Source: 2015 PUMaC

Solution: We will try to simplify $\text{lcm}(2n, n^2)$. Since n is included on both sides, we can take it out and we know the lcm is equivalent to $n \cdot \text{lcm}(2, n)$.

Now we will have two cases. Either n can be odd or even.

Case 1: n is odd

If n is odd, the lcm must be $2n$. The reason for this is that there are no common factors in between n and 2, so we can multiply both numbers. However, we can't just simply equate $2n$ to $14n - 24$. We have to multiply the LCM we found to n since we took that out in the beginning.

$$2n^2=14n-24$$

Dividing by 2 and bringing the terms to one side and factoring gives

$$(n-3)(n-4)=0$$

n must equal to 3 or 4. However, only 3 works because in this case we assumed that n is odd, so we must discard 4.

Case 2: n is even

If n is even, the $\text{lcm}(2, n)$ must simply be n . The reason for this is that n is already divisible by 2, so we don't need to multiply it by 2 like how we did for the first case. Now we will multiply n by n since we took it out in the beginning.

$$n^2 = 14n - 24$$

Bringing the terms to one side and factoring gives

$$(n - 2)(n - 12) = 0$$

n must be equal to 2 or 12. Both in this case are even, so both work as answers.

We add up our solutions $3+2+12$ to get **17**.

Problem 2.10.7 — Albert has a very large bag of candies and he wants to share all of it with his friends. At first, he splits the candies evenly amongst his 20 friends and himself and he finds that there are five left over. Ante arrives, and they redistribute the candies evenly again. This time, there are three left over. If the bag contains over 500 candies, what is the fewest number of candies the bag can contain?

Source: 2012 PUMAC

Solution: This problem is much easier than it looks at first sight. It looks long, but the problem is very simple. We will try to write this problem in math terms. Let's assume that the total number of candies are n . When the candy is split among his 20 friends and him (21 people in total), 5 are leftover. We can write this as

$$n \equiv 5 \pmod{21}$$

After another person joins, there are 22 people in total. This time, 3 are leftover.

$$n \equiv 3 \pmod{22}$$

We can solve both of the linear statements using Chinese Remainder Theorem to get $n \equiv 47 \pmod{462}$

Since we want the answer to be more than 500, we add 462 to 47 to get **509**.

Problem 2.10.8 — How many integer x are there such that $\frac{x^2-6}{x-6}$ is a positive integer?

Solution: We will combine an algebraic technique to this problem. We will first use long division to divide the fraction. Long division gives us $x + 6 + \frac{30}{x-6}$.

This will be an integer whenever $x - 6$ divides 30. We will write out the divisors of 30 and see what values of x will make $x - 6$ divide it.

Divisors of 30 are 1, 2, 3, 5, 6, 10, 15, and 30. It's important to note that $x - 6$ doesn't have to equal to a positive divisor. It can be that same divisor but negative!

Possible values of $x - 6$ so far are 1, 2, 3, 5, 6, 10, 15, 30, -1, -2, -3, -5, -6, -10, -15, and -30.

The respective values of x that allow $x - 6$ to equate to those values can be found by simply adding 6 to all of the numbers to get

7, 8, 9, 11, 12, 16, 21, 36, 5, 4, 3, 1, 0, -4, -9, and -24.

However, plugging in those values of x might not make the whole expression $x + 6$

$+\frac{30}{x-6}$ positive. Only the first 8 values make the entire term positive. Thus, the answer is **8**.

Problem 2.10.9 — How many positive integers n are there such that $n \leq 2012$, and the greatest common divisor of n and 2012 is a prime number?

Source: 2012 PUMAC

Solution: We will prime factorize 2012 into $2^2 \cdot 503$. The gcd of the two numbers in this problem must either be 2 or 503 as those are the only unique prime factors that divide 2012.

Case 1: The GCD of n and 2012 is 2

Any even value of n will have a gcd of 2 except the ones that are divisible by 4 and 503. The only even number divisible by 2 (not 4) and 503 that is less than 2012 is 1006. On top of this, we must subtract all multiples of 4. Thus, there are $\frac{2012}{4}$ values of n , but we have to subtract 1 to this to remove 1006. This boils down to 502

Case 1: The GCD of n and 2012 is 503

Now n must be divisible by 503 but not 2 or else the gcd will become 1006. There are only two possibilities for n here, 503 and 1509 giving us 2 cases.

The final answer is $502 + 2$ which is **504**.

Problem 2.10.10 — Let N be the greatest integer multiple of 36 all of whose digits are even and no two of whose digits are the same. Find the remainder when N is divided by 1000.

Source: 2010 AIME

Solution: In this problem, since we know that the digits are even, the only possible values for the digits are 0, 2, 4, 6, and 8. We also know that none of the digits can be repeated. Thus, we only have to work with these 5 digits without having to worry about repeats. Since we want to maximize our number, it's best to write the largest digit towards the left and smallest to the right to get 86420.

Now we need to check if it's divisible by 36. Clearly our number isn't divisible by 9 since the sum of the digits isn't. Thus, we have to remove one digit so that the sum of the digits becomes 9 or 18. Some inspection shows us that we simply have to remove the digit 2 to make the sum of the digits divisible by 9. After we do that, we get 8640. This number is clearly divisible by 36, so we just have to divide it by 1000 to find our final answer to be **640**.

Problem 2.10.11 — There are positive integers x and y that satisfy the system of equations

$$\begin{aligned}\log_{10} x + 2 \log_{10}(\gcd(x, y)) &= 60 \\ \log_{10} y + 2 \log_{10}(\text{lcm}(x, y)) &= 570.\end{aligned}$$

Let m be the number of (not necessarily distinct) prime factors in the prime factorization of x , and let n be the number of (not necessarily distinct) prime factors in the prime factorization of y . Find $3m + 2n$.

Source: 2019 AIME 1

Solution: In this problem, we need to find a way to simplify the algebraic equations to get rid of the lcm and gcds. We remember one rule that we learned

$$a \cdot b = \text{lcm}(a, b) \cdot \gcd(a, b).$$

To get a similar result as shown above, we will add (not multiply) the two equations to get

$$\log_{10} x + 2 \log_{10}(\gcd(x, y)) + \log_{10} y + 2 \log_{10}(\text{lcm}(x, y)) = 630$$

Using our logarithm rules, we get

$$\log(xy) + 2(\log(\gcd(x, y) \cdot \text{lcm}(x, y))) = 630$$

Using the rule $a \cdot b = \text{lcm}(a, b) \cdot \gcd(a, b)$. and simplifying, we get

$$\log_{10} xy = 210$$

Solving for xy gives us 10^{210}

However, now we are stuck. We will assume x and y are both powers of 10. x is less than y , and x is 10^a while y is 10^b .

We will plug this back into the equation given originally.

$$\log_{10} 10^a + 2 \log_{10}(\gcd(10^a, 10^b)) = 60$$

Since we know that x is less than y as stated before, the GCD simply is 10^a .

$$a + 2a = 60$$

$$a = 20$$

Now that we got the value of a which is the exponent of x , we can quickly find that b is 190 since $xy = 10^{210}$.

Now we need to find the number of not necessarily different prime factors in x and y . Since x is $2^{20} \cdot 5^{20}$ while y is $2^{190} \cdot 5^{190}$, x has 40 unique prime factors while y has 380.

Plugging this into $3m + 2n$ gives an answer of **880**.

Problem 2.10.12 — Positive integer n has the property such that $n64$ is a positive perfect cube. Suppose that n is divisible by 37. What is the smallest possible value of n ?

Source: 2018 SMT

Solution: Since we know that n is divisible by 37, let's assume that $n = 37a$ where a is also a positive integer.

Now we know that $37a - 64$ is a perfect cube. Let's assume our cube is x^3 .

$$n = 37a - 64 = x^3$$

$$n = 37a = x^3 + 64$$

Recalling the fact that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ allows us to factor $x^3 + 64$

$$37a = (x + 4)(x^2 - 4x + 16)$$

Now since 37 is a prime number, it can only either divide $x + 4$ or $x^2 - 4x + 16$

If it divides $x + 4$, then x has to be 33.

Now let's assume that 37 divides $x^2 - 4x + 16$ for our second case.

If we simply try assuming that this value equals to $x^2 - 4x + 16$, we can rearrange it before solving it to get $x^2 - 4x - 21 = 0$

This factors as $(x + 3)(x - 7) = 0$ which tells us that our value of x should be 7. We can plug this into $n = 37a = x^3 + 64$ to get that $n = 7^3 + 64$ which is **407**.

Problem 2.10.13 — Find the number of eight-digit positive integers that are multiples of 9 and have all distinct digits

Source: 2018 HMMT

Solution: In this problem, we have 10 possible digits from 0 to 9, and all 10 digits sum to 45. Thus, if we want a number with 8 distinct digits, we must remove 2 numbers from 0 to 9.

Since the sum of the 8 digits should be divisible by 9 for that number to be divisible by 9, we know that the 2 digits we remove must also sum to a multiple of 9. The only possible pairs of numbers that will work are (0, 9), (1, 8), (2, 7), (3, 6), and (4, 5).

If we remove 0 and 9, then there are simply $8!$ (8 factorial) ways to arrange those digits.

For the other 4 pairs of 2 digits, there are 7 places where we can put the 0 (every place except for the first digit), and there are $7!$ ways to rearrange the remaining numbers.

This means there are $7 \cdot 7!$ ways, and we multiply this by 4 since there are 4 pairs.

$$8! + 28 \cdot 7! = 36 \cdot 7! \text{ which is } \mathbf{181440}.$$

Problem 2.10.14 — What is the smallest positive integer n such that $2016n$ is a perfect cube?

Source: 2018 HMMT

Solution: In this problem, we'll first prime factorize 2016 to get $2^5 \cdot 3^2 \cdot 7$.

To make this a perfect cube, all exponents should be divisible by 3. For that, we should add one power of 2 so the exponent becomes 6. We do the same for 3. For 7, we multiply 2016 by 7^2 so the exponent will become 3.

Thus, n is $2 \cdot 3 \cdot 7^2$ which is **294**.

Problem 2.10.15 — What are the last two digits of $2022^{2022^{2022}}$?

Source: 2022 SMT

Solution: Finding the last two digits is equivalent to finding the number mod 100. Using the Chinese Remainder Theorem, we can separate 100 to 4 and 25 and find the number mod 4 and mod 25 separately.

$$\begin{aligned} 2022^{2022^{2022}} &\equiv ? \pmod{4} \\ 2022^{2022^{2022}} &\equiv ? \pmod{25} \end{aligned}$$

Clearly $2022^{2022^{2022}} \equiv 0 \pmod{4}$, because 2022 is divisible by 2 and it is raised to a very high power.

For the mod 25 equation, we can use Euler's Totient function and we know that $2022^{20} \equiv 1 \pmod{25}$

However, our exponent for base 2022 is 2022^{2022} , so we can take that exponent mod 20. We should split this up into mod 4 and 5 to use the Chinese Remainder Theorem.

$$\begin{aligned} 2022^{2022} &\equiv 0 \pmod{4} \\ 2022^{2022} &\equiv 2^{2022} \pmod{5} \end{aligned}$$

Clearly our expression in mod 5 is simply 4, and we can find that using our technique in the pattern for units digit.

Since we know that $2022^{2022} \equiv 0 \pmod{4}$ and $2022^{2022} \equiv 4 \pmod{5}$, we can use the Chinese Remainder Theorem to find that $2022^{2022} \equiv 4 \pmod{20}$.

Using that, we know that $2022^{2022^{2022}} \equiv 2022^4 \equiv (-3)^4 \pmod{25}$
We now get $81 \equiv 6 \pmod{25}$.

Since we know that

$$\begin{aligned} 2022^{2022^{2022}} &\equiv 0 \pmod{4} \\ 2022^{2022^{2022}} &\equiv 6 \pmod{25} \end{aligned}$$

Using this we know that our expression's last 2 digits are **56**.

Problem 2.10.16 — Digits H, M , and C satisfy the following relations where ABC denotes the number whose digits in base 10 are A, B , and C .

$$\begin{aligned}\overline{H} \times \overline{H} &= \overline{M} \times \overline{C} + 1 \\ \overline{HH} \times \overline{H} &= \overline{MC} \times \overline{C} + 1 \\ \overline{HHH} \times \overline{H} &= \overline{MCC} \times \overline{C} + 1\end{aligned}$$

Find $\overline{HMC}$.

Source: 2011 CHMMC

Solution: In this problem, we can expand $\overline{HH}$ as $10H + H$ which is $11H$. We can expand $\overline{HHH}$ as $100H + 10H + H$ which is $111H$.

$$\begin{aligned}\overline{MC} &= 10M + C \\ \overline{MCC} &= 100M + 10C + C = 100M + 11C\end{aligned}$$

We can convert those three equations into

$$\begin{aligned}H^2 &= MC + 1 \\ 11H^2 &= 10MC + C^2 + 1 \\ 111H^2 &= 100MC + 11C^2 + 1\end{aligned}$$

We can multiply the first equation by 11 and equate it to the second equation to get

$$10MC + C^2 + 1 = 11MC + 11$$

Rearranging and factoring this equation gives $C(C - M) = 10$

From here, we can plug in possible values for C since it must divide 10. The only possible values are 1, 2, 5

Testing all values gives that C must be 5 and plugging this into that equation gives $M = 2$. We now plug this into $H^2 = MC + 1$ to get that $H = 4$.

These digits gives that our number $\overline{HMC}$ is **435**.

Problem 2.10.17 — Given $k \geq 1$, let p_k denote the k -th smallest prime number. If N is the number of ordered 4-tuples (a, b, c, d) of positive integers satisfying $abcd = \prod_{k=1}^{2023} p_k$ with $a < b$ and $c < d$, find $N \bmod (1000)$

Source: 2022 PUMAC

Solution: In this problem, we will work separately with ab and cd .

From the 2023 prime factors of $abcd$, we can have as many as 1 to 2022 prime factors to divide ab , and the remaining ones will divide cd . The number of ways to choose i prime factors are $\binom{2023}{i}$.

Let's say that i prime factors divide ab while $2023 - i$ divide cd .

From here, we have 2^i ways to distribute the i prime factors for ab and 2^{2023-i} ways to distribute the $2023 - i$ prime factors for cd . However, we need to satisfy the case in which

$a < b$ and $c < d$. For each factor pair of a and b that multiplies to the i prime factors, since we know that there are 2^i ways to distribute them, we have to divide it by 2 to guarantee that the smaller one will be a .

Using that strategy, we get

$$\sum_{i=1}^{2022} \binom{2023}{i} \cdot \frac{2^i}{2} \cdot \frac{2^{2023-i}}{2}$$

This summation becomes $\sum_{i=1}^{2022} \binom{2023}{i} \cdot 2^{2021}$

This simplifies to $2^{2021}(\binom{2023}{1} + \binom{2023}{2} \dots + \binom{2023}{2022})$

$\sum_{x=0}^{2023} \binom{2023}{x} = 2^{2023}$ (this is an identity that you'll learn in the combinatorics section)

However, we must subtract $\binom{2023}{1}$ and $\binom{2023}{2023}$ from that to get $2^{2023} - 2$

We multiply that expression by 2^{2021} which gives $2^{2021}(2^{2023} - 2)$

This expression can be rewritten as $2^{2022}(2^{2022} - 1)$

Now we use the Chinese Remainder Theorem since we want to find this mod 1000. We can solve for it mod 8 and 125.

$$2^{2022}(2^{2022} - 1) \equiv 0 \pmod{8} \text{ (Since } 2^3 \text{ divides } 2^{2022})$$

$$2^{2022}(2^{2022} - 1) \equiv ? \pmod{125}$$

Using the Euler's Totient Theorem gives us that the totient of 125 is 100. This leads to $2^{100} \equiv 1 \pmod{125}$. Using that and simplifying the expression gives that the remainder mod 1000 is 112.

3 Algebra

§3.1 Word Problems

This section is mainly about reading the problem, and converting long paragraphs into equations.

Problem 3.1.1 — Points A , B , and C lie in that order along a straight path where the distance from A to C is 1800 meters. Ina runs twice as fast as Eve, and Paul runs twice as fast as Ina. The three runners start running at the same time with Ina starting at A and running toward C , Paul starting at B and running toward C , and Eve starting at C and running toward A . When Paul meets Eve, he turns around and runs toward A . Paul and Ina both arrive at B at the same time. Find the number of meters from A to B .

Source: 2018 AIME

Solution: In this problem, we are given the relationship between the three characters speeds: Ina, Eve, Paul. Since we know that Ina runs twice as fast as Eve, let's assume that Eve runs at a pace of v meters per unit of time. In that case, Ina runs at a pace of $2v$ meters per unit of time. Since Paul runs twice as fast as Ina, his speed is double Ina's speed so it is $4v$.

Eve's Speed = v

Ina's Speed = $2v$

Paul's Speed = $4v$

Now the problem says that the time it takes Ina to reach B without any interruptions is the same as the time it takes Paul to collide with Eve, and then reverse his direction to B .

Let's assume the distance between points A and B is d . This is the distance that Ina travels. The total time is simply distance/speed which in this case is $\frac{d}{2v}$.

We'll now find a similar equation using distance/speed but for Paul. Since Paul moves at a speed of 4 times of Eve, he will travel 4 times the distance that Eve does at the moment they both collide. That means he travels $\frac{4}{5}$ of the whole distance between B and C which is $1800 - d$. However, he travels that same distance back meaning he travels $\frac{8}{5}$ of $1800 - d$. We divide it by his speed which is $4v$ and solve.

$$\frac{(1800-d)8}{5 \cdot 4v} = \frac{d}{2v}.$$

Cancelling out the v in the denominator makes this a simple 1 variable equation. Solving it gives that $d = 800$.

Problem 3.1.2 — Alpha and Beta both took part in a two-day problem-solving competition. At the end of the second day, each had attempted questions worth a total of 500 points. Alpha scored 160 points out of 300 points attempted on the first day, and scored 140 points out of 200 points attempted on the second day. Beta who did not attempt 300 points on the first day, had a positive integer score on each of the two days, and Beta's daily success rate (points scored divided by points attempted) on each day was less than Alpha's on that day. Alpha's two-day success ratio was $300/500 = 3/5$. The largest possible two-day success ratio that Beta could achieve is m/n , where m and n are relatively prime positive integers. What is $m+n$?

Source: 2014 AIME

Solution: In many problems such as this one, it's good to convert the problem into an inequality. You have to read through the problem and assign variables where it's necessary.

Since we know that Beta's daily success ratio was less than Alpha's success ratio on both days, we can write out Beta's daily success ratios for both days with variables. Let's assume he attempted y problems on day 1 and got x of them right and on day 2 he got z number of problems right. That means his success ratio for day 1 and 2 respectively was $\frac{x}{y}$ and $\frac{z}{500-y}$

Writing out the inequalities gives us

$$\frac{x}{y} < \frac{160}{300} = \frac{8}{15}$$

$$\frac{z}{500-y} < \frac{140}{200} = \frac{7}{10}$$

Since we want to maximize the value of $x+z$ (total number solved) to ultimately maximize the total success ratio, we will isolate the left side for both inequalities to have only x and z on one side. After that, we'll simply add the two inequalities.

$$x < \frac{8y}{15}$$

$$z < \frac{7(500-y)}{10}$$

Adding both equations and simplifying the terms on the right side gives

$$x+z < 350 - \frac{y}{6}$$

Since we want to maximize $x+z$, this happens when the fraction is a super small number making the maximum possible value just slightly lower than 350 which is 349. We want the answer to be in the form of his two-day success ratio which is $\frac{349}{500}$. Adding the numerator and denominator gives **849**.

Problem 3.1.3 — Anh read a book. On the first day she read n pages in t minutes, where n and t are positive integers. On the second day Anh read $n+1$ pages in $t+1$ minutes. Each day thereafter Anh read one more page than she read on the previous day, and it took her one more minute than on the previous day until she completely read the 374 page book. It took her a total of 319 minutes to read the book. Find $n+t$.

Source: 2016 AIME

Solution: In this problem, we will assume that the total number of days he reads pages is x . Now we will sum up n to $n + x - 1$ and t to $t + n - 1$ and equate them to their corresponding values.

Using the arithmetic sequence sum formula gives us two separate equations.

$$\frac{(2n+x-1)x}{2} = 374$$

$$\frac{(2t+x-1)x}{2} = 319$$

Since the values n , t , and x are integers, x must divide both 374 and 319. The GCD of 374 and 319 is 11. Thus, we will test 11 for x which is the total number of days and see if it works. Plugging it into both equations gives

$$\frac{(2n+10)11}{2} = 374$$

$$\frac{(2t+10)11}{2} = 319$$

$$2n+10=68$$

$$2t+10=58$$

We get 29 for n and 24 for t . Summing n and t gives **53** as the final answer.

§3.2 Sequences And Series

Definition 3.2.1

In an arithmetic sequence, the difference between two consecutive terms is the same. There is a common difference which is often referred to as d .

Definition 3.2.2

In a geometric sequence, the ratio between two consecutive terms is the same, denoted by r .

Definition 3.2.3

An arithmetic-geometric sequence is made by multiplying each corresponding term of an arithmetic sequence to a geometric sequence.

Theorem 3.2.4

Sum of Arithmetic Sequence

Assuming that the first term is a , the difference is d , and there are n terms. The terms in order are a , $a+d$, $a+2d$, $a+3d$... , $a + (n - 1)d$.

The sum is $\frac{(2a+nd-d)n}{2}$

Theorem 3.2.5**Sum of Geometric Sequence**

Assuming the first term is a , the ratio is r , and there are n terms. The terms in order are $a, ar^1, ar^2, ar^3, ar^4 \dots ar^{n-1}$

The sum is $\frac{a(r^{n+1}-1)}{r-1}$

This is it about the formulas and definitions you need to know for this section. The best way to now master this topic is to practice problems.

Problem 3.2.6 — For each positive integer k , let S_k denote the increasing arithmetic sequence of integers whose first term is 1 and whose common difference is k . For example, S_3 is the sequence $1, 4, 7, 10, \dots$. For how many values of k does S_k contain the term 2005?

Source: 2005 AIME

Solution: In this problem, we already know the first term of any arithmetic sequence which is 1. Now we will assume that the n th term is 2005 and the difference between two consecutive terms is k . Now we can write an algebraic expression.

$$1 + k(n - 1) = 2005$$

Subtracting 1 gives

$$k(n - 1) = 2004$$

The equation above shows that k must divide 2004. Thus, we will prime factorize 2004.

$$2004 = 2^2 \times 3 \times 167$$

Now, k can be any factor of 2004. Thus, we use our formula to find the number of factors which gives us

$$(2 + 1)(1 + 1)(1 + 1) = \mathbf{12}.$$

Problem 3.2.7 — Let $P(x) = x^3 + x^2 - r^2x - 2020$ be a polynomial with roots r, s, t . What is $P(1)$.

Source: 2020 HMMT

Solution: In this problem, we will use Vieta's Theorem. From Vieta's, we know that:

$$r + s + t = -1$$

$$rs + rt + st = -r^2$$

$$rst = 2020$$

Rearranging the second equation gives $rs + r^2 + rt + st = 0$.

We can factor it to get

$$r(r + s + t) = -st$$

Now from Vieta's, we already know that $r + s + t = -1$, so we can plug that into $r(r + s + t) = -st$ to get $-r = -st$. Dividing by -1 gives $r = st$

Now we can plug this into $rst = 2020$ to get $r^2 = 2020$.

Now since we want to find $P(1)$, we first plug in 1 into $P(x)$ which is $x^3 + x^2 - r^2x - 2020$. We get $-r^2 - 2018$. Now we can plug in $r^2 = 2020$ into $-r^2 - 2018$ to get an answer of **-4038**.

Problem 3.2.8 — The degree measures of the angles in a convex polygon 18-sided polygon form an increasing arithmetic sequence with integer values. Find the degree measure of the smallest angle.

Source: 2011 AIME

Solution: In this problem, we first find the total sum of all the angles of this 18 sided figure. We use a simple formula for that which is that the sum of the angles for an n -sided shape is simply $180 \cdot (n - 2)$.

$$180 \cdot 16 = 2880$$

Since the angles form an arithmetic sequence, we assume that the first term is a , and the difference is d . This means that the terms in order are $a, a+d, a+2d, a+3d \dots, a+17d$.

Now we will sum up these 18 values to get

$$\begin{aligned} &18a + (1+2+3 \dots + 17)d \\ &= 18a + \frac{17 \cdot 18}{2}d \\ &= 18a + 153d \end{aligned}$$

We will assume the expression above to 2880 and solve.

$$18a + 153d = 2880$$

Dividing each term by 9 gives

$$2a + 17d = 320$$

We now know that d has to be even for 2 to divide $320 - 17d$. Thus, we will assume that d is 2. Then, we find that $a = 143$. We will now test to see if the largest angle ($a+17d$) is less than 180 degrees. Plugging in 143 and 2 gives 177.

On a side note, if we plugged in other values for d such as 4 or 6, then the largest angle would be more than 180 which isn't possible since our shape is **convex**. Thus, the answer is **143**.

Problem 3.2.9 — Two distinct, real, infinite geometric series each have a sum of 1 and have the same second term. The third term of one of the series is $\frac{1}{8}$, and the second term of both series can be written in the form $\frac{\sqrt{m-n}}{p}$, where m, n , and p are positive integers and m is not divisible by the square of any prime. Find $100m + 10n + p$.

Source: 2002 AIME

Solution: In this problem, we will assume that the second term in both series is x . Then, for one of the series, we know that the third term is $\frac{1}{8}$. Thus, we will find the value of r for this first series in terms of x . We simply get $\frac{1}{8x}$ for the ratio. Now we divide this ratio from the second term to find the first term. When we do this, we get $8x^2$. Now we will use our formula of the sum of an infinite geometric series.

The first term is $8x^2$ and the ratio is $\frac{1}{8x}$. We will plug this into the formula and equate it to the sum which is 1.

$$\frac{8x^2}{1 - \frac{1}{8x}}.$$

Simplifying this expression gives the equation

$$64x^3 - 8x + 1 = 0.$$

We can factor it to get

$$(4x - 1)(16x^2 + 4x - 1) = 0$$

Solving for the roots gives

$$\frac{1}{4} \text{ and } \frac{-1+\sqrt{5}}{8} \text{ and } \frac{-1-\sqrt{5}}{8}$$

Only the second root fits the format that we want our answer in, so we simply answer in the form of $100m+10n+p$ to get **518**.

§3.3 Polynomials

This is one of the most important topics of algebra that shows up on AMC. It is SUPER common on both the AMC and AIME. You need to know this well.

We'll start with **Vieta's Theorem**.

Vieta's is a great way to find the sum of the roots or the multiplication of it. In any polynomial that is in the form of ax^2+bx+c , Vieta says that the sum of the roots is $-\frac{b}{a}$. The product of the roots in this polynomial is just $\frac{c}{a}$. (Don't forget that in Vieta's you don't include the variables. You just use coefficients.)

Theorem 3.3.1**Vieta's Theorem**

For any polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, the theorem below applies.

$$\begin{aligned} r_1 + r_2 + \dots + r_{n-1} + r_n &= -\frac{a_{n-1}}{a_n} \\ (r_1 r_2 + r_1 r_3 + \dots + r_1 r_n) + (r_2 r_3 + r_2 r_4 + \dots + r_2 r_n) + \dots + r_{n-1} r_n &= \frac{a_{n-2}}{a_n} \\ \vdots & \\ r_1 r_2 \dots r_n &= (-1)^n \frac{a_0}{a_n}. \end{aligned}$$

For any polynomial with n complex roots, the sum of the roots will always be $-\frac{a_{n-1}}{a_n}$. A very common mistake made is finding the sum of the roots of a polynomial in the form ax^3+bx+c . Some people think that the sum of the roots is just $\frac{-b}{a}$ which in this case is wrong. We use the coefficient of the second highest degree divided by the coefficient with the highest degree of the variable. For those who don't know, a degree of the term is its exponent. For example, in $3x^2+4x+7$, the term $3x^2$ has a degree of 2 while the term $4x$ has a degree of 1. In the trick problem, the sum of the roots is 2. The reason is that there is an imaginary term. For example, ax^3+bx+c is the same thing as ax^3+0x^2+bx+c . The sum of the roots is actually 0. Don't fall for this trap!

It's also important to know what the discriminant is. The discriminant of a quadratic $ax^2 + bx + c$ is $b^2 - 4ac$

If the value of the discriminant is more than 0, the quadratic has two distinct real roots.

If the value of the discriminant is 0, the quadratic has one distinct root (but it's repeated).

If the discriminant is less than 0, the quadratic has no real roots (it has two complex roots)

Theorem 3.3.2**Remainder Theorem**

If we divide a polynomial $f(x)$ by $x - r$, then the remainder will be $f(r)$

Problem 3.3.3 — Find the smallest positive integer n with the property that the polynomial $x^4 - nx + 63$ can be written as a product of two nonconstant polynomials with integer coefficients.

Source: 2010 AIME

Solution: In this problem, since we know that two polynomials will multiply to this polynomial with a degree of 4, our two polynomials can either have degree 1 and 3 or 2 and 2. We'll work with each case separately.

Case 1: **degrees 1 and 3**

$$(x + a)(x^3 + bx^2 + cx + d) = x^4 - nx + 63$$

Expanding the left side gives

$$x^4 + (a + b)x^3 + (ab + c)x^2 + (ac + d)x + ad = x^4 - nx + 63$$

Now we equate the coefficients. This means for each term with the same power of x , we will equate the coefficient for that.

$$a + b = 0$$

$$ab + c = 0$$

$$ac + d = -n$$

$$ad = 63$$

Now we will first work with the first two equations and rearrange them. The first one gives $a = -b$ and plugging this into the second one gives

$$c = b^2$$

Now we will plug in $-b$ for a into the last two equations to get

$$-bc + d = -n$$

$$-bd = 63$$

Now we will plug in our b^2 for c in the new equation we made to get

$$-b^3 + d = -n$$

$$-bd = 63$$

Trying values for b and d is possible now since we know their product to be simply 63. After we do this a few times, we get 48 as the smallest answer for n for Case 1.

Case 2: **degrees 2 and 2**

$$(x^2 + ax + b)(x^2 + cx + d) = x^4 - nx + 63$$

Expanding the left side gives

$$x^4 + (a + c)x^3 + (b + d + ac)x^2 + (ad + bc)x + bd = x^4 - nx + 63$$

Now we equate the coefficients again to get a few equations

$$a + c = 0$$

$$b + d + ac = 0$$

$$ad + bc = -n$$

$$bd = 63$$

Solving this equation in a similar way to how we solved it in case 1 gives us $n = 8$.

Problem 3.3.4 — The quadratic equation $x^2 + mx + n$ has roots twice those of $x^2 + px + m$, and none of m, n , and p is zero. What is the value of n/p ?

(A) 1 (B) 2 (C) 4 (D) 8 (E) 16

Source: 2005 AMC

Solution: Assuming that the roots of the equation $x^2 + px + m$ are r and s , we know that the roots of the other quadratic is $2r$ and $2s$ because it says that the roots of it are double.

Now we will write down a few equations for the quadratic equations using Vieta's Formula.

Equations for the quadratic $x^2 + mx + n$

$$2r + 2s = -m$$

$$4rs = n$$

Equations for the quadratic $x^2 + px + m$

$$r + s = -p$$

$$rs = m$$

Now since we want to find $\frac{n}{p}$, we will substitute expressions for n and p using the equations we wrote to get

$$\frac{n}{p} = \frac{4rs}{-(r+s)}$$

Now simplifying this might look hard, but we notice that there are two equations involving m, r , and s . Thus, we want to try to replace rs and $r + s$ with something in terms of m to cancel the m out.

Since $rs = m$, we can multiply both sides by 4 to get $4rs = 4m$.

Since $2(r + s) = -m$, we can divide both sides by 2 to get $r + s = -\frac{m}{2}$.

Now we will substitute our new findings into the other fraction to get

$$-\frac{4m}{-\frac{m}{2}} = 8 \text{ (D)}.$$

Problem 3.3.5 — Suppose $f(x) = x^3 + x - 1$ has roots r, s, t .

What is $\frac{r^3}{1-r} + \frac{s^3}{1-s} + \frac{t^3}{1-t}$?

Source: 2013 CHMMC

Solution: In this problem, most people will straight up jump to applying Vieta's Theorem. However, after performing some manipulations and adding a few fractions, we get a complicated expression. Thus, we can't use Vieta's Theorem here.

However, we can rewrite $x^3 + x - 1 = 0$ as $x^3 = 1 - x$

Dividing both sides by $1 - x$ gives $\frac{x^3}{1-x} = 1$

Since we already know the solutions for the equations are r, s, t , we can plug them in to get that $\frac{r^3}{1-r}, \frac{s^3}{1-s}, \frac{t^3}{1-t}$ which is all equal to 1. Thus, our answer is simply the sum of the 3 expressions which is $3 \cdot 1$ leading to **3**.

Problem 3.3.6 — Let f be a function for which $f\left(\frac{x}{3}\right) = x^2 + x + 1$. Find the sum of all values of z for which $f(3z) = 7$.

- (A) $-1/3$ (B) $-1/9$ (C) 0 (D) $5/9$ (E) $5/3$

Source: 2000 AMC

Solution: In this problem, we can't just directly plug in $3z$ to the expression given for $f\left(\frac{x}{3}\right)$. The reason is that plugging in $3z$ simply makes it evaluate to $f(z)$. Thus, we must plug in $9z$ for x to make it equivalent to $f(3z)$ on the left side.

$$f\left(\frac{9z}{3}\right) = f(3z): (3z)^2 + 3z + 1 = 81z^2 + 9z + 1 = 7.$$

Subtracting 7 from both sides gives

$$81z^2 + 9z - 6 = 0$$

We can simply use Vieta's Theorem now to find the sum of the roots which gives us $-\frac{1}{9}$ **(B)**.

Problem 3.3.7 — Consider a polynomial $p(x)$ of degree 1 such that for a real number a , $p(a) = 2$, $p(p(a)) = 17$ and $p(p(p(a))) = 167$. What is the value of a ?

- (A) -1 (B) $-\frac{1}{2}$ (C) 0 (D) $\frac{1}{2}$ (E) 1

Source: 2019 CMC 10

Solution: In this problem, since we know that $p(x)$ is a 1 degree polynomial, we can assume that it is $cx + d$.

Since $p(a) = 2$, we can plug a in for x to get $ac + d = 2$

Using $p(p(a)) = 17$, we can plug in $p(a) = 2$ into this to get that $p(2) = 17$. We can plug in 2 for x into our expression for $p(x)$ to get $2c + d = 17$

Using $p(p(p(a))) = 167$, we can plug in $p(p(a)) = 17$ into $p(p(p(a))) = 167$ to get $p(17) = 167$

We plug in 17 for x into our expression for $p(x)$ to get $17c + d = 167$.

The 3 equations we get by plugging in values into $p(x) = cx + d$ are

$$ac + d = 2$$

$$2c + d = 17$$

$$17c + d = 167$$

We can subtract the second equation from the third one to get $15c = 150$ which gives us that $c = 10$. Plugging this into any of the equations (second or third) gives that $d = -3$.

This means that our polynomial $p(x) = 10x - 3$.

Now using the equation $p(a) = 2$, we get $10a - 3 = 2$, and our value of a is $\frac{1}{2}$ **(D)**.

Problem 3.3.8 — Suppose r, s, t are the roots of the polynomial $x^3 - 2x + 3$. Find

$$\frac{1}{r^3 - 2} + \frac{1}{s^3 - 2} + \frac{1}{t^3 - 2}$$

Source: 2018 CHMMC

Solution: Let's plug in r into the polynomial and this gives us $r^3 - 2r + 3 = 0$

We can do this for s and t to get

$$s^3 - 2s + 3 = 0$$

$$t^3 - 2t + 3 = 0$$

Rewriting these equations gives $r^3 - 2 = 2r - 5$. We do this for the other roots to get

$$s^3 - 2 = 2s - 5$$

$$t^3 - 2 = 2t - 5$$

We can plug this into $\frac{1}{r^3-2} + \frac{1}{s^3-2} + \frac{1}{t^3-2}$, and this expression becomes $\frac{1}{2r-5} + \frac{1}{2s-5} + \frac{1}{2t-5}$

We can sum up these fractions by taking the common denominator to get

$$\frac{-20(r+s+t)+4(rs+rt+st)}{50(r+s+t)-20(rs+rt+st)+8rst}$$

Using Vieta's theorem on $x^3 - 2x + 3$, we can find equations relating our roots to the coefficients.

$$r + s + t = 0$$

$$rs + rt + st = -2$$

$$rst = -3$$

We can plug this into our expression to find that the answer is $\frac{-67}{109}$

Problem 3.3.9 — For certain real numbers a, b , and c , the polynomial

$$g(x) = x^3 + ax^2 + x + 10$$

has three distinct roots, and each root of $g(x)$ is also a root of the polynomial

$$f(x) = x^4 + x^3 + bx^2 + 100x + c.$$

What is $f(1)$?

(A) -9009 (B) -8008 (C) -7007 (D) -6006 (E) -5005

Source: 2017 AMC

Solution: In this problem, since we know that all of the roots of the polynomial $g(x)$ are also roots of the polynomial $f(x)$, $g(x)$ divides $f(x)$. However, we must multiply it by a degree one polynomial to get $f(x)$ since we have to account for the fourth root in $f(x)$ that isn't a root of $g(x)$.

Assuming that r is the root that divides $f(x)$ but not $g(x)$, we can write this out as

$$g(x)(x - r) = f(x)$$

$$(x^3 + ax^2 + x + 10)(x - r) = x^4 + x^3 + bx^2 + 100x + c$$

Expanding the left side and equating both polynomials by equating the coefficients gives

$$x^4 + (a - r)x^3 + (1 - ar)x^2 + (10 - r)x - 10r = x^4 + x^3 + bx^2 + 100x + c$$

Equating coefficients gives the equations

$$a - r = 1$$

$$1 - ar = b$$

$$10 - r = 100$$

$$c = -10r$$

We can simply use the third equation to find r to be -90 . Then, plugging that into the fourth one gives c as 900 . Then, we can also find a from the first equation, and plug it into the second to get b .

After we find those values, we have our full form of the polynomial. We can substitute 1 into the polynomial to get **-7007 (C)**.

Problem 3.3.10 — Suppose a, b are nonzero integers such that two roots of $x^3 + ax^2 + bx + 9a$ coincide, and all three roots are integers. Find $|ab|$.

Source: 2013 PUMAC

Solution: Since the problem says that two roots coincide, that means they are the same. However, there is another root that's different from that common one. Let's assume the two unique roots are r, s and r will be our repeated root.

Using Vieta's Theorem, we can write out 3 equations relating the roots and the coefficients

$$2r + s = -a$$

$$2rs + r^2 = b$$

$$r^2s = -9a$$

Looking at these three equations, you should think of combining the first and the third one since they're both equal to a term that involves the variable a . We can equate them by multiplying the first one by 9 making the right side equivalent to $-9a$ for the first and third equation. Thus, we multiply the first equation by 9 and subtract it from the third to get:

$$r^2s - 18r - 9s = 0$$

We should realize that this is a polynomial in terms of r and s , -18 , and -9 are the coefficients. We can now use the quadratic formula to find the roots in terms of s . Doing so will give us a square root term which must be eliminated (meaning that the term inside the square root must be a perfect square). This is true because r is an integer.

The term inside the square root is $36(s^2 + 9)$

Since 36 is already a perfect square, we will try to find a value of s so that $s^2 + 9$ will also be one. We can equate $s^2 + 9$ to x^2 and solve.

$$s^2 + 9 = x^2$$

$$(x - s)(x + s) = 9$$

Testing all the various cases by writing the factors out and seeing if it fits all the conditions given in the problem statement gives us that s must be 4 or -4 .

We can now plug in our two values of s to $r^2s - 18r - 9s = 0$. After solving them, we see that the two possible solutions for (r, s) are $(-6, -4)$ and $(6, 4)$.

Testing out both solutions and plugging them in to find a and b shows us that they both lead to the same solution of **1344**.

Problem 3.3.11 — Suppose $P(x)$ is a quadratic polynomial with integer coefficients satisfying the identity

$$P(P(x)) - P(x)^2 = x^2 + x + 2016$$

for all real x . Find $P(1)$

Source: 2017 CMIMC Algebra

Solution: In this problem, let's assume $P(x) = ax^2 + bx + c$. Let's substitute this into $P(P(x)) - P(x)^2 = x^2 + x + 2016$.

$$\begin{aligned} aP(x)^2 + bP(x) + c - P(x)^2 &= x^2 + x + 2016 \\ (a-1)P(x)^2 + bP(x) + c &= x^2 + x + 2016 \end{aligned}$$

Since the term $(a-1)P(x)^2$ has degree 4, we must do something so that the terms get cancelled out since we equated it to a second degree polynomial. We can do that by making $a = 1$. This makes the equation become

$$bP(x) + c = x^2 + x + 2016 \text{ and } P(x) = x^2 + bx + c$$

We can plug in our polynomial $P(x)$ to get

$$bx^2 + b^2x + bc + c = x^2 + x + 2016$$

Equating coefficients gives

$$b = 1$$

$$b^2 = 1$$

$$bc + c = 2016$$

Clearly $b = 1$ and plugging this into the third equation gives that $c = 1008$.

This means our polynomial $P(x) = x^2 + x + 1008$

We want to find $P(1)$ which is $1^2 + 1 + 1008$, and our answer is **1010**.

Problem 3.3.12 — Let $P(x) = x^2 - 3x - 7$, and let $Q(x)$ and $R(x)$ be two quadratic polynomials also with the coefficient of x^2 equal to 1. David computes each of the three sums $P + Q$, $P + R$, and $Q + R$ and is surprised to find that each pair of these sums has a common root, and these three common roots are distinct. If $Q(0) = 2$, then $R(0) = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Source: 2020 AIME

Solution: Since we know that $Q(0) = 2$, it means that the constant term is 0 which gives us that $Q(x)$ can be represented as $x^2 + ax + 2$

We can write $R(x) = x^2 + bx + c$

$$P+Q: 2x^2 + x(a-3) - 5 = 0 \text{ (roots } r, s)$$

$$P+R: 2x^2 + x(b-3) + c - 7 = 0 \text{ (roots } r, t)$$

$$Q+R: 2x^2 + x(a+b) + c + 2 = 0 \text{ (roots } s, t)$$

We can write out the roots easily with variables since we know that each polynomial has a common root. Now we can use Vieta's Theorem to relate the coefficients for the roots for each polynomial separately.

$$1 : r + s = \frac{3-a}{2}$$

$$2 : rs = \frac{-5}{2}$$

$$3 : r + t = \frac{3-b}{2}$$

$$4 : rt = \frac{-7}{2}$$

$$5 : s + t = \frac{-a-b}{2}$$

$$6 : st = \frac{c+2}{2}$$

$$1 - 3 : s - t = \frac{-a+b}{2} \text{ (subtracting the third equation from the first)}$$

Now we can add the equation above that we found from subtracting to equation 5.

This gives us $2s = -a$ which is equivalent to $s = \frac{-a}{2}$. We plug this into the first equation to get that $r = \frac{3}{2}$.

Plugging this into the second equation gives us that $s = \frac{-5}{3}$.

We plug in $r = \frac{3}{2}$ into $rt = -7$ to get that $t = \frac{-7}{3}$.

If we plug in these values into equation 6 which is $st = \frac{c+2}{2}$, we can solve for c to get that it is $\frac{52}{19}$.

Since we know that we want to find $R(0)$ and we said that $R(x) = x^2 + bx + c$, $R(0) = c$ and our value of c is $\frac{52}{19}$. Our final answer is $52 + 19$ which is **71**.

§3.4 Logarithms: AMC 12 ONLY

Logarithms are a pretty simple topic on the AMC 12, and with proper practice these should be easier to attempt than other topics.

Where:

$b > 0$ but $b \neq 1$, and M , N , and k are real numbers but M and N must be positive!

Rule 1: $\log_b(M \cdot N) = \log_b M + \log_b N$

Rule 2: $\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$

Rule 3: $\log_b(M^k) = k \cdot \log_b M$

Rule 4: $\log_b(1) = 0$

Rule 5: $\log_b(b) = 1$

Rule 6: $\log_b(b^k) = k$

Rule 7: $b^{\log_b(k)} = k$

Memorize the logarithm rules above. They are super important.

Theorem 3.4.1

Change of Base Formula

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Logarithm Change of Base Formula

$$\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$$

c is the new log base

The above formula is super important. You can use it to rewrite any logarithm as a fraction of logarithms any new base of your choice.

Also, remember that $\log_a b = e$ is the same thing as $a^e = b$

Tip: A common way to solve logarithm problems is to rewrite it in exponential form. This is super important!

Now the best way to learn logarithms in depth is by simply practicing, so that is what we will do now.

Problem 3.4.2 — If $\log(xy^3) = 1$ and $\log(x^2y) = 1$, what is $\log(xy)$?

(A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{2}$ (D) $\frac{3}{5}$ (E) 1

Source: 2003 AMC 12

Solution: We will rewrite our logarithms in exponential form.

$$xy^3 = 10$$

$$x^2y = 10$$

We can write x in terms of y in the first equation to get

$$x = \frac{10}{y^3}$$

Now we plug in our value of x in terms of y into the second equation to get

$$\frac{100}{y^6} \cdot y = 10$$

Simplifying the above equation gives

$$y^5 = 10 \text{ which gives us the value of } y \text{ as } y = 10^{1/5}$$

Plugging in our value of y back gives us the value of x as $10^{2/5}$.

Now we simply want to find $\log_{10} 10^{3/5}$

The answer to the expression above is simply $\frac{3}{5}$ (D).

Problem 3.4.3 — Let x, y , and z be real numbers satisfying the system

$$\log_2(xyz - 3 + \log_5 x) = 5,$$

$$\log_3(xyz - 3 + \log_5 y) = 4,$$

$$\log_4(xyz - 3 + \log_5 z) = 4.$$

Find the value of $|\log_5 x| + |\log_5 y| + |\log_5 z|$.

Source: 2016 AIME

Solution: In this problem, we'll convert the logarithm equations to exponential form to get

$$xyz - 3 + \log_5 x = 32$$

$$xyz - 3 + \log_5 y = 81$$

$$xyz - 3 + \log_5 z = 256$$

Adding 3 to both sides of all equations gives

$$xyz + \log_5 x = 35$$

$$xyz + \log_5 y = 84$$

$$xyz + \log_5 z = 259$$

Adding all 3 of these equations gives

$$3xyz + \log_5 xyz = 378$$

Some inspection clearly gives that $xyz = 125$.

We can plug these into all 3 equations to give $\log_5 x = -90$, $\log_5 y = -41$, and $\log_5 z = 134$.

Since we want to find the sum of all of these terms but take the absolute value for each, our total sum is $90 + 41 + 134$ which is **265**.

Problem 3.4.4 — The sum of the base-10 logarithms of the divisors of 10^n is 792.

What is n ?

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Source: 2008 AMC 12

Solution: In this problem, we must use some number theory techniques also. We notice that if n is odd, then all the factors can be paired up when they multiply to 10^n .

Now it's important to know that when we add all the divisors such as

$$\log_{10} d_1 + \log_{10} d_2 + \log_{10} d_3 \text{ it simply becomes } \log_{10} d_1 \cdot d_2 \cdot d_3$$

Using that observation we already know all the factor pairs multiply to 10^n . When n is odd, there won't be a factor without a pair (meaning the square root of the number 10^n).

We know that 10^n can be written as $2^n \cdot 5^n$,

This number has $(n+1)(n+1)$ factors and we divide that by 2 to find the number of factor pairs. Combining this with the product 10^n (that will be the product for each factor pair) gives us an equation

$$\log_{10} 10^{n \cdot \frac{(n+1)^2}{2}} = 792$$

This simply simplifies to

$$n \cdot \frac{(n+1)^2}{2} = 792.$$

Plugging in values from the answer choices for n gives us **11 (A)**.

Problem 3.4.5 — Positive numbers x , y , and z satisfy $xyz = 10^{81}$ and $(\log_{10} x)(\log_{10} yz) + (\log_{10} y)(\log_{10} z) = 468$. Find $\sqrt{(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2}$.

Source: 2010 AIME

Solution: In this problem, you will learn a super important strategy. It is to substitute other variables for our logarithm expressions. In this case, we will make the following substitutions

$$\log_{10} x = a \text{ (also the same thing as } x = 10^a)$$

$$\log_{10} y = b \text{ (also the same thing as } y = 10^b)$$

$$\log_{10} z = c \text{ (also the same thing as } z = 10^c)$$

We can rewrite the second equation before substituting to get

$$(\log_{10} x)(\log_{10} y + \log_{10} z) + (\log_{10} y)(\log_{10} z) = 468$$

Now plugging in our variables a, b, c in replacement for our logarithm expression gives

$$a(b + c) + bc = 468$$

$$10^{a+b+c} = 10^{81}$$

The second equation we wrote simplifies to $a + b + c = 81$

Our equations so far are

$$ab + ac + bc = 468$$

$$a + b + c = 81$$

We want to find $\sqrt{a^2 + b^2 + c^2}$

We know that $(a + b + c)^2 - 2(ab + ac + bc) = a^2 + b^2 + c^2$

Plugging in our values gives $468^2 - 2 \cdot 468$ to get 5625 for the value of $a^2 + b^2 + c^2$.

We simply want to find the square root of that which is simply **75**.

Problem 3.4.6 — For each ordered pair of real numbers (x, y) satisfying

$$\log_2(2x + y) = \log_4(x^2 + xy + 7y^2)$$

there is a real number K such that

$$\log_3(3x + y) = \log_9(3x^2 + 4xy + Ky^2).$$

Find the product of all possible values of K .

Source: 2018 AIME

Solution: In this problem, we will have to rewrite both logarithm expressions and try to make them into the same base.

Just remember that we will now have to work backwards using our knowledge of the logarithm identities.

$$\log_{x^a} y = \frac{1}{a} \cdot \log_x y$$

$$\log_2(2x + y) = \frac{1}{2} \cdot \log_2(x^2 + xy + 7y^2)$$

Now we will simplify the right side again using our logarithmic identities to get

$$\log_2(2x + y) = \log_2(\sqrt{x^2 + xy + 7y^2})$$

We can do these exact steps again but for the other equation to get

$$\log_3(3x + y) = \log_3(\sqrt{3x^2 + 4xy + Ky^2})$$

Now for both of our equations, since the base of the logarithm is the same, the top parts

are also equal. We now get

$$2x + y = \sqrt{x^2 + xy + 7y^2}$$

$$3x + y = \sqrt{3x^2 + 4xy + Ky^2}$$

Squaring both sides of both equations and simplifying gives

$$x^2 + xy - 2y^2 = 0$$

$$6x^2 + 2xy + (1 - k)y^2$$

Factoring the top equation from the two seen above gives

$$(x - y)(x + 2y) = 0$$

Solving it gives $x = y$ and $-2y$

Now we plug in these values of x in terms of y into the second simplified equation that includes k . After that, we simply solve for k for both cases to get 9 and 21 as the possibilities.

$$9 \cdot 21 = \mathbf{189}.$$

Problem 3.4.7 — What is the value of

$$\left(\sum_{k=1}^{20} \log_{5^k} 3^{k^2} \right) \cdot \left(\sum_{k=1}^{100} \log_{9^k} 25^k \right)?$$

(A) 21

(B) $100 \log_5 3$

(C) $200 \log_3 5$

(D) 2,200

(E) 21,000

Source: 2021 AMC 12

Solution: In this problem, we need to use change of base to try to simplify our expression. Doing so gives

$$\left(\sum_{k=1}^{20} \frac{\log 3^{k^2}}{\log 5^k} \right)$$

Now simplifying the inside part further using our logarithm rules gives

$$\left(\sum_{k=1}^{20} \frac{k^2 \cdot \log 3}{k \cdot \log 5} \right)$$

The k cancels out to give

$$\left(\sum_{k=1}^{20} \frac{k \cdot \log 3}{\log 5} \right)$$

We do this same thing but for the other summation to get

$$\left(\sum_{k=1}^{100} \frac{\log 5}{\log 3} \right)$$

The first summation simply is the sum of 1 through 20 multiplied to $\frac{\log 3}{\log 5}$

That gives us $\frac{20 \cdot 21}{2}$ which is 210

Multiplying it to the log part of it gives $\frac{210 \cdot \log 3}{\log 5}$

We do the same for the summation from 1 to 100 and realize its just 100 times the log part which gives

$$\frac{100 \cdot \log 5}{\log 3}$$

Multiplying both of our simplified summations cancels out the log parts and simply gives $210 \cdot 100$ as **21000 (E)**.

Problem 3.4.8 — Compute the number of ordered pairs of integers (a, b) , with $2 \leq a, b \leq 2021$, that satisfy the equation $a^{\log_b a^{-4}} = b^{\log_a ba^{-3}}$
Source: 2021 HMMT

Solution: Sometimes a good strategy in logarithm problems is to take the log of both sides. In this case, we should take the log with base a of both sides since it simplifies the left side severely.

The a from the exponent on the left side cancels out and we get

$$\log_b a^{-4} = \log_a b^{\log_a ba^{-3}} = \log_a b \cdot \log_a ba^{-3}$$

We can rewrite $\log_a ba^{-3}$ as $\log_a a^{-3} + \log_a b$. This simplifies to $-3 + \log_a b$.

Plugging this in gives

$$-4 \log_b a = \log_a b \cdot (-3 + \log_a b).$$

We will substitute x for $\log_a b$ and this means that $\log_b a$ is $\frac{1}{x}$.

Plugging this in gives

$$\frac{-4}{x} = x(-3 + x)$$

Multiplying both sides by x and simplifying gives

$$x^3 - 3x^2 + 4 = 0$$

Factoring gives $(x - 2)(x + 1)(x - 2) = 0$

We get that x must be $2, -1, 2$ which means that $\log_a b$ must be those numbers.

Thus, $b = a^2$ or $b = a^{-1}$. However, the second case isn't possible because both of our numbers must be positive. Thus, the number of answers are simply the square root of 2021 and rounded down after that. However, we must subtract 1 from that since the case of $(1, 1)$ won't work because both numbers have to be at least 2. Thus, the answer is **43**.

§3.5 Factoring Equations

In many problems, it's extremely important to factor to simplify the problem.

An important way to factor is to complete the square of the equation. This can sometimes make an equation of the circle especially when there are terms like x^2 and y^2 .

Problem 3.5.1 — There are several pairs of integers (a, b) satisfying $a^2 - 4a + b^2 - 8b = 30$. Find the sum of the sum of the coordinates of all such points.
Source: 2019 BMT

Solution: We can complete the square to get

$$(a - 2)^2 - 4 + (b - 4)^2 - 16 = 30$$

$$(a - 2)^2 + (b - 4)^2 = 50$$

From here we know that perfect squares $(a - 2)$ and $(b - 4)$ that sum up to 50 are $(49, 1)$ and $(25, 25)$ and $(1, 49)$. In total, there are 12 points because for each square we have 2 options since for example $a - 2$ can equal to 7 and -7 for its squared value to be 49.

We can now write out all 12 points by equating them individually to $a - 2$ and $b - 4$. Adding the sum of the coordinates gives us a final answer of **72**.

Theorem 3.5.2

An important factoring method is

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz)$$

Problem 3.5.3 — Let r_1, r_2, r_3 be the (possibly complex) roots of the polynomial $x^3 + ax^2 + bx + \frac{4}{3}$. How many pairs of integers a, b exist such that $r_1^3 + r_2^3 + r_3^3 = 0$

Source: 2019 BMT

Solution: In this problem, we will use the theorem for factoring discussed above. We know that

$$r_1^3 + r_2^3 + r_3^3 - 3r_1r_2r_3 = (r_1 + r_2 + r_3)(r_1^2 + r_2^2 + r_3^2 - r_1r_2 - r_1r_3 - r_2r_3)$$

From Vieta's theorem, we know that

$$r_1 + r_2 + r_3 = -a$$

$$r_1r_2 + r_1r_3 + r_2r_3 = b$$

$$r_1r_2r_3 = \frac{-4}{3}$$

We will plug this into our factored equation that we know from Theorem 3.5.2.

We can directly plug in 0 for $r_1^3 + r_2^3 + r_3^3$, and this makes the left side equate to 4.

Some manipulation gives $(r_1 + r_2 + r_3)^2 - 2(r_1r_2 + r_1r_3 + r_2r_3) = r_1^2 + r_2^2 + r_3^2$. This means that $r_1^2 + r_2^2 + r_3^2$ equals to $a^2 - 2b$

Our factored equation now becomes

$$4 = -a(a^2 - 2b)$$

Since we know that a must be a factor of 4, the possible values are 1, 2, 4, -1, -2, -4. We try all of them to find equation pairs for (a, b) , and we get **3** pairs in total.

Theorem 3.5.4

Basic Factoring Techniques

Difference of Squares: $a^2 - b^2 = (a - b)(a + b)$

Difference of Cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Sum of Cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Difference of powers: $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

Sophie's Germain Identity: $a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$

Theorem 3.5.5**Simon's Favorite Factoring Trick (SFFT)**

SFFT is used whenever there is a product of two variables added to the sum of the linear terms of both of the variables and there is also a constant. For example, $xy + 3y + 4x = 18$ is an example of such an equation.

Let's assume that we have an equation $xy + rx + sy = a$ (r, s , and a are integer constants), then the following equation holds: $(x + s)(y + r) = a + rs$

Problem 3.5.6 — m, n are integers such that $m^2 + 3m^2n^2 = 30n^2 + 517$. Find $3m^2n^2$.

Source: 1987 AIME

Solution: In this problem, we will try to use SFFT (Simon's Favorite Factoring Trick). It might not seem obvious to you immediately what the two variables are. In this case, they are m^2 and n^2 . We'll substitute x for m^2 and y for n^2 .

This gives us: $x + 3xy = 30y + 517$ and we want to find $3xy$

Factoring this using SFFT gives

$$(3x + 1)(y - 10) = 507$$

We know that 507 is $3 \cdot 13^2$

Now we can substitute our original variables back and try to pair up the factors to them.

$$3m^2 + 1 = ?$$

$$n^2 - 10 = ?$$

Trying out factor pairs shows that $3m^2 + 1$ should be 13 while $n^2 - 10$ should be 39. Solving for m^2 and n^2 gives m^2 as 4 and n^2 as 49. We simply want to find $3m^2n^2$ and get **588**.

Problem 3.5.7 — Express $\sqrt{25 + 21 \cdot 22 \cdot 23 \cdot 24 \cdot 25}$ as an integer.

Source: 2016 MMATH

Solution: In this problem, we can factor out 25 from the expression to get

$$\sqrt{25(1 + 21 \cdot 22 \cdot 23 \cdot 24)}$$

We can bring out the 25 and take the square root of it to get 5.

$$5 \cdot \sqrt{1 + 21 \cdot 22 \cdot 23 \cdot 24}$$

We can now plug in x instead of 21, and this means $x + 1 = 22$, $x + 2 = 23$, and $x + 3 = 24$.

We can plug this into the expression to get $5 \cdot \sqrt{1 + x(x + 1)(x + 2)(x + 3)}$

We can expand this to get $5 \cdot \sqrt{x^4 + 4x^3 + 6x^2 + 4x + 1}$.

We can factor the polynomial as $(x^2 + 3x + 1)^2$. We take the square root of that and our expression becomes $5 \cdot (x^2 + 3x + 1)$. Our value of x is 21 so we can plug that in to get our answer which is **2525**.

Problem 3.5.8 — The parabolas $y = x^2 + 15x + 32$ and $x = y^2 + 49y + 593$ meet at one point (x_0, y_0) . Find $x_0 + y_0$.

Source: 2016 CMIMC

Solution: In this problem, we will first add the two equations to see if we can come up with something. It gives us

$$x + y = x^2 + y^2 + 15x + 49y + 625$$

Rearranging it makes it become

$$x^2 + 14x + y^2 + 48y + 625$$

$$\text{We'll factor it as } (x + 7)^2 + (y + 24)^2 = 0$$

From this equation, $x + 7$ and $y + 24$ are both clearly 0 meaning $x = -7$ and $y = -24$. We sum those two numbers up to get **-31**.

Problem 3.5.9 — Let a, b, c be real numbers such that

$$a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

and $abc = 5$. The value of $(a - \frac{1}{b})^3 + (b - \frac{1}{c})^3 + (c - \frac{1}{a})^3$ can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers.

Source: 2020 BMT

Solution: We will substitute variables for the terms like $a - \frac{1}{b}$, $b - \frac{1}{c}$, and $c - \frac{1}{a}$. We will substitute x, y, z for them respectively.

This means that we're finding $x^3 + y^3 + z^3$.

We will try to use our factoring trick of

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz).$$

$$a + b + c - \frac{1}{a} - \frac{1}{b} - \frac{1}{c} \text{ which is equivalent to } x + y + z = 0$$

We can substitute the fact that $x + y + z = 0$ to our factoring trick equation and we get that $x^3 + y^3 + z^3 = 3xyz$.

We will now substitute our original expressions back such as $a - \frac{1}{b}$ and we get $(a - \frac{1}{b})^3 + (b - \frac{1}{c})^3 + (c - \frac{1}{a})^3 = 3(a - \frac{1}{b})(b - \frac{1}{c})(c - \frac{1}{a})$

On the right side, the product after expanding out becomes $abc - \frac{1}{abc}$.

We can evaluate that by plugging in $abc = 5$ and it gives us a value of $\frac{24}{5}$. However, we must multiply this by 3 to get that $(a - \frac{1}{b})^3 + (b - \frac{1}{c})^3 + (c - \frac{1}{a})^3$ is equivalent to $\frac{72}{5}$. Adding the numerator and denominator gives an answer of **77**.

Problem 3.5.10 — Suppose a, b, c , and d are non-negative integers such that

$$(a + b + c + d)(a^2 + b^2 + c^2 + d^2)^2 = 2023$$

Find $a^3 + b^3 + c^3 + d^3$

Source: 2023 CMIMC

Solution: In this problem, we will first factor 2023.

$$2023 = 7 \cdot 17^2$$

We notice our equation follows the format of the factors we wrote out. For example, we have two prime factors, and each can represent either $a + b + c + d$ or $a^2 + b^2 + c^2 + d^2$. On top of this, one of the prime factors (17) is raised to the power of 2 just like the term $a^2 + b^2 + c^2 + d^2$.

Since $(a^2 + b^2 + c^2 + d^2)$ is squared, we can say that it is 17.

$$a + b + c + d = 7$$

$$a^2 + b^2 + c^2 + d^2 = 17$$

Since a, b, c, d are nonnegative integers, we can test out squares that sum up to 17. For (a^2, b^2, c^2, d^2) , the possible values for (a, b, c, d) are $(0, 0, 1, 16)$ and $(0, 4, 4, 9)$.

We can take the square root of those 2 sets of 4 numbers to find the possible values for (a, b, c, d) : $(0, 2, 2, 3)$ and $(0, 0, 1, 4)$.

Only $(0, 2, 2, 3)$ will work since those 4 numbers sum to 7 (which is what we want because $a + b + c + d = 7$).

We plug in those numbers to find $a^3 + b^3 + c^3 + d^3$, and we get **43**.

§3.6 Systems of Equations

Many systems of equations problems will be a word problem in which you have to write out the systems. In others you will have to directly compute it by substituting or using some type of technique.

Also, if you have the symmetric sums given just like we saw in Vieta's Formula, then you can make a polynomial out of it.

For example, if

$$x + y = -5$$

$$xy = 4$$

Then, we can write out the polynomial $a^2 + 5a + 4$ where x and y are the roots of it.

Problem 3.6.1 — Let a, b , and c be real numbers such that $a - 7b + 8c = 4$ and $8a + 4b - c = 7$. Then $a^2 - b^2 + c^2$ is

- (A) 0 (B) 1 (C) 4 (D) 7 (E) 8

Source: 2002 AMC

Solution: In this problem, we will try to eliminate one of the variables to simplify our system of equations. We can do this by multiplying the first one by 8 and subtracting that from the second one.

$$8a - 56b + 64c = 32$$

$$8a + 4b - c = 7$$

Multiplying the first one by 8 gives us the new pair of equations shown. Subtracting them and simplifying gives

$$13c - 12b = 5$$

From here, we can plug in values to notice it works when b and c are equal to 5. Now, plugging this back into our original equations gives -1 for a . Now we can use these numbers and plug it into $a^2 - b^2 + c^2$ to get **1 (B)**.

Problem 3.6.2 — The solutions to the system of equations

$$\log_{225} x + \log_{64} y = 4$$

$$\log_x 225 - \log_y 64 = 1$$

are (x_1, y_1) and (x_2, y_2) . Find $\log_{30}(x_1 y_1 x_2 y_2)$.

Source: 2002 AIME

Solution: In this problem, we will make a substitution to replace the logarithm expressions.

We already know that $\log_a b = \frac{1}{\log_b a}$

Using this, we already know that if we plug in a for $\log_{225} x$ and b for $\log_{64} y$, then $\log_x 225$ is $\frac{1}{a}$ and $\log_y 64$ is $\frac{1}{b}$

Replacing the original values with these values in terms of a and b gives

$$a + b = 4$$

$$\frac{1}{a} + \frac{1}{b} = 1$$

Substituting $b = 4 - a$ into the second equation and solving gives

$$a = 3 \pm \sqrt{5} \text{ and } b = 1 \pm \sqrt{5}$$

Now we want to find the values of x and y that work for this system.

We know that x is equal to 225^a and y is equal to 64^b (this comes from the substitutions we made).

Thus, after we multiply those values in the solution format, we notice that it simply adds up the exponents of 225 separately and 64 separately. This means that we add both solutions that we found for a separately, and that is the exponent for 225, and we do the same thing for b .

Doing that gives us $\log_{30} 225^6 \cdot 64^2$.

This simply simplifies to $\log_{30} 30^{12}$ which gives a final answer of **12**.

Problem 3.6.3 — For real numbers (x, y, z) satisfying the following equations, find all possible values of $x + y + z$.

$$x^2y + y^2z + z^2x = -1$$

$$xy^2 + yz^2 + zx^2 = 5$$

$$xyz = -2$$

Source: SMT 2012 Algebra Test

Solution: In this problem, we will try to factor our equations. We notice that we can't get anywhere if we try to factor the first or second equation alone. Thus, we add them to see if we can get anything out of it.

$$x^2y + y^2z + z^2x + xy^2 + yz^2 + zx^2 = 4$$

We notice that

$$(x + y)(x + z)(y + z) = x^2y + y^2z + z^2x + xy^2 + yz^2 + zx^2 + 2xyz$$

Now we can substitute our values on the right side to get

$$(x + y)(x + z)(y + z) = 0$$

Since we know that the product equals 0, we know that one of the following must be true: $x = -y$, $x = -z$, or $y = -z$.

From here, we will assume that $x = -y$. Now since we want to find the sum of $x + y + z$, substituting $-y$ instead of x gives us that the sum is simply z .

Now using our substitution of $x = -y$, we will plug this into the original 3 equations to get

$$1. \ y^3 + y^2z - z^2y = -1$$

$$2. \ -y^3 + yz^2 + zy^2 = 5$$

$$3. \ y^2z = 2$$

Plugging in 2 for y^2z (since $y^2z = 2$) into the second equation gives

$$2. \ -y^3 + yz^2 = 3$$

We will factor out the y from the first and the new second equation above and then divide the first equation from the second to get

$$\frac{y^2 - z^2}{y^2 + yz - z^2} = 3$$

Now, we will multiply both sides by the denominator to get

$$y^2 - z^2 = 3y^2 + 3yz - 3z^2$$

Moving all terms to one side and factoring gives

$$(2y - z)(y + 2z) = 0$$

Now we know that $z = 2y$ or $y = -2z$

Plugging both of these cases into $y^2z = 2$ gives us 2 final solutions of 2 and $\sqrt[3]{\frac{1}{2}}$

Problem 3.6.4 — Suppose that x , y , and z are three positive numbers that satisfy the equations $xyz = 1$, $x + \frac{1}{z} = 5$, and $y + \frac{1}{x} = 29$. Then $z + \frac{1}{y} = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Source: 2000 AIME

Solution: In this problem, since we are already given 3 equations and there are only 3 variables, we will directly solve it.

Using $xyz = 1$, we will rewrite x as $\frac{1}{yz}$ and substitute that in replace of x .
Doing so gives the equations

$$\begin{aligned}\frac{1}{yz} + \frac{1}{z} &= 5 \\ y + yz &= 29\end{aligned}$$

Multiplying both sides of the first equation by yz gives

$$1 + y = 5yz$$

We now multiply the other equation by 5 to get

$$5y + 5yz = 145$$

Since we know that $5yz = 1 + y$, we can substitute it into the other equation to get

$$5y + y + 1 = 145$$

Solving for y gives 24. Plugging this value into the other equations to find x and z gives z as $\frac{5}{24}$ and x as $\frac{1}{5}$

Plugging this value into the expression we want to find gives us an answer of $\frac{1}{4}$. Our answer is just the sum of the numerator and denominator ($1 + 4$) which is 5.

Problem 3.6.5 — Let (a, b, c) be the real number solution of the system of equations $x^3 - xyz = 2$, $y^3 - xyz = 6$, $z^3 - xyz = 20$. The greatest possible value of $a^3 + b^3 + c^3$ can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Source: 2010 AIME

Solution: First of all, since we know that $a, b, and c$ are the solutions to this, we will simply plug them into the equations in replace of the other variables to get

$$\begin{aligned}a^3 - abc &= 2 \\ b^3 - abc &= 6 \\ c^3 - abc &= 20\end{aligned}$$

In this problem, we will try to manipulate the equations and try to find something. We will add abc to both sides of all equations to give us the following new equations

1. $a^3 = 2 + abc$
2. $b^3 = 6 + abc$
3. $c^3 = 20 + abc$

Now we will multiply all 3 equations together to get

$$a^3b^3c^3 = a^3b^3c^3 + 28a^2b^2c^2 + 172abc + 240$$

Simplifying this equation and substituting the variable e in replace of abc gives the new equation

$$7a^2 + 43a + 60 = 0$$

Factoring it gives: $(7a + 15)(a + 4)$

Our possible values for a or abc are $\frac{-15}{7}$ and -4

We now add our original 3 equations that are also labelled to get

$$a^3 + b^3 + c^3 = 28 + 3abc$$

Substituting our 2 possible values of abc into the equation above gives us a maximum answer of $\frac{151}{7}$ which boils down to **158**.

Problem 3.6.6 — Let a and b be complex numbers such that

$$(a + 1)(b + 1) = 2$$

and $(a^2 + 1)(b^2 + 1) = 32$

Compute the sum of all possible values of $(a^4 + 1)(b^4 + 1)$

Source: 2021 CMIMC

Solution: In this problem, we will first expand both given equations to get

$$ab + a + b = 1$$

$$a^2b^2 + a^2 + b^2 = 31$$

We will substitute x for ab and y for $a + b$ to make the first equation into

$$x + y = 1$$

We also plug these variables into the other equation to get that a^2b^2 as x^2 . $a^2 + b^2 = (a + b)^2 - 2ab$ which is $y^2 - 2x$

We sum those terms up to get that $a^2b^2 + a^2 + b^2 = 31$ is equivalent to $x^2 + y^2 - 2x = 31$

We rearrange $x + y = 1$ to get that $y = 1 - x$ and plug that into the other equation.

$$x^2 + (1 - x)^2 - 2x = 31$$

This simplifies to $x^2 - 2x - 15 = 0$ which factors to $(x + 3)(x - 5)$.

The solutions to this are $x = -3, 5$.

x is equivalent to ab . We plug in our values of ab to $ab + a + b = 1$ to get that for $ab = -3, 5$, $a + b$ will be $4, -4$ respectively.

Now to evaluate $(a^4 + 1)(b^4 + 1)$, we expand it and get $a^4b^4 + a^4 + b^4 + 1$.

We plug in our values of ab and $a + b$ by manipulating $a^4 + b^4$ as $(a^2 + b^2)^2 - 2a^2b^2$ and we get two possible answers for our expansion which are 548 and 612.

They sum to **1160**.

§3.7 Inequalities

Inequalities is a topic that sometimes shows up on the AMC and involves a lot of inspection and practice to be able to solve the problems.

Theorem 3.7.1

The **trivial inequality** states that for any real number x , $x^2 \geq 0$ always.

Theorem 3.7.2

The AM-GM Inequality (Arithmetic Geometric) states

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

Equality occurs when $x_1 = x_2 = \dots = x_n$. (Equality means the left side of the inequality equals to the right side)

Example 3.7.3

We have non-negative real numbers x, y , and z . Prove that $x^3 + y^3 + z^3 \geq x^2y + y^2z + z^2x$

Solution: In many of these problems, you just have to continuously try combinations of numbers to use an inequality for. In this case, we will use the AM-GM Inequality. You might wonder how I got the numbers below, but trust me: if you practice more then you will notice the combinations that can solve these inequalities.

Using AM-GM Inequality, we will write out 3 equations.

$$\begin{aligned} \frac{x^3 + x^3 + y^3}{3} &\geq \sqrt[3]{x^3 x^3 y^3} \\ \frac{y^3 + y^3 + z^3}{3} &\geq \sqrt[3]{y^3 y^3 z^3} \\ \frac{z^3 + z^3 + x^3}{3} &\geq \sqrt[3]{z^3 z^3 x^3} \end{aligned}$$

Simplifying all of the three inequalities above gives

$$\begin{aligned} x^3 + x^3 + y^3 &\geq 3x^2y \\ y^3 + y^3 + z^3 &\geq 3y^2z \\ z^3 + z^3 + x^3 &\geq 3z^2x \end{aligned}$$

We notice that now we can add our 3 simplified inequalities to get

$$3(x^3 + y^3 + z^3) \geq 3(x^2y + y^2z + z^2x)$$

Dividing both sides by 3 gives

$$x^3 + y^3 + z^3 \geq x^2y + y^2z + z^2x$$

Now this completes the proof as we arrive to the initial part.

Theorem 3.7.4

This is another major theorem known as the **Cauchy-Schwarz inequality**

For any real numbers $a_1, a_2, a_3 \dots a_n$ and $b_1, b_2, b_3 \dots b_n$, the inequality below is true

$$(a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2)(b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n)^2$$

Problem 3.7.5 — Find the minimum value of $\frac{9x^2 \sin^2 x + 4}{x \sin x}$ for $0 < x < \pi$.

Source: 1983 AIME

Solution: In this problem, we want to minimize the given expression. This shows us that it might be an inequality problem, since many inequality problems are about finding the minimum or maximum possible value.

We can rewrite $\frac{9x^2 \sin^2 x + 4}{x \sin x}$ as

$$9x + \frac{4}{x}$$

We can now use AM-GM Inequality because we know that after multiplying both of those terms, the x part will cancel out.

$$\frac{9x + \frac{4}{x}}{2} \geq \sqrt{36}$$

Multiplying both sides by 2 gives

$$9x + \frac{4}{x} \geq 12$$

This shows us that our expression's minimum value is 12 since it must always be more than 12, so we can simply say it's 12. However, this now means that we want the equality condition of the AM-GM Inequality. To confirm this is possible, we need to know if all the terms x_1, x_2 are equal.

In this case, we simply equate

$$9x = \frac{4}{x}$$

Bringing the x to one side gives us that x is equivalent to $\frac{2}{3}$. We can now graph x and see that there will be a point where the term equals to $\frac{2}{3}$. Thus, this confirms that our final answer is simply 12.

Problem 3.7.6 — If $x^2 + 3xy + y^2 = 60$ where x and y are real, then determine the maximum possible value of xy .

Source: CEMC

Solution: Using the AM-GM inequality gives that $\frac{x^2 + y^2}{2} \geq \sqrt{x^2 y^2} = xy$

Multiplying both sides by 2 gives $x^2 + y^2 \geq 2xy$

To maximize xy , we should minimize $x^2 + y^2$ because $3xy = 60 - (x^2 + y^2)$

Since the minimum value for $x^2 + y^2$ is $2xy$ from the inequality we found through the

AM-GM inequality, we get that $3xy = 60 - 2xy$.
Adding $2xy$ to both sides gives that $xy = 12$.

Now we must check our equality condition and it gives us $x^2 = y^2$ since those are the terms we plugged into AM-GM. It means that $x = y$.

We can plug this into the original equation given in the problem statement to get $5x^2 = 60$ which means $x^2 = 12$. In addition, the xy that we found the value to is equivalent to x^2 which is also 12.

Since our equality condition meets, the maximum value for xy is 12.

Problem 3.7.7 — Let x and y be positive real numbers such that $x + y = 1$. Show that

$$(1 + \frac{1}{x})(1 + \frac{1}{y}) \geq 9$$

Source: CMO

Solution: Expanding the inequality gives $1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \geq 9$
Subtracting 1 from both sides gives $\frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \geq 8$

We can take the common denominator and rewrite the inequality as

$$\frac{x+y+1}{xy} \geq 8$$

We can plug in our $x + y = 1$ expression to the inequality to get $\frac{2}{xy} = 8$. This simplifies to $xy \leq \frac{1}{4}$.

From here, we can use the initial condition of $x + y = 1$ and apply AM-GM to it.

$$\frac{x+y}{2} \geq \sqrt{xy}$$

$$\frac{1}{2} \geq \sqrt{xy}$$

Squaring both sides of the inequality above gives $xy \leq \frac{1}{4}$

Both inequalities become the same after we simplified which means that we have proven the statement that $(1 + \frac{1}{x})(1 + \frac{1}{y}) \geq 9$ by using AM-GM inequality.

Problem 3.7.8 — If a, b, c are real numbers with $a - b = 4$, find the maximum value of $ac + bc - c^2 - ab$.

Source: SMT 2018 Tiebreakers

Solution: Our goal in this problem should be to try to find the maximum value using the given equation $a - b = 4$. To do this, we first try to factor the expression we want to find the max of.

$$ac + bc - c^2 - ab = (a - c)(c - b)$$

Now we will try to use AM GM inequality with our factored expression. We will assume that our x_1 is $(a - c)$ and our x_2 is $(c - b)$

AM-GM gives:

$$\frac{a-c+c-b}{2} \geq \sqrt{(a-c)(c-b)}$$

The $a - c + c - b$ on the left simplifies to $a - b$ which is already given to be 4. Di-

viding it by 2 gives us 2 on the left side of the inequality.

$$2 \geq \sqrt{(a-c)(c-b)}$$

We can now square both sides of the inequality to get

$$4 \geq (a-c)(c-b)$$

4 is our final answer because the right side of our equation is simply the factored version of $ac + bc - c^2 - ab$.

§3.8 Functional Equations

The functional equations that show up on the AMC are relatively on the easy side. You just need to plug in values and try to determine a relationship.

An example of a functional equation is $f(x) = f(xy)$

On functional equations at the AMC, you will not have to worry about proving is something is surjective, injective, or bijective.

In a functional equation, a good strategy is to plug in 0 for a variable and try to determine a relationship.

Example 3.8.1

If $f(x + y) = f(xy)$ for all real numbers x and y , and $f(2019) = 17$, what is the value of $f(17)$?

Source: BMT 2019

Solution: In this problem, we will try to use our strategy of plugging in 0. We will plug in $x = 0$, and this gives us that $f(0 + y) = f(0)$ which becomes $f(y) = f(0)$

$f(y) = f(0)$ means that regardless of what value we plug in for y , it will always be equivalent to $f(0)$ which is a constant. In this case, that constant is 17 since we already know that $f(2019)$ is 17. Thus, $f(17)$ must also be **17**.

Problem 3.8.2 — Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(x)f(y) = f(xy)$. Find all possible values of $f(2017)$.

Source: 2017 HMMT

Solution: In this problem, we will first plug in 0 for both x and y .

This gives $f(0)^2 = f(0)$. The solutions to that is $f(0) = 0$ or 1.

Now to make a term in the function f 0 on the right side, we test the case when $x = y$.

This gives $f(x)^2 = f(0)$.

From here we already know our values of $f(0)$. When we plug in 0 for it, we get that $f(x) = 0$. Thus, 0 is a possible value for $f(2017)$.

Now we test the case for $f(0) = 1$. We find out that possible values are $f(x) = -1, +1$. However, we must notice one idea. Since in both cases our function $f(x)$ is a constant (it's always one value no matter what we plug in for x , we can directly plug in both of those numbers in the equation given initially: $f(x)f(y) = f(xy)$.

Checking $f(x) = 1$ shows that it is possible because $1 \cdot 1 = 1$.
However, when we check $f(x) = -1$, it's not possible because $-1 \cdot -1 \neq -1$.

Thus, the only possible values for $f(2017)$ are **0 and 1**.

Problem 3.8.3 — For positive real numbers x and y , let $f(x, y) = x^{\log_2 y}$. The sum of the solutions to the equation

$$4096f(f(x, x), x) = x^{13}$$

can be written in simplest form as $\frac{m}{n}$. Compute $m + n$.

Source: 2016 PUMAC

Solution: In this problem, we will work with the given equation and plug in (x, x) . This gives us $4096f(x^{\log_2 x}, x) = x^{13}$

From here, we will again work with $f(x^{\log_2 x}, x)$ and this makes the equation

$$4096x^{(\log_2 x)^2} = x^{13}.$$

Now, we will temporarily get rid of the logarithm term by substituting a for $\log_2 x$. This means that $x = 2^a$.

Making the substitutions turns the equation into

$$4096 \cdot 2^{a^3} = 2^{13a}.$$

From here, we will rewrite 4096 as 2^{12}

$$2^{12} \cdot 2^{a^3} = 2^{13a}.$$

We can add the exponents on the left to get

$$2^{a^3+12} = 2^{13a}$$

Now, we can equate the exponents because the bases are the same.

$$a^3 + 12 = 13a$$

We can bring all terms to one side and find the roots to get $a = -3, 1, 4$

Since we know that $x = 2^a$, we can plug in each of those 3 numbers separately and find that the possible solutions are $x = 2, 8, \frac{1}{16}$.

Adding up all 3 solutions gives $\frac{161}{16}$ and the answer is $161 + 16$ which is **177**

Problem 3.8.4 — Let $f : R^+ \rightarrow R^+$ be a function such that for all $x, y \in R^+$, $f(x)f(y) = f(xy) + f(\frac{x}{y})$, where R^+ represents the positive real numbers. Given that $f(2) = 3$, compute the last two digits of $f(2^{2^{2020}})$

Source: 2020 BMT

Solution: In this problem, we first plug in $x = 2$ and $y = 1$ to get $f(2)f(1) = f(2) + f(2)$. Plugging in 3 for $f(2)$ gives that $f(1) = 2$

Now plugging in $x = 2$ and $y = 2$, we get that $f(2^2) = f(4) = 7$

Plugging in $x = 4$ and $y = 4$ into the functional equation gives that $f(2^{2^2}) = 47$. After that, we get that $f(2^{2^3}) = 7$. We notice that this pattern continues

Continuing this in mod 100 gives that $f(2^{2^{2020}}) = \mathbf{47}$

These problems make it clear that most functional equation problems are about plugging in numbers. It's almost always beneficial to plug in the number 0. In addition, if you have something like $f(x - y)$, then it's worth testing out the equation after plugging in $x = y$.

§3.9 Telescoping

Definition 3.9.1

Telescoping series is about noticing a pattern and how many of the numbers in something like a summation can cancel out. The best way to master this topic is to simply practice as there isn't much theory to go with it. A good tip is to write a few of the terms and see if anything cancels

Example 3.9.2

$$\prod_{i=1}^{10} \frac{i}{i+1}$$

The big Pi sign represents product symbol. It's similar to as ummation but instead you multiply the terms. In this case, we will write out each term since it might make it easier for you to recognize the pattern.

$$\prod_{i=1}^{10} \frac{i}{i+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdot \frac{6}{7} \cdot \frac{7}{8} \cdot \frac{8}{9} \cdot \frac{9}{10} \cdot \frac{10}{11}$$

Clearly, most of the denominators cancel out with the numerators since they are the same. The only numbers that are left are 1 in the numerator and 11 in the denominator. Thus, the product after cancelling out everything is $\frac{1}{11}$

Theorem 3.9.3

Fraction Decomposition

While decomposing a fraction, you first have to factor the denominator and then write out the terms according to the way shown below. This is an important technique as it can help you in telescoping often and finding other interesting results.

Linear Factors: $\frac{x+2}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$

Repetition of Linear Factors: $\frac{x-4}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$

Higher Order Factors: $\frac{x^2+3x-5}{(x-2)(x^2+16)} = \frac{A}{x-2} + \frac{Bx-C}{x^2+16}$

You can use the techniques above to decompose a fraction.

Example 3.9.4

What is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$$

Solution: In this problem, we will see a popular technique. $\frac{1}{1 \cdot 2}$ is the same thing as $\frac{1}{1} - \frac{1}{2}$ and $\frac{1}{2 \cdot 3}$ is the same thing as $\frac{1}{2} - \frac{1}{3}$. We notice that $\frac{1}{n \cdot (n+1)}$ is $\frac{1}{n} - \frac{1}{n+1}$.

Using that info, we can rewrite the question we're asked as

$$\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} \dots$$

Clearly everything cancels out since our sum is infinite, except for 1. Thus, our answer is 1.

Problem 3.9.5 — Consider the sequence defined by $a_k = \frac{1}{k^2+k}$ for $k \geq 1$. Given that $a_m + a_{m+1} + \dots + a_{n-1} = 1/29$, for positive integers m and n with $m < n$, find $m + n$.

Source: 2002 AIME

Solution: We will use fraction decomposition techniques that we learned.

Clearly, $\frac{1}{k^2+k}$ can be simplified to

$$\frac{1}{k^2+k} = \frac{1}{k(k+1)} \text{ which is } \frac{1}{k} - \frac{1}{k+1}$$

From here, we know that we simply sum all these numbers from a_m to a_{n-1} and most of the terms cancel out except for $\frac{1}{m} - \frac{1}{n}$.

We can equate that to $\frac{1}{29}$

$$\frac{1}{m} - \frac{1}{n} = \frac{1}{29}$$

Multiplying both sides by $29mn$ gives

$$29n - 29m = mn.$$

Bringing all terms to one side to apply SFFT (Simon's Favorite Factoring Trick) gives us $mn + 29m - 29n$ which simplifies to $(m - 29)(n + 29) = -29 \cdot 29$

Clearly we must work separately with $m - 29$ and $n + 29$ and try factor pairs of -841 to see which one will work since we know that both m, n are positive and m is less than n

Trying all values slowly shows that $m - 29$ must be -1 while $n + 29$ must be 841 .

$$m - 29 = -1 \text{ (this simplifies to } m = 28)$$

$$n + 29 = 841 \text{ (this simplifies to } n = 812)$$

Now we simply add up m and n to get our final answer as **840**.

Problem 3.9.6 — An infinite sequence of real numbers $a_1, a_2, \dots$ satisfies the recurrence $a_{n+3} = a_{n+2} + 2a_{n+1} + a_n$ for every positive integer n . Given that $a_1 = a_3 = 1$ and $a_{98} = a_{99}$, compute $a_1 + a_2 + \dots + a_{100}$.

Source: 2016 HMMT

Solution: In this problem, we will try to cancel out terms again as that is the way to telescope many summations. We will ignore a_1, a_2, a_3 for now.

$$a_4 = a_3 - 2a_2 + a_1$$

$$a_5 = a_4 - 2a_3 + a_2$$

$$a_6 = a_5 - 2a_4 + a_3$$

$$a_7 = a_6 - 2a_5 + a_4$$

Now when we add the four equations and combine it with the there terms we skipped initially (a_1, a_2, a_3 , we get most things to cancel out. We are only left with
 $a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = a_6 - a_5 + 2a_1 + a_3$

We should be able to realize that when we continue to write all of those numbers till a_{100} , then the summation will simply equate to $a_{99} - a_{98} + 2a_1 + a_3$. We have all four of these variables given, and plugging them in simply gives **3**.

Problem 3.9.7 — Compute $\sum_{n=1}^{\infty} \frac{2^{n+1}}{8 \cdot 4^n - 6 \cdot 2^n + 1}$
Source: 2016 CHMMC

Solution: In this problem, since many of our terms in the summation involve 2^n , let's plug in $a = 2^n$. This turns the inside part of the summation into

$$\frac{2a}{8a^2 - 6a + 1}$$

We can factor this as $\frac{2a}{(4a-1)(2a-1)}$

We can use our fraction decomposition method to get

$$\frac{2a}{(4a-1)(2a-1)} = \frac{x}{4a-1} + \frac{y}{2a-1}$$

$$\text{This becomes } \frac{2a}{(4a-1)(2a-1)} = \frac{2ax - x + 4ay - y}{(4a-1)(2a-1)}$$

We can equate the numerators and we get $a(2x + 4y) - (x + y) = 2a$

We can equate the coefficients to get

$$2x + 4y = 2$$

$$x + y = 0$$

The second equation gives $x = -y$ and when we plug this into the first we get that $y = 1$. This also gives that $x = -1$

Our fractional decomposition now becomes $\frac{-1}{4 \cdot 2^n - 1} + \frac{1}{2 \cdot 2^n - 1}$

We can now plug this back into the summation to get

$$\sum_{n=1}^{\infty} \frac{1}{2 \cdot 2^n - 1} - \frac{1}{4 \cdot 2^n - 1}$$

Now we can begin writing the terms out to get

$$\frac{1}{3} - \frac{1}{7} + \frac{1}{7} - \frac{1}{15} + \dots$$

All the terms in the summation cancel out except for the first one, and our answer is $\frac{1}{3}$.

Problem 3.9.8 — $\log_2 9 \cdot \log_3 16 \cdot \log_4 25 \cdot \log_5 36 \cdot \dots \cdot \log_{999} 1000000$

Evaluate the expression above

Source: 2013 CHMMC

Solution: Since we know that $\log_a b = \frac{\log b}{\log a}$, we can apply this to the product we want to find.

$$\frac{\log 9}{\log 2} \cdot \frac{\log 16}{\log 3} \cdot \frac{\log 25}{\log 4} \cdot \frac{\log 36}{\log 5} \cdot \dots \cdot \frac{\log 1000000}{\log 999}$$

We can use the identity $\log x^2 = 2 \cdot \log x$. We do this for the numerators to get

$$\frac{2 \log 3}{\log 2} \cdot \frac{2 \log 4}{\log 3} \cdot \frac{2 \log 5}{\log 4} \cdot \frac{2 \log 6}{\log 5} \cdot \dots \cdot \frac{2 \log 1000}{\log 999}$$

Most of the terms cancel out in this and our expression becomes $2^{998} \cdot \log_2 1000$.

§3.10 Floor Functions

Definition 3.10.1

Pretend I have a number 6.2.

$\lfloor 6.2 \rfloor$: If you put a number the way I did on the left, then it means it's asking for the greatest integer less than that number which is obviously 6.

$\{6.2\}$: If you put the number the way I did on the left, then it's asking for the fractional part of the number which is 0.2.

Theorem 3.10.2

This is a crucial theorem that you need to know if you want to solve many floor function problems.

For any number x , $\{x\} + \lfloor x \rfloor = x$

It's very helpful to plug in the formula above. It can sometimes allow you to simplify the expression.

Since $0 \leq \{x\} < 1$, it's sometimes useful to plug in $\{x\} = x - \lfloor x \rfloor$ and bound the expression.

Problem 3.10.3 — Find the smallest real number x such that $\frac{x}{\lfloor x \rfloor} = \frac{2002}{2003}$

Source: ARML

Solution: In this problem, we substitute $\lfloor x \rfloor + \{x\}$ for x to get

$$\frac{\lfloor x \rfloor + \{x\}}{\lfloor x \rfloor} = \frac{2002}{2003}$$

Then, we multiply both sides by $2003 \cdot \lfloor x \rfloor$ to get

$$2003 \cdot \lfloor x \rfloor + 2003 \cdot \{x\} = 2002 \cdot \lfloor x \rfloor$$

Simplifying gives $\lfloor x \rfloor = -2003 \cdot \{x\}$

Dividing both sides by -2003 gives $\{x\} = \frac{-\lfloor x \rfloor}{2003}$.

Since $0 < \{x\} < 1$ always,

$$0 \leq \frac{-\lfloor x \rfloor}{2003} < 1$$

We multiply both sides by -2003 to get

$$-2003 < x \leq 0.$$

Clearly, since we want to minimize x , the least value for $\lfloor x \rfloor$ is -2002 .

We plug in $\lfloor x \rfloor$ which is -2002 to the original equation which is $\frac{\lfloor x \rfloor + \{x\}}{\lfloor x \rfloor} = \frac{2002}{2003}$

Plugging it in gives $x = \frac{-2002^2}{2003}$

Problem 3.10.4 — For each real number x , let $\lfloor x \rfloor$ denote the greatest integer function— that does not exceed x . For how many positive integers n is it true that $n < 1000$ and that $\lfloor \log_2 n \rfloor$ is a positive even integer?

Source: 1996 AIME

Solution: In this problem, we will work in cases. Since $\lfloor \log_2 n \rfloor$ is a even number, we can test cases assuming that the expression is 2, 4, 6, 8, 10, etc.

If $\lfloor \log_2 n \rfloor$ is 2, then n has to be between 2^2 and 2^3 . There are $2^3 - 2^2$ or 4 numbers in between.

If $\lfloor \log_2 n \rfloor$ is 4, then n has to be between 2^4 and 2^5 . There are $2^5 - 2^4$ or 16 numbers in between.

If $\lfloor \log_2 n \rfloor$ is 6, then n has to be between 2^6 and 2^7 . There are 64 numbers in between.

If $\lfloor \log_2 n \rfloor$ is 8, then n has to be written between 2^8 and 2^9 . There are 256 numbers in between.

We notice that $\lfloor \log_2 n \rfloor$ can't be 10 or higher because then n exceeds the bound of 1000. Thus, for the 4 cases, we sum up all four numbers to get $4 + 16 + 64 + 256$ which is **340**.

Problem 3.10.5 — Let x be a real number selected uniformly at random between 100 and 200. If $\lfloor \sqrt{x} \rfloor = 12$, find the probability that $\lfloor \sqrt{100x} \rfloor = 120$. ($\lfloor v \rfloor$ means the greatest integer less than or equal to v .)

- (A) $\frac{2}{25}$ (B) $\frac{241}{2500}$ (C) $\frac{1}{10}$ (D) $\frac{96}{625}$ (E) 1

Source: 1989 AHSME

Solution: In this problem, since $\lfloor \sqrt{x} \rfloor = 12$, we know that $12 \leq \lfloor \sqrt{x} \rfloor < 13$

We can square both sides to get $144 \leq x < 169$ which gives a total possible size of 25 for x in the denominator.

From $\lfloor \sqrt{100x} \rfloor = 120$, we get that $120 \leq \lfloor \sqrt{100x} \rfloor < 121$. Squaring the entire inequality and dividing by 100 gives

$$144 \leq x < 146.41$$

In the numerator now, we get $146.41 - 144$ which is 2.41.

The probability is $\frac{2.41}{25}$ which is equivalent to $\frac{241}{2500}$ (B)

Problem 3.10.6 — Find the number of positive integers n satisfying

$$\lfloor \frac{n}{2014} \rfloor = \lfloor \frac{n}{2016} \rfloor$$

Source: 2015 CHMMC

Solution: Let's assume that $\lfloor \frac{n}{2014} \rfloor = \lfloor \frac{n}{2016} \rfloor = a$ where a is an integer. This means that

$$a \leq \lfloor \frac{n}{2014} \rfloor < a + 1$$

$$a \leq \lfloor \frac{n}{2016} \rfloor < a + 1$$

We can multiply the first inequality by 2014 and the second one by 2016 which gives

$$2014a \leq n < 2014a + 2014$$

$$2016a \leq n < 2016a + 2016$$

Since we know that n is at least both $2014a$ and $2016a$, we can simply say that it is at least $2016a$ since any number that is at least $2016a$ will also automatically be at least $2014a$. We can apply this similar logic to the numbers $2014a + 2014$ and $2016a + 2016$ to get

$$2016a \leq n < 2014a + 2014$$

Value of a	Range of Values of n	Number of Values of n
0	$[0, 2013]$	2014
1	$[2016, 4027]$	2012
2	$[4032, 6041]$	2010
3	$[4048, 6055]$	2008
...	...	...

Now we notice that the number of values for n are all even numbers from 2 to 2014.

The sum of the first n even numbers is $n(n + 1)$ and we have 1007 even numbers.

Our sum is $1007 \cdot 1008$. However, we must subtract 0 since in the case for $a = 0$, we included the value when n is 0, but our problem statement wants n to be a **positive** integer.

Our answer is $(1007 \cdot 1008) - 1$ which is **1015055**

Problem 3.10.7 — How many positive integers n satisfy

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor?$$

(Recall that $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

(A) 2 (B) 4 (C) 6 (D) 30 (E) 32

Source: 2020 AMC 10

Solution: Since $\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor$, we know that $\frac{n + 1000}{70} \leq \lfloor \sqrt{n} \rfloor < \frac{n + 1070}{70}$

We can solve each inequality separately.

$$n - 70\sqrt{n} + 1000 \leq 0$$

$$n - 70\sqrt{n} + 1070 > 0$$

From the first inequality, we can factor it to get $(\sqrt{n} - 20)(\sqrt{n} - 50)$

Using the quadratic formula on the second inequality, we get that $\sqrt{n} > 35 + \sqrt{155}$ or

$\sqrt{n} < 35 - \sqrt{155}$. Since n is an integer, it's easy to bound n in this case.

After using all 4 inequalities that we made after factoring and using the quadratic formula, we simply find that there are **6 (C)** solutions in total.

Problem 3.10.8 — Compute the sum of the real solutions to $\lfloor x \rfloor \{x\} = 2020x$. Here, $\lfloor x \rfloor$ is defined as the greatest integer less than or equal to x , and $\{x\} = x - \lfloor x \rfloor$.

Source: 2022 BMT

Solution: In this problem, we will plug in $x = \lfloor x \rfloor + \{x\}$ into the given equation. Doing so gives $\lfloor x \rfloor \{x\} = 2020(\lfloor x \rfloor + \{x\})$.

Simplifying gives $\lfloor x \rfloor \{x\} = 2020 \lfloor x \rfloor + 2020 \{x\}$.

We should notice that this looks like the form in which we apply Simon's Favorite Factoring Trick to. Doing so gives

$$(\lfloor x \rfloor - 2020)(\{x\} - 2020) = 2020^2$$

Since we know that $0 \leq \{x\} < 1$, we can subtract 2020 from both sides to get $\{x\} - 2020$ which is bounded by -2010 and -2009 .

Dividing this from 2020^2 to get $\lfloor x \rfloor - 2020$ gives us our bounds for this expression.

$$\frac{-2020^2}{2019} + 2020 < \lfloor x \rfloor \leq 0$$

Since we know that $\lfloor x \rfloor$ is always an integer, using the two bounds we get that the only possible values for it are 0 and -1 . We can find this by rounding the expression on the left and we simply find that those two numbers are the only possible values.

From there, we can plug the fact that $\lfloor x \rfloor$ is either 0 or -1 to $\lfloor x \rfloor \{x\} = 2020 \lfloor x \rfloor + 2020 \{x\}$. Doing this for 0 gives that the fractional part must also equal to 0, so our solution for this case is $0 + 0$ which is 0.

Doing the same for $\lfloor x \rfloor = -1$ gives that the fractional part must be $\frac{2020}{2021}$. We now add up the floor and the fractional part to get $-1 + \frac{2020}{2021}$ to get $\frac{-1}{2021}$ as our real solution. We sum up our real solutions to get $0 + \frac{-1}{2021}$ which becomes $\frac{-1}{2021}$.

§3.11 Recursive Sequences

In this topic, you often get an expression involving values like x_n, a_n, b_n . You can plug in previous terms and continue to find more using the previously found ones. Recursive sequences are clearly infinite sequences.

Sometimes a helpful strategy is to add or multiply the recursive sequences that you have. Also, in many problems, you can simply brute force it out by continuing to plug in numbers until you find the desired value.

Problem 3.11.1 — The sequence of numbers $t_1, t_2, t_3, \dots$ is defined by $t_1 = 2$ and

$$t_{n+1} = \frac{t_n - 1}{t_n + 1}$$

for every positive integer n . Determine the value of t_{999} .

Source: COMC

Solution: We can try to find a pattern in our terms by finding out the next few terms.

$$\begin{aligned} t_2 &= \frac{t_1 - 1}{t_1 + 1} = \frac{1}{3} \\ t_3 &= \frac{t_2 - 1}{t_2 + 1} = \frac{-1}{2} \\ t_4 &= \frac{t_3 - 1}{t_3 + 1} = -3 \\ t_5 &= \frac{t_4 - 1}{t_4 + 1} = 2 \end{aligned}$$

Since we get that $t_1 = t_5 = 2$, the remaining terms will repeat after. The period of this sequence is $5 - 1 = 4$.

This means that the terms will repeat itself every 4 numbers. Clearly $t_4 = t_8 = t_{12} = t_{16} = \dots = t_{996} = -3$

Using that, we can go forward 3 more terms to see that $t_{999} = \frac{-1}{2}$.

Problem 3.11.2 — A sequence of numbers is defined recursively by $a_1 = 1$, $a_2 = \frac{3}{7}$, and

$$a_n = \frac{a_{n-2} \cdot a_{n-1}}{2a_{n-2} - a_{n-1}}$$

for all $n \geq 3$. Then a_{2019} can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

(A) 2020 (B) 4039 (C) 6057 (D) 6061 (E) 8078

Source: 2019 AMC

Solution: In this problem, we will simply try to directly find the value of a_{2019} by finding the values of a_3 and a_4 . We will find those earlier terms to try to look for a pattern.

$$a_3 = \frac{a_1 \cdot a_2}{2a_1 - a_2} = \frac{3}{11}$$

$$a_4 = \frac{a_2 \cdot a_3}{2a_2 - a_3} = \frac{3}{15}$$

We notice a pattern with the denominator. Our numerator always remains 3, but the denominator goes up by 4 each time.

It's safe to assume that this pattern will continue. From observation, it's easy to tell that $a_n = \frac{3}{4n-1}$

We can use that equation above to get that $a_{2019} = \frac{3}{8075}$, and our final answer is $3 + 8075$ which is **8078 (E)**.

Problem 3.11.3 — Let $a_0 = 2, b_0 = 1$, and for $n \geq 0$, let

$$a_{n+1} = a_n + b_n + \sqrt{a_n^2 + b_n^2}$$

$$b_{n+1} = a_n + b_n - \sqrt{a_n^2 + b_n^2}$$

Source: 2012 HMMT

Solution: In many cases, a good thing to do when you have multiple recurrences is to add them, subtract them, or multiply.

$$a_{n+1} + b_{n+1} = 2(a_n + b_n)$$

$$a_{n+1} \cdot b_{n+1} = 2(a_n \cdot b_n)$$

$$a_0 + b_0 = -1$$

We simply multiply 2 to this and this continues 2012 times to get that $a_{2012} + b_{2012} = -2^{2012}$

We know that $a_0 b_0$ is -2 . To find $a_{2012} b_{2012}$, we notice that -2^{2013} .

We can make a polynomial with roots a_{2012} and b_{2012} to get $x^2 + 2^{2012}x - 2^{2013}$

Since a_{2012} must be more than b_{2012} because you constantly subtract $\sqrt{a_n^2 + b_n^2}$ to get b_n while you add it for a_n .

Using the quadratic formula we find that the larger root is $-2^{2011} + 2^{2006}\sqrt{2^{2010} + 2}$ which is a_{2012}

§3.12 Cumulative Problems

Problem 3.12.1 — Let a, b, c be the solutions of the equation $x^3 - 3 \cdot 2021^2 x = 2 \cdot 2021^3$.

Compute $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

Source: 2021 BMT

Solution: In this problem, we will first rewrite $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab+ac+bc}{abc}$$

Using Vieta's Theorem to write equations for the roots using the given polynomial, we get that

$$a + b + c = 0$$

$$ab + ac + bc = -3 \cdot 2021^2$$

$$abc = 2 \cdot 2021^3$$

Plugging in the expressions for $ab + ac + bc$ and abc gives us $\frac{-3 \cdot 2021^2}{2 \cdot 2021^3}$ which simplifies to $\frac{-3}{4042}$

Problem 3.12.2 — Distinct prime numbers p, q, r satisfy the equation

$$2pqr + 50pq = 7pqr + 55pr = 8pqr + 12qr = A$$

for some positive integer A . What is A ?

Source: 2018 HMMT

Solution: In this problem, it's obvious that A is divisible by pqr . Thus, we can divide everything by pqr and $\frac{A}{pqr}$ will still be an integer.

$$2 + \frac{50}{r} = 7 + \frac{55}{q} = 8 + \frac{12}{p} = \frac{A}{pqr}$$

From here, we know that in each of the individual equations, r must divide 50, q must divide 55, and p must divide 12. Since all three of those numbers are prime, we can write out the unique prime factors for each.

$$r : 2, 5$$

$$q : 5, 11$$

$$p : 2, 3$$

Using the information that the three numbers p, q, r are distinct and trying out all cases gives that $p = 3$, $q = 11$, $r = 5$. Plugging this into any one of the equations such as $2pqr + 50pq = A$ gives us that A is **1980**.

Problem 3.12.3 — Compute the sum of all real numbers x which satisfy the following equation $\frac{8^x - 19 \cdot 4^x}{16 - 25 \cdot 2^x}$

Source: 2021 PUMAC

Solution: In this problem, we notice how terms like 8^x which is $(2^x)^3$ and 4^x which is $(2^x)^2$ and 2^x are constantly used. Thus, we substitute a for 2^x . This makes the equation become

$$\frac{a^3 - 19a^2}{16 - 25a} = 2$$

Multiplying both sides by $16 - 25a$ and bringing all terms to one side gives

$$a^3 - 19a^2 + 50a - 32 = 0$$

Factoring it gives $(a - 1)(a - 2)(a - 16) = 0$ and the solutions are $a = 1, 2, 16$.

Since we know that $2^x = a$, we can solve for the value of x for each case and get $x = 0, 1, 4$ respectively for $a = 1, 2, 16$.

Thus, our answer is simply $0 + 1 + 4$ which is **5**

Problem 3.12.4 — For real numbers a, b , and c , the roots of the polynomial

$$x^5 - 10x^4 + ax^3 + bx^2 + cx - 320$$

form an arithmetic progression. Find $a + b + c$.

Source: 2023 Purple Comet

Solution: Since our roots are in arithmetic progression, let's say that our roots are $x - 2y, x - y, x, x + y, x + 2y$.

Now we can apply Vieta's Theorem to these roots to get

$$\begin{aligned} x - 2y + x - y + x + x + y + x + 2y &= 5x = 10 \\ (x - 2y)(x + 2y)(x - y)(x + y)x &= 320 \end{aligned}$$

From the equation which involved the sum of the roots, we get that $x = 2$.

We plug this into the equation in which we multiplied the roots and related it to the last coefficient to find that $y = 3$. This means that our roots are $-4, -1, 2, 5, 8$.

This means that our polynomial is $(x + 4)(x + 1)(x - 2)(x - 5)(x - 8)$

Since we know that the sum of all coefficients in a polynomial can be found by simply plugging in 1, the sum of the roots in this case are $(1 + 4)(1 + 1)(1 - 2)(1 - 5)(1 - 8)$ which is -280 .

Now we can find the sum of the coefficients in $x^5 - 10x^4 + ax^3 + bx^2 + cx - 320$ to get that it is $1 + (-10) + a + b + c + (-320)$. We equate this to -280 .

$$-329 + a + b + c = -280$$

Adding 329 to both sides gives $a + b + c = 49$ which is our answer.

Problem 3.12.5 — Find the unique real number c such that the polynomial

$$x^3 + cx + c$$

has exactly two real roots.

Source: 2020 CMIMC

Solution: In this problem, since there are two real roots, we know that there is one more root left. However, there should only be an even number of nonreal roots, but 1 is odd. Thus, that third root must also be *real*.

Our repeated root is r and the one not repeated is s .

We will now apply Vieta's Theorem to get the equations

$$2r + s = 0$$

$$r^2 + 2rs = c$$

$$r^2s = -c$$

From the first equation, we get $s = -2r$.

Now we can equate the second and third equation by multiplying the third one by -1 to get $-r^2s = c$.

Now we can plug this into the second equation to get $r^2 + 2rs = -r^2s$. Bringing all

terms to one side gives

$$r^2 + s(2r + r^2)$$

Then we plug in $s = -2r$ to the equation above to get $3r^2 = 2r^3$ which gives r as $\frac{3}{2}$.

We can plug this into our first equation to find s , and then use that to find the value of c which simply is $\frac{-27}{4}$

Problem 3.12.6 — Let a, b, c , and d be positive real numbers which satisfy the system of equations

$$(a + b)(c + d) = 143,$$

$$(a + c)(b + d) = 150,$$

$$(a + d)(b + c) = 169.$$

Find the smallest possible value of $a^2 + b^2 + c^2 + d^2$

Source: 2016 CMIMC

Solution: In this problem, we can first rewrite the term that we want to find which is $a^2 + b^2 + c^2 + d^2$

$$a^2 + b^2 + c^2 + d^2 = (a + b + c + d)^2 - 2(ab + ac + ad + bc + bd + cd)$$

Expanding all three of the given equations gives

$$ab + ac + bc + bd = 143$$

$$ab + ad + bc + cd = 150$$

$$ab + ac + bd + cd = 169$$

Adding all the equations gives $2(ab + ac + ad + bc + bd + cd) = 462$

We substitute this into the equation we manipulated to get

$$a^2 + b^2 + c^2 + d^2 = (a + b + c + d)^2 - 462$$

To minimize $a^2 + b^2 + c^2 + d^2$, we should minimize $(a + b + c + d)^2$

Now we can apply AM-GM ($\frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2}$) to the three given equations

For example, on the first one we can say $x_1 = a + b$ and $x_2 = c + d$

$$a + b + c + d \geq 2\sqrt{(a + b)(c + d)} \text{ which becomes } a + b + c + d \geq 2\sqrt{143}$$

We can now square the inequality to get $(a + b + c + d)^2 \geq 572$

We can now do the same for all the other equations and we get that $(a + b + c + d)^2$ must be greater than or equal to 572, 600, and 676.

The minimum value of $(a + b + c + d)^2$ is 676 because it must be that for it to satisfy the other two inequalities which involve 572 and 600. For example, if we say the minimum is 600, then it won't satisfy the inequality of being larger than 676.

Now we simply plug in 676 for $(a + b + c + d)^2$ in $a^2 + b^2 + c^2 + d^2 = (a + b + c + d)^2 - 462$ to get that the answer we're looking for is simply $676 - 462$ which is **214**

Problem 3.12.7 — Let a, b , and c be positive integers satisfying

$$a^4 + a^2b^2 + b^4 = 9633$$

$$2a^2 + a^2b^2 + 2b^2 + c^5 = 3605$$

What is the sum of all distinct values of $a + b + c$.

Source: 2012 PUMAC

Solution: We will try to add both of these equations as it is a good strategy in many algebra problems.

Adding gives

$$a^4 + 2a^2b^2 + b^4 + 2a^2 + 2b^2 + c^5 = 13238$$

We notice this factors as $(a^2 + b^2 + 1)^2 + c^5 = 13239$

From here, since we know that a, b, c are all integers, $a^2 + b^2 + 1$ is also one. Thus, we can simply plug in integer values for c since there aren't many that we can plug in.

We can only plug in 1, 2, 3, 4, 5 for c and we plug in each individually into the equation $(a^2 + b^2 + 1)^2 + c^5 = 13239$

Doing this for all values of c gives us that $(a^2 + b^2 + 1)^2$ is 13238, 13207, 12996, 12215, 10114 respectively for $c = 1, 2, 3, 4, 5$.

Only 12996 is a perfect square which confirms that $c = 3$. We get that $a^2 + b^2 + 1 = 114$ which simplifies to $a^2 + b^2 = 113$.

The only two perfect squares that sum up to 113 are when both are 7 and 8.

Since we want to find $a + b + c$, our numbers are 7, 8, 3 which sums to **18**.

Problem 3.12.8 — If f is a monic cubic polynomial with $f(0) = 64$, and all roots of f are non-negative real numbers, what is the largest possible value of $f(1)$? (A polynomial is monic if it has a leading coefficient of 1.)

Source: 2012 SMT

Solution: In this problem, let's assume that the roots of this polynomial are r_1, r_2, r_3 . Our polynomial now is

$$f(x) = (x - r_1)(x - r_2)(x - r_3)$$

Plugging in -1 gives us the value we want to find in terms of our roots which is $(-1 - r_1)(-1 - r_2)(-1 - r_3)$

Plugging in 0 gives that $-r_1r_2r_3 = -64$ which means $r_1r_2r_3 = 64$

we can now expand our expression that we found of $f(-1)$ to get $-1 - (r_1 + r_2 + r_3) - (r_1r_2 + r_1r_3 + r_2r_3 - r_1r_2r_3)$. We can substitute $r_1r_2r_3 = 64$ to get $-65 - (r_1 + r_2 + r_3) - (r_1r_2 + r_1r_3 + r_2r_3)$

Since we are subtracting our values $r_1r_2r_3$ and $r_1r_2 + r_1r_3 + r_2r_3$, to maximize the entire expression we should minimize the values we subtract.

We can now apply AM-GM to r_1, r_2, r_3 to get

$$\frac{r_1+r_2+r_3}{3} \geq \sqrt[3]{r_1 r_2 r_3}$$

Plugging in 64 for it gives $\frac{r_1+r_2+r_3}{3} \geq \sqrt[3]{64}$

This becomes $r_1 + r_2 + r_3 \geq 12$

Now this means we plug in $r_1 + r_2 + r_3$ as 12 since we want to minimize it.

Now we use AM-GM on $r_1 r_2, r_1 r_3, r_2 r_3$ to get

$$\frac{r_1 r_2 + r_1 r_3 + r_2 r_3}{3} \geq \sqrt[3]{(r_1 r_2 r_3)^2}$$

Now we can plug in 64 for $r_1 r_2 r_3$ again and our inequality simplifies to

$$r_1 r_2 + r_1 r_3 + r_2 r_3 \geq 48$$

Now this means our minimum value of $r_1 r_2 + r_1 r_3 + r_2 r_3$ is 48.

We can plug in our values to $-65 - (r_1 + r_2 + r_3) - (r_1 r_2 + r_1 r_3 + r_2 r_3)$ and our minimum value at the end is $-65 - 12 - 48$ which is **-125**.

Problem 3.12.9 — Suppose x, y , and z are nonzero complex numbers such that

$$(x + y + z)(x^2 + y^2 + z^2) = x^3 + y^3 + z^3$$

Compute $(x + y + z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

Source: 2016 CMIMC Team

Solution: Expanding the given equation on the left side gives

$$x^3 + y^3 + z^3 = x^2y + x^2z + y^2x + y^2z + z^2x + z^2y = x^3 + y^3 + z^3$$

Subtracting $x^3 + y^3 + z^3$ from both sides gives $x^2y + x^2z + y^2x + y^2z + z^2x + z^2y = 0$

Now since we want to compute $(x + y + z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$, we can rewrite this as $(x + y + z)\left(\frac{xy + xz + yz}{xyz}\right)$

We continue to multiply the value we want to compute to get $\frac{x^2y + x^2z + y^2x + y^2z + z^2x + z^2y + 3xyz}{xyz}$

We can plug in $x^2y + x^2z + y^2x + y^2z + z^2x + z^2y = 0$ into that and we're left with $\frac{3xyz}{xyz}$ to get **3**.

4 Combinatorics

§4.1 Basic Combinatorics Principles

One important thing we need to know is the process of casework. Casework is about organizing our counting into different cases and counting each case separately. At the end of that, we add all the numbers. This is super common on both the AMC and AIME.

Definition 4.1.1

Complementary Counting is about counting the opposite of what we want and subtracting that from the total. For example, if we want to find the number of boys in a class of 40 students, we can simply count the number of girls and subtract that from 40 which gives us the number of boys.

$\binom{n}{k}$ represents the amount of k -sized subsets from a set of numbers from 1 to n . It also represents the number of ways to choose k people from a group of n .

The number of permutations of the set $1, 2, \dots, n$ is $n!$ (! represents factorial)

Theorem 4.1.2

How many ways are there to rearrange the letters of a word when some letters are repeated?

Let's assume that there are n letters in total, x_a amount of the letter a , x_b amount of the letter b , x_c amount of the letter c continuing all the way to x_z amount of the letter z .

This gives us the equation $x_a + x_b + x_c + \dots + x_z = n$
The number of ways to rearrange the letters are $\frac{n!}{x_a! \cdot x_b! \cdot x_c! \cdot \dots \cdot x_z!}$

Example 4.1.3

- How many ways are there to rearrange the letters of the word SANDWICH?
- How many ways are there to rearrange the letters of the word SYNONYMS?

Solution to a. In the word sandwich, we have 8 letters. None of the letters are repeated, so we don't have to divide it by anything. Thus, the number of ways to rearrange it is $8!$ which is 40320

Solution to b. In the word SYNONYMS (8 letters in total), S is repeated (2 occurrences), Y is repeated (2 occurrences), and N is repeated (2 occurrences).

The number of ways to rearrange the letters is simply $\frac{8!}{2! \cdot 2! \cdot 2!}$ which is $\frac{8!}{8}$ which becomes $7!$.

Problem 4.1.4 — A mathematical organization is producing a set of commemorative license plates. Each plate contains a sequence of five characters chosen from the four letters in AIME and the four digits in 2007. No character may appear in a sequence more times than it appears among the four letters in AIME or the four digits in 2007. A set of plates in which each possible sequence appears exactly once contains N license plates. Find $\frac{N}{10}$.

Source: 2007 AIME

Solution: In this problem, we have to choose 5 characters from *AIME* and 2007. We know that each letter can only appear once in our plate except for 0 which can appear 2 times at max. We will have *three* cases for the number of occurrences of 0.

Case 1: There are zero 0s.

In this case, the 5 characters must come from AIME and 27. Thus, we can simply choose 5 characters from the pool of 6.

$$\binom{6}{5} = 6.$$

However, we must multiply this to $5!$ which is the number of rearrangements. Multiplying that to 6 gives 720

Case 2: There is one 0.

In this case, we must now choose 4 characters from AIME and 27 since we already know that one is 0. Then, we have to multiply that by $5!$ which is the number of rearrangements.

$$\binom{6}{4} \cdot 5! = 1800$$

Case 3: There are two 0s

Now since there are two 0s, we only need to choose 3 more digits from AIME and 27.

$$\binom{6}{3}$$

However, this time we can't directly multiply by $5!$ since not all of the numbers that we will rearrange are the same! 0 has been repeated so we must divide it by the number of times 0 occurs and take the factorial of that which gives us $2!$.

$$\binom{6}{3} \cdot \frac{5!}{2!} = 1200$$

Adding up our numbers gives $720 + 1800 + 1200$ which is 3720.

We divide that by 10 to get **372**.

Problem 4.1.5 — How many four-digit positive integers have at least one digit that is a 2 or a 3?

(A) 2439 (B) 4096 (C) 4903 (D) 4904 (E) 5416

Source: 2006 AMC 10

Solution: In this problem, we will now employ the technique of complementary counting. A good way to recognize when to use complementary counting is through the phrase **at least one**.

Now instead of finding the number of digits that have at least one 2 or 3, we can simply find the opposite of that which means they have no 2 or 3. Then, we can subtract

that from the total amount of 4 digit numbers.

Since there are 10 digits from 0 to 9, we know that for the thousands digit 0 won't be possible. However, we also know that neither 2 nor 3 will work giving us 7 cases for that digit.

For the next 3 digits, anything from 0 to 9 will work except 2 and 3 giving us 8 possible digits.

Since each digit is independent of the other, we can simply multiply these numbers to get $7 \cdot 8 \cdot 8 \cdot 8$ which is 3584.

The number of 4 digits number can be found in a similar way. There are 9 possible digits for the thousands digit, and 10 for the remaining ones giving us 9000.

The answer is thus $9000 - 3584 = \mathbf{5416}$ (E).

Problem 4.1.6 — How many ordered sequences of 1's and 3's sum to 16? (Examples of such sequences are 1, 3, 3, 3, 3, 3 and 1, 3, 1, 3, 1, 3.)

Source: 2012 SMT

Solution: In this problem, let's assume the number of occurrences of 1 are x and the number of occurrences of 3 are y . This means that we can rewrite our sum as $x + 3y = 16$

In total, we have $x + y$ letters and each 1 has x of occurrences while 3 has y occurrences. The number of ways to rearrange this is $\frac{(x+y)!}{x! \cdot y!}$ which is equivalent to $\binom{x+y}{y}$.

Using the equation $x + 3y = 16$, our possible solutions are

x	y	Number of rearrangements
1	5	$\binom{1+5}{5} = \binom{6}{5}$
4	4	$\binom{4+4}{4} = \binom{8}{4}$
7	3	$\binom{7+3}{3} = \binom{10}{3}$
10	2	$\binom{10+2}{2} = \binom{12}{2}$
13	1	$\binom{13+1}{1} = \binom{14}{1}$
16	0	$\binom{16+0}{0} = \binom{16}{0}$

Now we add up all of the values on the rightmost column to get **277**.

Problem 4.1.7 — Define the sum of a finite set of integers to be the sum of the elements of the set. Let D be the set of positive divisors of 700. How many nonempty subsets of D have an even sum? (Simplify as reasonably as possible)

Source: 2015 MMATHS

Solution: In this problem, we first factor out 700 to get $2^2 \cdot 5^2 \cdot 7$

Let's divide all factors as even or odd. An even factor must have a power of 2 meaning the exponent of 2 must either be 1 or 2 (2 possibilities). We multiply that by the number of factor of $5^2 \cdot 7$ which is 6 to get that there are 12 even factors and 6 odd factors of 700.

From here, we know that there must be an even amount of odd factors for it to sum to an even number. This means that there must be 0, 2, 4, or 6 odd factors in our subset.

Also, there's no restriction for even numbers since any amount of even numbers always sum to an even number.

Case 1: 0 odd factors: Now we just choose find the number of ways to choose our odd factors which is $\binom{6}{0}$ multiplied by the number of ways to choose our even factors which is 2^{12} . This gives us $2^{12} \cdot \binom{6}{0}$

Case 2: 2 odd factors: Now we just choose find the number of ways to choose our odd factors which is $\binom{6}{2}$ multiplied by the number of ways to choose our even factors which is 2^{12} . This gives us $2^{12} \cdot \binom{6}{2}$

Case 3: 4 odd factors: Now we just choose find the number of ways to choose our odd factors which is $\binom{6}{4}$ multiplied by the number of ways to choose our even factors which is 2^{12} . This gives us $2^{12} \cdot \binom{6}{4}$

Case 4: 6 odd factors: Now we just choose find the number of ways to choose our odd factors which is $\binom{6}{6}$ multiplied by the number of ways to choose our even factors which is 2^{12} . This gives us $2^{12} \cdot \binom{6}{6}$

We sum up all of these values to get $2^{12}(\binom{6}{0} + \binom{6}{2} + \binom{6}{4} + \binom{6}{6})$ which is 2^{17} .

However, since our subsets must be **nonempty**, we must subtract the case where there are no odd factors nor no even factors in the subset, and our final answer is $2^{17} - 1$ which is **131071**.

Problem 4.1.8 — Find the number of subsets of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ that are subsets of neither $\{1, 2, 3, 4, 5\}$ nor $\{4, 5, 6, 7, 8\}$.

Source: 2017 AIME

Solution: Now in this problem, we will again use the technique of complementary counting.

The number of subsets in any set of size N are 2^N . Thus, in this case, there are 2^8 number of subsets in the set of 1 to 8 which is 256.

Now we will subtract all the subsets from the other 2 sets. Since they each have 5 terms, there are 2^5 or 32 possible subsets that can be formed from each. We multiply that by 2 to get 64.

We now subtract 64 from 256 to get 192. However, we are still not done. We need to remember that both sets of size 5 that are listed have 2 common terms (4 and 5). We have to now add back all possible sets that can be made with 4 and 5 since we subtracted it two times. There are 4 subsets that can be made with those 2 terms, so we add that back.

Our final answer is $192 + 4$ which is **196**.

Problem 4.1.9 — Let a, b, c, d be four not necessarily distinct integers chosen from the set $0, 1, \dots, 2019$. What is the probability that $(a + b)(c + d) + (a + c)(b + d) + (a + d)(b + c)$ is divisible by 4?

Source: 2019 CMC 10

Solution: In this problem, let's first expand the expression $(a + b)(c + d) + (a + c)(b + d) + (a + d)(b + c)$ to get $2(ab + ac + ad + bc + bd + cd)$.

From here, if we want 4 to divide that expression, then we just have to check whether or

not 2 divides $ab + ac + ad + bc + bd + cd$ since 2 already divides the entire expression.

We can plug in numbers mod 2 for the variables a, b, c, d and see when it will be 0 mod 2.

Case 1: All 4 variables are 0 mod 2

In this case, we can plug in 0 for those 4 variables and we get that the expression is 0 mod 2.

The probability for all numbers to be 0 mod 2 is simply $(\frac{1}{2})^4$ which is $\frac{1}{16}$. We found $\frac{1}{2}$ as the probability for one to be 0 mod 2 since we have 2020 numbers in total, and 1010 of them are even which means the probability is $\frac{1}{2}$

Case 2: 3 variables are 0 mod 2 and one is 1 mod 2

In this case, we can assume that a, b, c are 0 mod 2 while d is 1 mod 2. If we plug this into the expression we get that $ab + ac + ad + bc + bd + cd$ is 0 mod 2.

The probability for this to occur is simply $\frac{1}{16}$. However, we need to multiply this by the number of ways to rearrange it since either a, b, c , or d can be 0 mod 2. The number of ways to rearrange it is 4 so our total probability is $4 \cdot \frac{1}{16}$ which is $\frac{1}{4}$.

Case 3: Now for the case when either 2 of the numbers are 0 mod 2 or when 1 of the numbers is 0 mod 2, then the final expression for $ab + ac + ad + bc + bd + cd$ will be 1 mod 2, so the probability will be 0 for this case.

Case 4: Now for the case when all 4 variables are 1 mod 2, our final expression $ab + ac + ad + bc + bd + cd$ is 0 mod 2.

The probability of any individual variable being 1 mod 2 is simply $\frac{1}{2}$ and we raise this to the power of 4 since there are 4 variables to get $\frac{1}{16}$.

The total probability is $\frac{1}{16} + \frac{1}{4} + \frac{1}{16} = \frac{3}{8}$ (C).

Problem 4.1.10 — A choir director must select a group of singers from among his 6 tenors and 8 basses. The only requirements are that the difference between the number of tenors and basses must be a multiple of 4, and the group must have at least one singer. Let N be the number of different groups that could be selected. What is the remainder when N is divided by 100?

(A) 47 (B) 48 (C) 83 (D) 95 (E) 96

Source: 2021 AMC

Solution: In this problem, our main cases will be regarding the difference between the number of tenors and basses.

Case 1: The difference between the number of tenors and basses is 0. This means that the number of tenors and basses must be the same. The sum of the number of cases for this situation is $\binom{6}{1} \cdot \binom{8}{1} + \binom{6}{2} \cdot \binom{8}{2} + \binom{6}{3} \cdot \binom{8}{3} + \binom{6}{4} \cdot \binom{8}{4} + \binom{6}{5} \cdot \binom{8}{5} + \binom{6}{6} \cdot \binom{8}{6}$.

The above sums up to 3002.

Case 2: The difference between the number of tenors and basses is 4.

This means that the number of tenors and basses respectively can be (0, 4), (1, 5), (2, 6), (3, 7), (4, 8), (4, 0), (5, 1), and (6, 2).

The sum for the number of cases for the situation above is $\binom{6}{0} \cdot \binom{8}{4} + \binom{6}{1} \cdot \binom{8}{5} + \binom{6}{2} \cdot \binom{8}{6} +$

$\binom{6}{3} \cdot \binom{8}{7} + \binom{6}{4} \cdot \binom{8}{8} + \binom{6}{4} \cdot \binom{8}{0} + \binom{6}{5} \cdot \binom{8}{1} + \binom{6}{6} \cdot \binom{8}{2}.$
 The above sums up to 1092.

Case 3: The difference between the number of tenors and basses is 8.

This means that the number of tenors must be 0 while the number of basses is 8. The number of cases for this is simply $\binom{6}{0} \cdot \binom{8}{8}$ which is 1.

Thus, the final answer is $3002 + 1092 + 1$ which is 4095. Our answer (mod 100) is simply **95 (D)**.

Extension to this problem: The solution method shown above was a good example of casework. However, that's not the only way to solve this problem. Find a neater way to solve this problem using your combinatorial identities (once you learn that topic later!)

§4.2 Sets and PIE

Definition 4.2.1

PIE is a super important topic that shows up on the AMC. It stands for the Principal of Inclusion and Exclusion.

Let's pretend that we have 2 sets that are A and B. The numbers in A are 3, 4, 5, and 6 while the numbers in B are 5, 6, 8, 9. The way to find the union of these two sets is

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

The $||$ represents the number of terms in those sets. The symbol on the very right $\cap$ (intersection sign) means the number of terms that are common in both sets. For the example I showed, it means that the union of the two sets is the number of terms in set A added to the number of terms in set B. After that, you subtract the number of terms that are common between both of the sets to find the union.

This also works if there are more than two sets. Below is the way to represent it when there are 3 sets.

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_2 \cap A_3| - |A_3 \cap A_1| + |A_1 \cap A_2 \cap A_3|$$

Problem 4.2.2 — Let $\mathcal{S}$ be the set $\{1, 2, 3, \dots, 10\}$. Let n be the number of sets of two non-empty disjoint subsets of $\mathcal{S}$. (Disjoint sets are defined as sets that have no common elements.) Find the remainder obtained when n is divided by 1000.

Source: 2002 AIME

Solution: In this problem, since any of the numbers from 1 through 10 can either be in our first set, second set, or in none; there are three locations where each number can go. Since each number's placement is independent of the other, we can simply raise that to the power of 10.

However, 3^{10} includes the cases when either the first set is empty or both are empty. If the first set is empty, then there are 2^{10} ways of placing the other numbers since they can only go in the second set or not be there.

$$3^{10} - 2 \cdot 2^{10}$$

I multiplied 2 to 2^{10} because 2^{10} is the number of rearrangements when either the first set or the second one is empty.

However, because of PIE, we must add 1 to this answer because we subtracted the case of both being empty twice.

$$3^{10} - 2 \cdot 2^{10} + 1 = 57002 \text{ which is } 2 \pmod{100}.$$

§4.3 Path Counting + Bijections

In counting the number of paths on a square grid where we have to go m units up and n units to the right, we can use a bijection technique to find the number of ways to do so.

Example 4.3.1

If we want to move from $(0, 0)$ to (m, n) , how many ways are there to do so when we can only move up and to the right.

Since we know that we need to go to (m, n) , we know that in our path we have to go up 1 unit n times and 1 unit right m times. Thus, we have $UUU\dots(n \text{ times})$ and $RRRRR\dots(m \text{ times})$ where U represents up and R represents right.

This is the same thing as rearranging the letters of a word. We have $m+n$ letters in total and we divide this by the number of letters that are repeated, and take the factorial of it.

Using the formula of rearranging a letter and accounting for the repetition of the letters, we get

$$\frac{(m+n)!}{m! \cdot n!} = \binom{m+n}{m}$$

Bijections is about taking a complicated problem and boiling it down to a simpler idea which is much easier to solve. We will now do many problems to apply this topic since the best way to practice this topic is to simply practice. There are no simple formulas to make it easy.

Problem 4.3.2 — A fair die is rolled four times. The probability that each of the final three rolls is at least as large as the roll preceding it may be expressed in the form m/n , where m and n are relatively prime positive integers. Find $m + n$.

Source: 2001 AIME

Solution: In this problem, since each roll must be greater than the other, it means our number must go "up" as our roll number goes up "to the right". This is the same thing as having a grid that is 4 units wide (from 1 to 4) and it is 6 units high (from 1 to 6) representing the amount on the roll.

We will start at the point $(0, 1)$ which represents the 0th roll number and a minimum roll of 1. Now we simply find the number of paths to 4, 6 which is the rectangle we saw above. It is 5 units long and 4 units wide which means there are $\binom{9}{5}$ or (126) ways of moving.

Since each dice roll has 6 possible results, there are 6^4 results for the rolls which is our denominator.

Our answer is $\frac{126}{6^4} = \frac{7}{72}$ and the sum of the numerator and denominator is **79**.

In this problem we saw how we made the bijection of the rolls on the dice to a simple grid. These problems often involve thinking deeper than usual and making connections between important math ideas.

Problem 4.3.3 — Let $(a_1, a_2, a_3, \dots, a_{12})$ be a permutation of $(1, 2, 3, \dots, 12)$ for which

$$a_1 > a_2 > a_3 > a_4 > a_5 > a_6 \text{ and } a_6 < a_7 < a_8 < a_9 < a_{10} < a_{11} < a_{12}.$$

An example of such a permutation is $(6, 5, 4, 3, 2, 1, 7, 8, 9, 10, 11, 12)$. Find the number of such permutations.

Source: 2006 AIME

Solution: In this problem, since a_6 is the smallest in both inequalities, it must be the smallest number from the set which is 1.

From here, we notice that we simply need to choose 5 numbers from the remaining 11 numbers from 2 to 12. Regardless of what 5 number combination we choose, there will always be a way to arrange it exactly once to make it satisfy $a_1 > a_2 > a_3 > a_4 > a_5 > a_6$

Thus, the answer is simply $\binom{11}{5}$ which is **462**.

Problem 4.3.4 — A moving particle starts at the point $(4, 4)$ and moves until it hits one of the coordinate axes for the first time. When the particle is at the point (a, b) , it moves at random to one of the points $(a - 1, b)$, $(a, b - 1)$, or $(a - 1, b - 1)$, each with probability $\frac{1}{3}$, independently of its previous moves. The probability that it will hit the coordinate axes at $(0, 0)$ is $\frac{m}{3^n}$, where m and n are positive integers such that m is not divisible by 3. Find $m + n$.

Source: 2019 AIME

Solution: We have 3 types of moves.

The move to $(a - 1, b)$ is L (left)

The move to $(a, b - 1)$ is D (down)

The move to $(a - 1, b - 1)$ is C (cross meaning we go down and to the left)

Overall, from $(4, 4)$ to $(0, 0)$, we go down 4 times and to the left 4 times. However, we can have cross moves to $(a - 1, b - 1)$ to decrease the number of times we have to go to choose an L or D move.

Since we want the first time it hits the x or y axis to be at point $(0, 0)$, it can't

hit the point $(1, 0)$ or $(0, 1)$. This means that it must come from the point $(1, 1)$ to $(0, 0)$ using the C move.

Now we need to find the probability of going from $(4, 4)$ to $(1, 1)$, and we simply multiply that by $\frac{1}{3}$ since there's a $\frac{1}{3}$ chance of going from $(1, 1)$ to $(0, 0)$ in 1 move.

We can do casework to find the probability of going from $(4, 4)$ to $(1, 1)$. Cases can include something like 3L and 3D moves or 2L, 2D, and 1C move.

Case 1: Probability of 3 L and 3 D moves

This case means we simply go to the left and down 3 times each. It's like rearranging the letters of a word as we talked before in which there are 6 letters and 2 unique ones, each which is repeated 3 times (LLLDDD).

The number of ways to rearrange it is $\frac{6!}{3!3!}$. The probability of one arbitrary move is $\frac{1}{3}$, and since we have 8 moves, the total probability becomes $\frac{1}{3^6}$. We can multiply this to the number of ways to rearrange it.

$$\frac{6!}{3!3!} \cdot \frac{1}{3^6} = \frac{20}{3^6}$$

Case 2: Probability of 2 L and 2 D moves and 1 C move

This is again the same thing as rearranging the 5 letter word LLDDC.

The number of ways to do so is $\frac{5!}{2!2!1!}$. The probability of one rearrangement is $\frac{1}{3^5}$

The total probability is $\frac{30}{3^5}$ which is $\frac{10}{3^4}$

Case 3: Probability of 1 L and 1 D moves and 2 C move

This is the same thing as rearranging the 4 letter word LDCC.

The number of ways to do this is $\frac{4!}{2!1!1!}$. The probability of one rearrangement is $\frac{1}{3^4}$.

The total probability is $\frac{12}{3^4}$ which is $\frac{4}{3^3}$

Case 4: Probability of 0 L and 0 D moves and 3 C move

This is the same thing as rearranging the 3 letter word CCC.

The number of ways to do this is $\frac{3!}{3!0!0!}$ which is 1. The probability of one rearrangement is $\frac{1}{3^3}$ (since there are 3 moves to get to $(1, 1)$). The total probability for this case is $\frac{1}{3^3}$.

The sum of the 4 probabilities in these 4 cases is $\frac{20}{3^6} + \frac{10}{3^4} + \frac{4}{3^3} + \frac{1}{3^3} = \frac{245}{3^6}$.

We multiply this by $\frac{1}{3}$ which is the probability to go from $(1, 1)$ to $(0, 0)$. Our probability $\frac{1}{3} \cdot \frac{245}{3^6}$ is $\frac{245}{3^7}$

Our answer in the format that the problem wants is $245 + 7$ which is **252**.

§4.4 Stars and Bars

This is one of the most important combinatorics topics with an extremely high yield.

A basic stars and bars problem is to find the number of ways to distribute n **identical objects** to k people.

Example 4.4.1

Let's pretend we have 10 identical pieces of candies. We want to find the number of ways to give it out to 4 children so that each student receives at least one piece of candy. How will we proceed from here?

Solution: Our identical candy pieces are our stars. The number of stars are the number of candies.

In the 10 stars above, we want to split it to 4 people. We notice that if we insert 3 bars anywhere, then it splits the number of candy pieces into 4 separate groups (one for each person).

Now we need to find the number of places where we can place a bar. We notice that we can't put a bar at any of the ends because then it will create a group with 0 candies which isn't at least 1. Thus, there are only 9 places to put the bars in between the stars (everywhere except to the left of the leftmost star and to the right of the rightmost star).

From those 9 places, we simply need to choose 3 meaning the number of ways to hand the candies out is $\binom{9}{3}$ which is **84**

Stars and bars now gives us two formulas

Theorem 4.4.2

The number of **positive** integer solutions $(x_1, x_2, \dots, x_k)$ to $x_1 + x_2 + \dots + x_k = n$ is $\binom{n-1}{k-1}$

The number of non-negative integer solutions $(x_1, x_2, \dots, x_k)$ to $x_1 + x_2 + \dots + x_k = n$ is $\binom{n+k-1}{k-1}$

Problem 4.4.3 — Given eight distinguishable rings, let n be the number of possible five-ring arrangements on the four fingers (not the thumb) of one hand. The order of rings on each finger is significant, but it is not required that each finger have a ring. Find the leftmost three nonzero digits of n .

Source: 2000 AIME

Solution: In this problem, we have eight rings and we need five to put on four fingers. The first thing we do is choose 5 from the 8 to get $\binom{8}{5}$ number of ways to choose our 5 rings. Now, we know that there are $5!$ ways to rearrange the rings.

From the 5 rings we have, we know that they all have to go on 4 fingers. We can use stars and bars assuming there are x_1, x_2, x_3, x_4 number of rings on the 4 fingers

$$x_1 + x_2 + x_3 + x_4 = 5$$

This is stars and bars for nonnegative integers which means the total is $\binom{5+4-1}{4-1} = \binom{8}{3} = 56$

We multiply all our 3 numbers together which are $\binom{8}{5}$ and $5!$ and 56 to get

376320 , and the first 3 digits are 376

Problem 4.4.4 — When 7 fair standard 6-sided dice are thrown, the probability that the sum of the numbers on the top faces is 10 can be written as

$$\frac{n}{6^7},$$

where n is a positive integer. What is n ?

(A) 42 (B) 49 (C) 56 (D) 63 (E) 84

Source: 2018 AMC 10

Solution: In this problem, assuming that our roll amounts are $x_1, x_2, x_3, x_4, x_5, x_6$, and x_7 , we know that

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 10$$

Using the information above, since each roll is of course a positive number (we can't roll 0), from stars and bars the number of solutions to this equation is $\binom{10-1}{7-1} = \binom{9}{6}$ which is 84.

From here, we'll see if there are any solutions to subtract. That will occur if any of our rolls is more than 6, since that isn't possible. We notice that even if one roll is 7, then that case isn't possible because the remaining 6 rolls must be a minimum of 1. That makes the sum of that combination 13 which is clearly greater than 10. Thus, we don't have any cases to subtract.

Our final answer is **84**.

Problem 4.4.5 — For some particular value of N , when $(a + b + c + d + 1)^N$ is expanded and like terms are combined, the resulting expression contains exactly 1001 terms that include all four variables a, b, c , and d , each to some positive power. What is N ?

(A) 9 (B) 14 (C) 16 (D) 17 (E) 19

Source: 2016 AMC 10

Solution: In this problem, we need to know that for any term in an expression raised to any power, the sum of the power of each individual variables part of that term must sum up to the main power of the expression.

This might seem confusing at first, so I will explain that with a simpler example.

Pretend we have $(a + b + c)^3$, the power of all variables in any term in this expression will always sum to 3. We can test that on our own since

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3b^2a + 3b^2c + 3c^2a + 3c^2b + 6abc$$

In the expanded form above, we can see that for any individual term such as $3a^2$ or $6abc$, the sum of the powers of each variable is equivalent to the power we raised $a + b + c$ to

Using these observations, if the exponents of a, b, c, d , and 1 in this problem are x_1, x_2, x_3, x_4, x_5 respectively, then

$$x_1 + x_2 + x_3 + x_4 + x_5 = N$$

Since we want to find the number of terms that always include the variables a, b, c, d , that means x_1, x_2, x_3, x_4 must be greater than or equal to 1. However, x_5 can be anything greater than or equal to 0.

When such a situation arises, we can make a clever substitution. We can make 4 substitutions now.

$$x'_1 = x_1 - 1$$

$$x'_2 = x_2 - 1$$

$$x'_3 = x_3 - 1$$

$$x'_4 = x_4 - 1$$

Now we can plug this into our original equation $x_1 + x_2 + x_3 + x_4 + x_5 = N$ and make it $x'_1 + x'_2 + x'_3 + x'_4 + x_5 = N - 4$

For the new equation, all 5 variables can now be greater than or equal to 0.

The number of solutions to this is $\binom{N-4+5-1}{5-1} = \binom{N}{4}$

Since $\binom{N}{4}$ is 1001, plugging in a few numbers from N from the answer choices gives **14 (B)**.

Problem 4.4.6 — How many sets of positive integers (a, b, c) satisfies $a \leq b \leq c \leq 103$ and $a + b + c = 103$?

Source: 2014 PUMAC

Solution: In this problem, we know that since the three numbers are positive, the number of solutions from stars and bars after ignoring the condition of $a \leq b \leq c \leq 103$ is simply $\binom{102}{2}$ which is 5151.

However, now we need to subtract cases that we overcounted. We notice that as long as all 3 numbers that sum up to 102 are DISTINCT, then there is exactly one way to rearrange it so that it satisfies $a > b > c > 0$. However, since we ignored all conditions when finding the amount through stars and bars, we need to subtract the case when a number is repeated (all numbers can't be repeated since 103 isn't divisible by 3)

Assuming our repeated number is x , and the other number is y , we want to find the number of cases in which $2x + y = 103$

Since x can be anything from 1 to 51, and for all of those numbers y is also greater than or equal to 1, that means there are 51 cases. However, we must multiply by 3 to account for rearrangements of x, x, y .

In total, we must subtract by 153.

$$5151 - 153 = 4998$$

From here, we are only left with numbers a, b, c that sum to 103 and are distinct. However, we must divide by 6 because of overcounts. This is because numbers such as 1, 2, 100 can be rearranged 6 times even though it only gives 1 solution due to our limitation: $a > b > c > 0$.

Thus, the final answer is $\frac{4998}{6}$ which is **833**.

Problem 4.4.7 — A gardener plants three maple trees, four oaks, and five birch trees in a row. He plants them in random order, each arrangement being equally likely. Let $\frac{m}{n}$ in lowest terms be the probability that no two birch trees are next to one another. Find $m + n$.

Source: 1984 AIME

Solution: In this problem, we can treat the nonbirch trees as the same. In that case, we have 7 non-birch trees and 5 birch trees. We can treat the non-birch trees like our stars, and the birch trees as the bars.

If we first lay out the non-birch trees, we simply draw out 7 stars

Now there are 8 spots in between and on the left and right of each star. From these, we want to choose 5 spots for the birch trees. This gives us $\binom{8}{5}$ for the amount of ways to rearrange them so that no two birch trees are next to each other.

The total number of combinations (for the denominator) is simply $\binom{12}{5}$ which represents choosing 5 spots at random from 12 to place the birch trees, and in the remaining spots we'll place the non-birch trees.

Our answer is $\binom{8}{5}$ divided by $\binom{12}{5} = \frac{7}{99}$. The final answer is 106.

Problem 4.4.8 — Connie finds a whiteboard that has magnet letters spelling MISSISSIPPI on it. She can rearrange the letters, in which identical letters are indistinguishable. If she uses all the letters and does not want to place any Is next to each other, how many distinct rearrangements are possible?

Source: 2019 SMT

Solution: In this problem, since we want to separate the Is and not have any be next to each other, let's assume that our 4 I's are the "separators." We can write it out as IIII. It creates 5 sections for other letters to go (in between each pair and at the very leftmost and rightmost point).

From here, we can use stars and bars and assume that there are a, b, c, d, e amount of letters in each place.

aIbIcIdIe (where a, b, c, d, e represent the number of letters while I is the actual letter)

From here, we know that $a + b + c + d + e = 7$ because there are 7 letters that aren't an I (there is 1 M, 4 S, and 2 P's).

b, c, d must be at least 1. We can now use the same technique we saw before.

$$b' = b - 1$$

$$c' = c - 1$$

$$d' = d - 1$$

We plug this into the equation we saw that involved a, b, c, d, e and our equation becomes $a + b' + c' + d' + e = 4$

From here we know that our n is 4, our k is 5. These numbers have to be nonnegative due to our substitution since previously b, c, d needed to be at least 1 because if it would be 0, then the I's would be consecutive.

We substitute this into $\binom{n+k-1}{k-1}$ to get $\binom{8}{4}$ which is 70.

Our remaining letters not including I are MSSSSPP. We have 7 letters in this so there are $7!$ ways to reorganize it. We must divide this by $4!$ and $2!$ since S is repeated 4 times while P is repeated 2 times.

Our total answer is $70 \cdot \frac{7!}{4!2!}$ which is **7350**

§4.5 Binomial

You need to know the binomial theorem to be able to derive these coefficients.

We know that $(x + y)^2$ is simply $x^2 + 2xy + y^2$. I'm sure that to find the answer, you would multiply each term with each other. That method is great, but what if you want to compute $(x + y)^6$. Calculating that can be painful by hand. That's when the binomial theorem comes in.

Theorem 4.5.1

Binomial Theorem $(x + y)^n = x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \dots + \binom{n}{n-1}xy^{n-1} + y^n$

It's important to note how the coefficient of each term changes in a pattern.

Theorem 4.5.2

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n-2} + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

You can sometimes use the combinatorial identity shown above in Algebra problems. For example, we already know that if we have a polynomial $f(x)$, and we want to find the sum of the coefficients of it, we can simply plug in 1 and find $f(1)$

Sometimes you will use that idea combined with the combinatorial identity shown in this theorem to find the sum of the coefficients.

Theorem 4.5.3

Pascal's Identity

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$$

The above theorem works for positive integer n and when $0 \leq k \leq n - 1$

Theorem 4.5.4**Hockey Stick Identity**

$$\sum_{i=k}^n \binom{i}{k} = \binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \dots + \binom{n}{k} = \binom{n+1}{k+1}$$

The above theorem works as long as $0 \leq k \leq n$

Theorem 4.5.5**Vandermonde's Identity**

This identity is on the rarer side, but it's still important to know just in case it shows up.

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Problem 4.5.6 — Given that $\frac{1}{2!17!} + \frac{1}{3!16!} + \frac{1}{4!15!} + \frac{1}{5!14!} + \frac{1}{6!13!} + \frac{1}{7!12!} + \frac{1}{8!11!} + \frac{1}{9!10!} = \frac{N}{1!18!}$ find the greatest integer that is less than $\frac{N}{100}$.

Source: 2000 AIME

Solution: In this problem, we notice that the denominator on the left side of the equation has two factorials, and the numbers without the factorial sign all sum to 19. Thus, to try some manipulation we multiply both sides by $19!$. Doing so gives

$$\frac{19!}{2!17!} + \frac{19!}{3!16!} + \frac{19!}{4!15!} + \frac{19!}{5!14!} + \frac{19!}{6!13!} + \frac{19!}{7!12!} + \frac{19!}{8!11!} + \frac{19!}{9!10!} = \frac{19!N}{1!18!}$$

We recognize that terms like $\frac{19!}{2!17!}$ are simply $\binom{19}{2}$. Using this we rewrite the remaining terms to get

$$\binom{19}{2} + \binom{19}{3} + \binom{19}{4} + \binom{19}{5} + \binom{19}{6} + \binom{19}{7} + \binom{19}{8} + \binom{19}{9} = N \cdot \binom{19}{1}$$

From the identity $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n-2} + \binom{n}{n-1} + \binom{n}{n} = 2^n$, we know that $\binom{19}{0} + \binom{19}{1} + \binom{19}{2} + \binom{19}{3} + \dots + \binom{19}{17} + \binom{19}{18} + \binom{19}{19} = 2^{19}$

We know that all the terms are symmetrical (meaning the first term equals the last), so we can divide both sides of it by 2 to get

$$\binom{19}{0} + \binom{19}{1} + \binom{19}{2} + \binom{19}{3} + \dots + \binom{19}{7} + \binom{19}{8} + \binom{19}{9} = 2^{18}$$

We subtract the first two terms which sum up to 20 to get $2^{18} - 20$

We can now rewrite our equation as

$$2^{18} - 20 = N \cdot \binom{19}{1}$$

Simplifying gives us $\frac{262124}{19}$ for N , and we write the answer in the form that we want to get **137**.

Problem 4.5.7 — The expression

$$(x + y + z)^{2006} + (x - y - z)^{2006}$$

is simplified by expanding it and combining like terms. How many terms are in the simplified expression?

- (A) 6018 (B) 671,676 (C) 1,007,514 (D) 1,008,016 (E) 2,015,028

Source: 2006 AMC 12

Solution: In this problem, we first have to use stars and bars. We already know that for any algebraic expression, the number of terms can be found through stars and bars since all the degrees of each term sum up to the power we raised our original expression up to.

Assuming that x_1 , x_2 , and x_3 are the coefficients of x , y , z respectively, then $x_1 + x_2 + x_3 = 2006$. The three numbers are non nonnegative.

Applying stars and bars gives $\binom{2006+3-1}{3-1} = \binom{2008}{2} = 2015028$

However, we now must subtract a few cases since some of the terms will cancel out when we subtract the two algebraic terms $(x + y + z)^{2006}$ and $(x - y - z)^{2006}$

A term will only cancel out if the coefficient is negative for the right term (the coefficient is always positive for the terms from $(x + y + z)^{2006}$

Assuming that y_1 , y_2 , and y_3 are the exponents of a term that comprises $x, -y, -z$, then we know that $y_1 + y_2 + y_3 = 2006$

However, for a term to be negative for this part, $y_2 + y_3$ must be odd since any negative term raised to the odd power still remains to be negative.

This means that y_1 must also be odd if the combined sum of $y_2 + y_3$ is odd.

y_1	Number of Cases
1	$\binom{2006}{1}$
3	$\binom{2004}{1}$
5	$\binom{2002}{1}$
7	$\binom{2000}{1}$
9	$\binom{1998}{1}$
11	$\binom{1996}{1}$
...	...
2005	$\binom{2}{1}$

We can simply sum up the right side of the columns which is the number of cases. The sum is simply the sum of the odd numbers from 2 to 1006 which is 1007012 (using the sum of the first n even numbers).

We subtract 1007012 from 2015028 to get **1008016 (D)**

Problem 4.5.8 — Find the sum of all real numbers x such that $5x^4 - 10x^3 + 10x^2 - 5x - 11 = 0$

Source: 2014 HMMT

Solution: In this problem, seeing the coefficients such as 5, 10, 10, and 5 should make us think of some of our pascal numbers such as $\binom{5}{1}$ and $\binom{5}{2}$.

We want to try to get $(x - 1)^5$ to be part of the problem since that polynomial has the coefficients that we also have.

We can rewrite the polynomial that we have as

$$x^5 - x^5 + 5x^4 - 10x^3 + 10x^2 - 5x - 1 + 12 = 0$$

$$x^5 + 5x^4 - 10x^3 + 10x^2 - 5x - 1 - x^5 + 12 = 0$$

The first 5 terms simply are $(x - 1)^5$

$$(x - 1)^5 - x^5 + 12 = 0$$

In the polynomial above, if r is a root, then $1 - r$ must also be due to the symmetry of x and $x - 1$. Also, some inspection tells us that there are only 2 possible real roots for this problem. We already know that if r is a root, then $1 - r$ must be the second root. Thus, we only have to find the value of r . However, if we look back at the problem, it wants us to find the sum of the roots. We can simply sum up $1 - r$ and r to get **1** as the answer because the r cancels out.

Problem 4.5.9 — A club consisting of 11 men and 12 women needs to choose a committee from among its members so that the number of women on the committee is one more than the number of men on the committee. The committee could have as few as 1 member or as many as 23 members. Let N be the number of such committees that can be formed. Find the sum of the prime numbers that divide N .

Source: 2020 AIME

Solution: In this problem, since we know that there must be exactly one more women than men, we can write it out as cases. If there is 1 woman, there must be 0 men. If there are 2 women, there must be 1 men. If there are 3 women, there must be 2 men.

We know that the number of ways to choose k number of women is $\binom{12}{k}$ while the number of ways to choose $k - 1$ number of men (one less than amount of women) is $\binom{11}{k-1}$. We must multiply these two numbers together for each value of k . k can clearly range from 1 (when there's one women) to 12 (when we choose all the women)

We can write this out as a summation

$$\sum_{k=1}^{12} \binom{11}{k-1} \binom{12}{k}$$

Writing out the terms gives (I'll be writing out all, but you don't have to notice the next observation):

$$\begin{aligned} & \binom{11}{0} \cdot \binom{12}{1} + \binom{11}{1} \cdot \binom{12}{2} + \binom{11}{2} \cdot \binom{12}{3} + \binom{11}{3} \cdot \binom{12}{4} + \binom{11}{4} \cdot \binom{12}{5} + \binom{11}{5} \cdot \binom{12}{6} + \binom{11}{6} \cdot \binom{12}{7} \\ & + \binom{11}{7} \cdot \binom{12}{8} + \binom{11}{8} \cdot \binom{12}{9} + \binom{11}{9} \cdot \binom{12}{10} + \binom{11}{10} \cdot \binom{12}{11} + \binom{11}{11} \cdot \binom{12}{12} \end{aligned}$$

We can recognize this as the Vandermonde's identity which states $\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$

Applying Vandermonde's identity after converting a few of the terms such as $\binom{n}{k}$ to $\binom{n}{n-k}$ gives us $\binom{23}{12}$ and writing out the answer in the form that it wants gives us **81**.

Problem 4.5.10 — Let S be the set of all 1000 element subsets of the set $1, 2, 3, \dots, 2018$. What is the expected value of the minimum element of a set chosen uniformly at random from S ?

Source: 2018 SMT

Solution: Let's assume the minimum value is 1. Then, the number of subsets in that case is $\binom{2017}{999}$.

Now if the minimum value is 2, there are $\binom{2016}{999}$ subsets. We can continue this until the minimum value is 1019, and then there are $\binom{999}{999}$ subsets.

Since the expected value can be found by the number of ways for each possibility to occur times the value of that possibility divided by the total number of ways for something to occur.

The value in our denominator will simply be $\binom{2018}{1000}$ as we just want to choose 1000 numbers from 2018 to be in our subset.

The numerator will be $1 \cdot \binom{2017}{999} + 2 \cdot \binom{2016}{999} \dots + 1019 \cdot \binom{999}{999}$.

We can write this numerator as

$$\binom{2017}{999} + \binom{2016}{999} + \dots + \binom{999}{999} = \binom{2018}{1000}$$

$$\binom{2016}{999} + \binom{2015}{999} + \dots + \binom{999}{999} = \binom{2017}{1000}$$

$$\binom{2015}{999} + \binom{2014}{999} + \dots + \binom{999}{999} = \binom{2016}{1000}$$

.

.

.

$$\binom{1000}{999} + \binom{999}{999} = \binom{1001}{1000}$$

$$\binom{999}{999} = \binom{1000}{1000}$$

We will apply Vandermonde's Identity ($\sum_{i=k}^n \binom{i}{k} = \binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \dots + \binom{n}{k} = \binom{n+1}{k+1}$) to these individual summations to get

$$\binom{2018}{1000} + \binom{2017}{1000} + \binom{2016}{1000} + \dots + \binom{1001}{1000} + \binom{1000}{1000}$$

We can now apply Vandermonde's identity to this summation again. Our k is 1000 and n is 2018 to get $\binom{2019}{1001}$.

This is the sum of all possible values, but we must divide it by the number of subsets which is $\binom{2018}{1000}$.

Doing so gives

$$\frac{2019!}{1001!1018!} \cdot \frac{1000!1018!}{2018!} \text{ and this gives } \frac{2019}{1001}$$

§4.6 Recursion

Recursion is a crucial topic on AMC and especially the AIME. For the elite programmers, they might find this topic similar to dynamic programming or many algorithms that they have seen.

Recursion reduces a harder problem into a smaller problem and relates it to larger numbers by setting up a relationship.

The Fibonacci sequence is a popular example of a recurrence relation in which the previous two terms are summed up to find your current term.

In the Fibonacci Sequence, F_n represents the n th term, and in this sequence $F_0 = 0$ and $F_1 = 1$. The recurrence relation is $F_n = F_{n-1} + F_{n-2}$.

Example 4.6.1

Bob wants to climb 8 stairs. He can climb 1 or 2 stairs at a time. How many ways are there for him to climb the 8 stairs?

Solution: In this problem, an example of a path that would work would be to climb 2 stairs a total of four times. Or, he could climb 1, 2, 2, 2, and 1.

However, counting in that way will take a long time. That's when recursion comes in.

Let's assume that a_n represents the number of ways to climb n stairs. We want to try to relate this terms that precede it such as a_{n-1} , a_{n-2} , a_{n-3} , etc.

We notice that if Bob is at $a_n - 1$ or $a_n - 2$ a step before when he's at a_n , then he can always reach a_n .

Thus, this means that $a_n = a_{n-1} + a_{n-2}$. We now need to find the first few terms to be able to find the rest using the relation above.

Clearly a_1 is 1 since there is only 1 way to climb 1 stairs. Similarly, a_2 is 2 since there are 2 ways to climb 2 stairs (either use the 2 stairs move or use the 1 stairs move twice).

$$a_1 = 1$$

$$a_2 = 2$$

$$a_3 = a_2 + a_1 = 3$$

$$a_4 = a_3 + a_2 = 5$$

$$a_5 = a_4 + a_3 = 8$$

$$a_6 = a_5 + a_4 = 13$$

$$a_7 = a_6 + a_5 = 21$$

$$a_8 = a_7 + a_6 = 34$$

Since we found a_8 to be 34, that means our final answer is 34.

Problem 4.6.2 — Everyday at school, Jo climbs a flight of 6 stairs. Jo can take the stairs 1, 2, or 3 at a time. For example, Jo could climb 3, then 1, then 2. In how many ways can Jo climb the stairs?

(A) 13 (B) 18 (C) 20 (D) 22 (E) 24

Source: 2010 AMC 8 Problem 25

Solution: In this problem, there are many simpler ways to solve the problem, but we will use recursion.

a_n will represent the number of ways for Jo to climb n stairs.

Clearly a_1 is 1.

a_2 is 2 because he can either climb 1 stairs two times or 2 stairs once.

In a similar manner, a_3 is 4 (3), (1, 2), (2, 1), (1, 1, 1)

We notice that this is similar to the example covered before. We can again relate it to smaller terms such as a_{n-1}

As long as Joe's location before he comes to the n th step is $n-1$, $n-2$, or $n-3$, then there will always be a way for him to make it to the n th step.

This means that $a_n = a_{n-1} + a_{n-2} + a_{n-3}$

We can use our previous terms to find the remaining terms now. We simply find a_6 to be **24**.

Problem 4.6.3 — Call a set of integers "spacy" if it contains no more than one out of any three consecutive integers. How many subsets of $\{1, 2, 3, \dots, 12\}$, including the empty set, are spacy?

(A) 121 (B) 123 (C) 125 (D) 127 (E) 129

Source: (2007 AMC 12)

Solution: In this problem, we will again try to use recursion.

a_n represents the number of subsets that satisfy the requirement of being spacy from numbers 1 through n .

We will find the first few terms. a_1 is clearly 2 since the subset can either have 1 or be empty.

Similarly, a_2 is 3 since the subset can have 1, 2, or be empty.

Similarly for a_3 , the subset can have 1, 2, 3, or be empty. Thus, a_3 is 4

Now to find a relationship for a_n , we will take two cases. One will be to count how many subsets are possible assuming that subset contains n , and the other will be to find the subsets that are possible assuming the subset doesn't have n .

Case 1: Our subset includes n

In this case, the numbers $n - 1$ and $n - 2$ can't qualify as a candidate for the subset. However, any number before $n - 3$ and inclusive of that can. Thus, the number of possible cases here is a_3

Case 2: Our subset doesn't include n

In this case, $n - 1$ is the highest possible element of the subset. Thus, our total count here is simply a_{n-1} representing the number of subsets we can make from 1 to $n - 1$.

This tells us that $a_n = a_{n-1} + a_{n-3}$

Now using the terms we found in the beginning for a_1 , a_2 , a_3 , we can find the remaining terms and find a_{12} to be **129**.

Problem 4.6.4 — How many sequences of 0s and 1s of length 19 are there that begin with a 0, end with a 0, contain no two consecutive 0s, and contain no three consecutive 1s?

(A) 55 (B) 60 (C) 65 (D) 70 (E) 75

Source: 2019 AMC 10

In this problem, we will assume that a_n represents the number of sequences of length n that start with 0 and end with 0. We know that after 0, we can't have another 0 because that violates our conditions of not containing any consecutive 0s. Thus, we must have 1 appear after.

Some inspection shows us that we can add 10 or 110 to our 0. Since those strings are 2 and 3 characters long, this means that

$a_n = a_{n-2} + a_{n-3}$ (if you're still confused about this step then go back and review this section)

We know that a_3 must be 1 since the only possible sequence is 010

a_4 must be 1 since the only possible sequence is 0110

a_5 must be 1 since the only possible sequence1 are 01010

Using these 3 terms, we can calculate the rest and find that a_{19} is **65 (C)**.

Problem 4.6.5 — Define a sequence recursively by $t_1 = 20$, $t_2 = 21$, and

$$t_n = \frac{5t_{n-1} + 1}{25t_{n-2}}$$

for all $n \geq 3$. Then t_{2020} can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.

Source: 2020 AIME

Solution: In this problem, although we are given a recursive formula, sometimes you can find the answer by looking for repetition in the terms. The terms will start to repeat, and that will simplify the problem for you.

In this case, we will try to find a few of the numbers and see what the period is.

$$t_1 = 20$$

$$t_2 = 21$$

$$t_3 = \frac{5t_2+1}{25t_1} = \frac{53}{250}$$

$$t_4 = \frac{5t_3+1}{25t_2} = \frac{103}{26250}$$

$$t_5 = \frac{5t_4+1}{25t_3} = \frac{101}{525}$$

$$t_6 = \frac{5t_5+1}{25t_4} = 20$$

$$t_7 = \frac{5t_6+1}{25t_5} = 21$$

From here we notice that the term at t_1 repeats at t_6 . This means that the period is 5. Clearly $t_1 = t_6 = t_{11} \dots = t_{2016}$.

From there, we just count up 4 numbers from t_1 since that is equivalent to t_{2016} , and counting up 4 gives us t_5 which is the answer. We add the numerator and denominator to get **626**.

§4.7 Conditional Probability

Conditional probability is an important topic that shows up a lot on the AMC and AIME. It's about finding the probability of something given that a condition is already true.

Example 4.7.1

Given that the number we roll on a dice is even, what is the probability that our number is 2.

Solution: In a way, this problem is asking for $P(\text{Even Number that is 2})/P(\text{even number})$

The given condition is the number being even, and there are 3 even numbers that are possible which are 2, 4, 6. Thus, our probability is simply $\frac{1}{3}$

Theorem 4.7.2

Mathematical Notation of Bayes Theorem $P(A|B) = \frac{P(A \cap B)}{P(B)}$

The statement above is trying to say the Probability of event A happening given that event B happened already is $P(A \text{ and } B)/P(B)$

If A and B are independent events, then $P(A|B) = P(A)$.

We can confirm the validity of the statement above because if A and B are independent, then whether B occurs or not won't affect the chance of A occurring.

Problem 4.7.3 — Let S be the set of permutations of the sequence 1, 2, 3, 4, 5 for which the first term is not 1. A permutation is chosen randomly from S . The probability that the second term is 2, in lowest terms, is a/b . What is $a + b$?

- (A) 5 (B) 6 (C) 11 (D) 16 (E) 19

Source: 2003 AMC 12

Solution: In this problem, we are already given that the first term is not 1. Our condition is that the second term is 2. We can use the $P(A|B) = \frac{P(A \cap B)}{P(B)}$ statement here and rewrite it in our terms.

$P(\text{Second term is 2})$ given that the first term is not 1

$P(\text{Second term is 2 and first term is not 1}) / P(\text{first term is not 1})$

The number of ways for the second term to be 2 and first term to not be 1 is simply $3 \cdot 3!$ since we have 3 spots to place 1

Similarly, the number of ways that the first term is not 1 is $4 \cdot 4!$ since we have 4 places to place 1 (any place that isn't in the first term) and $4!$ ways to rearrange the others

Our answer is now simply $\frac{3 \cdot 3!}{4 \cdot 4!} = \frac{3}{16}$ and our answer is $3 + 16$ which is 19 **(E)**

Problem 4.7.4 — A bug starts at a vertex of an equilateral triangle. On each move, it randomly selects one of the two vertices where it is not currently located, and crawls along a side of the triangle to that vertex. Given that the probability that the bug moves to its starting vertex on its tenth move is m/n , where m and n are relatively prime positive integers, find $m + n$.

Source: 2003 AIME

Solution: In this problem, our given condition is that there are 10 moves while our condition is that it comes back to the starting position.

We first calculate the total number of ways there are for there to be 10 moves which goes in the denominator. Since in each move there are 2 places where you can move, there are a total of 2^{10} ways to move.

Now we can assume that CW represents a clockwise move in our triangle while CCW represents a counterclockwise move. Let's assume that X is the number of CW moves while Y is the number of CCW moves.

We know that $X + Y = 10$

We also know that $X - Y \equiv 0 \pmod{3}$ since after all the moves we need to count the number of ways it returns to the starting point.

$$X \equiv Y \pmod{3}$$

Let's assume that both X and Y are $0 \pmod{3}$. However, the sum of that is $0 \pmod{3}$ but we know that the two numbers sum to 10 which is $1 \pmod{3}$. Similarly, both cannot

be 1 mod 3, but they have to be 2 mod 3.

Since we know that both X and Y are 2 mod 3 and that they sum to 10, the possible values for (X, Y) are (2, 8), (5, 5), and (8, 2). From here, we calculate the number of possible ways for each case. We realize it's simply rearranging the 10 letters (such as XXXYYYYYYY) so we get

$$\binom{10}{2} + \binom{10}{5} + \binom{10}{8} = 342$$

Our total probability is $\frac{342}{2^{10}} = \frac{171}{512}$. Our answer is $171 + 512$ which is **683**

§4.8 Expected Value

Theorem 4.8.1

Expected value is the sum of all probabilities times value that you get with that probability. It is the average of all outcomes in a way. You might be confused with this, so just continue reading to make sense of this.

Expected value is what the average of a certain outcomes will be. For example, pretend that I have a probability of $\frac{1}{3}$ of winning one dollar. I also have a probability of $\frac{1}{2}$ of losing 2 dollars. I also have a probability of $\frac{1}{6}$ of not gaining or losing any money. Then, we find the expected value by multiplying the chance of the outcome to the outcome. After that, you add all of those numbers. In this case, the first outcome is winning a dollar and there is $\frac{1}{3}$ chance of that. Furthermore, the second outcome states you have a probability of $\frac{1}{2}$ when it comes to losing 2 dollars. You write the equation out for all 3 outcomes. $(\frac{1}{3} \cdot 1) + (\frac{1}{2} \cdot (-2)) + (\frac{1}{6} \cdot 0)$. After writing the expression for the expected value, now you compute it. Remember that I put negative 2 because you LOSE two dollars with a probability of $\frac{1}{2}$. After multiplying the terms, you get $\frac{1}{3} + -1 + 0$. Adding them up gives you $-\frac{2}{3}$ which is the expected value.

Example 4.8.2

If we have a dice with 6 sides. What is the expected value of one roll?

The possible rolls for us to roll are 1, 2, 3, 4, 5, and 6. Each roll has a $\frac{1}{6}$ probability of showing up. Thus, we simply multiply the probability of each roll to the roll value to find the expected value of the roll.

$$\frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = \frac{1}{6} \cdot 21 = \frac{7}{2}$$

Thus, the expected of one roll is simple $\frac{7}{2}$

What if we want to find the expected value of the sum of the numbers that show up on like 50 dices or 25 dices. How will we do that now? If we individually write out all possible sums that will take a LONG time.

Theorem 4.8.3**Linearity of Expectation**

If we have random variables X and Y , then the $E[X + Y] = E[X] + E[Y]$ (even if they are dependent).

You can do this for a large amount of events also, it doesn't have to be just two.

Example 4.8.4

What is the expected value of the sum of numbers on six dice if we roll them all at once.

In this problem, we will now use linearity of expectation.

$$E[X_1 + X_2 + X_3 + X_4 + X_5 + X_6] = E[X_1] + E[X_2] + E[X_3] + E[X_4] + E[X_5] + E[X_6]$$

$$= 6 \cdot E[\text{One roll}] = 6 \cdot \frac{7}{2} = 21$$

Thus, from linearity of expectation, the expected value of the sum of 6 rolls is 21

Problem 4.8.5 — A 13-digit binary number with four 1's is chosen at random. What is its expected value?

Solution: The leftmost digit must obviously be 1. This means that all the numbers will have a 2^{12} part to it.

Since expected value is simply the sum of the probabilities multiplied to the possible value, we can find the probability for each digit to be 1. Since there are 12 remaining digits not including the leftmost one, the probability for any of those digits to be 1 is simply $\frac{3}{12}$ which is $\frac{1}{4}$ (since there are 3 more digits that must be 1).

This means that the expected value is $2^{12} + \frac{1}{4}(2^{11} + 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0)$.

We can use the sum of a geometric sequence to get that $2^{11} + 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$ is $\frac{2^{12}-1}{2-1}$ which is equivalent to 4095.

Our expected value is $2^{12} + \frac{4095}{4}$ which is $\frac{20479}{4}$.

Problem 4.8.6 — Let $\mathcal{S}$ be the set of real numbers that can be represented as repeating decimal decimals of the form $0.\overline{abc}$ where a, b, c are distinct digits. Find the sum of the elements of $\mathcal{S}$.

Source: 2006 AIME

Solution: In this problem, we will first convert $0.\overline{abc}$ to a fraction.
 $x = 0.\overline{abc}$

Multiplying both sides by 1000 gives

$$1000x = abc.\overline{abc}$$

Subtracting both equations gives

$$999x = abc$$

Finally, we get $x = \frac{100a+10b+c}{999}$ (and x is also $0.\overline{abc}$)

From here, we know that a, b, c can be any digit from 0 to 9. Instead of summing up each possible number, we can simply find the expected value of each digit. The expected value of each digit is 4.5. Using that, we know that

$$E[x] = \frac{100 \cdot 4.5 + 10 \cdot 4.5 + 4.5}{999}.$$

We must multiply this to the number of possible numbers which are $10 \cdot 9 \cdot 8$ (since all digits must be distinct) which is 720.

$$720 \cdot \frac{100 \cdot 4.5 + 10 \cdot 4.5 + 4.5}{999} = \mathbf{360}.$$

§4.9 States (Markov Chains)

In markov chain equations, you write an equation for a current point based on previous points that you could visit. You represent the probability of each state based on the probability of previous ones.

I highly recommend watching this video to get a basic introduction:

<https://www.youtube.com/watch?v=i3AkT09HLXo>

Problem 4.9.1 — A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square?

(A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{7}{8}$

Source: 2020 AMC 10

Solution: From point $(1, 2)$, the frog can move to $(1, 1)$, $(1, 3)$, $(0, 2)$, and $(2, 2)$.

There's a $\frac{1}{4}$ chance of the frog landing at each of those positions. If it lands at $(0, 2)$, then it reaches a vertical side which is what we want. The probability of that happening is $\frac{1}{4}$.

If it goes to either points $(1, 1)$ or $(1, 3)$, then by symmetry the chance of hitting a vertical side is $\frac{1}{2}$ (think about symmetry to find out why).

The probability of the frog landing at one of those points is $\frac{1}{4} \cdot \frac{1}{2}$ which is $\frac{1}{8}$. We must multiply this by 2 since that probability works for 2 points. We get $\frac{1}{4}$.

The last case to look at is when the frog jumps to $(2, 2)$ from its starting location. It is again easy to see that from symmetry the probability of landing on a vertical side is $\frac{1}{2}$. This means the probability for this case is $\frac{1}{4} \cdot \frac{1}{2}$ which is $\frac{1}{8}$.

Adding up our probabilities gives $\frac{1}{4} + \frac{1}{4} + \frac{1}{8}$ which is $\frac{5}{8}$ (B).

Problem 4.9.2 — Dave rolls a fair six-sided die until a six appears for the first time. Independently, Linda rolls a fair six-sided die until a six appears for the first time. Let m and n be relatively prime positive integers such that $\frac{m}{n}$ is the probability that the number of times Dave rolls his die is equal to or within one of the number of times Linda rolls her die. Find $m + n$.

Source: 2009 AIME

Solution: Let's assume that p_n represents the probability of getting a 6 for the first time on the n -th roll. It's easy to see that $p_n = (\frac{5}{6})^{n-1} \cdot \frac{1}{6} = \frac{5^{n-1}}{6^n}$.

It's easy to see that our total probability is $p_1(p_1 + p_2) + p_2(p_1 + p_2 + p_3) + p_3(p_2 + p_3 + p_4) \dots$

Plugging in a few terms gives us that every term starting from $p_2(p_1 + p_2 + p_3)$ is part of an infinite geometric series. The ratio of it is $\frac{25}{36}$.

Our total probability can be found by using the infinite geometric series.

Our total probability is thus $p_1(p_1 + p_2) + \frac{p_2(p_1 + p_2 + p_3)}{1 - \frac{25}{36}}$.

Evaluating it gives us an answer of $\frac{8}{33}$, and we get $8 + 33 = 41$.

Problem 4.9.3 — Lily pads $1, 2, 3, \dots$ lie in a row on a pond. A frog makes a sequence of jumps starting on pad 1. From any pad k the frog jumps to either pad $k + 1$ or pad $k + 2$ chosen randomly with probability $\frac{1}{2}$ and independently of other jumps. The probability that the frog visits pad 7 is $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.

Source: 2019 AIME

Solution: Let's say that the probability of reaching pad n is p_n .

Since we know that from pad k the frog can only jump to $k + 1$ or $k + 2$, it's easy to find that $p_n = \frac{1}{2}(p_{n-1} + p_{n-2})$.

We can write the equations out for $n = 3, 4, 5, 6$, and 7 .

$$p_3 = \frac{1}{2}(p_1 + p_2)$$

$$p_4 = \frac{1}{2}(p_2 + p_3)$$

$$p_5 = \frac{1}{2}(p_3 + p_4)$$

$$p_6 = \frac{1}{2}(p_4 + p_5)$$

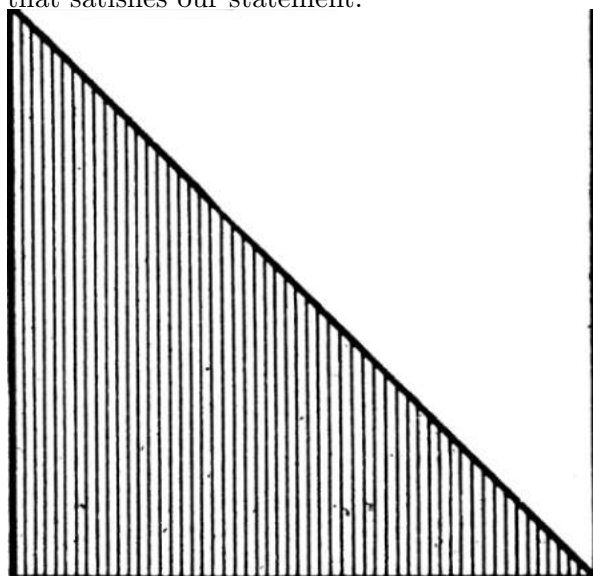
$$p_7 = \frac{1}{2}(p_5 + p_6)$$

Since we start at pad 1, $p_1 = 1$. Also, $p_2 = \frac{1}{2}$. We can plug these values into the 5 equations above and find that $p_7 = \frac{43}{64}$, and our answer is $43 + 64 = 107$.

§4.10 Geometric Probability

If you have ever seen a problem in which the number of cases you want is infinite, then all you do is “graph” it or draw it in a shape. For example, pretend that I have two numbers x and y which are both in the range of 0 and 1. We want to know the probability of getting $x + y < 1$. There are an infinite amount of options for this. However, we can make a unit square for this problem and graph out $x + y = 1$. Then, we take the region

that satisfies our statement.



This shows that whenever the desired outcomes can't be counted, you graph it out and find the area of the desired region divided by the area of the entire region.

Problem 4.10.1 — Two mathematicians take a morning coffee break each day. They arrive at the cafeteria independently, at random times between 9 a.m. and 10 a.m., and stay for exactly m minutes. The probability that either one arrives while the other is in the cafeteria is 40%, and $m = a - b\sqrt{c}$, where a, b , and c are positive integers, and c is not divisible by the square of any prime. Find $a + b + c$.

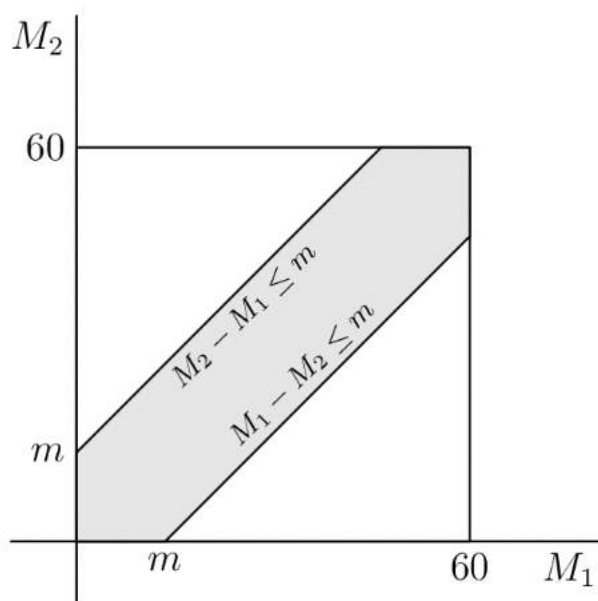
Source: 1998 AIME

Solution: We can make a graph in which the x and y -axis range from 0 to 60 minutes.

We can have two cases. The first mathematician will arrive at time M_1 while the second will arrive at M_2 . If M_1 comes before M_2 , then $M_1 - M_2 \leq m$.

Now if M_2 comes before M_1 , then $M_2 - M_1 \leq m$.

We can graph our two inequalities to get



Clearly the two lines bound an area which is the area we want to find. Instead of finding the shaded area, we can find the area of the white part and subtract that out from the total area.

We have two isosceles right triangles with legs of length $60 - m$. The area of one right triangle is $\frac{(60-m)(60-m)}{2}$. We multiply that by 2 to find the area of both to get $(60 - m)(60 - m)$.

The total area of the entire square is $60 \cdot 60$ which is 3600. We know that the fraction of area occupied by the unshaded part is $1 - 0.4 = 0.6$.

Writing this as an equation gives $\frac{(60-m)(60-m)}{3600} = 0.6$

Solving gives that $m = 60 - 12\sqrt{15}$.

Thus, our answer is $60 + 12 + 15$ which is **87**.

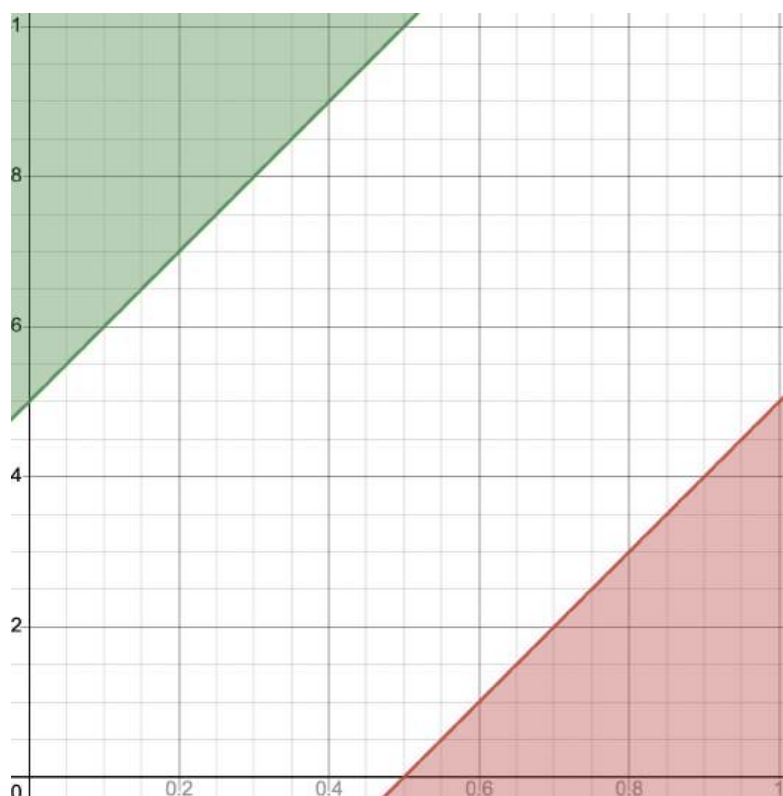
Problem 4.10.2 — Let S be a square of side length 1. Two points are chosen independently at random on the sides of S . The probability that the straight-line distance between the points is at least $\frac{1}{2}$ is $\frac{a - b\pi}{c}$, where a , b , and c are positive integers with $\gcd(a, b, c) = 1$. What is $a + b + c$?

(A) 59 (B) 60 (C) 61 (D) 62 (E) 63

Source: 2015 AMC

Source: In this problem, we can fix one point on one of the sides of the square. Let's say point A is fixed on one of the sides. Then, point B can either go on an adjacent side, same side, or the opposite side in that square.

Case 1: The two points are on the same side. Since there are infinite locations it go on the same side, we can graph this out to see the probability of this occurring.



We can easily see that the probability of both lying on the same side and creating a side length of more than $\frac{1}{2}$ is $\frac{1}{4}$. However, we must multiply this by $\frac{1}{4}$ since that's the probability for point B to lie on the side that point A is on. The total probability for this case is $\frac{1}{16}$.

Case 2: Point B lies on an adjacent side to A. Let's assume that point A lies on $(x, 0)$ while point B lies on $(0, y)$. Then, the distance between them is $\sqrt{x^2 + y^2}$. We know that this must be $\geq \frac{1}{2}$.

Squaring both sides gives $x^2 + y^2 \geq \frac{1}{4}$. We know that this circle has a radius of $\frac{1}{2}$ and it's total area is $\frac{\pi}{4}$. However, we only want the area of the quarter circle since we must subtract that from the area of a square. We can see this through graphing. The area of the quarter circle is one fourth the area of the entire circle which is $\frac{\pi}{16}$. Subtracting this from 1 gives $\frac{16-\pi}{16}$.

Now we multiply that by $\frac{1}{2}$ which is the probability of choosing an adjacent side to the side that point A is on.

We get $\frac{16-\pi}{32}$ as the total probability for this case.

Case 3: Now point B lies on the side opposite to point A. The chance of us choosing a side opposite to the side point A is on is $\frac{1}{4}$. No matter where point B lies on the side opposite to the side with point A, the distance between them will always be $\geq \frac{1}{2}$. Thus, the total probability for hits case is $\frac{1}{4}$.

Adding up all the probabilities gives $\frac{1}{16} + \frac{16-\pi}{32} + \frac{1}{4}$ which is $\frac{26-\pi}{32}$. Our final answer is $26 + 1 + 32$ which is **59 (A)**.

5 Geometry

Geometry is all about practice. Throughout this section, you will find various formulas and concepts. Your goal shouldn't be to memorize all the formulas. It should be to learn the application of those formulas and where to apply them.

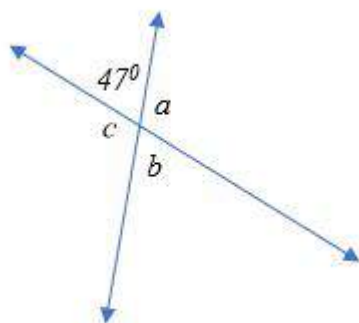
§5.1 Angles

Angles is a very common topic on both the AMC 10 and AMC 12 along with many other contests. It is important to have a strong base in this concept to do well on these contests. You can find the answers to some angle problems by using a protractor. However, that method won't always work, and this section will give you a strong base for angles and related problems.

Definition 5.1.1

Straight angles are angles that have a measure of 180 degrees. All lines are straight because their angle measure is 180 degrees.

Problem 5.1.2 — Using the diagram below and the concept that you just learned, find the measures of angles a , b , and c .



Hopefully you tried the problem. Maybe some of you know that you could simply find all of those angles by the vertical angle theorem (which we'll learn about later). That's perfectly right. However, we'll demonstrate the way to find the measures using the fact that straight lines have an angle measure of 180 degrees.

Since we know that all the lines are straight, $47 + \angle c$ has to be 180° . Thus, we know that $\angle c$ is simply $180 - 47$ which is 133° . You can do the same for the other lines to find that $\angle c + \angle b = 180^\circ$ and $\angle a + \angle b = 180^\circ$. Clearly $\angle b$ is 47° and $\angle a$ is 133° .

Definition 5.1.3

A right angle is an angle with a measure of 90 degrees.

Definition 5.1.4

Complementary angles are two angles that sum to 90 degrees.

Problem 5.1.5 — Find the complement of 30 degrees.

Now we know that if two angles are complementary, then the sum of the measures will be 90 degrees. Let's pretend that the measure of the complement of 30 degrees is x . We know that $30 + x = 90$. We can easily solve this by subtracting 30 degrees from both sides to find that the answer is 60 degrees.

Definition 5.1.6

Supplementary angles are two angles that sum to 180 degrees. This is an important definition to know.

Problem 5.1.7 — Find the supplement of 85 degrees.

The problem above can easily be solved. Let's pretend that the supplement is x . We know that $x + 85 = 180$ because supplementary angles sum to 180 degrees. We can solve the equation by subtracting 85 from both sides to get 95 degrees as our answer.

Definition 5.1.8

An **acute** angle has an angle measure that is strictly less than 90 degrees. An **obtuse** angle has an angle measure that is always strictly greater than 90 degrees.

Now let's hope that you have learned all of the definitions and key terms. There will be super challenging problems at the end of the chapter. Before skipping to them, read all the sections in this chapter.

Interior and Exterior Angles in Any Polygon

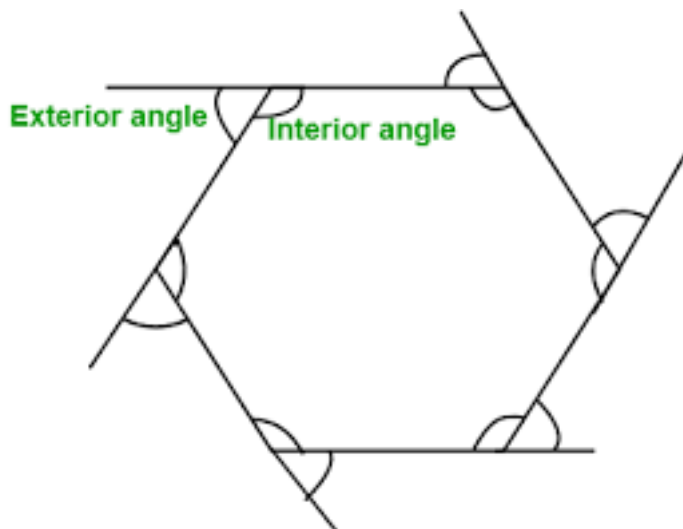
In geometry, there are lots of 2D shapes that include triangles, squares, rectangles, parallelograms, pentagons, hexagons, octagons, etc. Memorizing the sum of the angle measures for all the shapes can be hard. However, you can memorize one easy formula that will help you derive the sum of the angle measures in any polygon. Is it better to memorize one formula, or the angle sum for lots of shapes?

Theorem 5.1.9

In any polygon, the sum of the angle measures is $180(n - 2)$ where n represents the number of sides in the shape.

Definition 5.1.10

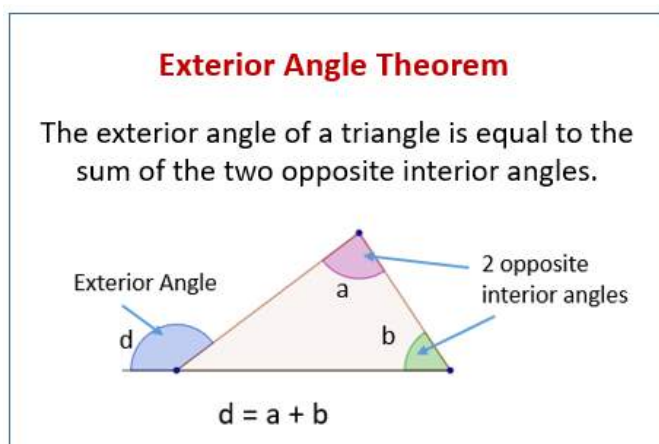
An exterior angle is an angle outside the polygon. To find the measure of any exterior angle, extend the side in the same direction. Then, subtract the measure of the interior angle from 180 degrees to find the exterior angle of that side.

**Theorem 5.1.11**

In any **regular** polygon, all the exterior angles are $\frac{360}{n}$ where n represents the number of sides. Remember that the sum of the exterior angles for any polygon is **always** 360 degrees.

Theorem 5.1.12

In all triangles, the exterior angle of any side is the sum of the two opposite remote interior angles. If this doesn't make sense, then the picture below will clear the questions that you have.



Theorem 5.1.13

A cyclic quadrilateral is a quadrilateral that can be inscribed in a circle with all the vertices on the circumference. A property states that opposite angles always sum to 180 degrees. (This is just the basics of cyclic quadrilaterals. In future chapters, you will learn more and go into the depth.)

Parallel Lines and Angles

Whenever you see parallel lines, then a lot of the angles will be congruent. If you don't know why, then this section will help you understand why.

Definition 5.1.14

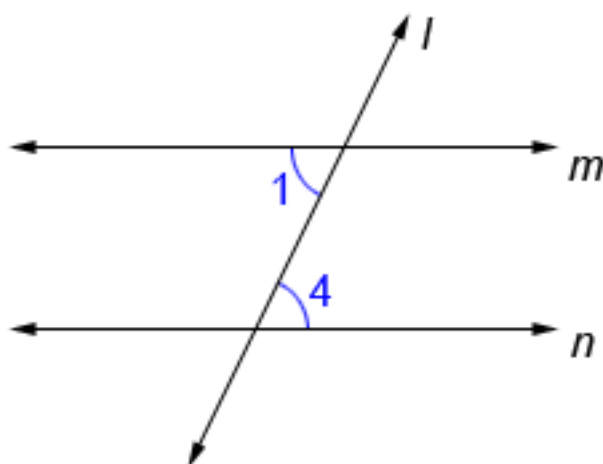
A transversal is a line that intersects other lines. You will see the importance in a bit.

Definition 5.1.15

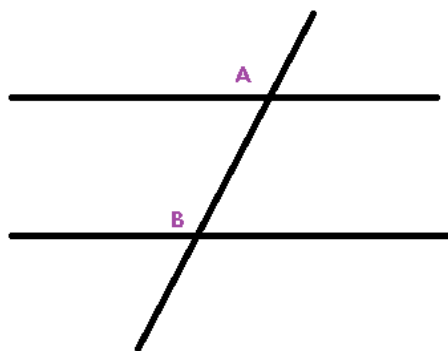
Parallel lines are lines that **never** intersect.

Theorem 5.1.16

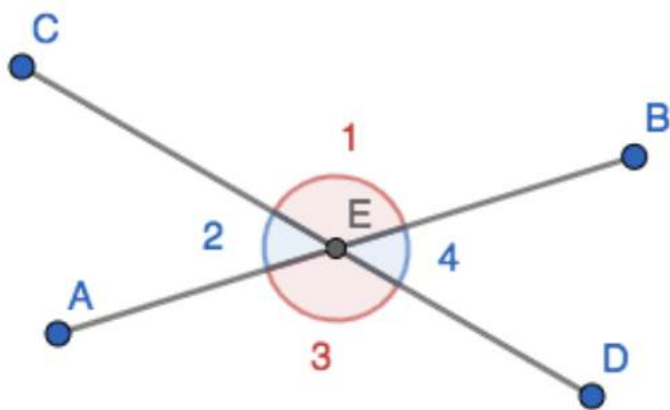
If you have two parallel lines and a transversal intersecting both, then alternate interior angles are always congruent. In the diagram below, angles 1 and 4 are congruent because of this theorem.

**Theorem 5.1.17**

If you have two parallel lines and a transversal intersects both, then corresponding angles are always congruent. In the picture below, the two angles that are A and B are an example of two congruent corresponding angles.

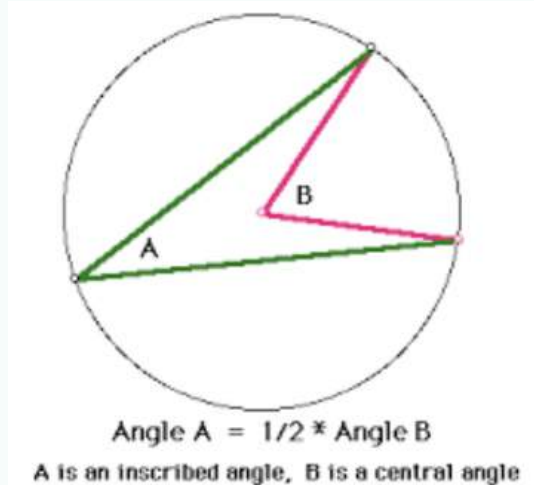
**Theorem 5.1.18**

If there are two intersecting lines, then vertical angles (pair of opposite angles that are created by two intersecting lines) will always be congruent no matter what. In the picture below, angle 2 is congruent to angle 4 while angle 1 is congruent to angle 3 because of this theorem.

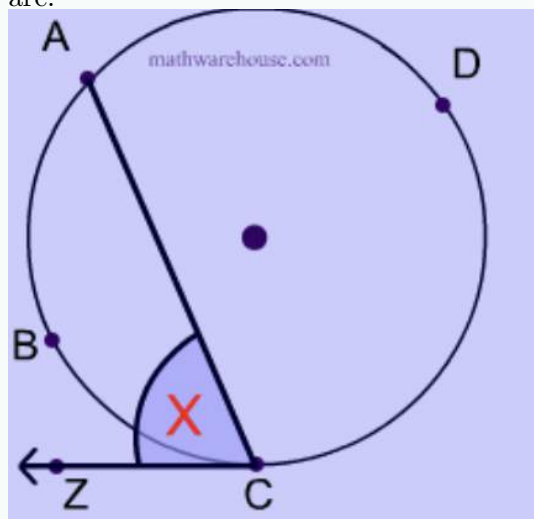


Theorem 5.1.19

An inscribed arc has an angle measure half of the central angle measure.

**Theorem 5.1.20****Inscribed Angle with Tangent Line Theorem**

The angle that is cut by the side tangent to the circle is equivalent to the inscribed arc.

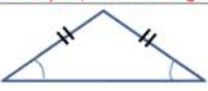
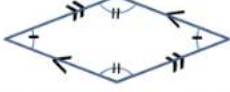
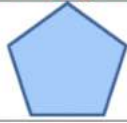
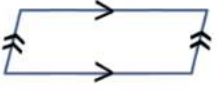
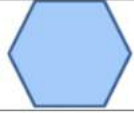
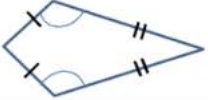
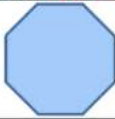
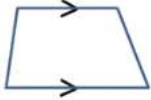
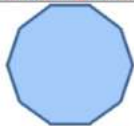
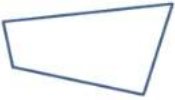


In the diagram above, $\angle ACZ = \angle ADC$

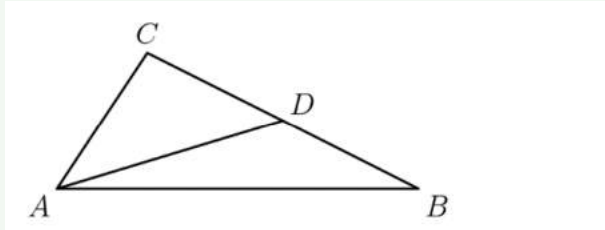
Common Shapes and their Angles

LIST OF GEOMETRIC SHAPES 2D



TRIANGLES	QUADRILATERALS	REGULAR POLYGONS
 Equilateral triangle All sides equal; interior angles 60°	 Square All sides equal; all angles 90°	 Equilateral triangle 3 sides; angles 60°
 Isosceles triangle 2 sides equal; 2 congruent angles	 Rectangle Opposite sides equal, all angles 90°	 Square 4 sides; angles 90°
 Scalene triangle No sides or angles equal	 Rhombus All sides equal; 2 pairs of parallel lines; opposite angles equal	 Regular Pentagon 5 sides; angles 108°
 Right triangle 1 right angle	 Parallelogram Opposite sides equal, 2 pairs of parallel lines	 Regular Hexagon 6 sides; angles 120°
 Acute triangle All angles acute	 Kite Adjacent sides equal; 2 congruent angles	 Regular Octagon 8 sides; angles 135°
 Obtuse triangle 1 obtuse angle	 Trapezoid 1 pair of parallel sides	 Regular Decagon 10 sides; angles 144°
	 Trapezium No pairs of parallel sides	

Problem 5.1.21 — In triangle ABC , $AC = CD$ and $\angle CAB - \angle ABC = 30$ degrees. What is $\angle BAD$.



Source: 1957 AHSME

Solution: In this problem, we'll assume that $\angle ABC = b^\circ$

Since we know that $\angle CAB - \angle ABC = 30$, $\angle CAB = b + 30^\circ$

Since the sum of the angles of a triangle sum to 180, we can find $\angle ACB$ in terms of b to be $180 - (b + b + 30) = 150 - 2b$.

Now since we know that $\angle ACB = \angle ACD = 150 - 2b$, we can use this to find $\angle CAD$ and $\angle CDA$ since both of those angles are equal. The reason is that $\triangle ACD$ is isosceles. This means that using the equation,

$$2 \cdot \angle CAD + \angle ACD = 180,$$

We get that $\angle CAD = b + 15^\circ$

Now since we know that $\angle CAB = b + 30^\circ$ and $\angle CAD = b + 15^\circ$, we can find $\angle DAB$ since

$$\angle DAB + \angle CAD = \angle CAB$$

$$\angle DAB + b + 15 = b + 30$$

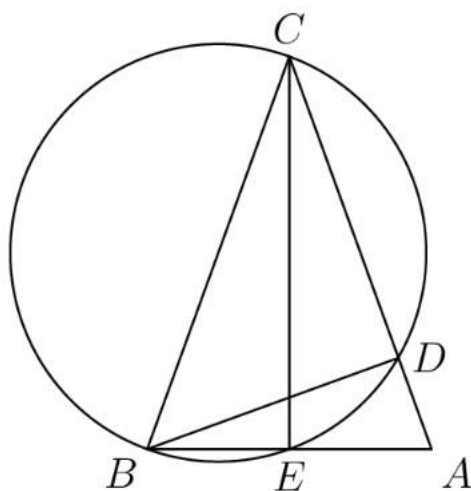
This means that our answer for $\angle DAB$ is 15° .

Problem 5.1.22 — Let $\triangle ABC$ be an isosceles triangle with $BC = AC$ and $\angle ACB = 40^\circ$. Construct the circle with diameter $\overline{BC}$, and let D and E be the other intersection points of the circle with the sides $\overline{AC}$ and $\overline{AB}$, respectively. Let F be the intersection of the diagonals of the quadrilateral $BCDE$. What is the degree measure of $\angle BFC$?

(A) 90 (B) 100 (C) 105 (D) 110 (E) 120

Source: 2019 AMC 10

Solution: In this problem, we will first make a diagram.



In this problem, since we know that arc measure of CB is simply 180 degrees, then the inscribed angles that correspond with arc CB are $\angle CDB$ and $\angle CEB$.

Since we know that the inscribed angle is half the measure of the arc, that means $\angle CDB$ and $\angle CEB$ are 90° .

We know that $\angle ACB = 40^\circ$. This means that $\angle CAB$ and $\angle CBA$ are 70° since $\triangle ABC$ is isosceles.

Since we know that $\triangle CEB$ is a right triangle since the measure of $\angle CEB$ is 90° , $\angle ECB$ must be 20° because $\angle EBC$ is 70° .

Similarly, for $\triangle CDB$: since $\angle CDB = 90^\circ$ and $\angle ACB = \angle DCB = 40^\circ$, $\angle CBD$ must be 50° .

Since we know that $\angle ECB = \angle FCB = 20^\circ$ and $\angle CBD = \angle CBF = 50^\circ$, $\angle BFC$ must be $180 - 20 - 50 = 110^\circ$.

§5.2 Similar Triangles

The AMC and other contests are full of problems that involve triangles. Many of them look the same, and you might assume that all the side lengths are the same even though you aren't sure. This section will help you prove that the triangles are actually similar to each other.

Definition 5.2.1

Congruent triangles are triangles that have all 3 angles and sides that are exactly the same.

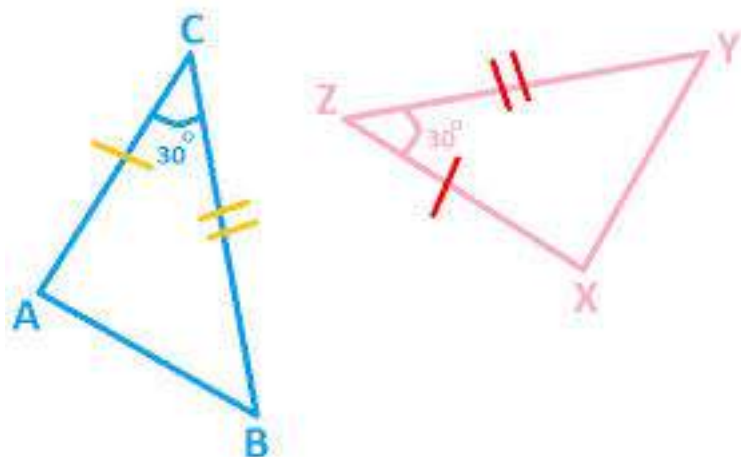
Theorem 5.2.2

SSS Congruence: This theorem stands for side side side. This theorem states that if two triangles have all 3 sides that have the same length, then both of them are congruent. (The picture below shows an example of two triangles that can be proved congruent using SSS Congruence.)

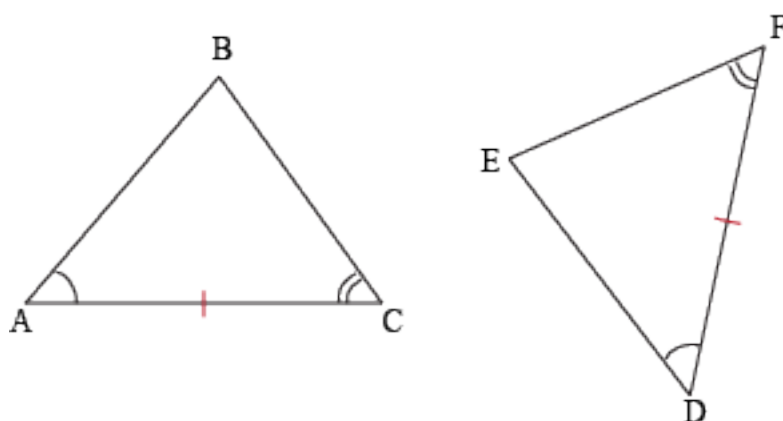


Theorem 5.2.3

SAS Congruence: This stands for side angle side. It states that if two sides of one triangle and the angle between those two sides are equal to the corresponding sides and angles of a different triangle, then both of the triangles are congruent.

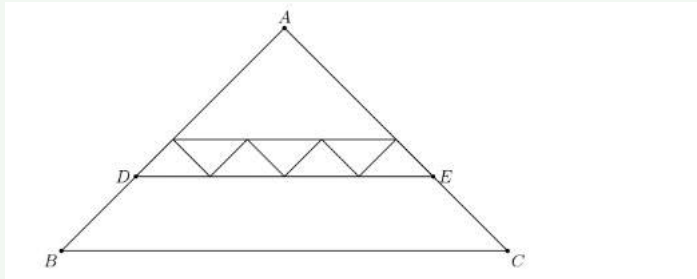
**Theorem 5.2.4**

ASA Congruence: This theorem stands for angle side angle. This theorem can prove two triangles congruent if both of the triangles have two angles that are congruent and a side between those two angles that is also congruent. (The picture below will help you understand this theorem better.)

**Theorem 5.2.5**

AAS Congruence: This theorem stands for angle angle side. This theorem works if two angles and a side of one triangle is equal to the corresponding angles and sides of a different triangle. If this is true, then both of the triangles are congruent.

Problem 5.2.6 — All of the triangles in the diagram below are similar to isosceles triangle ABC , in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$?



(A) 16 (B) 18 (C) 20 (D) 22 (E) 24

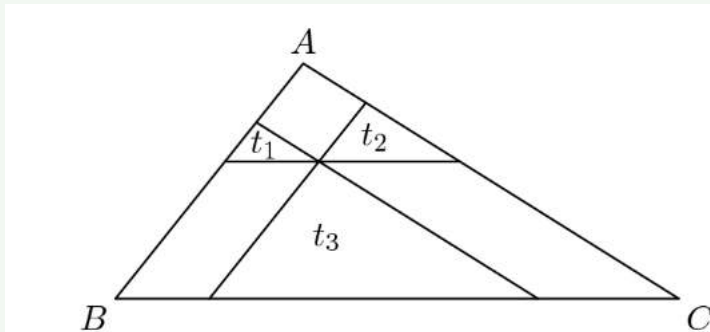
Source: 2018 AMC 10

Solution: In this problem, since we know that all triangles are similar, that means that $\triangle ADE$ is similar to each of the 7 smaller triangles. Clearly we need 4 bases of the small triangles to make the base of $\triangle ADE$ which is DE

Since the ratio is 4 for the base, it means the ratio is 4 for all sides. Since the area of the small triangle is 1, it means the area of $\triangle ADE$ is 4^2 times the area of the small triangle meaning it is 16.

Since the area of the entire shape is 40, the area of the trapezoid is simply $40 - 16$ which is **24(E)**.

Problem 5.2.7 — A point P is chosen in the interior of $\triangle ABC$ such that when lines are drawn through P parallel to the sides of $\triangle ABC$, the resulting smaller triangles t_1 , t_2 , and t_3 in the figure, have areas 4, 9, and 49, respectively. Find the area of $\triangle ABC$.



Source: 1984 AIME

Solution: Since the lines drawn from point P are all parallel to the sides of $\triangle ABC$, the triangles with area 4, 9, 49 are similar to $\triangle ABC$.

We know that if the ratio of the areas between two similar triangles is r^2 , then the ratio of the side lengths is r .

We can take the square root of all the areas for triangles t_1, t_2, t_3 to get 2, 3, 7 for the ratios between the side lengths.

Let's assume that the bases for each of those triangles is $2x, 3x, 7x$.

We want to find the length of the base of $\triangle ABC$.

Since we have a parallelogram below triangle t_1 and t_2 , the side opposite to the bases of triangle t_1 and t_2 (that are also side lengths of the parallelogram) will have the same lengths of $2x, 7x$. This means that the total length of base $\overline{BC}$ is $2x + 3x + 7x = 12x$.

Since the value $12x$ is 6 times the amount of the base of triangle t_1 , it means that the ratio between those two triangles is 6. Since the ratio of the areas is the square of the ratio of side lengths for similar triangles, the area of $\triangle ABC$ is $6^2 \cdot 4$ which is **144**.

Problem 5.2.8 — Triangle GRT has $\overline{GR} = 5$, $\overline{RT} = 12$, and $\overline{GT} = 13$. The perpendicular bisector of $\overline{GT}$ intersects the extension $\overline{GR}$ at O. Find $\overline{TO}$.

Source: 2018 HMMT

Solution: In this problem, first draw a diagram. We know that $\overline{MG} = \overline{MT}$, $\overline{MO}$ is the same and a shared side for $\triangle MOT$ and $\triangle MOG$, and $\angle TMO = \angle GMO$ because both are right angles. This means that $\triangle MOT$ and $\triangle MOG$ are congruent because of SAS congruency.

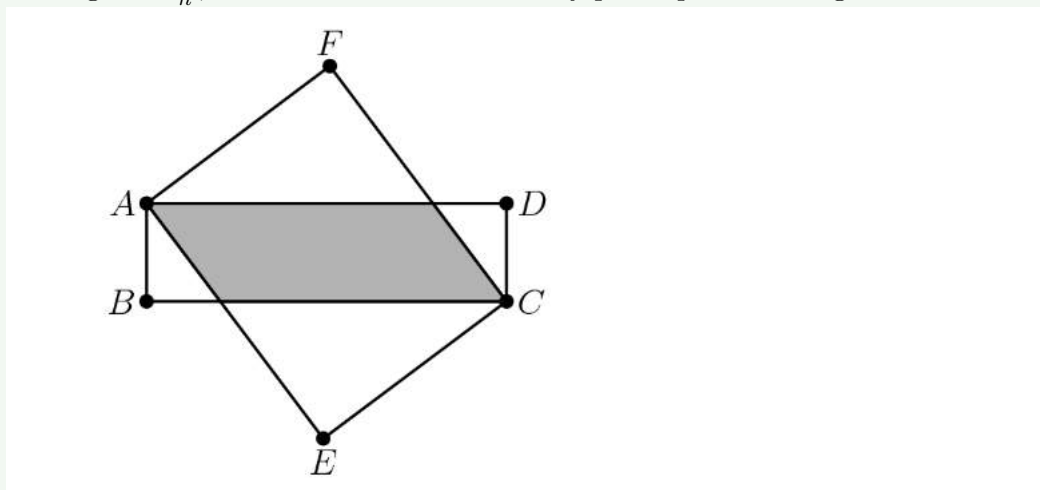
Since both triangles are congruent, $TO = GO$ (both have the same length).

Let's assume that $\angle MGO = a$ degrees. Then, we know that $\angle MTO$ also has a measure of a degrees since we found that the two triangles are congruent. From the AA similarity theorem, we know that $\triangle TMO$ is similar to $\triangle GRT$.

Equating the ratios of two of the sides gives $\frac{\overline{TO}}{\overline{TM}} = \frac{\overline{TG}}{\overline{RG}}$.

Since we know that $\overline{TM} = \frac{13}{2}$, $\overline{TG} = 13$, and $\overline{RG} = 5$, we can plug in those numbers into the similarity equation to get that $\overline{TO} = \frac{169}{10}$.

Problem 5.2.9 — In the diagram below, $ABCD$ is a rectangle with side lengths $AB = 3$ and $BC = 11$, and $AECF$ is a rectangle with side lengths $AF = 7$ and $FC = 9$, as shown. The area of the shaded region common to the interiors of both rectangles is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.



Source: 2021 AIME

Solution: In this problem, let's assume that $\overline{AE}$ intersects $\overline{BC}$ at point Z .

Since $\overline{AE}$ and $\overline{BC}$ intersect, it means opposite angles are equal. Thus, $\angle AZB$ and $\angle EZC$ are equal. Also, in $\triangle AZB$ and $\triangle EZC$ we already know that both have a right angle. Thus, they both are similar due to AA similarity.

Let's assume that the length of $\overline{AZ}$ is x which means that the length of $\overline{EZ}$ is $9 - x$.

We can write similarity equations to get

$$\frac{\overline{AB}}{\overline{BZ}} = \frac{\overline{CE}}{\overline{EZ}}.$$

Substituting values for the side lengths gives $\frac{3}{\overline{BZ}} = \frac{7}{9-x}$

Solving for $\overline{BZ}$ in terms of x gives us that it's length is $\frac{27-3x}{7}$

We can use $\triangle AZB$ and $\triangle EZC$ again to write out the ratios

$$\frac{\overline{AB}}{\overline{AZ}} = \frac{\overline{CE}}{\overline{CZ}}$$

Substituting values for the side lengths gives $\frac{3}{x} = \frac{7}{\overline{CZ}}$

Solving for $\overline{CZ}$ in terms of x gives that it's length is $\frac{7x}{3}$.

Since we know that $\overline{BZ} + \overline{CZ} = 11$, we can plug in our expressions for the two side lengths and solve for the value of x .

$$\frac{27-3x}{7} + \frac{7x}{3} = 11$$

Solving this expression gives that $x = \frac{15}{4}$.

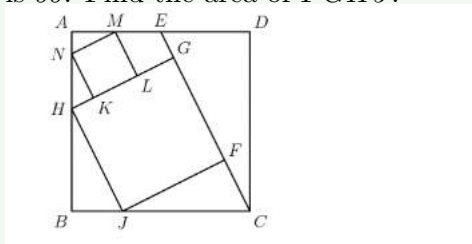
Now to find the area of the shaded region, we can simply find the area of rectangle $AECF$ and subtract $2 \cdot$ the area of $\triangle CEZ$ (since both triangles that we subtract to get the

shaded region are congruent).

The area of one triangle is $\frac{(9-x)7}{2}$ which is $\frac{147}{8}$. We multiply this by 2 to give $\frac{147}{4}$. We must subtract this from the area of the rectangle which is 63 to get $\frac{105}{4}$.

Our answer is $105 + 4 = 109$.

Problem 5.2.10 — In the diagram below, $ABCD$ is a square. Point E is the midpoint of $\overline{AD}$. Points F and G lie on $\overline{CE}$, and H and J lie on $\overline{AB}$ and $\overline{BC}$, respectively, so that $FGHJ$ is a square. Points K and L lie on $\overline{GH}$, and M and N lie on $\overline{AD}$ and $\overline{AB}$, respectively, so that $KLMN$ is a square. The area of $KLMN$ is 99. Find the area of $FGHJ$.



Source: 2015 AIME

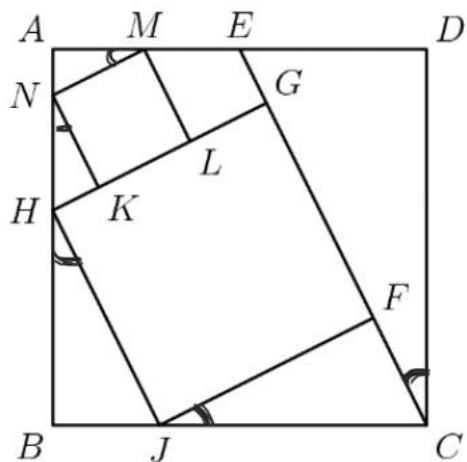
Solution: In this problem, we'll first try to find similar triangles since many angles are equal.

Since $\angle FCD$ and $\angle FCJ$ are complements, we can assume that $\angle FCD$ is a while $\angle FCJ$ is $90 - a$.

Clearly since $\triangle FCJ$ is a right triangle and we know that $\angle FCJ$ is $90 - a$, $\angle FJC$ must be a degrees.

Continuing this for all the right triangles $\triangle DCE$, $\triangle FJC$, $\triangle BHJ$, $\triangle KNH$, and $\triangle AMN$ gives us that $\angle DCE$, $\angle FJC$, $\angle BHJ$, $\angle KNH$, and $\angle AMN = a^\circ$.

Since the angles for the 5 right triangles are the same, from AA similarity we get that $\triangle DCE$, $\triangle FJC$, $\triangle BHJ$, $\triangle KNH$, and $\triangle AMN$ are all similar.



Let's assume that the side length of the square $ABCD$ is s . Since we know that E is the midpoint of $\overline{AD}$, $DE = \frac{s}{2}$.

We can use the Pythagorean theorem on $\triangle DCE$ to find that CE has a length of $s\sqrt{2}$.

For the right triangles that we found are similar, the ratios of the side lengths are $1 : 2 : \sqrt{5}$.

We also know that $\overline{MN} = \overline{NK} = 3\sqrt{11}$ because the area of square NMKL is 99 (given in the problem statement).

Using similarity equations for $\triangle AMN$ and $\triangle KNH$ and the ratio we found for the side lengths gives that $\overline{AN} = \frac{3\sqrt{55}}{5}$ and $\overline{NH} = \frac{3\sqrt{55}}{2}$.

Since we know that FGHJ is also a square, we can assume that its side length is x . Since this means that the side lengths $\overline{HJ} = \overline{JF} = x$, we can use similar triangles and the ratios we found for our side lengths to get that $\overline{HB} = \frac{2x\sqrt{5}}{5}$, $\overline{BJ} = \frac{x\sqrt{5}}{5}$, and $\overline{JC} = \frac{x\sqrt{5}}{2}$.

For those who are confused about what happened above so far, then we used similar triangles to compute the lengths of $\overline{AN}$, $\overline{NH}$, $\overline{HB}$, $\overline{BJ}$, and $\overline{JC}$.

We find that $\overline{AN} + \overline{NH} + \overline{HB}$ is the side length of the square of ABCD, and $\overline{BJ} + \overline{JC}$ is also the length of the square of ABCD.

We can equate the sum for those lengths since the 4 sides of a square have the same length.

$$\frac{3\sqrt{55}}{5} + \frac{3\sqrt{55}}{2} + \frac{2x\sqrt{5}}{5} = \frac{x\sqrt{5}}{5} + \frac{x\sqrt{5}}{2}$$

Solving for x (the side length of square FGHJ) gives us $x = 7\sqrt{11}$

Since the problem wants us to find the area of FGHJ, we simply square $7\sqrt{11}$ (since the area of a square is simply the side length squared) to get **539** as the final answer.

§5.3 Special Triangles

The main triangles that you need to know are isoceses, equilateral, $30 - 60 - 90$, and $45 - 45 - 90$ triangles.

Definition 5.3.1

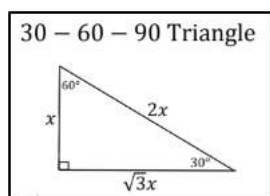
In an equilateral triangle, all angles of a triangle are equal and 60°

All the side lengths of the triangle are equal to s .

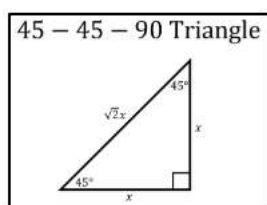
The height of an equilateral triangle is $\frac{s\sqrt{3}}{2}$

Definition 5.3.2

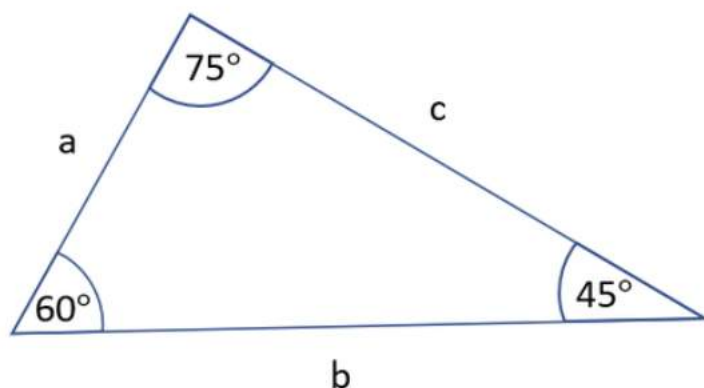
In a $30-60-90$ triangle, the ratio of all sides will always stay the same no matter what. We can prove this ratio by dropping a perpendicular from one vertex of an equilateral triangle to the base that's opposite of it.

**Definition 5.3.3**

In a $45-45-90$ triangle, the ratio of all sides will also stay the same no matter what. The image below shows that ratio.

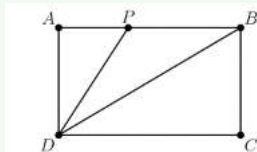


It might also be useful to know the $45-60-75$ triangle.



Challenge: Try dropping an altitude in this problem to separate the $45-60-75$ triangle into a $30-60-90$ and $45-45-90$ triangle.

Problem 5.3.4 — In rectangle $ABCD$, $AD = 1$, P is on $\overline{AB}$, and $\overline{DB}$ and $\overline{DP}$ trisect $\angle ADC$. What is the perimeter of $\triangle BDP$?



Source: 2000 AMC

Source: In this problem, since we know that $\overline{DP}$ and $\overline{DB}$ trisect the right angle, $\angle ADP = \angle PDB = \angle BDC = 30^\circ$

From here, we know that $\triangle ADP$ and $\triangle CDB$ are both $30 - 60 - 90$ triangles.

Since we know that the ratios of side lengths in a $30 - 60 - 90$ triangle for sides opposite to $30, 60, 90^\circ$ respectively are $1 : \sqrt{3} : 2$

Since we know that the side length in $\triangle ADP$ opposite to the 60° angle is 1, $\overline{AP} = \frac{\sqrt{3}}{3}$ and $\overline{DP} = \frac{2\sqrt{3}}{3}$.

Since we know that $\overline{BC} = 1$, we can use our ratios for the $30 - 60 - 90$ triangle to get that $\overline{BD} = 2$ and $\overline{DC} = \sqrt{3}$.

Since $\overline{AP} + \overline{PB} = \overline{AB} = \overline{DC}$, we get that $\overline{PB} = \frac{2\sqrt{3}}{3}$.

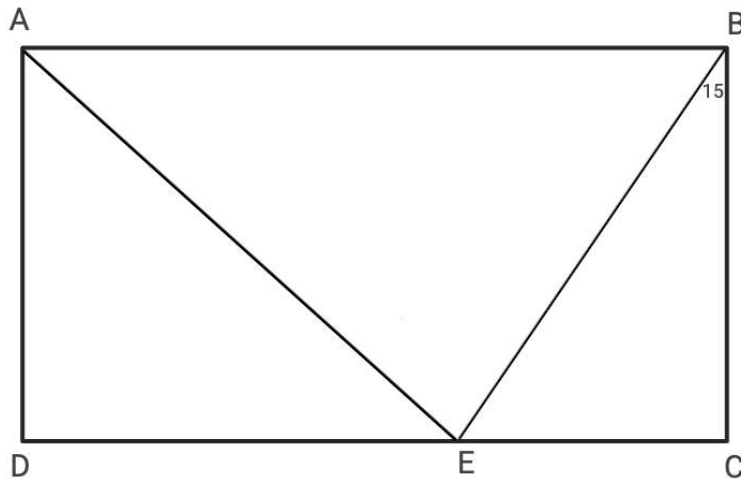
Now that we know the lengths of $\overline{PB}$, $\overline{PD}$, and $\overline{DB}$, we can sum them up to get $\frac{2\sqrt{3}}{3} + \frac{2\sqrt{3}}{3} + 2 = 2 + \frac{4\sqrt{3}}{3}$ **(B)**.

Problem 5.3.5 — In rectangle $ABCD$, $\overline{AB} = 20$ and $\overline{BC} = 10$. Let E be a point on $\overline{CD}$ such that $\angle CBE = 15^\circ$. What is $\overline{AE}$?

- (A) $\frac{20\sqrt{3}}{3}$ (B) $10\sqrt{3}$ (C) 18 (D) $11\sqrt{3}$ (E) 20

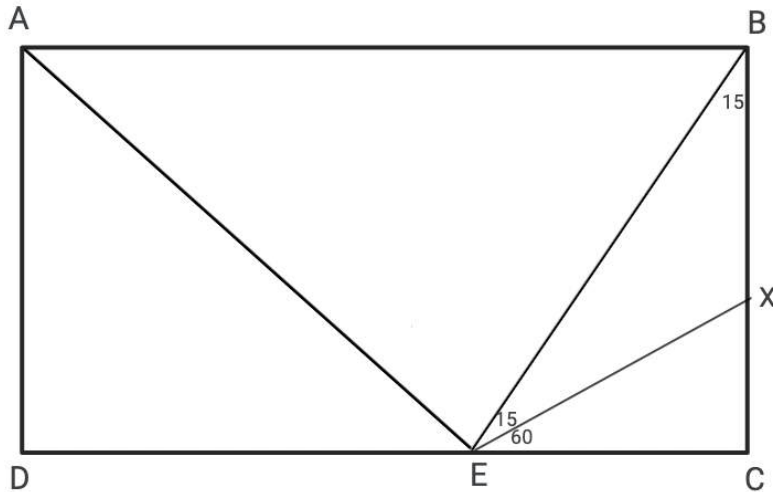
Source: 2014 AMC 10

Like always, we first draw a diagram.



Now, since the angles that we have include 75° , we should try to get angles of 30° and 45° since those are easy to work with.

We can draw a line from E to intersect $\overline{BC}$ at X so that $\angle CEX = 60^\circ$.



Since $\angle XBE = \angle XEB = 15^\circ$, $\triangle XEB$ is isosceles. Let's assume that $\overline{XB} = \overline{XE} = a$ since the triangle is isosceles. Now since we know the length of $\overline{XE}$ to be a , we can use our ratios for the $30 - 60 - 90$ triangle to get that $\overline{EC} = \frac{a}{2}$ and $\overline{CX} = \frac{a\sqrt{3}}{2}$.

Since we know that $\overline{BX} = a$ and $\overline{CX} = \frac{a\sqrt{3}}{2}$, $\overline{BX} + \overline{CX} = \overline{BC} = 10$.

We can solve this equation for a to get that $a = 40 - 20\sqrt{3}$

Since $\overline{EX} = a = 40 - 20\sqrt{3}$, we can use our $30 - 60 - 90$ triangle ratios to get $\overline{EC} = 20 - 10\sqrt{3}$.

Since we know that $\overline{EC} + \overline{DE} = \overline{DC} = 20$, we can plug in the value for the length of $\overline{EC}$ to get that the length of $\overline{DE} = 10\sqrt{3}$.

Since we know that $\overline{AD} = 10$ and $\overline{DE} = 10\sqrt{3}$, we notice that this is a $30 - 60 - 90$ triangle because the legs of this triangle are in the ratio of $1 : \sqrt{3}$. This means that the hypotenuse which is $\overline{AE}$ has a length of 20 since it must be 2 times the side length opposite to the 30° angle.

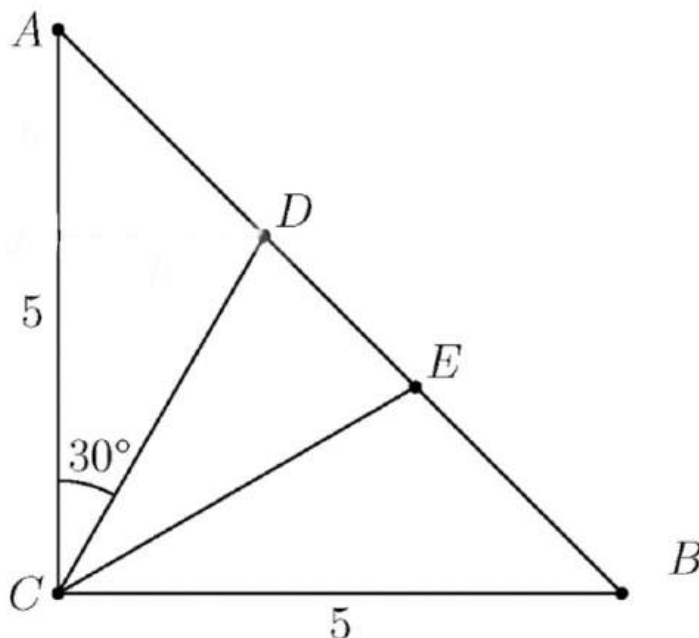
Our final answer is just 20 **(E)**.

Problem 5.3.6 — The isosceles right triangle ABC has right angle at C and area 12.5. The rays trisecting $\angle ACB$ intersect AB at D and E . What is the area of $\triangle CDE$?

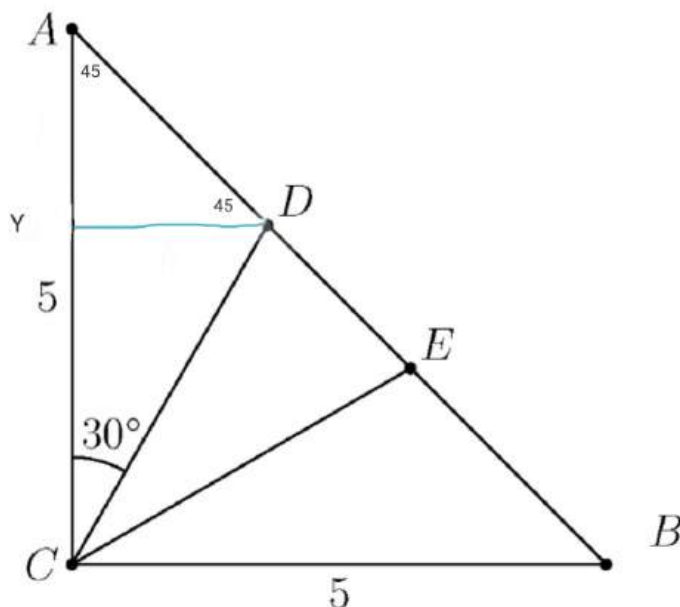
- (A) $\frac{5\sqrt{2}}{3}$ (B) $\frac{50\sqrt{3} - 75}{4}$ (C) $\frac{15\sqrt{3}}{8}$ (D) $\frac{50 - 25\sqrt{3}}{2}$ (E) $\frac{25}{6}$

Source: 2015 AMC 10

Solution: In this problem, since we know that the triangle is right and isosceles, it is a $45 - 45 - 90$ triangle. This means that the length of the legs are the same. Thus, since the area is 12.5, the length of $\overline{AC} = \overline{BC} = 5$.



From here, we know that $\triangle ADC$ is a $30 - 45 - 105$ triangle. We've already seen that there's a simple way to split this into $45 - 45 - 90$ and $30 - 60 - 90$ triangles. We can do that here by drawing a perpendicular from D to $\overline{AC}$ intersecting it at point Y .



We can assume that the length of $\overline{AY} = x$ which means that $\overline{YC} = 5 - x$. Since triangle $\triangle AYD$ is supposed to be isosceles, that means that the length of $\overline{YD} = x$.

Now we know the lengths of $\overline{YD}$ and $\overline{YC}$ in terms of x to be x and $5 - x$ respectively. Both these lengths are the legs of the $30 - 60 - 90$ $\triangle YDC$. We know that the ratios of the side length opposite to the $30, 60, 90$ degree angles must be $1 : \sqrt{3} : 2$. Similarly, our side lengths of x and $5 - x$ must be in the ratio $1 : \sqrt{3}$.

This means that $\frac{x}{5-x} = \frac{1}{\sqrt{3}}$

Solving this equation for x gives that $x = \frac{5\sqrt{3}-5}{2}$

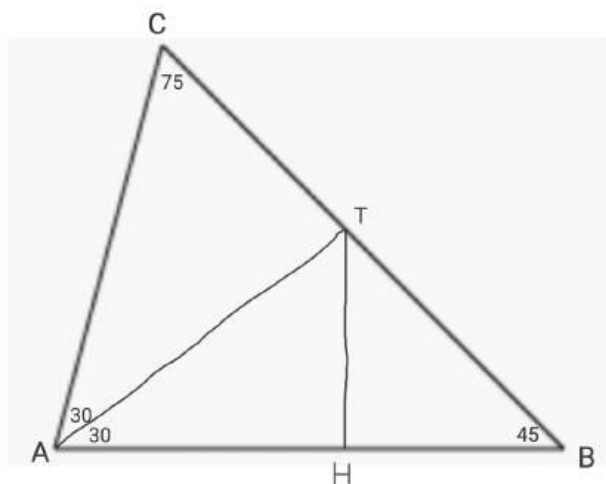
Since we want to find the area of $\triangle CDE$, we can simply find $[\triangle ABC] - [\triangle ADC] - [\triangle BEC]$ (note: the $[\]$ represents area)

Also, the diagram makes it obvious that the area of $\triangle ADC$ and $\triangle BEC$ are the same which means we just need to find the area of one of those triangles, multiply it by two, and then subtract it from the area of $\triangle ABC$.

We already know the area of $\triangle ABC$ to be 12.5. We can find the area of $\triangle ADC$ to simply be $\frac{AC \cdot YD}{2} = \frac{5 \cdot (5\sqrt{3}-5)}{4}$. We multiply this by 2 as previously stated and expand it to get $\frac{25\sqrt{3}-25}{2}$. Subtracting this expression from the total area of the triangle which is 12.5 gives $\frac{50-25\sqrt{3}}{2}$ (D).

Problem 5.3.7 — In triangle ABC , angles A and B measure 60 degrees and 45 degrees, respectively. The bisector of angle A intersects $\overline{BC}$ at T , and $AT = 24$. The area of triangle ABC can be written in the form $a + b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. Find $a + b + c$.
Source: 2001 AIME

Solution: We first draw a line perpendicular from T to H intersecting $\overline{AB}$.



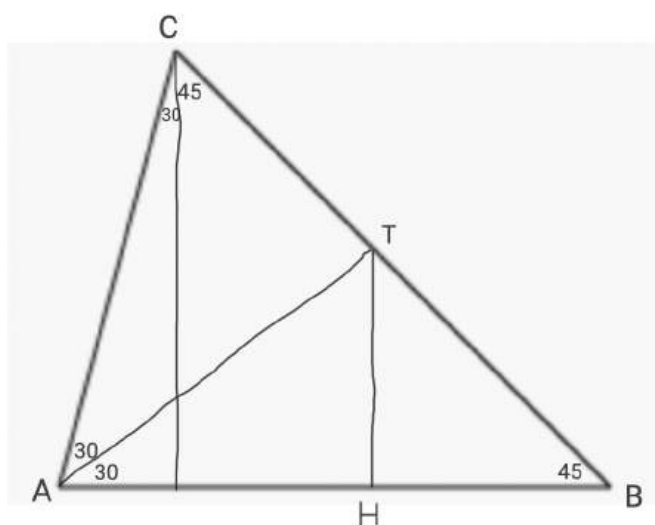
Now, since we know that the length of $\overline{AT} = 24$, we can use our $30 - 60 - 90$ triangle side length ratios to find that $\overline{AH} = 12\sqrt{3}$ and $\overline{HT} = 12$.

From here, doing some angle chasing in $\triangle ATB$ gives that $\angle ATB = 105^\circ$. Since we know that $\angle HTA = 60^\circ$, we can find that $\angle HTB = 45^\circ$. Since $\angle THB = 90^\circ$, we automatically know that $\triangle HTB$ is a $45 - 45 - 90$ triangle which gives us that $\overline{HB} = 12$ and $\overline{TB} = 12\sqrt{2}$.

From here, we know the total length of $\overline{AB}$ because it is equivalent to $\overline{AH} + \overline{HB}$ which is $12\sqrt{3} + 12$.

Now we can find $\overline{AT}$ since we notice that $\angle ATC = 75^\circ$ because we already know that the other two angles in $\triangle ATC$ are 30 and 75 . This means that this triangle is isosceles so $\overline{AC} = \overline{AT}$. Since we already know that $\overline{AT} = 24$, this means that the length of $\overline{AC}$ is also 24 .

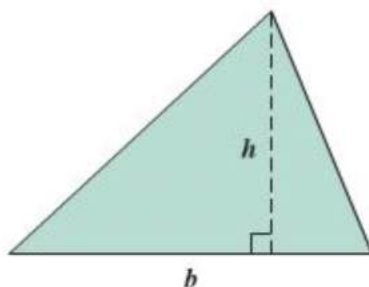
From here, we now drop a perpendicular from point C to $\overline{AB}$ intersecting $\overline{AB}$ at point M . We will now find the value of $\overline{CM}$ which is the altitude of this triangle with base $\overline{AB}$.



Now we can easily see that our altitude can simply be found by looking at $\triangle ACM$ since it is a $30 - 60 - 90$ triangle. We use our side ratios for a $30 - 60 - 90$ triangle to find that $\overline{CM}$ to be $12\sqrt{3}$.

Since we know the length of altitude $\overline{CM}$ and base $\overline{AB}$, we can easily find the area of this triangle now to be $\frac{12\sqrt{3} \cdot (12\sqrt{3} + 12)}{2}$ which is $72\sqrt{3} + 216$. Writing our answer in the form that the problem wants gives $216 + 72 + 3$ which is **291**.

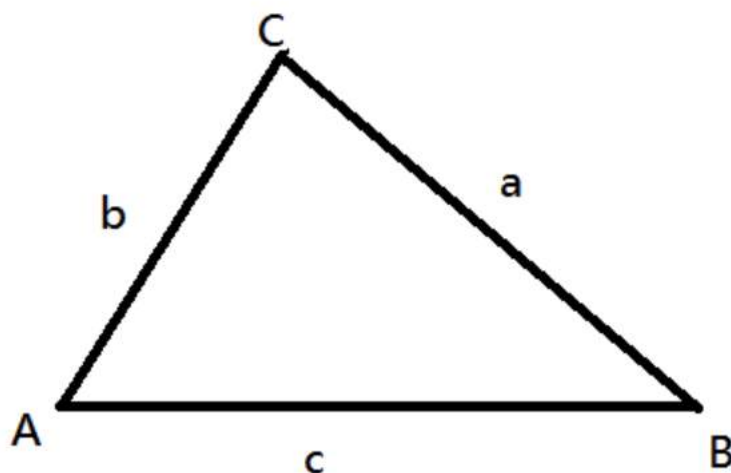
§5.4 Area



One of the most common ways to find area is to simply compute $\frac{bh}{2}$ where b represents base and h represents height.

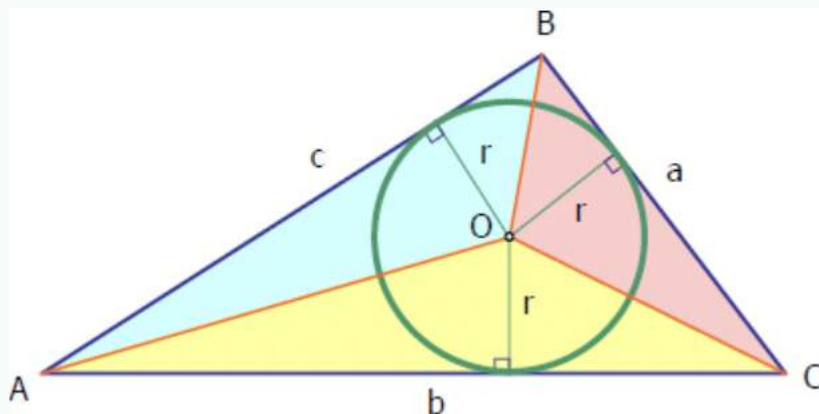
However, it is time to move up a step above this and find area in other ways that are crucial to your success on these competitive exams.

Theorem 5.4.1



For any triangle with side lengths a, b, c and angles A, B, C opposite to those side lengths respectively, then the following theorem will hold.

$$\text{Area} = \frac{ab \sin C}{2}$$

Theorem 5.4.2

Another way of finding the area of the triangle is to first find the semi-perimeter which is represented with the symbol s .

$s = \frac{a+b+c}{2}$ where a, b, c are side lengths of the triangle.

At the same time, r represents the inradius of the triangle. Then, the area of the triangle is $r \cdot s$.

Theorem 5.4.3

Also, since we know that the semi-perimeter of a triangle is $s = \frac{a+b+c}{2}$ where a, b, c are side lengths of the triangle, then another way to find it's area is by using the

Heron's Formula which gives

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

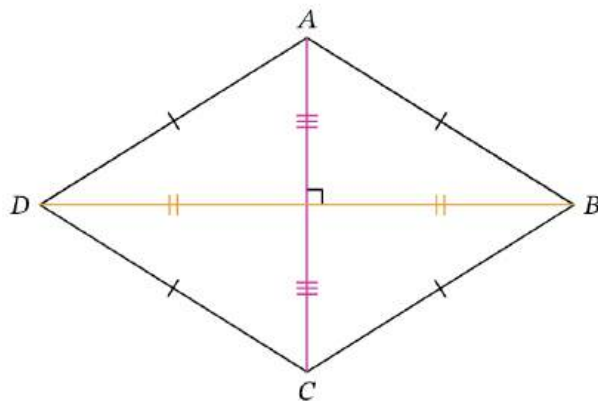
Theorem 5.4.4

For a triangle with circumradius R , the area can also be represented as $\frac{abc}{4R}$ where a, b, c are the side lengths of the triangle.

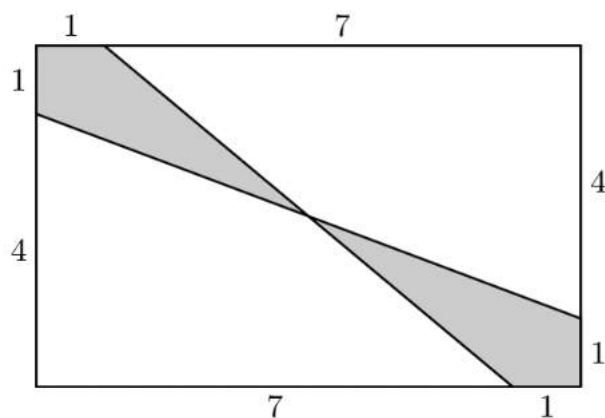
Theorem 5.4.5

The area of a trapezoid with bases a and b and a height of h is $\frac{a+b}{2} \cdot h$.

The area of the rhombus can be found by calculating the length of the perpendicular diagonals of length a and b . The area is $\frac{ab}{2}$.

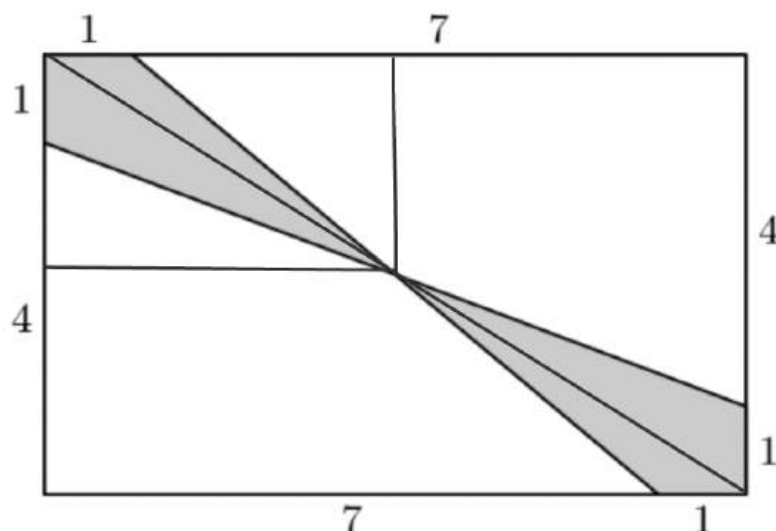


Problem 5.4.6 — Find the area of the shaded region.



Source: 2016 AMC 10

Solution: In this problem, we can find out the area of the 2 shaded triangles by splitting each up into something simpler.



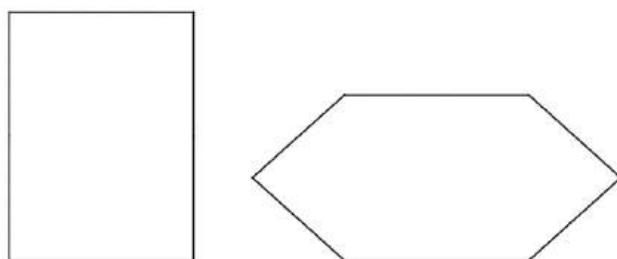
We now get 4 triangles. We can work with the ones on the left side of the rectangle and simply multiply the sum of those by 2.

We notice that the two triangles both have base 1. The height of one of them is simply half of the side length of the rectangle of 8 ($7 + 1$). Half of it is $\frac{8}{2} = 4$. Similarly, for the other triangle the height is half of the other side length in the rectangle: $5 (4 + 1)$. Half of it is $\frac{5}{2}$.

We can now sum up the areas using our area formula for a triangle $\frac{bh}{2}$.
 $\frac{1 \cdot 5}{4} + \frac{1 \cdot 4}{2} = \frac{13}{4}$.

We multiply our value by 2 to get a final answer of $\frac{13}{2}$ (D).

Problem 5.4.7 — A rectangle has sides of length a and 36. A hinge is installed at each vertex of the rectangle, and at the midpoint of each side of length 36. The sides of length a can be pressed toward each other keeping those two sides parallel so the rectangle becomes a convex hexagon as shown. When the figure is a hexagon with the sides of length a parallel and separated by a distance of 24, the hexagon has the same area as the original rectangle. Find a^2 .



Source: 2014 AIME

Solution: Since we know that the area of the rectangle and the area of the hexagon equates, we can first find the area of the rectangle. It is simply the length times the width which is $36a$.

Now we work on the area of the hexagon. To do this, we first notice that the side length of 36 is divided into two parts of an equal length of 18.



Since our hexagon is clearly made of a rectangle and two isosceles triangles around it, we can separately find the area of each. Working with the isosceles triangle shows that the base length of it is 24 while the other two sides are 18. We can simply drop a perpendicular as shown below to find the area of the isosceles triangles.



This splits the base of 24 into 12. Now we can use the pythagoreon theorem to find that the length of that altitude is $\sqrt{18^2 - 12^2} = 6\sqrt{5}$

Through this, we know that the area of one isosceles triangle is $\frac{bh}{2}$ which for us is $\frac{24 \cdot 6\sqrt{5}}{2} = 72\sqrt{5}$.

Since we have two isosceles triangles, we multiply this by 2 to get $144\sqrt{5}$.

Now, the area of the rectangle in the hexagon is simply $24a$ because 24 and a represent the length and the width clearly.

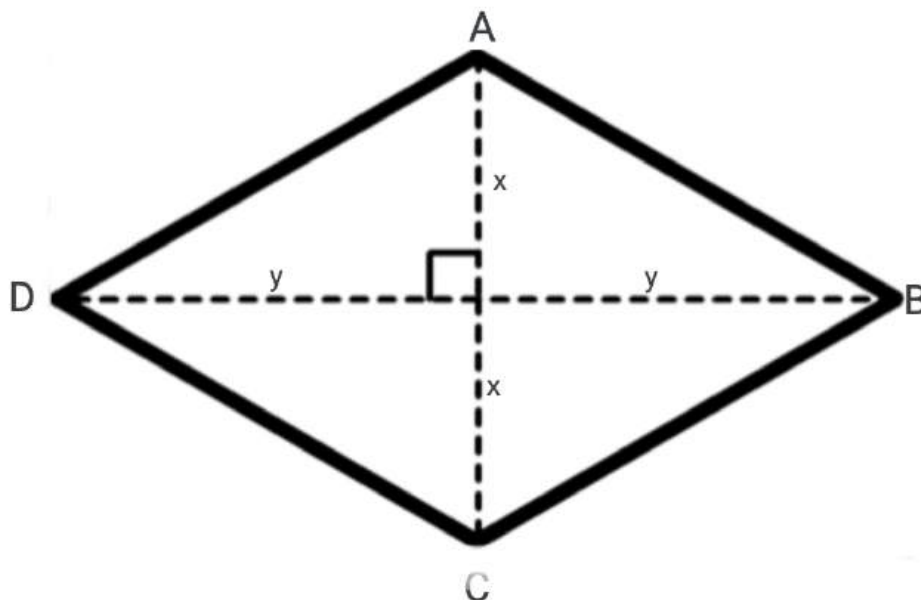
Our total area of the hexagon is $144\sqrt{5} + 24a$ and we know that this equates to the area of the rectangle which was $36a$.

Now we can solve $144\sqrt{5} + 24a = 36a$ which gives us that $a = 12\sqrt{5}$. Since we want to find a^2 for the final answer, it is simply $144 \cdot 5$ which is **720**.

Problem 5.4.8 — Find the area of rhombus $ABCD$ given that the circumradii of triangles ABD and ACD are 12.5 and 25, respectively.

Source: 2003 AIME

Solution: In this problem, since we are given the circumradii of the triangles, we should think of the area formula $\frac{abc}{4R}$.



As shown in the diagram above, let's assume that the diagonals are $2a$ and $2b$ long (each half part will be a and b long). We already know that the diagonals of a rhombus intersect at right angles. We can use this to apply our area formula.

We will work with $\triangle ABD$ first. It's base is $\overline{DB}$ which we already know has a length of $2y$. Now we find the length of $\overline{AD}$ and $\overline{AB}$ through the pythagoreon theorem and we get $\sqrt{x^2 + y^2}$.

Now plugging in those values into the formula $\frac{abc}{4R}$ gives $\frac{2y(x^2+y^2)}{4R}$. We know the circumradius of this triangle is 12.5 as given in the problem statement. We plug this in to further simplify our expression to $\frac{(x^2+y^2)y}{25}$

We will now work with $\triangle ACD$. We can look at our diagram and find that two side lengths are again $\sqrt{x^2 + y^2}$ and the other length is $2x$. Plugging this into our area formula along with the fact that it's circumradius is 25 gives that the area is $\frac{(x^2+y^2)x}{50}$

Since we know that the expression for both areas represent the area of half the entire rhombus, we can equate the two areas we found for $\triangle ACD$ and $\triangle ABD$.

$$\frac{(x^2+y^2)y}{25} = \frac{(x^2+y^2)x}{50}$$

This simplifies to $x = 2y$.

Now our diagonals simply become $2y$ and $4y$. Since we know that the area of a rhombus is simply $\frac{ab}{2}$ where a and b are the diagonals, we can plug in our values to get that our area is $4y^2$.

Now, we can use our area expressions for either $\triangle ABD$ or $\triangle ACD$, multiply it by 2, and then equate it to the area of the total rhombus. Then, we can solve for our value of y and eventually find the area.

We will use the expression the area for $\triangle ABD$ which we found to be $\frac{(x^2+y^2)y}{25}$. We will plug in $x = 2y$ to get that the area is $\frac{y^3}{5}$. We now multiply it by 2 to get $\frac{2y^3}{5}$. We equate this to $4y^2$ now to get that $y = 10$.

Since we know that the area of the rhombus is $4y^2$, we plug in our value $y = 10$ to get that the area is **400**.

§5.5 Special Points

Definition 5.5.1

A **cevian** is a line drawn from one vertex of a triangle to the opposite side.

Definition 5.5.2

Angle Bisector

An angle bisector is a line that goes through a vertex to divide that angle in half.

Definition 5.5.3

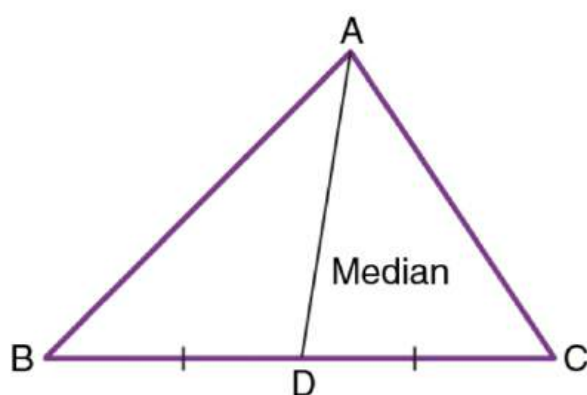
Incenter

An incenter is the point where the angle bisectors of all vertices meet. You will often see incenters being used in triangle problems.

Definition 5.5.4

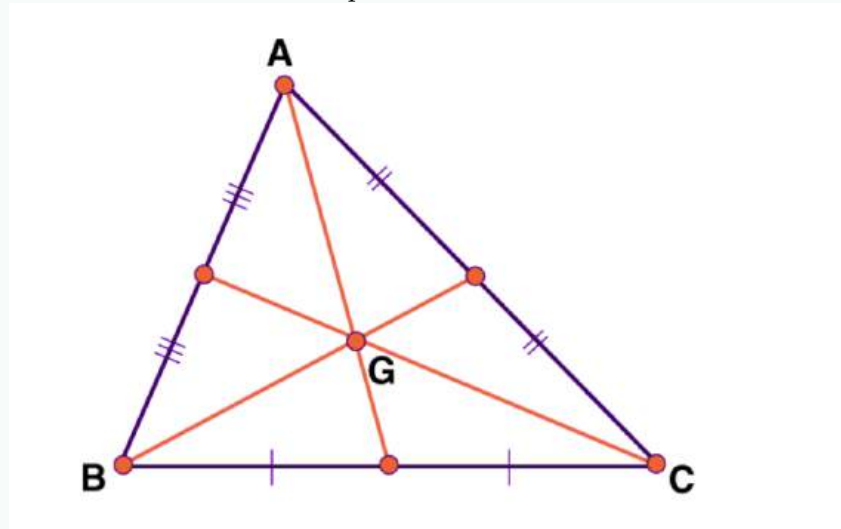
Median

A median is a line segment that connects the vertex and divides the side opposite to that in half.

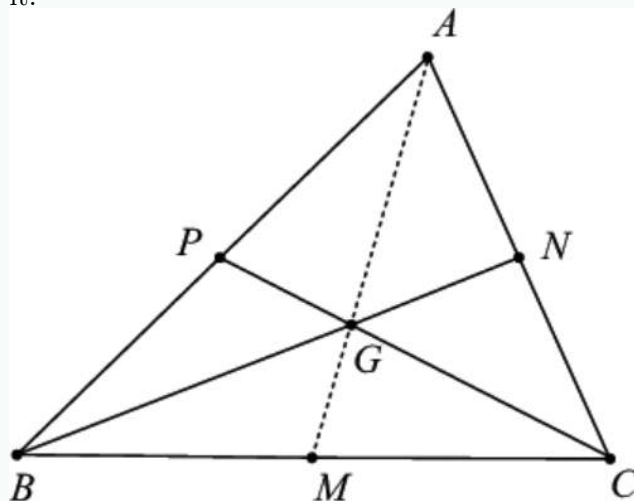


Definition 5.5.5**Centroid**

A centroid is the common point of intersection for the 3 median lines of a triangle.

**Theorem 5.5.6****Median Theorem**

The distance from each vertex to the centroid is double the distance of that centroid to the midpoint of the side opposite to that vertex. The image below should simplify it.



The points P, M, and N are midpoints for the side lengths of the triangles. G is the centroid of the triangle. The median theorem gives the following three relations.

$$AG = 2 \cdot GM$$

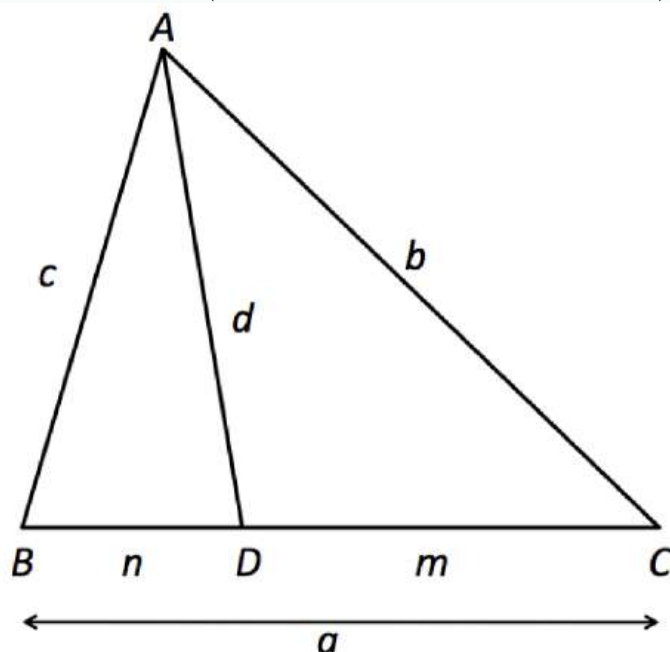
$$BG = 2 \cdot GN$$

$$CG = 2 \cdot GP$$

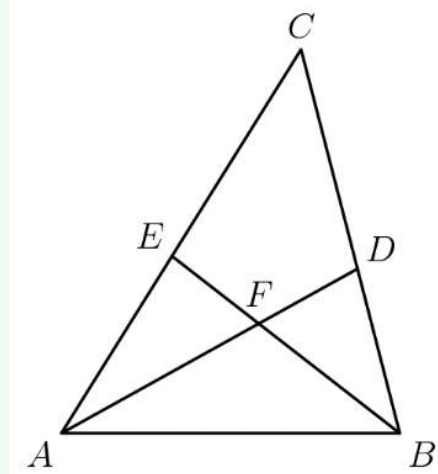
Theorem 5.5.7

Stewart's Theorem For any $\triangle ABC$ with cevian $\overline{AD}$ (where point D lies on $\overline{BC}$, the following theorem applies:

$man = bnb + cmc$ (each letter represents a side length)



Problem 5.5.8 — In $\triangle ABC$, $AB = 6$, $BC = 7$, and $CA = 8$. Point D lies on $\overline{BC}$, and $\overline{AD}$ bisects $\angle BAC$. Point E lies on $\overline{AC}$, and $\overline{BE}$ bisects $\angle ABC$. The bisectors intersect at F . What is the ratio $AF : FD$?



- (A) 3 : 2 (B) 5 : 3 (C) 2 : 1 (D) 7 : 3 (E) 5 : 2

Source: 2016 AMC 12

Solution: In this problem, we should use the angle bisector theorem. Using it first for $\angle ABC$ gives

$$\frac{\overline{AB}}{\overline{AE}} = \frac{\overline{CB}}{\overline{CE}}$$

Substituting our known side lengths gives $\frac{\overline{CE}}{\overline{AE}} = \frac{7}{6}$

We can assume that the length of $\overline{CE}$ is $7a$ while the length of $\overline{AE}$ is $6a$. They sum to $13a$ which is equivalent to the length of $\overline{AC}$ which is 8. This means that $\overline{CE}$ has a length of $\frac{56}{13}$ while $\overline{AE}$ has a length of $\frac{48}{13}$.

We can do the same for $\angle CAB$ to get that $\overline{BD}$ has a length of 3 while $\overline{CD}$ has a length of 4.

Also, by using the angle bisector theorem again for $\triangle ABD$ we get that $\frac{\overline{AB}}{\overline{AF}} = \frac{\overline{BD}}{\overline{FD}}$. Rearranging the equation gives $\frac{\overline{AB}}{\overline{BD}} = \frac{\overline{AF}}{\overline{FD}}$.

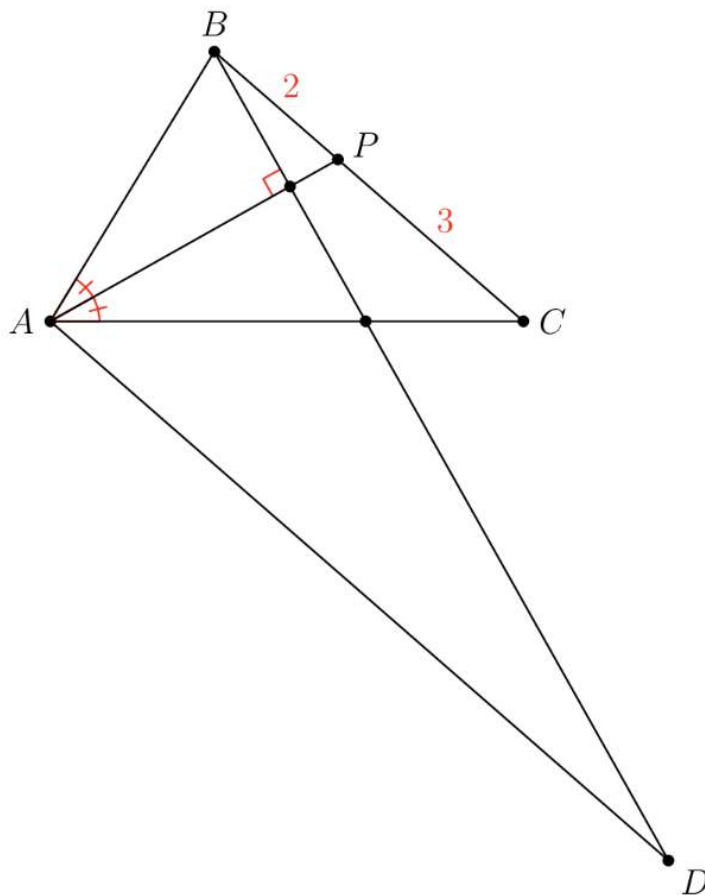
That means instead of finding $\frac{\overline{AF}}{\overline{FD}}$, we can simply find $\frac{\overline{AB}}{\overline{BD}}$. We already know the two lengths $\overline{AB}$ and $\overline{BD}$ to be 6 and 3 respectively. The ratio that we want to find is simply $\frac{6}{3}$ which simplifies to 2 or **2:1 (C)**.

Problem 5.5.9 — Let $\triangle ABC$ be a scalene triangle. Point P lies on $\overline{BC}$ so that $\overline{AP}$ bisects $\angle BAC$. The line through B perpendicular to $\overline{AP}$ intersects the line through A parallel to $\overline{BC}$ at point D . Suppose $BP = 2$ and $PC = 3$. What is AD ?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Source: 2022 AMC

Solution: As always, we draw a diagram.



Since we know that $\overline{AP}$ is an angle bisector for $\angle A$ in $\triangle ABC$, we can use the angle bisector theorem to get

$$\frac{\overline{AB}}{\overline{BP}} = \frac{\overline{AC}}{\overline{CP}}$$

We can equate both of the ratios above to a and using our known lengths ($BP = 2$ and $CP = 3$), we get that $\overline{AB}$ has a length of $2a$ and $\overline{AC}$ has a length of $3a$.

Now let's assume that $\overline{BD}$ intersects $\overline{AP}$ at point X and it intersects $\overline{AC}$ at point Y . It's obvious that $\triangle ABX$ and $\triangle AYX$ are both congruent right triangles. We can prove this from SAS congruency.

This means that the length of AB is equivalent to the length of AY which is $2a$. This means that CY has a length of a .

Also, since lines $\overline{BC}$ and $\overline{AD}$ are parallel and $\overline{BD}$ is a transversal going through them, $\angle ADY = \angle CBY$.

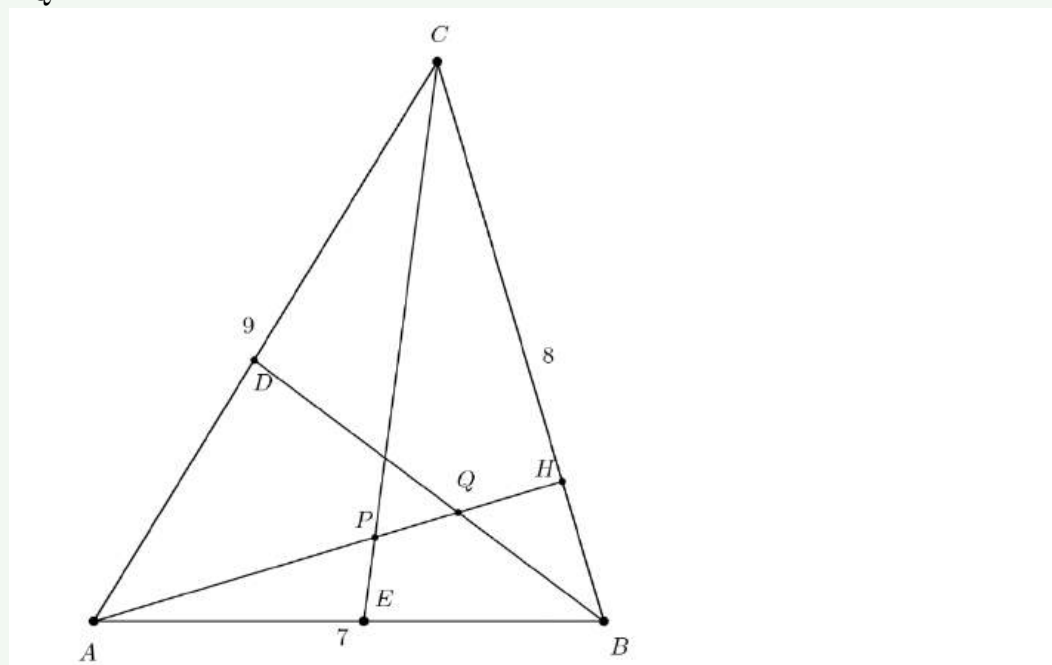
It's obvious that $\angle AYD = \angle CYB$.

Since all angles for $\triangle DYA$ and $\triangle BYC$ are equal, they both are similar due to AA similarity.

Clearly the scale factor for the side lengths is 2 because $\frac{\overline{AY}}{\overline{YC}} = 2$.

This means that the length of $\overline{AD}$ is simply 2 times the length of $\overline{BC}$ which means that it is **10 (C)**.

Problem 5.5.10 — In $\triangle ABC$ shown in the figure, $AB = 7$, $BC = 8$, $CA = 9$, and $\overline{AH}$ is an altitude. Points D and E lie on sides $\overline{AC}$ and $\overline{AB}$, respectively, so that $\overline{BD}$ and $\overline{CE}$ are angle bisectors, intersecting $\overline{AH}$ at Q and P , respectively. What is PQ ?



- (A) 1 (B) $\frac{5}{8}\sqrt{3}$ (C) $\frac{4}{5}\sqrt{2}$ (D) $\frac{8}{15}\sqrt{5}$ (E) $\frac{6}{5}$

Source: 2016 AMC

Solution: In this problem, we will first try to compute the length of $\overline{AH}$ (the height of

the triangle). We can do this by using the pythagoreen theorem two times.

Assuming that the length of $\overline{CH}$ is x , we immediately get that the length of $\overline{HB}$ is $8 - x$. We can assume that the length of the altitude $\overline{AH}$ is h . Applying pythagoreen theorem twice gives

$$h^2 + x^2 = 9^2$$

$$h^2 + (8 - x)^2 = 7^2$$

Now we can expand both equations and simply subtract them to get that $x = 6$.

This means that the length of $\overline{CH}$ is 6 and the length of $\overline{HB}$ is 2. It also means that the length of $\overline{AH}$ is $3\sqrt{5}$.

Now we can apply the angle bisector theorem twice on $\triangle BAH$ and $\triangle CAH$.

Applying it first on $\triangle BAH$ gives $\frac{\overline{AB}}{\overline{AQ}} = \frac{\overline{HB}}{\overline{HQ}}$

We can plug in our side lengths to simplify it to $\frac{7}{\overline{AQ}} = \frac{2}{\overline{HQ}}$

Since we know that $\overline{AQ} + \overline{HQ} = 3\sqrt{5}$, we can plug in $\overline{AQ} = 3\sqrt{5} - \overline{HQ}$ to the equation we got from the angle bisector theorem.

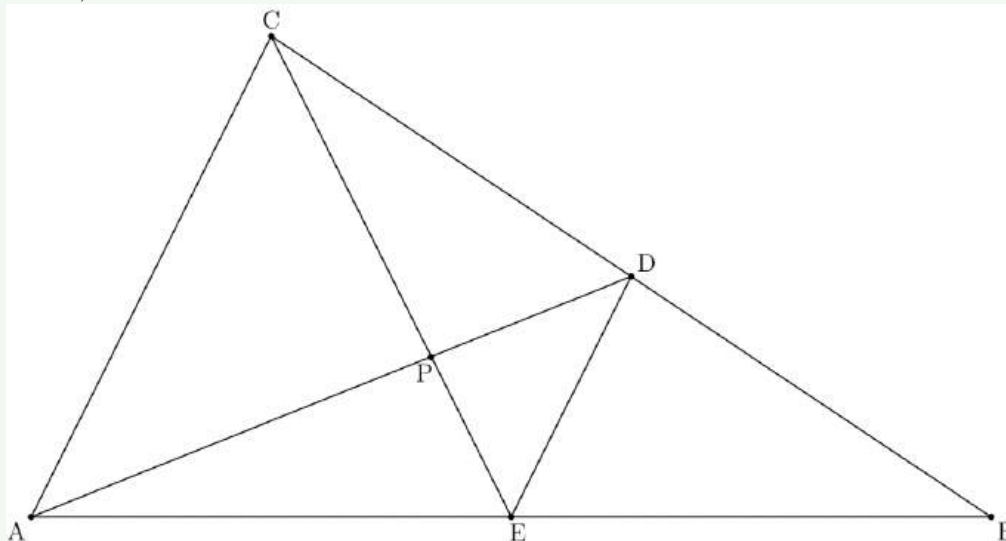
Solving it gives that the length of $\overline{AQ}$ is $\frac{7\sqrt{5}}{3}$.

We can apply the angle bisector theorem again in a similar fashion to $\triangle ACH$ since it has an angle bisector that is $\overline{CP}$.

Doing so gives us that the lengths of $\overline{AP}$ is $\frac{9\sqrt{5}}{5}$.

Now it's obvious that $\overline{PQ} + \overline{AP} = \overline{AQ}$. We can plug in the lengths we found to get that the length of $\overline{AP}$ is $\frac{8\sqrt{5}}{15}$ (D).

Problem 5.5.11 — In triangle ABC , medians AD and CE intersect at P , $PE = 1.5$, $PD = 2$, and $DE = 2.5$. What is the area of $AEDC$?



- (A) 13 (B) 13.5 (C) 14 (D) 14.5 (E) 15

Source: 2013 AMC

Solution: In this problem, we will use the median theorem.

Since P is the centroid, we know that the following two equations listed below are true:

$$CP = 2 \cdot PE$$

$$AP = 2 \cdot PD$$

Since we already know the lengths of $\overline{PE}$ and $\overline{PD}$, we can plug them in to get that $CP = 3$ and $AP = 4$

Also, we know that $\triangle CBA$ is similar to $\triangle DBE$. The reason is that the line connecting the two midpoints of a triangle is always parallel to one of the sides of the triangle.

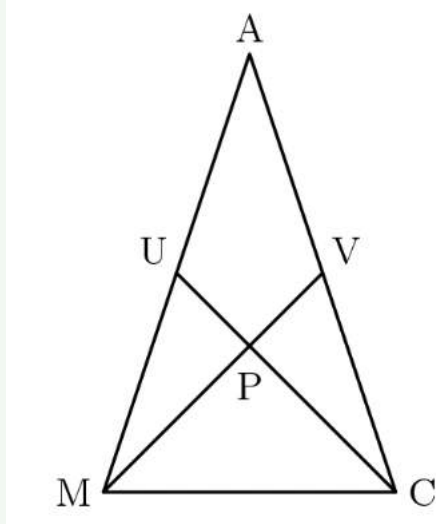
$\triangle CBA$ has a scale factor of 2 when compared to $\triangle DBE$. That means that $CA = 5$.

We know that $CP = 3$, $AP = 4$, and $CA = 5$. We could be able to recognize that $\triangle CPA$ is a right triangle of configuration $3-4-5$. Since $\angle CPA = 90^\circ$, it's obvious that $\overline{CE}$ and $\overline{AD}$ are perpendicular to each other.

This means that the quadrilateral $AEDC$'s area can simply be found by finding the area of the 4 right triangles we split it into.

The total area is $\frac{3 \cdot 4}{2} + \frac{3 \cdot 2}{2} + \frac{2 \cdot 1.5}{2} + \frac{1.5 \cdot 4}{2} = \mathbf{13.5}$ (B).

Problem 5.5.12 — Triangle AMC is isosceles with $AM = AC$. Medians $\overline{MV}$ and $\overline{CU}$ are perpendicular to each other, and $MV = CU = 12$. What is the area of $\triangle AMC$?



- (A) 48 (B) 72 (C) 96 (D) 144 (E) 192

Source: 2020 AMC

Solution: In this problem, we use the median theorem. We know that the medians of a triangle divide the lengths into a ratio of $2:1$.

This means that $CP = MP = 8$ and $PU = PV = 4$.

We know that $\angle MPC = 90^\circ$ which means that $\triangle MPC$ is a right triangle. The hypotenuse of that triangle is the base of the isosceles triangle $\triangle AMC$. The hypotenuse is $\sqrt{8^2 + 8^2}$ which is $8\sqrt{2}$.

We can use the pythagorean theorem on the two right triangles: $\triangle UPM$ and $\triangle VPC$. We get that $UM = VC = \sqrt{8^2 + 4^2} = 4\sqrt{5}$.

Since we know that U and V are the midpoints of $\overline{AM}$ and $\overline{AC}$ respectively, the length of AM and AC is $8\sqrt{5}$.

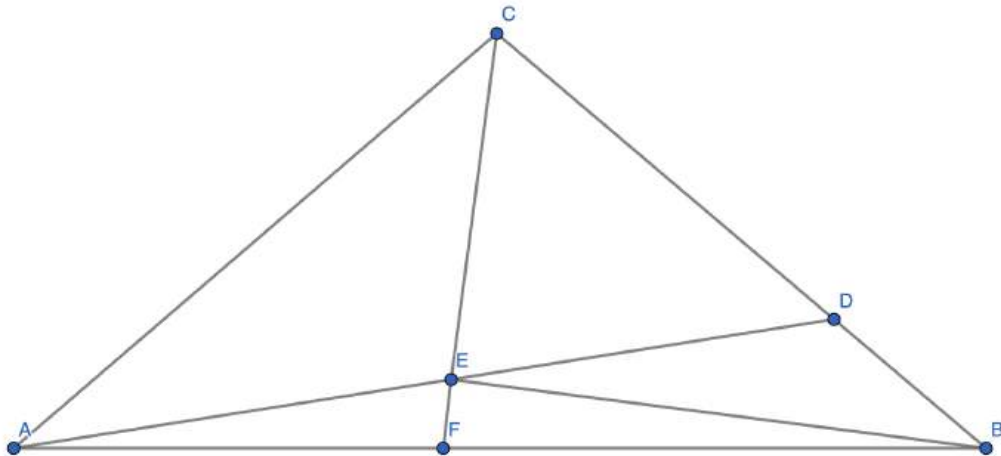
We now have an isosceles triangle ($\triangle AMC$) that has a common side length of $8\sqrt{5}$ and a base of $8\sqrt{2}$. We can find it's area by dropping a perpendicular to it's base.

Doing so gives us that the base is $8\sqrt{2}$ and the height is $12\sqrt{2}$. The area is simply $\frac{bh}{2}$ which is **96 (C)**.

Problem 5.5.13 — In $\triangle ABC$, $AC = BC$, and point D is on $\overline{BC}$ so that $CD = 3 \cdot BD$. Let E be the midpoint of $\overline{AD}$. Given that $CE = \sqrt{7}$ and $BE = 3$, the area of $\triangle ABC$ can be expressed in the form $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

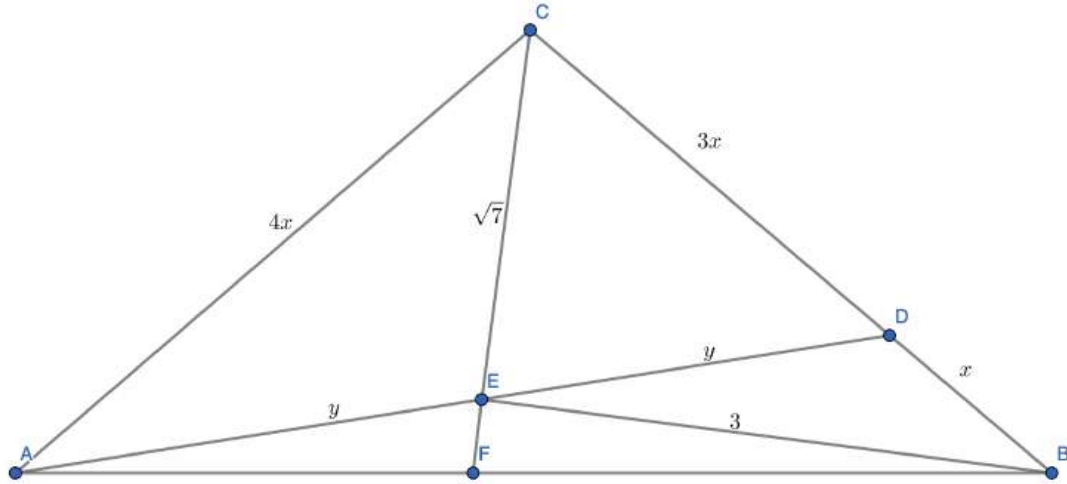
Source: 2013 AIME

Solution: This problem will demonstrate how powerful Stewart's Theorem is.



From the problem statement, we already know some relationships between side lengths: $AC = BC$, $CD = 3 \cdot BD$, and $AE = DE$.

Let's assume that CD has a length of $3x$ which means that BD has a length of x . This means that BC has a length of $4x$ so AC also has a length of $4x$.



We will first apply Stewart's Theorem on $\triangle BEC$ with cevian $\overline{ED}$.

Doing so gives: $CD \cdot BD \cdot BC + BC \cdot DE^2 = BD \cdot CE^2 + CD \cdot BE^2$.

Since $AE = DE$, let's assume that both have a length of y .

Now we can plug all of these side lengths into our equation above to get $12x^3 + 4xy^2 = 7x + 27x$

Simplifying the equation gives $12x^2 + 4y^2 = 34$

Now we will apply Stewart's Theorem again on $\triangle ACD$ with cevian $\overline{CE}$.

Doing so gives: $AE \cdot ED \cdot AD + AD \cdot CE^2 = ED \cdot AC^2 + AE \cdot CD^2$.

We will plug in our side lengths to get $2y^3 + 14y = 25x^2y$.

The equation above simplifies to $25x^2 - 2y^2 = 14$.

Now, we have two equations that we found from apply Stewart's Theorem twice.

$$12x^2 + 4y^2 = 34$$

$$25x^2 - 2y^2 = 14$$

Solving both equations give that $x = 1$ and $y = \frac{\sqrt{22}}{2}$.

Since $x = 1$ we can write out a few side lengths: $AB = BC = 4$, $CD = 3$, and $DB = 1$.

Using that $y = \frac{\sqrt{22}}{2}$, we get that $AE = ED = \frac{\sqrt{22}}{2}$ and $AD = \sqrt{22}$.

To find the area of $\triangle ABC$, we can find the length of the base of the triangle which is $\overline{AB}$.

To find the length of AB , we can apply Stewart's Theorem on $\triangle ABD$.

Doing so gives: $AE \cdot ED \cdot AD + AD \cdot BE^2 = AE \cdot DB^2 + ED \cdot AB^2$

We can plug in our side lengths to get $\frac{11\sqrt{22}}{2} + 9\sqrt{22} = \frac{\sqrt{22}}{2} + \frac{AB^2\sqrt{22}}{2}$

Solving the above equation gives that $AB^2 = 28$ which means AB has a length of $2\sqrt{7}$.

Since our isosceles triangle has a common side length of 4 and a base length of $2\sqrt{7}$, we

can simply drop an altitude from C to $\overline{AB}$. Solving for the height gives that it has a length of 3.

Since the isosceles triangle has a base length of $2\sqrt{7}$ and height of 3, the area of the triangle is $\frac{2\sqrt{7} \cdot 3}{2}$ which is $3\sqrt{7}$.

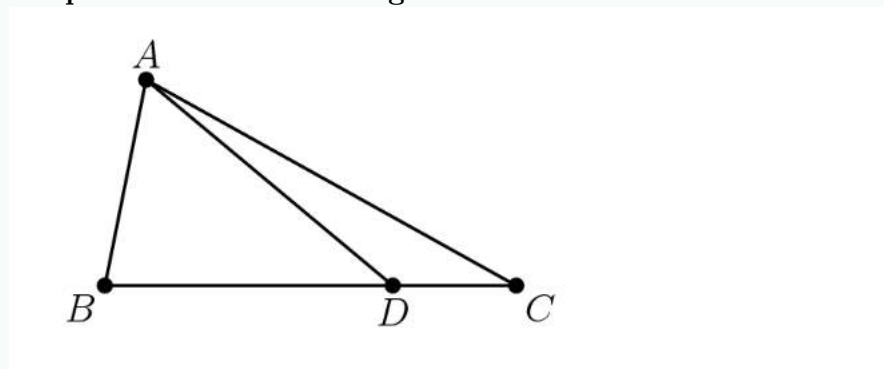
The answer in the format we want is $3 + 7$ which is **10**.

§5.6 Area Ratios Related to Side Length Ratios

This is a topic growing in popularity on more recent exams. There are some common strategies that you have to know to be able to solve these types of problems.

Theorem 5.6.1

Simple Area Ratio for height

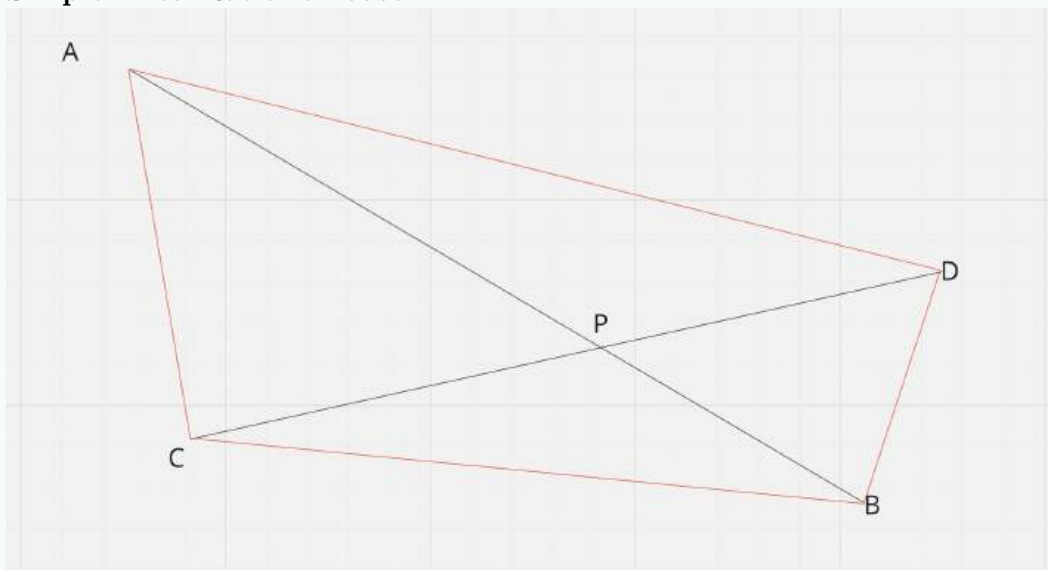


The theorem applies for any cevian in a triangle.

$\overline{AD}$ is our cevian that connects vertex A to $\overline{BC}$.

The theorem states: $\frac{BD}{DC} = \frac{[ABD]}{[ADC]}$

The reason for the above theorem being true is that both triangles have the same height. It is only the base that differs which is why the ratio of the areas is simply equivalent to the ratio of the bases.

Theorem 5.6.2**Simple Area Ratio for base**

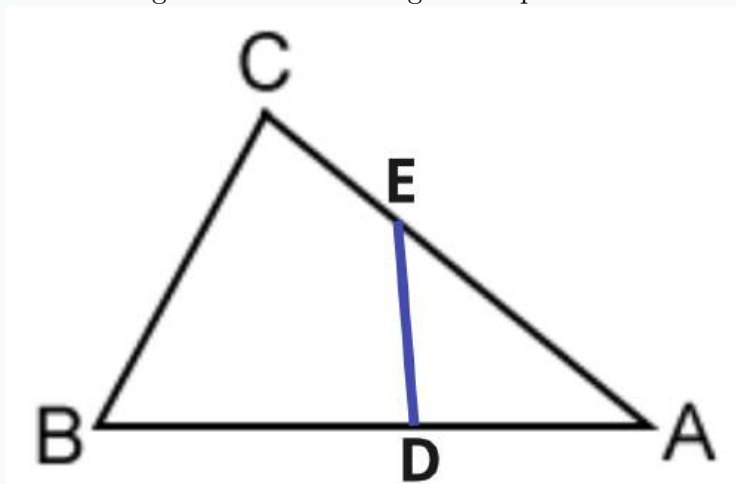
If two lines $\overline{AB}$ and $\overline{CD}$ intersect at a point P, then the theorem below applies.

$$\frac{AP}{BP} = \frac{[ACD]}{[BCD]}$$

The reason for this theorem being true is that the bases of both triangles is the same ($\overline{CD}$). Thus, the ratio of the areas is simply equivalent to the ratio of the heights.

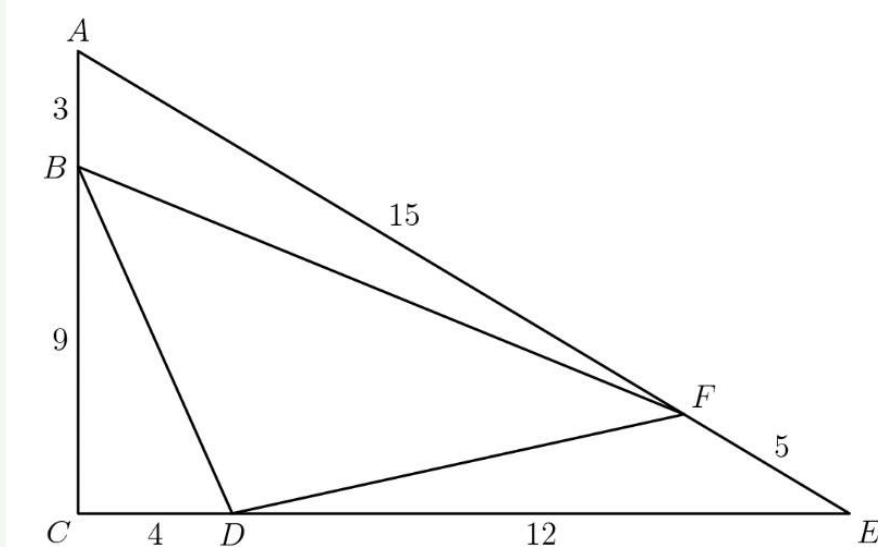
Theorem 5.6.3

When two triangles share a same angle, the ratio of the areas is simply the ratio of the side lengths around that angle multiplied.



Since $\angle A$ are equal for both the triangles, $\frac{[ADE]}{[ABC]} = \frac{AD \cdot AE}{AB \cdot AC}$.

Problem 5.6.4 — In the right triangle $\triangle ACE$, we have $AC = 12$, $CE = 16$, and $EA = 20$. Points B , D , and F are located on AC , CE , and EA , respectively, so that $AB = 3$, $CD = 4$, and $EF = 5$. What is the ratio of the area of $\triangle DBF$ to that of $\triangle ACE$?



- (A) $\frac{1}{4}$ (B) $\frac{9}{25}$ (C) $\frac{3}{8}$ (D) $\frac{11}{25}$ (E) $\frac{7}{16}$
Source: 2004 AMC

Solution: In this problem, instead of computing $\frac{[DBF]}{[ACE]}$, we can simply find:

$$1 - \left(\frac{[ABF]}{[ACE]} + \frac{[BCD]}{[ACE]} + \frac{[DEF]}{[ACE]} \right)$$

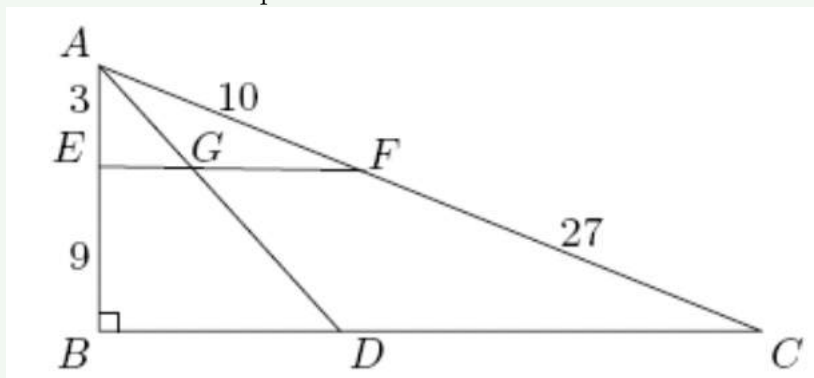
We can first try to find $\frac{[ABF]}{[ACE]}$. We notice that $\angle A$ is simply equal for both of these triangles. Thus, the ratio of both of the triangles is simply $\frac{AB \cdot AF}{AC \cdot AE}$.

The fraction above simply becomes $\frac{3 \cdot 15}{12 \cdot 20}$ which is $\frac{3}{16}$.

Using this same method and relating the common angle, we can find that $\frac{[BCD]}{[ACE]} = \frac{3}{16}$. $\frac{[DEF]}{[ACE]}$ is $\frac{15}{80}$.

We can use these ratios to get that $\frac{[DBF]}{[ACE]} = 1 - \frac{3}{16} - \frac{3}{16} - \frac{3}{16} = \frac{7}{16}$ **(E)**.

Problem 5.6.5 — In the diagram below, angle ABC is a right angle. Point D is on $\overline{BC}$, and $\overline{AD}$ bisects angle CAB . Points E and F are on $\overline{AB}$ and $\overline{AC}$, respectively, so that $AE = 3$ and $AF = 10$. Given that $EB = 9$ and $FC = 27$, find the integer closest to the area of quadrilateral $DCFG$.



Source: 2002 AIME

Solution: In this problem, we will use the idea that if two angles of a triangle are equal, then the ratio of the areas is simply the ratio of the side lengths around that angle multiplied.

Since $\triangle ABC$ is a right triangle, we can use the pythagorean base to find the length of $\overline{BC}$.

$$BC = \sqrt{37^2 - 12^2} = 35$$

Since we know that $BC = 35$ and $AB = 12$, we can use those two lengths to find the area of the entire triangle to be $\frac{35 \cdot 12}{2} = 210$.

To find our answer which is the area of quadrilateral $DCFG$, we can simply subtract the area of the smaller triangles from $\triangle ABC$.

Since both $\triangle AEF$ and $\triangle ABC$ share $\angle A$, $\frac{[AEF]}{[ABC]} = \frac{3 \cdot 10}{12 \cdot 37}$

This means that $[AEF] = \frac{5 \cdot [ABC]}{74} = \frac{525}{37}$.

Using the angle bisector theorem (since $\angle A$ is bisected), we know that $\frac{AB}{AC} = \frac{BD}{CD}$. The equation above simplifies to $\frac{BD}{CD} = \frac{12}{37}$.

At the same time, we know that since $\overline{AD}$ is a cevian in $\triangle ABC$ that connects the vertex A to $\overline{BC}$ at point D , $\frac{[ABD]}{[ADC]} = \frac{BD}{CD}$

We already know that $\frac{BD}{CD} = \frac{12}{37}$, which means that $\frac{[ABD]}{[ADC]} = \frac{12}{37}$.

Since we know that $[ABD] + [ADC] = [ABC] = 210$, it's obvious that $[ABD] = \frac{360}{7}$ and $[ADC] = \frac{1110}{7}$.

If we now sum up the areas of $[ABD]$ and $[AEF]$, we will overcount the area of $\triangle AEG$. Thus, we now need to find the area of that triangle.

Applying the angle bisector theorem on $\triangle AEF$, we get that $\frac{AE}{AF} = \frac{EG}{FG}$. This gives us that $\frac{EG}{FG} = \frac{3}{10}$. This also means that $\frac{[AEG]}{[AGF]} = \frac{3}{10}$.

Since we already know the area of $\triangle AEF$, we get that $[AEG] = \frac{1575}{481}$ and $[AGF] = \frac{5250}{481}$.

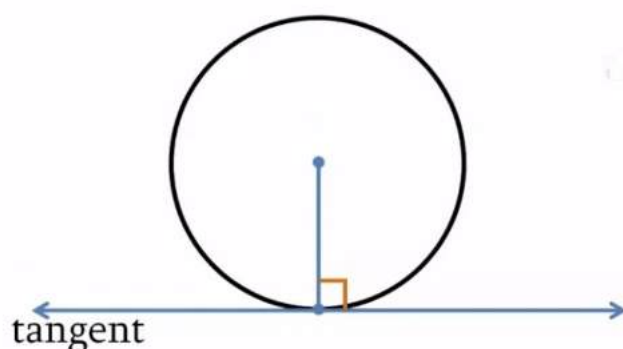
Now we can find the area of the entire triangle ABC not including the quadrilateral DCFG to be: $[ABD] + [AEF] - [AEG] = \frac{360}{7} + \frac{525}{37} - \frac{1575}{481} = \frac{209910}{3367}$.
 $[DCFG] = 210 - \frac{209910}{3367}$ rounds up to **148**.

Note: In this problem, we found the actual fractional value of the area, but you could have avoided that because you only need to find the closest integer to the actual area.

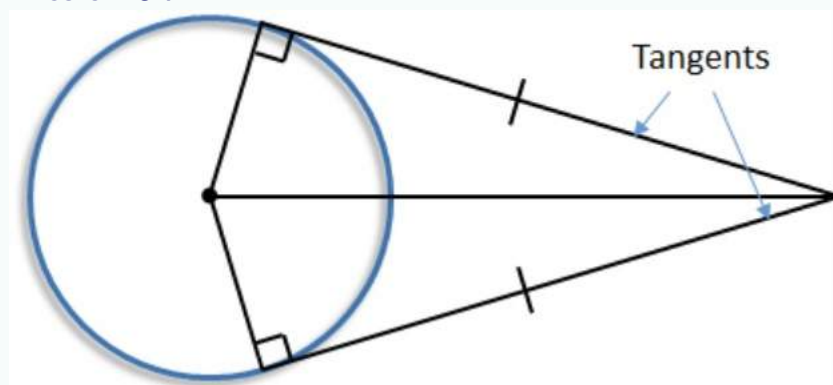
§5.7 Circles Part 1

Theorem 5.7.1

If there is a line that's tangent to a circle, the line connecting the tangency point and the radius will be perpendicular to the tangent line.



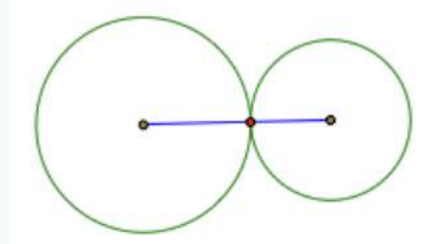
Theorem 5.7.2



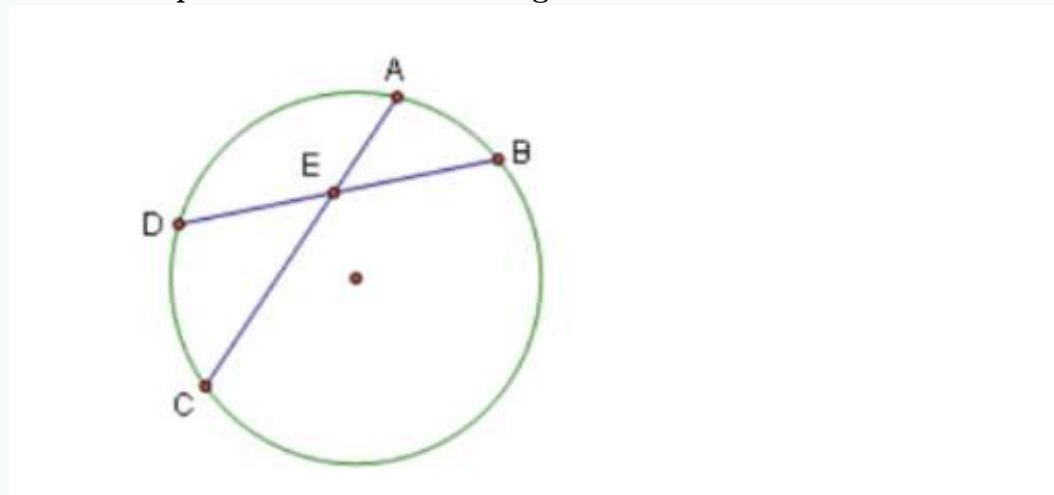
The theorem above states that if two tangents of the same circle intersect at one point, then the distance between the points where the tangent line intersects the circle to the intersection point of the two tangents will be the same.

Theorem 5.7.3

If two circles intersect at one point, then the line connecting their centers will go through that intersection point.

**Theorem 5.7.4**

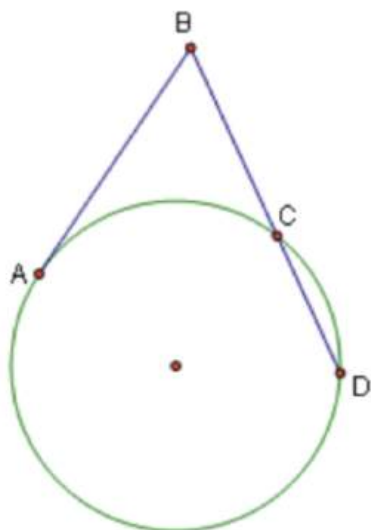
Power of a point for two intersecting chords



For two intersecting chords as shown above, power of a point gives $\overline{AE} \cdot \overline{CE} = \overline{BE} \cdot \overline{DE}$

Theorem 5.7.5

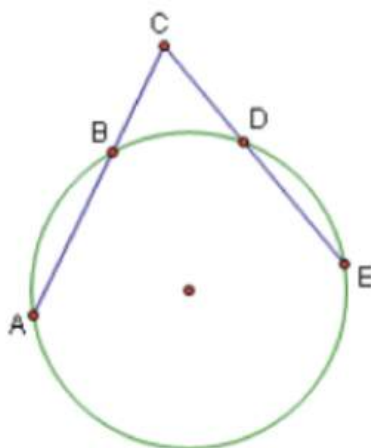
Power of a point for a tangent intersecting a secant



For a tangent intersecting a secant as shown above, power of a point gives $\overline{AB}^2 = \overline{BC} \cdot \overline{BD}$

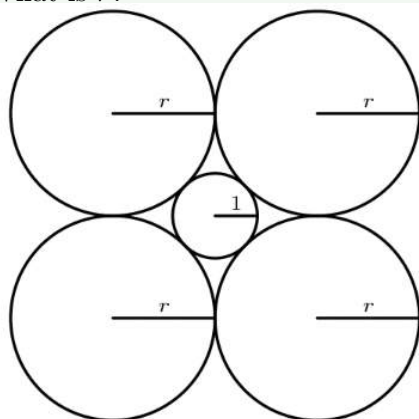
Theorem 5.7.6

Power of a Point for two secants intersecting outside of a circle



For two secants intersecting outside the circle as shown above, power of a point gives $\overline{CB} \cdot \overline{CA} = \overline{CD} \cdot \overline{CE}$.

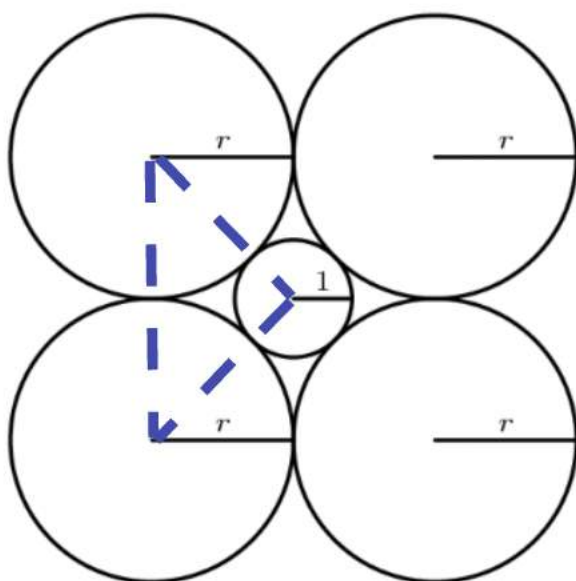
Problem 5.7.7 — A circle of radius 1 is surrounded by 4 circles of radius r as shown. What is r ?



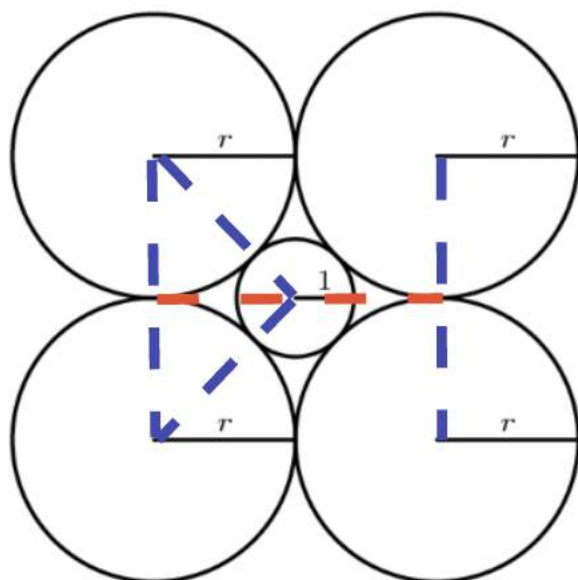
- (A) $\sqrt{2}$ (B) $1 + \sqrt{2}$ (C) $\sqrt{6}$ (D) 3 (E) $2 + \sqrt{2}$

Source: 2007 AMC

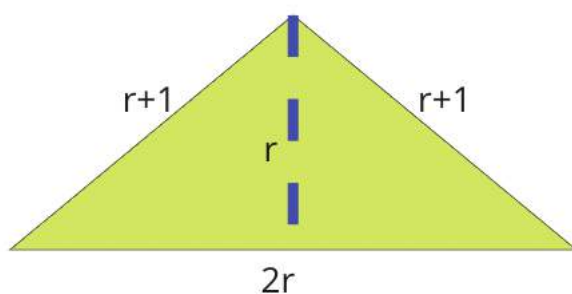
Solution: One helpful strategy for problems with many circles is to connect the centers. This is very helpful especially if the circles intersect with each other.



This should immediately make it obvious that the triangle drawn out by connecting the 3 centers is isosceles. The common side length is $r + 1$, and the base is $2r$. We can always split an isosceles triangle into two congruent right triangles with the use of an altitude.



It should be obvious from the diagram that the red line has a length of $2r$. Half of it is the length of the altitude due to the symmetry that exists.

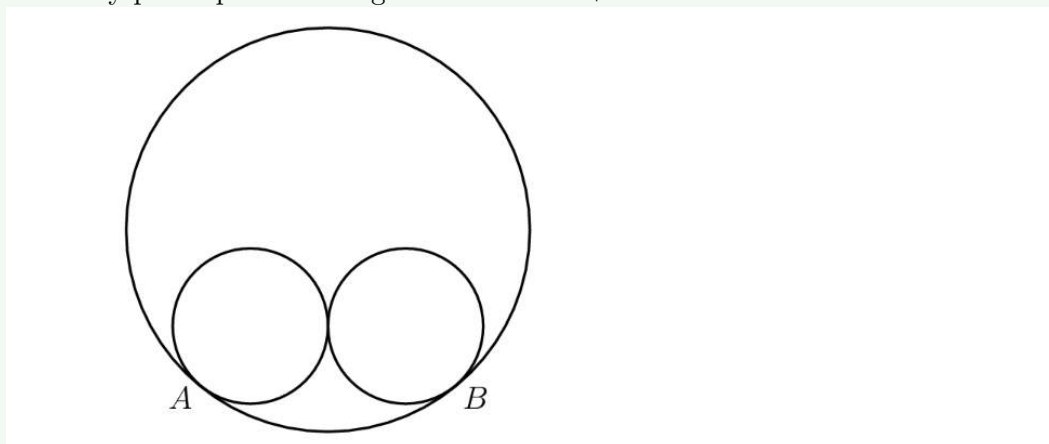


The diagram makes it obvious that the two identical right triangles have legs of r and r while the hypotenuse is $r + 1$. Using the pythagoreon theorem, we know that $r^2 + r^2 = (r + 1)^2$

We can expand the equation above and subtract like terms to get $r^2 - 2r - 1 = 0$.

We can use the quadratic formula to solve for the roots to get that $r = 1 \pm \sqrt{2}$. r must be $1 + \sqrt{2}$ (**B**) since $1 - \sqrt{2}$ is a negative value which isn't possible for a side length (length can't be a negative number).

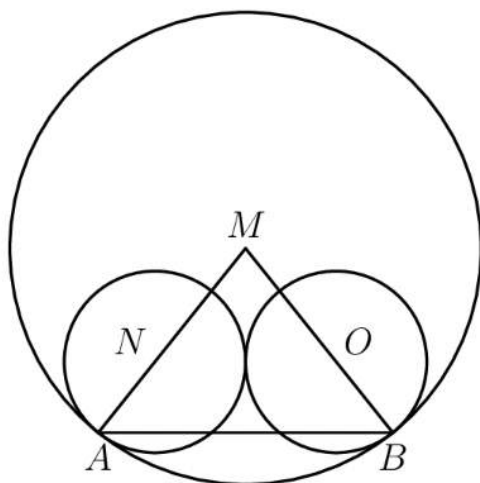
Problem 5.7.8 — Two circles of radius 5 are externally tangent to each other and are internally tangent to a circle of radius 13 at points A and B , as shown in the diagram. The distance AB can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?



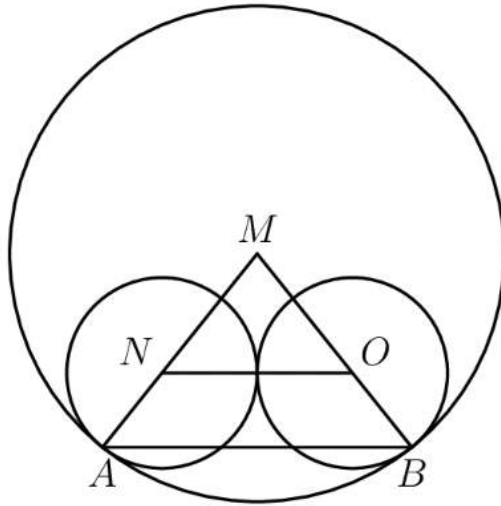
- (A) 21 (B) 29 (C) 58 (D) 69 (E) 93

Source: 2018 AMC

Solution: In this problem, we will again connect our centers. We know that if two circles intersect at one point, then the lines connecting their center must pass through their intersection point. Let's first assume that the center of the larger circle is M and the centers of the two smaller circles are N and O . Then, points M , N , and A will lie on the same line. At the same time points M , O , and B will also lie on the same line.



Now we can also draw a line connecting centers N and O , and it must go through their circles' intersection point.



From here, we should notice that $\triangle MNO$ is similar to $\triangle MAB$. To prove this, we already know that $\overline{MN} = \overline{MO}$ and $\overline{MA} = \overline{MB}$. Dividing both equations relating the side lengths gives that $\frac{\overline{MN}}{\overline{MA}} = \frac{\overline{MO}}{\overline{MB}}$. We also know that $\angle AMB = \angle NMO$. Since the two side lengths have the same ratio between them and the angle between them is the same, both triangles are similar due to SAS similarity.

Now we can use our side lengths ratios and our given side lengths to find $\overline{AB}$.

Due to similarity, we get $\frac{\overline{MN}}{\overline{NO}} = \frac{\overline{MA}}{\overline{AB}}$.

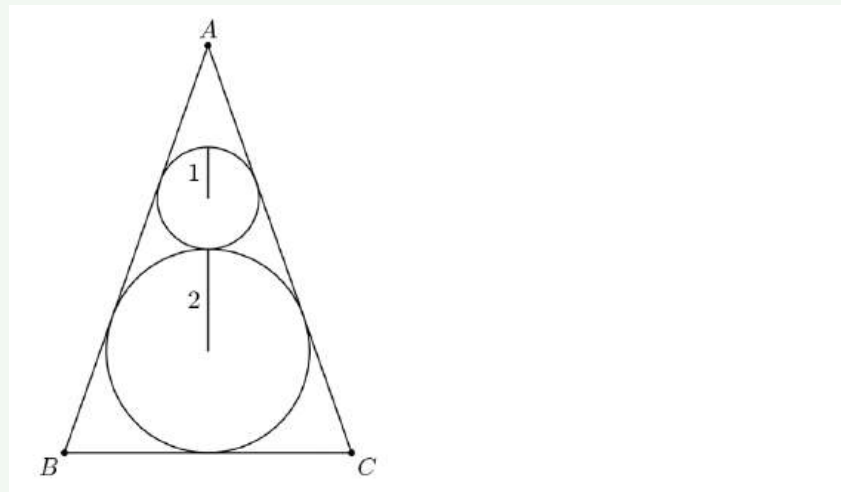
Since we know that the radius of the larger circle is 13, we automatically know that $\overline{MA}$ and $\overline{MB}$ both have a length of 13. In addition, since we know that the radius of the smaller circle is $\overline{NA}$ is 5, we can find $\overline{MN}$ because $\overline{MN} + \overline{NA} = \overline{MA}$.

This gives us that $\overline{MN}$ has a length of 8. We also know that $\overline{NO}$ has a length of 10 because it is the sum of the radii of the two small circles.

Since we know 3 of our side lengths from our similarity equation except $\overline{AB}$, we can plug in these values to evaluate our last length.

$\frac{8}{10} = \frac{13}{\overline{AB}}$. This gives us that $\overline{AB} = \frac{65}{4}$ and our final answer is $65 + 4 = \mathbf{69}$ (D).

Problem 5.7.9 — A circle of radius 1 is tangent to a circle of radius 2. The sides of $\triangle ABC$ are tangent to the circles as shown, and the sides $\overline{AB}$ and $\overline{AC}$ are congruent. What is the area of $\triangle ABC$?



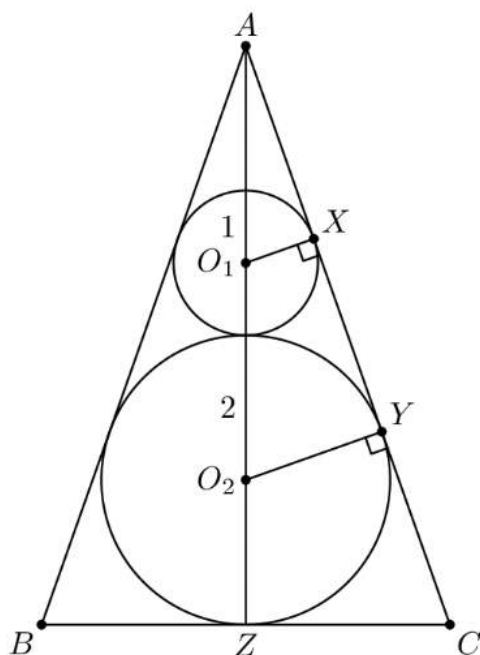
- (A) $\frac{35}{2}$ (B) $15\sqrt{2}$ (C) $\frac{64}{3}$ (D) $16\sqrt{2}$ (E) 24

Source: 2006 AMC

Solution: In this problem, we can first label the centers of the top circle as O_1 and the one on the bottom as O_2 . We know that for any line tangent to a circle, the line connecting the radius and the tangency point will be perpendicular to it.

We can apply the idea above to this problem. We can label the intersection (tangency point) of $\overline{AC}$ with the top circle as X and the bottom one as Y . At the same time, we can draw an altitude from point A to $\overline{BC}$ intersecting it at point F .

We immediately know that $AF \perp BC$, $O_1X \perp AX$, and $O_2Y \perp AY$.



In this problem, we can angle chase to find that $\triangle AXO_1$ is similar to $\triangle AYO_2$ which is similar to $\triangle AZC$.

We can write the ratios of the side lengths for $\triangle AXO_1$ and $\triangle AYO_2$ to get

$$\frac{\overline{AO_1}}{\overline{O_1X}} = \frac{\overline{AO_2}}{\overline{O_2Y}}.$$

We know the lengths of $\overline{O_1X}$ and $\overline{O_2Y}$ because they are the radii of the circle (1 and 2). However, we don't know the length of $\overline{AO_1}$ and $\overline{AO_2}$. Let's assume that the length of $\overline{AO_1}$ is x . Then, this means that $\overline{AO_2}$ has a length of $x + 3$, the sum of the two radii which is 3. Thus, our two lengths are x and $x + 3$. We can plug in these values into our similarity equations to get

$$\frac{x}{1} = \frac{x+3}{2}$$

Solving the equation for x (the side length $\overline{AO_1}$ gives that it is 3).

Now from the Pythagorean Theorem, we can find that $\overline{AX}$ has a length of $2\sqrt{2}$ and $\overline{AY}$ has a length of $4\sqrt{2}$.

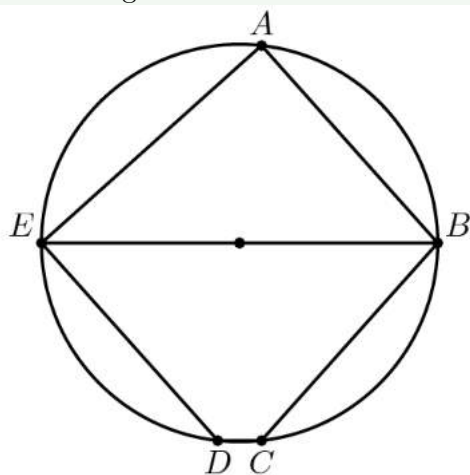
Since we know that $\triangle AXO_1$ is similar to $\triangle AZC$, we can write out similarity ratios to get

$$\frac{\overline{AX}}{\overline{O_1X}} = \frac{\overline{AZ}}{\overline{CZ}}$$

Plugging in our side lengths gives $\frac{2\sqrt{2}}{1} = \frac{8}{\overline{CZ}}$. We can solve for the length of $\overline{CZ}$ to get that it is $2\sqrt{2}$.

Now since we know that the length of $\overline{CZ}$ is half of the length of $\overline{BC}$, $\overline{BC}$ has a length of $4\sqrt{2}$. We can now find the area of $\triangle ABC$ since it's simply $\frac{\overline{AZ} \cdot \overline{BC}}{2}$. The area is $16\sqrt{2}$.

Problem 5.7.10 — In the given circle, the diameter $\overline{EB}$ is parallel to $\overline{DC}$, and $\overline{AB}$ is parallel to $\overline{ED}$. The angles $\angle AEB$ and $\angle ABE$ are in the ratio 4 : 5. What is the degree measure of angle $\angle BCD$?



- (A) 120 (B) 125 (C) 130 (D) 135 (E) 140

Source: 2011 AMC 10

Solution: In this problem, since we know that inscribed angles are half the measure of the intercepted arc, it becomes obvious that $\angle EAB$ is 90° . (In general any angle whose vertex is at the center of a circle and whose sides are chords is a central angle.)

intercepted arc contains the chord that is the diameter has angle 90° .)

Since we know that $\triangle EAB$ is right, we know that $\angle AEB + \angle ABE = 90^\circ$.

Since we know that they are in the ratio $4 : 5$, $\angle AEB = 40^\circ$ and $\angle ABE = 50^\circ$.

Also, since lines AB and ED are parallel and EB is a transversal that goes through them, we know that $\angle ABE = \angle BED = 50^\circ$.

We know that quadrilateral $BCDE$ is cyclic because all of its vertices lie on the circle. Since opposite angles must sum to 180° for a cyclic quadrilateral, $\angle BED + \angle BCD = 180^\circ$.

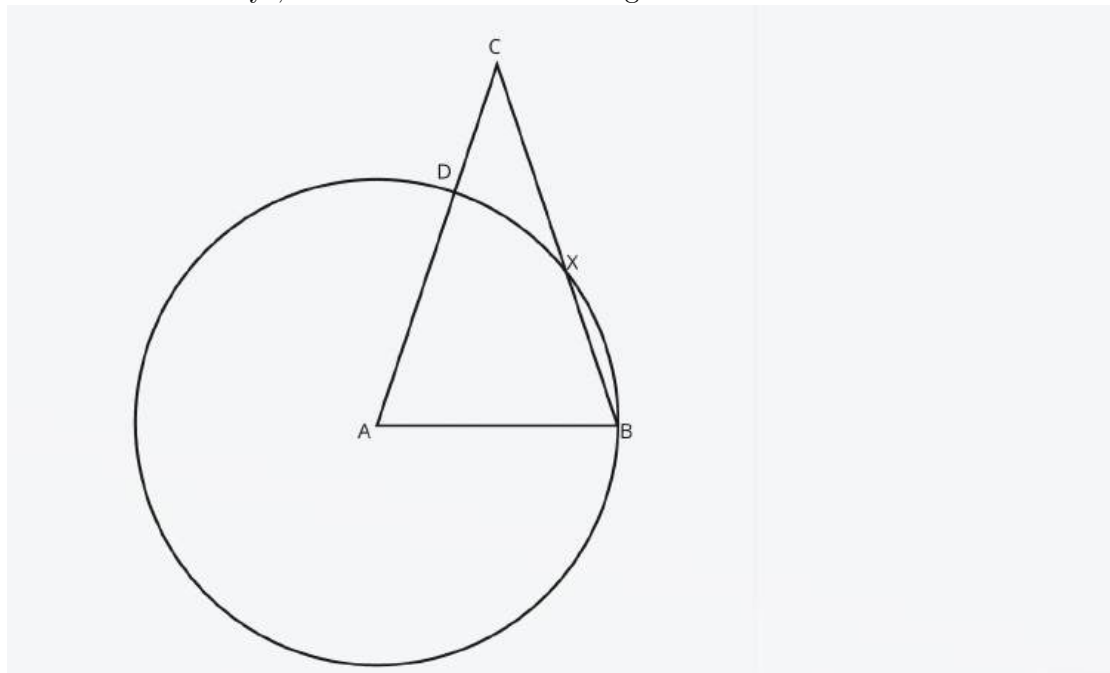
Since we know that $\angle BED = 50^\circ$, we can plug that in to get that $\angle BCD = 130^\circ$.

Problem 5.7.11 — In $\triangle ABC$, $AB = 86$, and $AC = 97$. A circle with center A and radius AB intersects $\overline{BC}$ at points B and X . Moreover $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC ?

(A) 11 (B) 28 (C) 33 (D) 61 (E) 72

Source: 2013 AMC 10

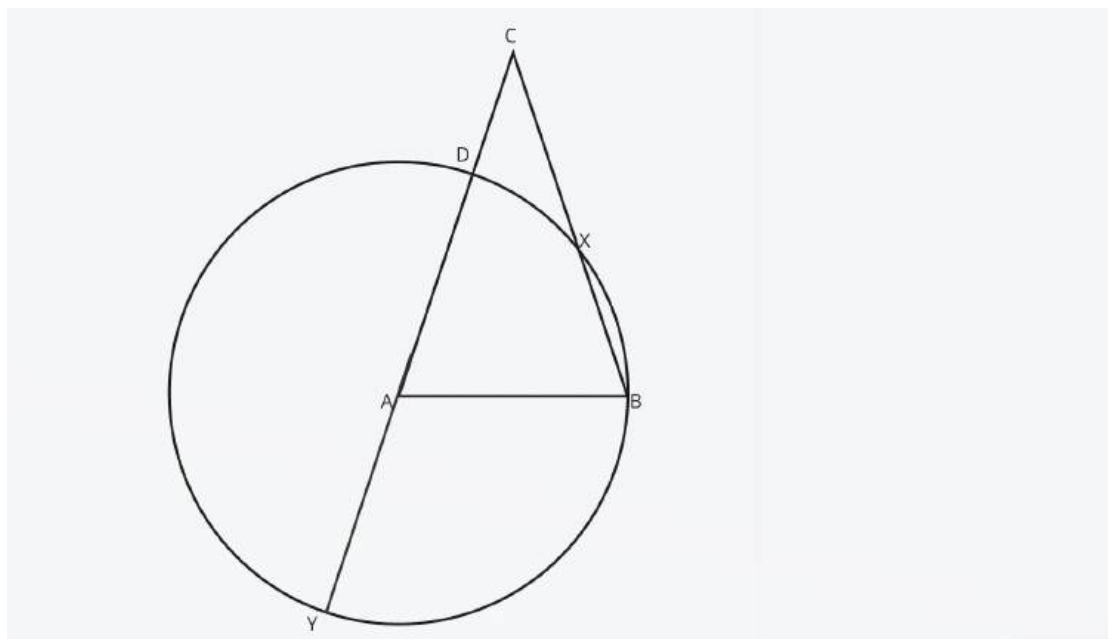
Solution: As always, we should first draw a diagram.



Let's assume that D is the intersection point of the line AC to the circle.

It's evident that $\overline{AD} = \overline{AB} = 86$ (radius of the circle). This means that $\overline{DC}$ has a length of $97 - 86 = 11$.

We can extend line AD to hit the circle again on the other side to make a full diameter.



We can use power of a point on the figure above because we have two secant lines that intersect outside of the circle. It's obvious that $\overline{CD} \cdot \overline{CF} = \overline{CX} \cdot \overline{CB}$

The equation simplifies to $\overline{CX} \cdot \overline{CB} = 2013 = 3 \cdot 11 \cdot 61$.

Now since we know that $\overline{CX}$ and $\overline{BX}$ have integer lengths, $\overline{CB}$ also has an integer length. We can try out values for each side by using the fact that $\overline{CX}$ has a length that is less than the length of $\overline{CB}$.

Before we try out values, we can apply the triangle inequality on $\triangle ACX$ to get that the length of $\overline{CX} > 11$.

It's obvious that the length of $\overline{CX}$ must be 33 since while $\overline{CB}$ must be 61. The reason for this is that $\overline{CX}$ can't have a length of either 3 or 11. If we pair up any factor then the length of it will be greater than the length of $\overline{CB}$ which isn't possible. Since the length of $\overline{CB}$ is 61, the length of $\overline{BC}$ is obviously 61 (**D**).

§5.8 Circles Part 2: Cyclic Quadrilaterals

This topic of cyclic quadrilaterals is much more important than it seems. It is extremely common and is starting to have a higher yield on more recent exams.

Definition 5.8.1

Cyclic quadrilaterals are quadrilaterals that can be inscribed in a circle. This means that all of its vertices will lie on a circle.

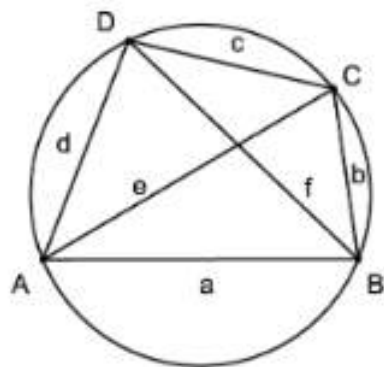
Theorem 5.8.2

Opposite angles of a cyclic quadrilateral always sum to 180° .

Theorem 5.8.3**Ptolemy's Theorem**

Ptolemy's theorem is about relating the side lengths to the diagonals in a cyclic quadrilateral.

Ptolemy's Theorem states that if you multiply the diagonals, then that will be equivalent to the sum of the multiplication of the opposite sides. If you don't understand what I mean, then the picture and example below should help you understand it.



Ptolemy's gives that $\overline{CB} \cdot \overline{DA} + \overline{DC} \cdot \overline{AB} = \overline{DB} \cdot \overline{AC}$

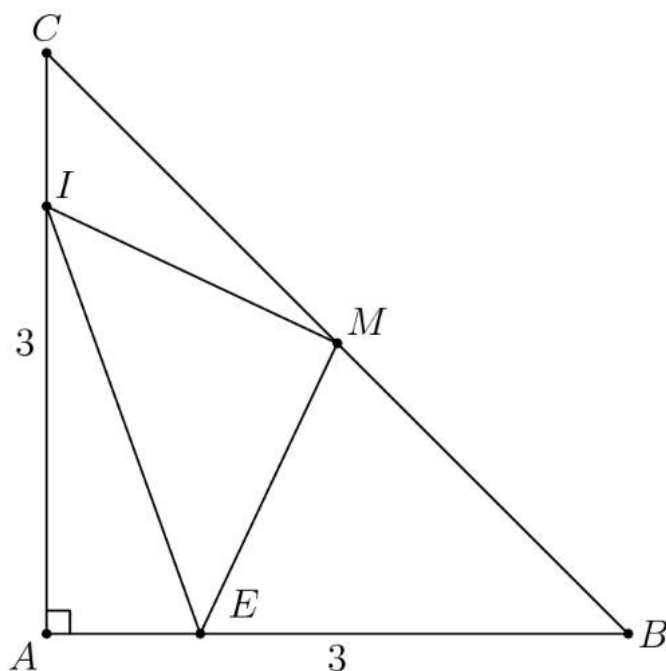
We can substitute our side lengths (in terms of the variables a, b, c, d, e , and f) to get that the same theorem is equivalent to $ac + bd = ef$.

Problem 5.8.4 — Triangle ABC is an isosceles right triangle with $AB = AC = 3$. Let M be the midpoint of hypotenuse $\overline{BC}$. Points I and E lie on sides $\overline{AC}$ and $\overline{AB}$, respectively, so that $AI > AE$ and $AIME$ is a cyclic quadrilateral. Given that triangle EMI has area 2, the length CI can be written as $\frac{a-\sqrt{b}}{c}$, where a, b , and c are positive integers and b is not divisible by the square of any prime. What is the value of $a + b + c$?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Source: 2018 AMC 12

Solution: As always, we should make our diagram.



In this problem, we should first know that if we draw a line connecting the midpoint of the hypotenuse of an isosceles right triangle to the vertex that has an angle of 90° , then that line will be perpendicular to the hypotenuse and it will bisect the right angle.

Using that information, it's obvious that $\angle MAC$ and $\angle MAB$ are both 45° .

We know that if two angles subtend the same arc they have the same angle. Since quadrilateral AIME is cyclic, $\angle IAM$ and $\angle IEM$ clearly subtend the same arc IM. This means that the two angles are both equal to 45° .

At the same time, since opposite angles of a cyclic quadrilateral add to 180° , it's obvious that $\angle IME$ is 90° and $\triangle IME$ is a right triangle.

This means that $\angle MIE$ is also 45° . This means that $\triangle IME$ is an isosceles right triangle. If the legs of it have a side length of s , then the area will be $\frac{s^2}{2}$. We know that the area of this triangle is 2 since it was given in the problem statement. This means that the length of the leg is 2.

Using the Pythagorean theorem on $\triangle IME$ we get that $\overline{IE}$ has a length of $2\sqrt{2}$. Also, since $\triangle AMC$ is an isosceles right triangle with a hypotenuse of 3, the legs such as $\overline{AM}$ have a length of $\frac{3\sqrt{2}}{2}$.

Now we can apply Ptolemy's Theorem to cyclic quadrilateral AIME.

$$\overline{AE} \cdot \overline{IM} + \overline{ME} \cdot \overline{AI} = \overline{AM} \cdot \overline{IE}$$

Plugging in the values of our lengths gives $\overline{AE} \cdot 2 + \overline{AI} \cdot 2 = \frac{3\sqrt{2}}{2} \cdot 2\sqrt{2}$.

Simplifying the equation above gives $\overline{AE} + \overline{AI} = 3$

We also know that $\overline{AE}^2 + \overline{AI}^2 = \overline{IE}^2$

Since we know that $\overline{IE}$ has a length of $2\sqrt{2}$, our equation becomes $\overline{AE}^2 + \overline{AI}^2 = 8$.

We can plug in x for the length of $\overline{AE}$ and y for the length of $\overline{AI}$ to get the two equations:

$$\begin{aligned}x + y &= 3 \\x^2 + y^2 &= 8\end{aligned}$$

Solving these equations gives that the two lengths must be either $\frac{3+\sqrt{7}}{2}$ or $\frac{3-\sqrt{7}}{2}$.

However, since we know that $\overline{AI} > \overline{AE}$, it's obvious that $\overline{AI}$ must be $\frac{3+\sqrt{7}}{2}$ since it's the bigger value.

Now we can subtract that length from 3 to get the length of $\overline{CI}$ which is $\frac{3-\sqrt{7}}{2}$ (**Think about why the length of $\overline{CI}$ is the same as the length of $\overline{AE}$**). The answer is $3 + 7 + 2$ which is **12**.

§5.9 Analytical Geometry

We can solve many geometry problems by "coord-bashing" which is about assigning coordinates to our geometrical points.

Theorem 5.9.1

Slope Intercept Form

$y = mx + b$ is the equation, and m represents the slope while b represents the y-intercept.

Theorem 5.9.2

Let (x_1, y_1) and (x_2, y_2) be points on a plane.

The distance between these two points is simply $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Theorem 5.9.3

Relating parallel and perpendicular lines through slopes

Parallel lines have the exact same slope.

If two lines are perpendicular, then their slopes multiply to -1 .

Also, it's common to see problems that involve a circle with the given equation. It's important to know the format of the equation.

Theorem 5.9.4

The equation of a circle is

$(x - a)^2 + (y - b)^2 = r^2$ and r is the radius of the circle and the center of this circle is (a, b) .

What if you want to find the area of a shape with multiple points that we have the coordinates for. It can be hard to break down the shape to find the area. However, we have a special theorem that can help us find the area for any number of coordinates in the plane.

Theorem 5.9.5

Shoelace Theorem

If a polygon has vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) ... and (x_n, y_n) .

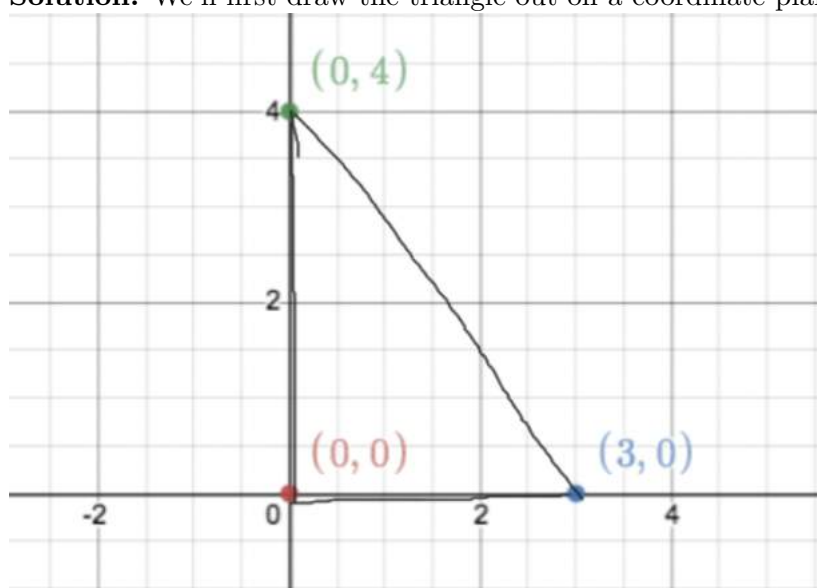
$$A = \frac{1}{2} |(x_1y_2 + x_2y_3 + \cdots + x_ny_1) - (y_1x_2 + y_2x_3 + \cdots + y_nx_1)|$$

It might be hard to understand the theorem above. Thus, I'll simplify it in an example below.

Example 5.9.6

We have a right triangle with side lengths 3, 4, 5. What is the area of this triangle. Find it using the Shoelace Theorem.

Solution: We'll first draw the triangle out on a coordinate plane.



Now all you do is start at any one point and write the x coordinate and the y coordinate next to it. Go through all the vertices in **one direction** (either counterclockwise or clockwise around the triangle). After you write all the vertices of the shape, write the point you started with again. In this case, we will go in the order of $(0,0)$ and $(3,0)$ and $(0,4)$ and $(0,0)$. You could also start at the point of $(3,0)$. However, once you start at a point and go in one direction, then you have to maintain that direction for the entire time.

X	Y
0	0
0	4
3	0
0	0

Now you will multiply and add in a shoelace pattern.

X	Y
0	0
0	4
3	0
0	0

We don't connect the last x coordinate to anything yet. So using these values, the shoelace theorem states that you multiply them. $(0x4) + (0x0) + (3x0) = 0$. Now we do this again but start with the y coordinate at the top.

X	Y
0	0
0	4
3	0
0	0

Now, the shoelace theorem states that $(0x0) + (4x3) + (0x0) = 12$. Now, you subtract the second value from the first one to get $0 - 12$ which is -12 . You take the absolute value of the number which is 12 and divide by 2 to get the area of the triangle to be **6**.

Now to confirm the validity of this theorem, we already know that the area of a 3, 4, 5 right triangle is 6 because the area of a triangle is $\frac{bh}{2}$ where b is the length of the base and h is the length of the height.

Key Takeaway: Use the shoelace theorem in problems with complicated shapes and when the coordinates are given. It can find the area relatively quickly.

Problem 5.9.7 — In the right triangle $\triangle ACE$, we have $AC = 12$, $CE = 16$, and $EA = 20$. Points B , D , and F are located on AC , CE , and EA , respectively, so that $AB = 3$, $CD = 4$, and $EF = 5$. What is the ratio of the area of $\triangle DBF$ to that of $\triangle ACE$?

Source: 2004 AMC

Solution: In this problem, we'll use coordinate geometry to write out all the points.

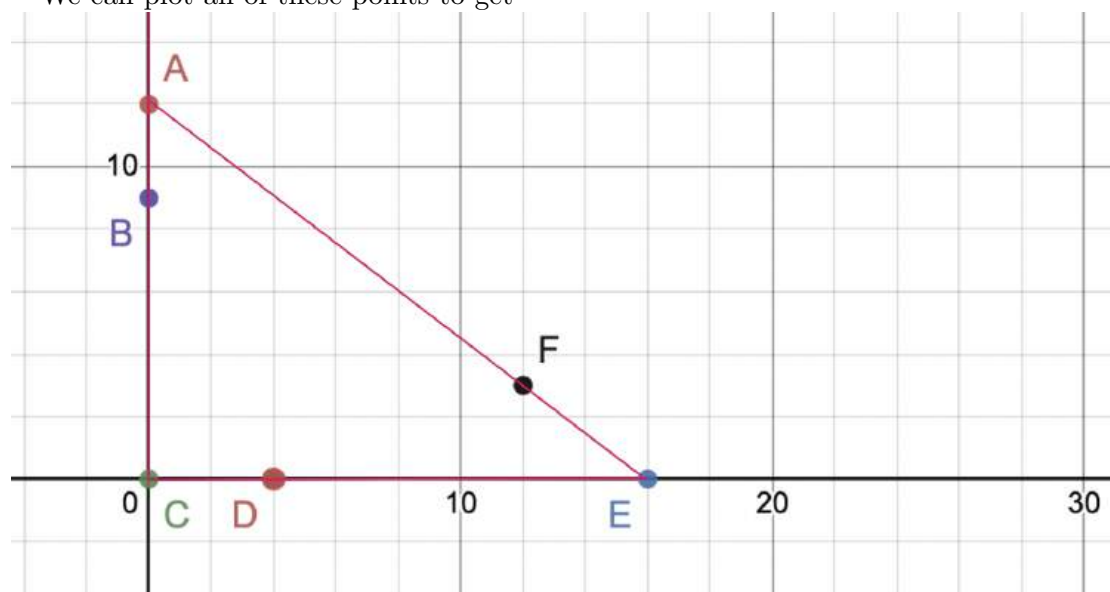
One important tip is to put the right angle at the origin. This almost always simplifies your coordinates and makes it easy to work with.

We can assume that point A is $(0, 12)$, point C is $(0, 0)$, and point E is $(16, 0)$.

Now since we know the lengths of AB , CD , EF , we can find the coordinates of B, D, and F using our coordinates of point A, C, and E.

We find that point B is $(0, 9)$, point D is $(4, 0)$, and point F is $(12, 3)$.

We can plot all of these points to get

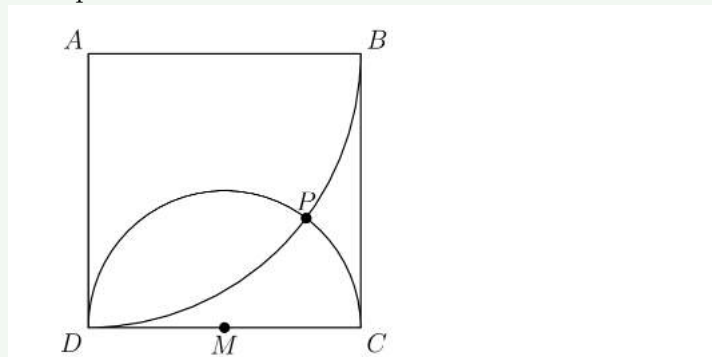


We can easily find the area of triangle ACE for the ratio we want to find. It is simply a right triangle with legs 12 and 16, and its area is $\frac{12 \cdot 16}{2}$ which is 96.

Now all we're left with is to find the area of $\triangle DBF$. We can do this easily using shoelace theorem (apply the theorem on your own to find the answer).

The area of $\triangle DBF$ is simply 42, and the ratio we want to find is $\frac{42}{96}$ which reduces to $\frac{7}{16}$ (E).

Problem 5.9.8 — Square $ABCD$ has sides of length 4, and M is the midpoint of $\overline{CD}$. A circle with radius 2 and center M intersects a circle with radius 4 and center A at points P and D . What is the distance from P to $\overline{AD}$?

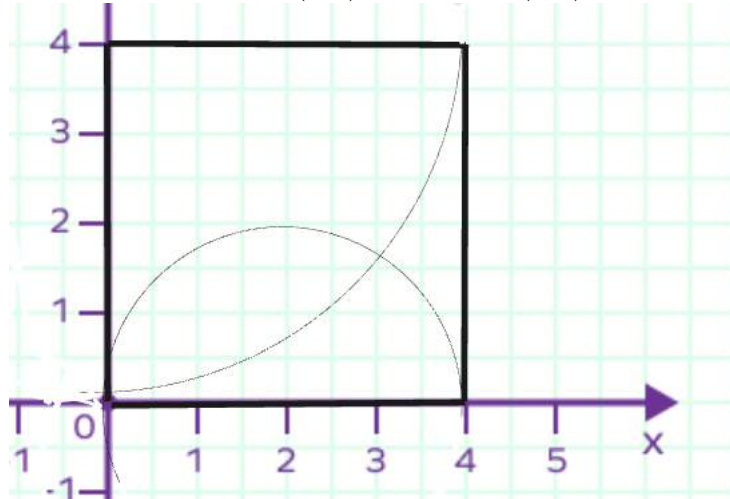


- (A) 3 (B) $\frac{16}{5}$ (C) $\frac{13}{4}$ (D) $2\sqrt{3}$ (E) $\frac{7}{2}$

Source: 2003 AMC

Solution: There are good ways to solve this problem synthetically (using techniques without bashing it through coordinate geometry). However, we'll use coordinate geometry to practice.

Let's label point D as $(0,0)$, point C as $(4,0)$, point A as $(0,4)$, and point B as $(4,4)$.



In this problem, we can write out the equations of the semicircle and quarter-circle. Since the center of the quarter-circle is at point A $(0,4)$ with a radius of 4, its equation is simply

$$x^2 + (y - 4)^2 = 16$$

The semicircle's center is at the midpoint of D and C which is at point M. Its coordinates are $(2,0)$ and it has a radius of 2. The equation of the full circle is simply

$$(x - 2)^2 + y^2 = 4$$

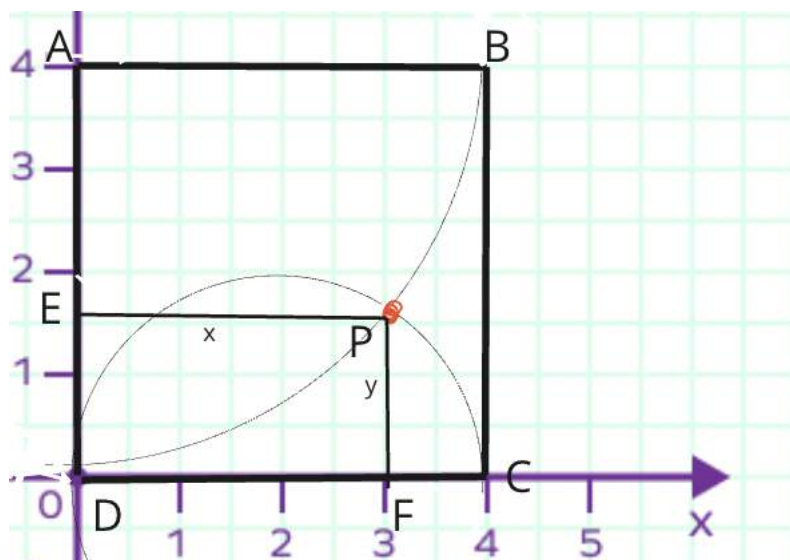
Since we know that these two circles intersect at point P, we can find it by solving for the two equations.

$$\begin{aligned} x^2 + (y - 4)^2 &= 16 \\ (x - 2)^2 + y^2 &= 4 \end{aligned}$$

Subtracting the two equations gives $4x - 8y = 0$ which gives that $x = 2y$.

Since we know that for any point with coordinates (x, y) , the distance of that point from the x -axis is simply y while the distance of that point from the y axis is x .

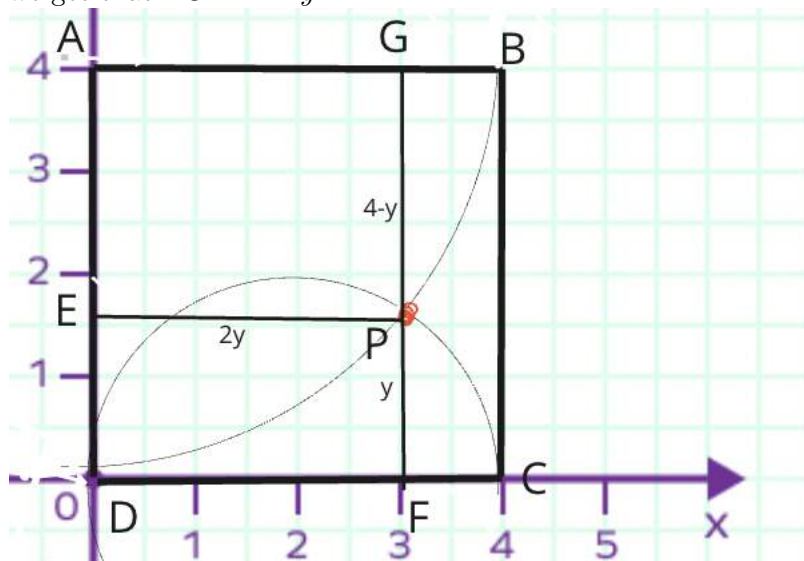
We can draw perpendicular lines from point P to intersect lines DC and AD.



Assuming that the coordinates of P are (x, y) , we know that PF has a length of y and PE has a length of x .

From the equations of the circles that we calculated, we found that $x = 2y$. We can plug in $2y$ for the length of $\overline{EP}$.

Now if we extend a line from point P to be perpendicular to $\overline{AB}$ at point G , then we get that $\overline{PG} = 4 - y$.



We now observe that the length of $\overline{EP} = \overline{AG} = 2y$.

From the Pythagorean theorem, we know that $\overline{AP}^2 = \overline{AG}^2 + \overline{PG}^2$

We can plug in our values for the length of $\overline{AG}$ and $\overline{PG}$.

We already know that $\overline{AP} = 4$ because it is the radius of the quarter-circle. We can now write the equation to be

$$4^2 = (2y)^2 + (4 - y)^2$$

We can expand the equation and solve for y to get that $y = \frac{8}{5}$. Since the distance from P to $\overline{AD}$ is simply $2y$, we get that the answer is $\frac{16}{5}$ (B).

§5.10 3D Geometry

Majority 3D geometry problems are about visualizing the problem. It's very important to practice a lot of problems to be able to master this. 3D geometry is all about finding a key observation to find the problem, and to be able to do so quickly and successfully is to practice a lot.

Theorem 5.10.1

For any 3D shape where V represents the number of vertices, E represents the number of edges, and F represents the number of faces, then $V + F - E = 2$.

Theorem 5.10.2

The distance between two points with coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Theorem 5.10.3

The volume of any pyramid with area of the base being b and the height being h is $\frac{bh}{3}$.

Theorem 5.10.4

Theorems of a Sphere

For a sphere with radius r , then the following formulas are always true

The surface area of the sphere is $4\pi r^2$

The volume of the sphere is $\frac{4\pi r^3}{3}$

Theorem 5.10.5

For any **cylinder** with radius r and height h , then the following theorems will work.

The surface area of this cylinder is $2\pi r^2 + 2\pi rh$

The volume of the cylinder is $\pi r^2 h$

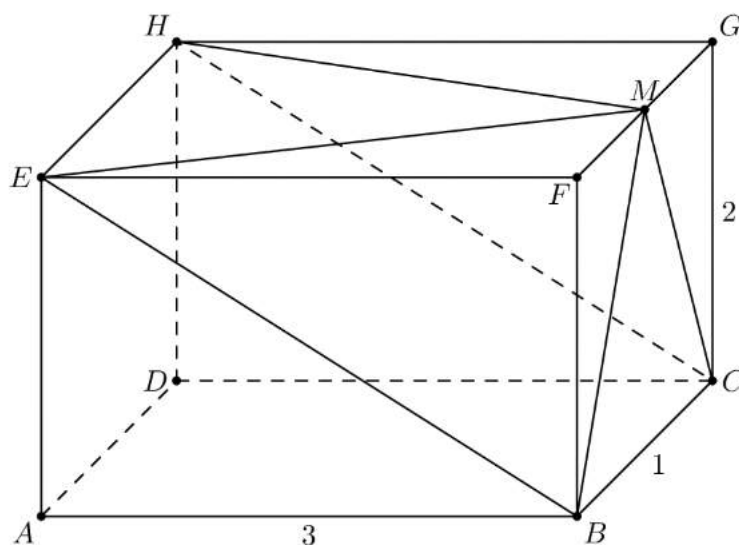
Theorem 5.10.6

A cone is a **pyramid** but it has a circular base instead. For a cone with radius r and height h , then the following theorems always apply.

The surface of the cone is $\pi r^2 + \pi r \sqrt{r^2 + h^2}$

The volume of the cone is $\frac{\pi r^2 h}{3}$

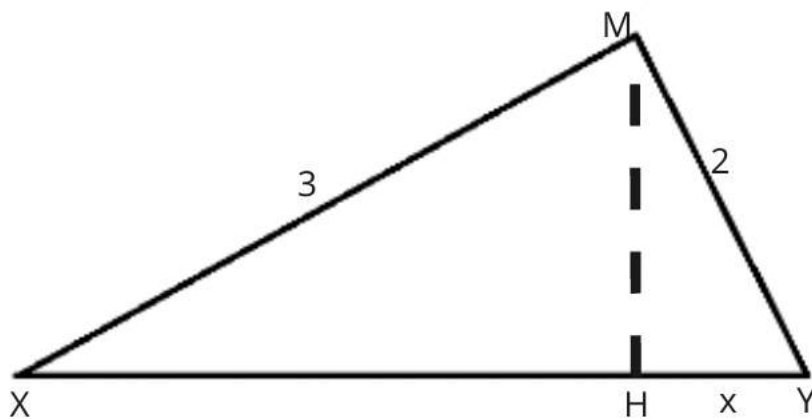
Problem 5.10.7 — In the rectangular parallelepiped shown, $AB = 3$, $BC = 1$, and $CG = 2$. Point M is the midpoint of $\overline{FG}$. What is the volume of the rectangular pyramid with base $BCHE$ and apex M ?



- (A) 1 (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$ (E) 2

Source: 2018 AMC

Source: In this problem, we will raw out a cross section in pyramid MBCHE. Let's assume that point X is the midpoint of $\overline{HE}$ and point Y is the midpoint of $\overline{BC}$. Then, we get a $\triangle MXY$ with side lengths 3, 2, and $\sqrt{13}$ (we can find this one by using the pythagoreen theorem to get $\overline{AB}^2 + \overline{AE}^2 = \overline{BE}^2 = \overline{XY}^2 = 13$)



In this problem, we know that $\overline{XY}$ has a length of $\sqrt{13}$. We'll drop an altitude from point A to $\overline{XY}$ intersecting it at point H .

Let's assume that $\overline{HY}$ has a length of x which means that $\overline{XH}$ has a length of $\sqrt{13} - x$

Now let's assume that our height $\overline{MH}$ has a length of h . We can now write out two equations for $\triangle MHY$ and $\triangle MHX$ using the pythagoreen theorem.

$$h^2 + x^2 = 4$$

$$h^2 + (\sqrt{13} - x)^2 = 9$$

Expanding both equations and then subtracting them from each other gives that $x = \frac{4\sqrt{13}}{13}$.

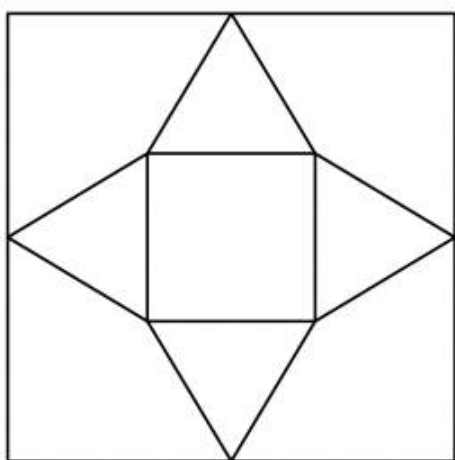
We can plug this back in to get that $h = \frac{6}{\sqrt{13}}$.

We know that the volume of a pyramid is $\frac{bh}{3}$ where b is the area of the base of the pyramid and h is the height of the pyramid.

We can find the area of the base easily since it's a simple rectangle with sides $\sqrt{13}$ and 1.

This means that the total volume is simply $\frac{1}{3} \cdot \sqrt{13} \cdot \frac{6}{\sqrt{13}} = 2$ (E).

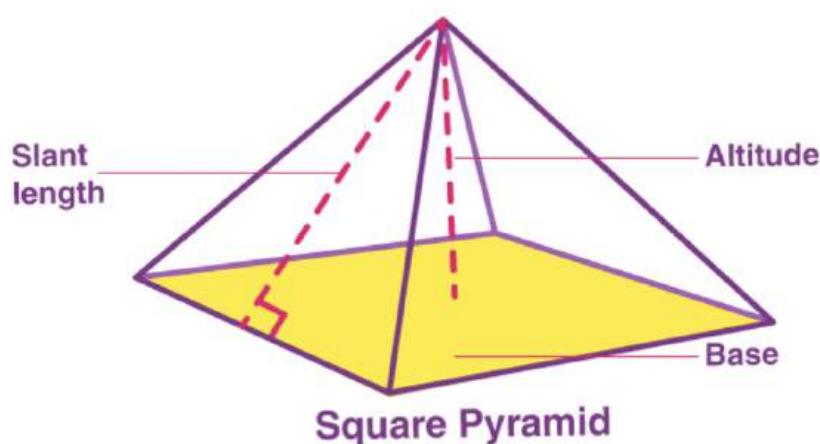
Problem 5.10.8 — In the accompanying figure, the outer square S has side length 40. A second square S' of side length 15 is constructed inside S with the same center as S and with sides parallel to those of S . From each midpoint of a side of S , segments are drawn to the two closest vertices of S' . The result is a four-pointed starlike figure inscribed in S . The star figure is cut out and then folded to form a pyramid with base S' . Find the volume of this pyramid.



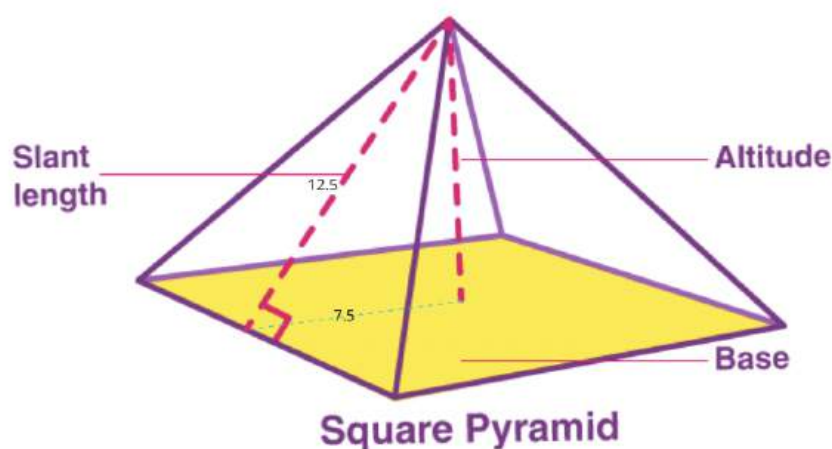
Source: 2012 AIME

Solution: We know that the volume of a pyramid is $\frac{bh}{3}$ where b represents the area of the base and h represents the height.

The base of this pyramid is simply the square with side length 15. The area of it is 225. Thus, now we simply have to find the height of this pyramid.



We need to be able to visualize the pyramid above. Clearly if we unravel it, then we should notice that the length of the base added to two times the slant height is equivalent to the side length of the larger square. This means that the slant height is simply $\frac{25}{2}$.



We can picture a right triangle in which the hypotenuse is the slant height, and the legs are: the height of this pyramid and half of the side length of square S' .

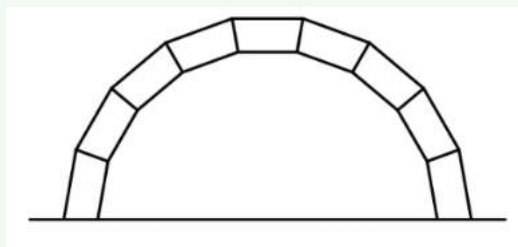
We can clearly calculate the height using Pythagorean theorem to get that $7.5^2 + h^2 = 12.5^2$

Our value of h (the height of the pyramid) is 10.

Thus, this means that the volume of our pyramid is simply $\frac{225 \cdot 10}{3}$ which simplifies to **750**.

§5.11 Cumulative Problems

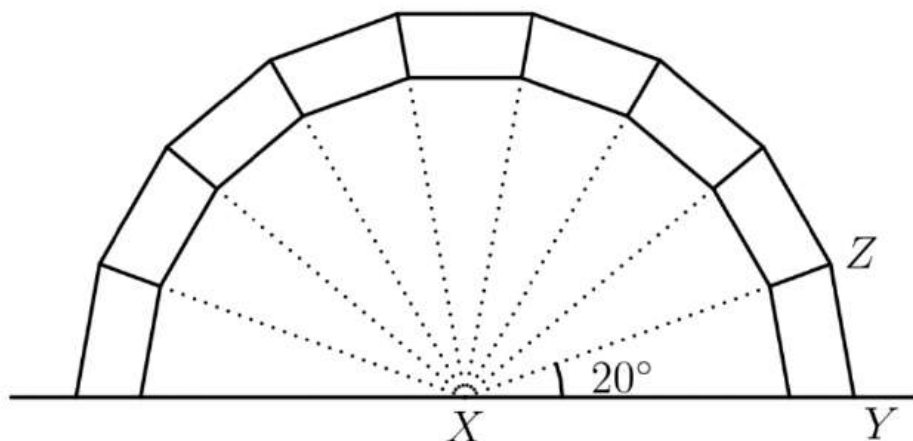
Problem 5.11.1 — The keystone arch is an ancient architectural feature. It is composed of congruent isosceles trapezoids fitted together along the non-parallel sides, as shown. The bottom sides of the two end trapezoids are horizontal. In an arch made with 9 trapezoids, let x be the angle measure in degrees of the larger interior angle of the trapezoid. What is x ?



- (A) 100 (B) 102 (C) 104 (D) 106 (E) 108

Source: 2009 AMC

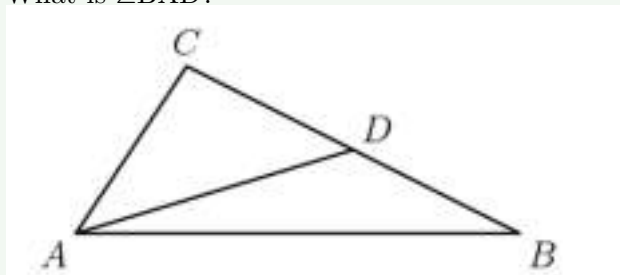
Solution: Due to the 9 trapezoids being arranged in the shape of a semicircle, we can use the symmetry to find a crucial angle.



Our $\angle YXZ$ is simply 20° . Since $\triangle YXZ$ is also an isosceles triangle, we know that the two base angles will be the same. Since the sum of the angles of any triangle is 180, we subtract 20 from that to get 160° . Then, we divide that by two to find one of them which boils down to 80° .

However, this is not our answer. We have to find the LARGER interior angle of the trapezoid. Since they are isosceles trapezoids, the two unique angles add up to 180° . Thus, we simply subtract 80 from 180 to get 100° (A).

Problem 5.11.2 — In $\triangle ABC$, $\overline{AC} = \overline{CD}$ and $\angle CAB - \angle ABC = 30^\circ$. What is $\angle BAD$?



Solution: Using variables to represent angles in angle chasing problems is a useful strategy. In this problem, since $AC = CD$, $\triangle ACD$ is an isosceles triangle in which $\angle CAD = \angle CDA$. We will label that angle as x .

We will also label $\angle DAB$ as y° . $\angle CAB$ is equivalent to $\angle CAD + \angle DAB$ which also equals to $x + y$. This means that $\angle ABC$ is $x + y - 30$.

Since $\angle CDA$ and $\angle BDA$ are supplementary, $\angle BDA$ is equivalent to $180 - x$. We will equate the sum of all 3 angles of $\triangle BDA$ to 180 in terms of x and y . The sum is $y + 180 - x + x + y - 30$. Equating this to 180 gives us $2y = 30$. Solving for y gives us 15° . Thus, $\angle BAD$ is 15° .

Problem 5.11.3 — The y -intercepts, P and Q , of two perpendicular lines intersecting at the point $A(6, 8)$ have a sum of zero. What is the area of $\triangle APQ$?

(A) 45 (B) 48 (C) 54 (D) 60 (E) 72

Source: 2014 AMC

Solution: This problem is perfect to coordinate bash. Since we know that the two lines had y -intercepts that summed to 0, we can assume that one line intersects $(0, y)$ (P) while the other intersects $(0, -y)$ (Q).

Now we will calculate the slope of $\overline{AP}$ and $\overline{AQ}$ to get $\frac{8-y}{6}$ and $\frac{8+y}{6}$ respectively. Since we know that the product of the slopes of two perpendicular lines is -1 , we can multiply both of those slopes and equate it to -1 .

$$\frac{8-y}{6} \cdot \frac{8+y}{6} = -1$$

Solving the equation gives $y^2 = 100$ which means that $y = 10$.

Now to find the area of right triangle APQ , we need to find the lengths of the legs.

We can use the formula that tells us the distance between two points on a coordinate plane.

We get that $AQ = 6\sqrt{10}$ and $AP = 2\sqrt{10}$.

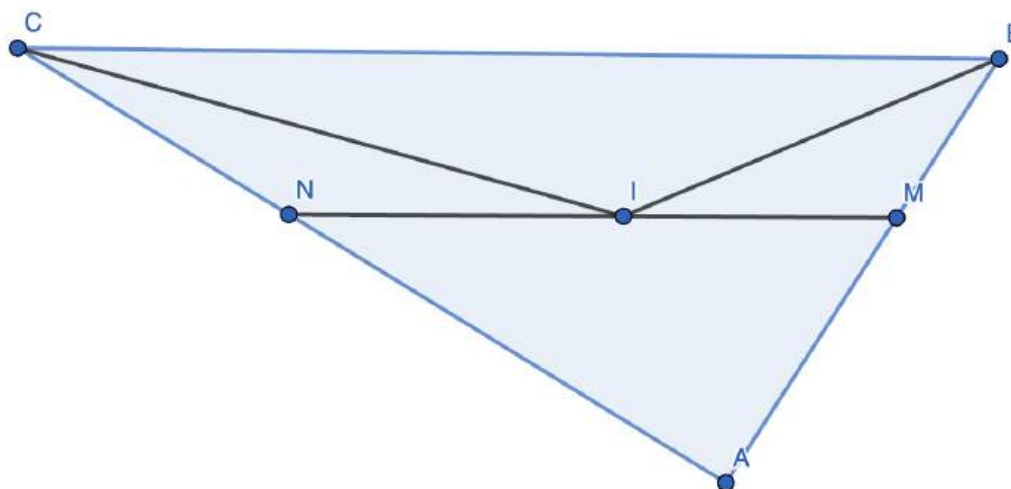
The area of the triangle is simply $\frac{6\sqrt{10} \cdot 2\sqrt{10}}{2}$ which is **60(D)**.

Problem 5.11.4 — Triangle ABC has side-lengths $AB = 12$, $BC = 24$, and $AC = 18$. The line through the incenter of $\triangle ABC$ parallel to $\overline{BC}$ intersects $\overline{AB}$ at M and $\overline{AC}$ at N . What is the perimeter of $\triangle AMN$?

(A) 27 (B) 30 (C) 33 (D) 36 (E) 42

Source: 2011 AMC

Solution: In this problem, we will first draw a diagram.



Since we know that $\overline{BI}$ and $\overline{CI}$ are angle bisectors, we can say that $\angle ABC$ has a measure of $2b^\circ$ while $\angle ACB$ has a measure of $2c^\circ$.

It's obvious that $\angle MBI$ has a measure of b° while $\angle NCI$ has a measure of c° .

We also know that $\overline{MN}$ is parallel to $\overline{BC}$. $\overline{AB}$ and $\overline{AC}$ are transversals that go through both of them. This means that $\angle AMN = \angle ABC = 2b$ and $\angle ANM = \angle ACB = 2c$.

This means that $\angle BMI = 180 - 2b$ and $\angle CNI = 180 - 2c$. From this, we now get that $\angle BIM = b^\circ$ and $\angle CIN = c^\circ$.

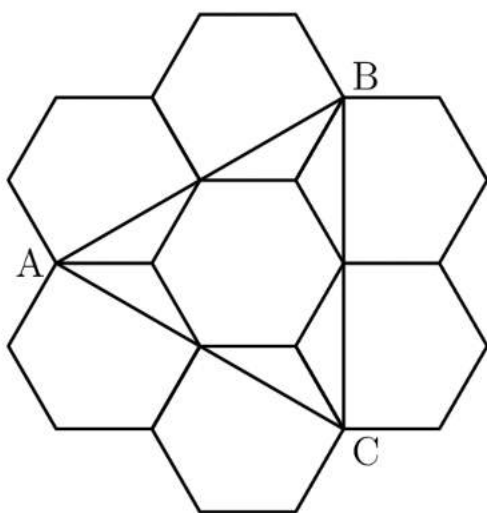
This means that $\triangle BIM$ and $\triangle CIN$ are isosceles triangles. This means that the length of $MI = MB$ and $NI = NC$.

We can assume that MI and MB have a length of x while NI and NC have a length of y .

Since CN has a length of y and BM has a length of x , we immediately know that AM has a length of $12 - x$ while AN has a length of $18 - y$.

Now to find the perimeter of $\triangle AMN$, we simply add it's side lengths that are in terms of x and y . The side lengths are $12 - x$, $18 - y$, and $x + y$. Adding up all 3 of those expressions cancels out the variables giving us an answer of **30 (B)**.

Problem 5.11.5 — Six regular hexagons surround a regular hexagon of side length 1 as shown. What is the area of $\triangle ABC$?



- (A) $2\sqrt{3}$ (B) $3\sqrt{3}$ (C) $1 + 3\sqrt{2}$ (D) $2 + 2\sqrt{3}$ (E) $3 + 2\sqrt{3}$

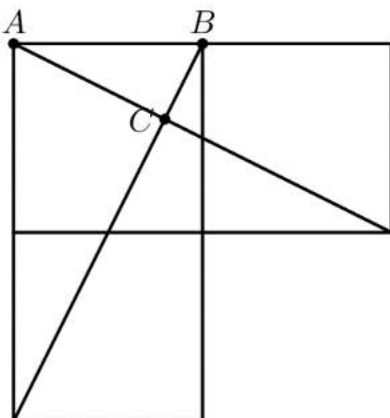
Source: 2014 AMC

Solution: The height of the triangle clearly includes one side length of the hexagon and one diagonal. It's obvious that the diagonal that it includes has a length of 2. (The length of the longest diagonal of a regular hexagon is 2 times its side length).

This means that the height of the equilateral triangle is 3. We can use our $30 - 60 - 90$ ratios to get that half the base of the equilateral triangle has a length of $\sqrt{3}$.

This means that the full base has a length of $2\sqrt{3}$. Thus, the area of our triangle is $\frac{3 \cdot 2\sqrt{3}}{2}$ which is $3\sqrt{3}$ **(B)**.

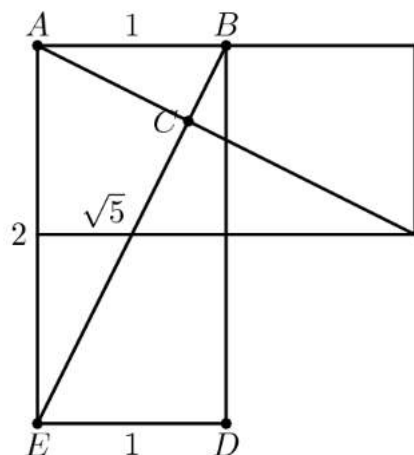
Problem 5.11.6 — Three unit squares and two line segments connecting two pairs of vertices are shown. What is the area of $\triangle ABC$?



- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{2}{9}$ (D) $\frac{1}{3}$ (E) $\frac{\sqrt{2}}{4}$

Source: 2012 AMC

Solution: In this problem, we will use similar triangles to solve it. To do so, we will first label a few sides.



From the diagram above, it's easy to notice that the slope of $\overline{BE}$ is 2 while it is $-\frac{1}{2}$ for $\overline{AC}$.

Since both of them multiply to -1 , it means that they are perpendicular to each other.

Now one can easily show that $\triangle ACB$ is similar to $\triangle BDE$.

Writing out a similarity equation gives $\frac{AB}{BC} = \frac{BE}{ED}$. Plugging in our side lengths gives $\frac{1}{BC} = \frac{\sqrt{5}}{1}$. This gives the length of BC as $\frac{\sqrt{5}}{5}$.

Now we can use the pythagorean theorem on $\triangle ABC$ to get that AC has a length of $\frac{2\sqrt{5}}{5}$.

Thus, the area of our triangle is simply $\frac{bh}{2}$ which is $\frac{1}{5}$ (B).

Problem 5.11.7 — A rectangular box has a total surface area of 94 square inches. The sum of the lengths of all its edges is 48 inches. What is the sum of the lengths in inches of all of its interior diagonals?

(A) $8\sqrt{3}$ (B) $10\sqrt{2}$ (C) $16\sqrt{3}$ (D) $20\sqrt{2}$ (E) $40\sqrt{2}$

Source: 2014 AMC

Solution: Let's assume that the width, length, and height of the rectangular box is x, y, z respectively.

This means that the total surface area in terms of those variables is $2(xy + xz + yz)$. The length of all the edges is $4(x + y + z)$.

Using the expressions we found, we can write out two equations.

$$2(xy + xz + yz) = 94$$

$$4(x + y + z) = 48$$

We get that $xy + xz + yz = 47$ and $x + y + z = 12$.

We also know that the rectangular box has 4 interior diagonals (this can be seen through visualizing the shape). All 4 of them have the same length.

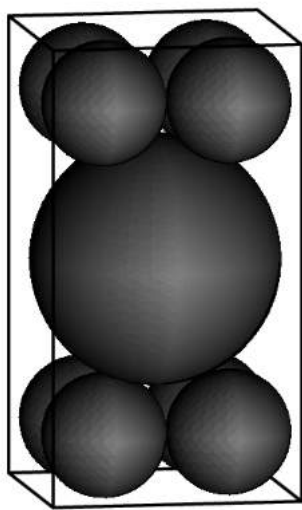
The length of one of them is $\sqrt{x^2 + y^2 + z^2}$

This means that the length of all 4 of them is $4\sqrt{x^2 + y^2 + z^2}$.

Since $x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + xz + yz)$, we can substitute our values to get that $x^2 + y^2 + z^2 = 12^2 - 2 \cdot 47$. This gives us that $x^2 + y^2 + z^2 = 50$.

We can plug that in to find that the sum of the lengths of all diagonals is $4\sqrt{50}$ which is $20\sqrt{2}$ (D).

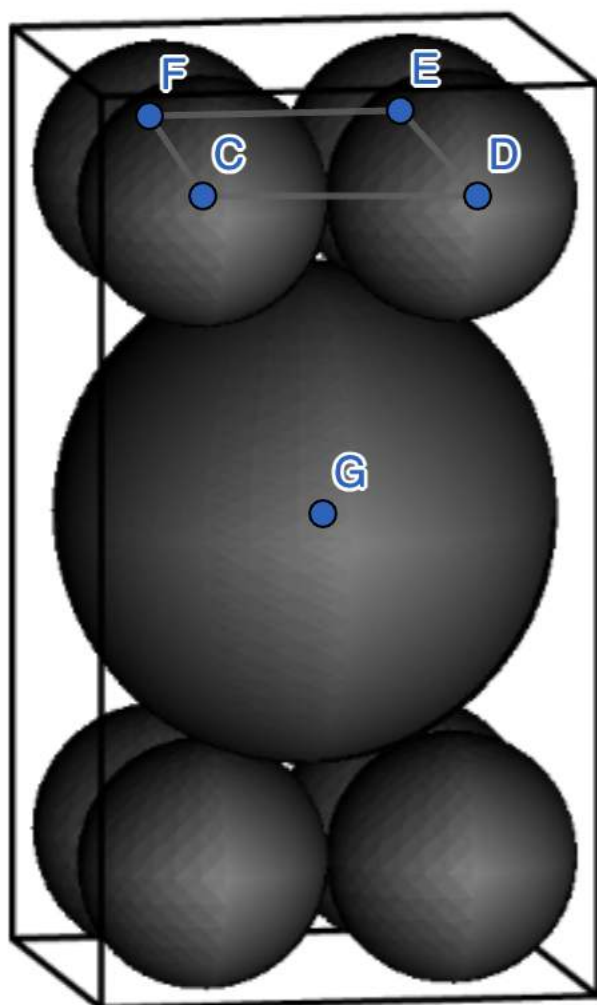
Problem 5.11.8 — A $4 \times 4 \times h$ rectangular box contains a sphere of radius 2 and eight smaller spheres of radius 1. The smaller spheres are each tangent to three sides of the box, and the larger sphere is tangent to each of the smaller spheres. What is h ?



(A) $2 + 2\sqrt{7}$ (B) $3 + 2\sqrt{5}$ (C) $4 + 2\sqrt{7}$ (D) $4\sqrt{5}$ (E) $4\sqrt{7}$

Source: 2014 AMC

Solution: We can first connect the centers of the smaller squares to get a square.



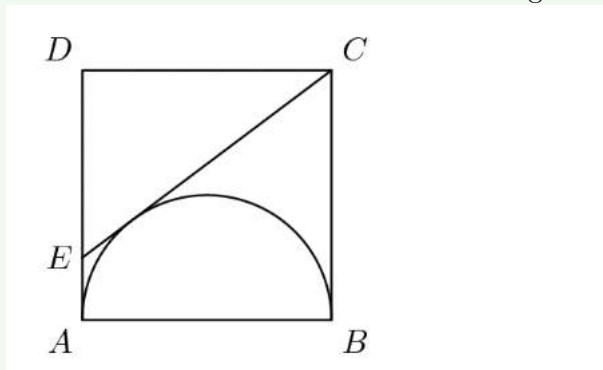
This square has a side length of 2. If we connect the center of the larger circle to all the vertices of the square, then we will get a pyramid with a square base. We now have to make one careful and key observation to solve this problem. The triangle connecting centers G, C, and the center of the square will make a right triangle.

The hypotenuse of this right triangle will be the distance between the two centers which is 3 (the sum of the two radii). One of the legs will simply be half the diagonal of the square which is $\sqrt{2}$.

Now using the pythagoreon theorem on that gives us that the height of this triangle is $\sqrt{7}$. Because of symmetry, this applies to the other pyramid from the 4 other spheres that weren't labelled in the diagram above. This means that the height just from the pyramids is $2\sqrt{7}$. However, we must add 2 to this because this doesn't account for the sum of the 2 radii we missed.

Thus, the answer is $2 + 2\sqrt{7}$ (A).

Problem 5.11.9 — Square $ABCD$ has side length 2. A semicircle with diameter $\overline{AB}$ is constructed inside the square, and the tangent to the semicircle from C intersects side $\overline{AD}$ at E . What is the length of $\overline{CE}$?

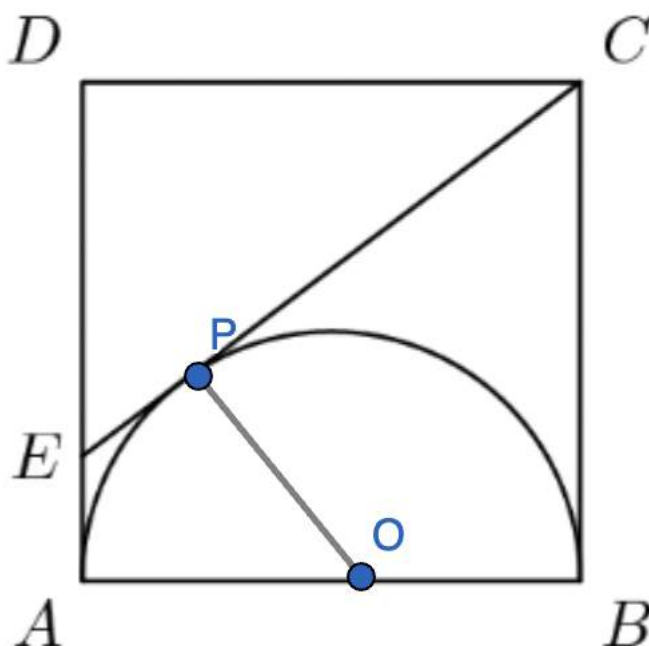


- (A) $\frac{2+\sqrt{5}}{2}$ (B) $\sqrt{5}$ (C) $\sqrt{6}$ (D) $\frac{5}{2}$ (E) $5 - \sqrt{5}$

Source: 2004 AMC

Solution: In this problem, we will use the idea that the line connecting the tangency point to the radius is perpendicular to the tangent line.

We will connect the center of the semicircle to its tangency point on $\overline{EC}$.



The Two Tangent Theorem tells us that the distance from a point (where two tangents meet) to the tangency points are equal. This means that $AE = AP$ and $PC = BC$.

Let's assume that the length of AE is a . This means that DE has a length of $2 - a$. Since BC is a side of the square, it has a length of 2. This means that PC also has a length of 2. Since $EC = EP + PC$, we instantly know that the length of EC is $a + 2$.

From pythagoreon theorem, we know that $DE^2 + DC^2 = EC^2$

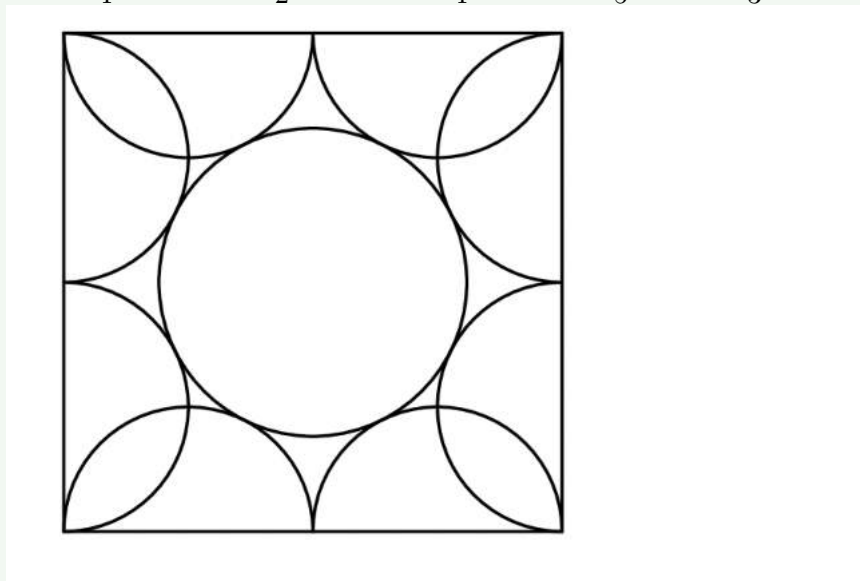
Plugging in our side lengths gives $(2 - a)^2 + 2^2 = (2 + a)^2$

Expanding the equation above gives $a^2 - 4a + 4 + 4 = a^2 + 4a + 4$ which simplifies to $8a = 4$.

Clearly, $a = \frac{1}{2}$. Since $CE = CP + PE$, the length of CE is simply $2 + \frac{1}{2}$ which is $\frac{5}{2}$ (**D**).

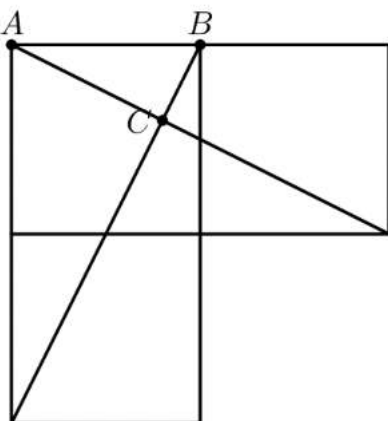
Problem 5.11.10 — Eight semicircles line the inside of a square with side length 2 as shown. What is the radius of the circle tangent to all of these semicircles?

- (A) $\frac{1 + \sqrt{2}}{4}$ (B) $\frac{\sqrt{5} - 1}{2}$ (C) $\frac{\sqrt{3} + 1}{4}$ (D) $\frac{2\sqrt{3}}{5}$ (E) $\frac{\sqrt{5}}{3}$



Source: 2014 AMC

Solution: In this problem, we will connect the centers of the circle to the center of the semicircles on one side of the square.



From the diagram, it's obvious that $\triangle LKM$ is isosceles. $\overline{NK}$ is the altitude of the triangle and $\overline{LM}$ is the base.

The length of LK is simply the sum of the 2 radii. The sum of the radius of the semicircle is simply $\frac{1}{2}$ since $4r = 2$ (since it lies on the side of the square).

We will assume that the radius of the large circle is r . This means the length of LK is $r + \frac{1}{2}$.

Also, NK has a length that is half the length of the square which is 1. We can easily see that LN has the same length as the radius of the semicircle which means that it is $\frac{1}{2}$.

We can use the pythagore theorem to get that $LN^2 + NK^2 = LK^2$
 Plugging in our side lengths gives $\frac{1}{2}^2 + 1^2 = (r + \frac{1}{2})^2$

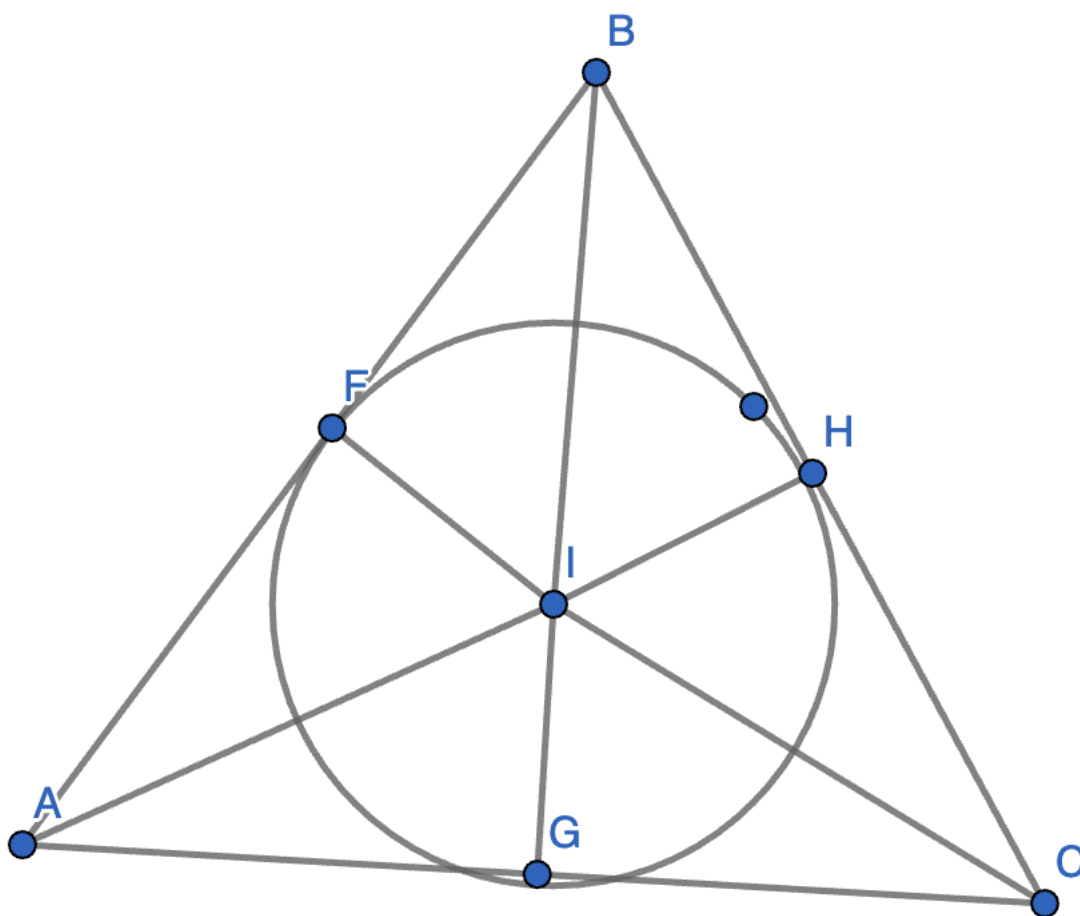
Expanding the equation and solving for r gives that the radius is $\frac{\sqrt{5}-1}{2}$ (B).

Problem 5.11.11 — Triangle ABC has $AB = 27$, $AC = 26$, and $BC = 25$. Let I be the intersection of the internal angle bisectors of $\triangle ABC$. What is BI ?

(A) 15 (B) $5 + \sqrt{26} + 3\sqrt{3}$ (C) $3\sqrt{26}$ (D) $\frac{2}{3}\sqrt{546}$ (E) $9\sqrt{3}$

Source: 2012 AMC

Solution: We first draw a diagram like always.



We know that that the area of a triangle is $r \cdot s$ where r represents the inradius and s represents the semiperimeter.

We know that the semiperimeter for this triangle is simply $\frac{25+26+27}{2}$ which is 39. We plug this in for s to get that the area is $39r$.

Now we can also find the area of this triangle again using Heron's Formula. Heron's formula is $\sqrt{s(s-a)(s-b)(s-c)}$ where s represents the semiperimeter. Plugging in these values gives that the area is $78\sqrt{14}$.

We can now set up the equation $39r = 78\sqrt{14}$. This gives us that $r = 2\sqrt{14}$ which is our inradius.

From the diagram, it's obvious that $BI^2 = HI^2 + HB^2$ from pythag theorem. Since we know that HI has a length of $2\sqrt{14}$ since it's simply the inradius, we just need to find the length of HB .

From the Two Tangent theorem applied to the triangle, we know that $AF = AG = x$, $BF = BH = y$, and $CG = CH = z$.

This gives us that $x + y = 27$, $x + z = 26$, and $y + z = 25$.

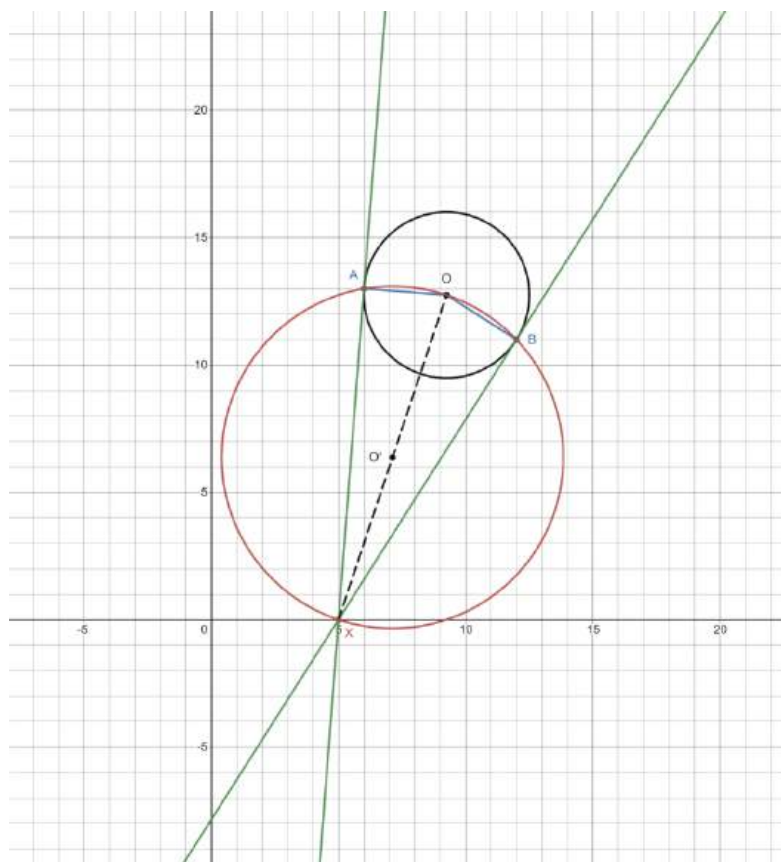
Solving gives that $y = BF = BH = 13$. We can plug this into our pythagorean theorem equation along with the length of the inradius to get $BI^2 = 169 + 56$. This gives us that BI has a length of **15**.

Problem 5.11.12 — Points $A = (6, 13)$ and $B = (12, 11)$ lie on circle ω in the plane. Suppose that the tangent lines to ω at A and B intersect at a point on the x -axis. What is the area of ω ?

(A) $\frac{83\pi}{8}$ (B) $\frac{21\pi}{2}$ (C) $\frac{85\pi}{8}$ (D) $\frac{43\pi}{4}$ (E) $\frac{87\pi}{8}$

Source: 2019 AMC 10

Solution: Let's assume that the intersection point of the two tangent lines that have tangency points at A and B intersect the x -axis at point $(x, 0)$.



From the two tangent theorem, we know that the distance between this intersection point

and A along with this same intersection point and B must be equal. Thus, we use the distance formula and equate the two distances to get:

$$\sqrt{(x-6)^2 + 13^2} = \sqrt{(x-12)^2 + 11^2}$$

Solving for x gives that $x = 5$.

Now let's assume that the distance between the circles center and the intersection point is y . Since we know that the radius (the one connecting the center to the tangency point) is always perpendicular to the tangent line, we can use the Pythagorean theorem. In terms of the diagram above, we can write the equation $AO^2 + AX^2 = OX^2$.

Substituting our known variable gives $r^2 + \sqrt{170}^2 = d^2$.

Now we use the Ptolemy's theorem on quadrilateral AXBO. We can do this because the opposite angles of it sum to 180° . The reason for that is the fact that $\angle OAX = \angle OBX = 90^\circ$.

Applying the Ptolemy's theorem gives $AO \cdot BX + BO \cdot AX = AB \cdot OX$

Plugging in our side lengths gives $2r\sqrt{170} = 2d\sqrt{10}$.

Simplifying the equation above gives $r\sqrt{17} = d$

We plug this expression for d into $r^2 + \sqrt{170}^2 = d^2$. This gives us that $r^2 + 170 = 17r^2$.

This expression gives us that $r^2 = \frac{85}{8}$.

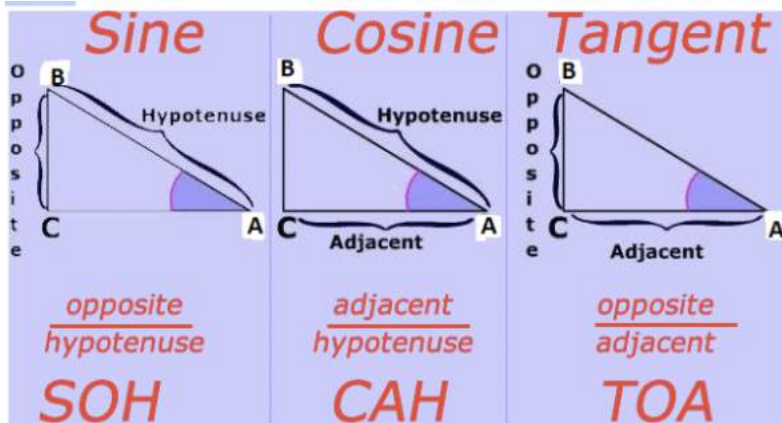
Since the area of a circle is $\pi \cdot r^2$, we plug our value in for r^2 to get that the area is $\frac{85\pi}{8}$ (C).

6 Trigonometry and Complex Numbers

§6.1 Trigonometry Basics

Definition 6.1.1

Sin, Cos, and Tan are just ratios in triangles. The picture below will explain it to you.



SOH CAH TOA is an easy way to remember what sine, cosine, and tangents are. I think you can tell that these ratios work for RIGHT triangles because it includes the hypotenuse. You are completely right if you think that, but sometimes you have to make right triangles. This can be done by drawing a perpendicular line to one of the bases. After that, you could use trigonometry.

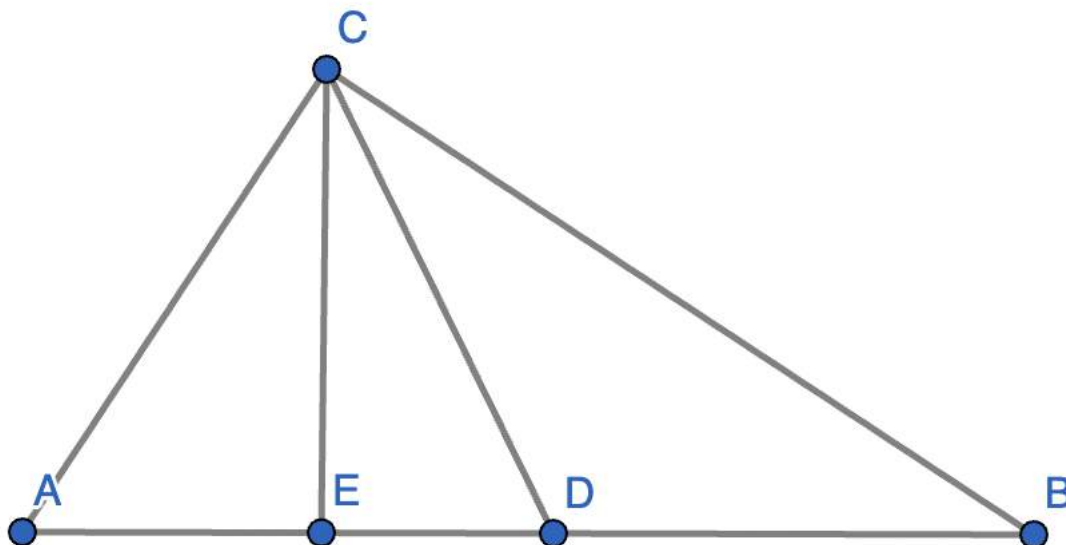
I recommend memorizing the sin and cosines of 15, 30, 45, 60, 75, 90, and 120. Also memorize the tan of 15, 30, 45, 60, 75, and 90 degrees. This would be really helpful for people taking the AMC 12. For people taking the AMC 10, this is an optional step. However, some of the geometry problems can be solved using trigonometry. If you know trigonometry and are taking the AMC 10, then it's very helpful to memorize it because it can be your backup plan for a problem that you don't know the solution for.

Problem 6.1.2 — A triangle has area 30, one side of length 10, and the median to that side of length 9. Let θ be the acute angle formed by that side and the median. What is $\sin \theta$?

- (A) $\frac{3}{10}$ (B) $\frac{1}{3}$ (C) $\frac{9}{20}$ (D) $\frac{2}{3}$ (E) $\frac{9}{10}$

Source: 2012 AMC

Solution: As always, we will first make a diagram.



$\overline{AB}$ has a length of 10, and $\overline{CD}$ is the median. $\overline{CE}$ is the altitude of the triangle. Since we know that the area of a triangle is $\frac{bh}{2}$, we can easily see that the height of this triangle is 6 since the base is 10. This means that the length of CE is 6.

Since we want to find the sin of the acute angle formed by the side and the median, it's obvious that we want to find $\sin \angle CDE$.

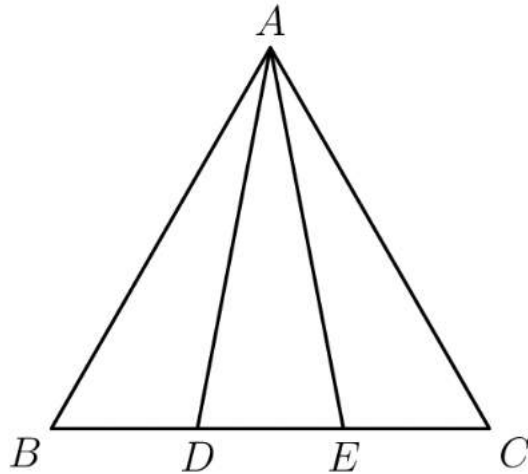
$$\sin(\angle CDE) = \frac{CE}{CD}.$$

We know that CE has a length of 6 and CD has a length of 9. Thus, $\sin(\angle CDE) = \frac{2}{3}$ (D).

Problem 6.1.3 — In equilateral $\triangle ABC$ let points D and E trisect $\overline{BC}$. Then $\sin(\angle DAE)$ can be expressed in the form $\frac{a\sqrt{b}}{c}$, where a and c are relatively prime positive integers, and b is an integer that is not divisible by the square of any prime. Find $a + b + c$.

Source: 2013 AIME

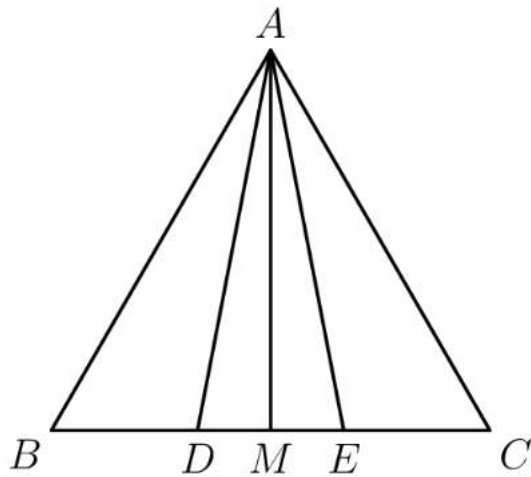
Solution: In all geometry problems with trigonometry applications, we should draw the diagram out.



In this problem, since $\overline{AB} = \overline{BC} = \overline{AC} = s$ where s is a variable representing the side length of the equilateral triangle

Since points D and E trisect BC , it means that $\overline{BD} = \overline{DE} = \overline{EC} = \frac{s}{3}$.

Since drawing an altitude in a triangle is a helpful strategy especially when we have an equilateral triangle, let's assume that we draw an altitude from A to $\overline{BC}$ and it intersects it at point M .



Since M is the midpoint of $\overline{BC}$, then $\overline{BM} = \overline{MC} = \frac{s}{2}$.

$$\overline{DM} = \overline{BM} - \overline{BD} = \frac{s}{2} - \frac{s}{3} = \frac{s}{6}$$

Now since we want to find the $\sin(\angle DAE)$, we can do so by first finding it for half the angle which is $\sin(\angle DAM)$. After that, we can apply the double sin identity. Using that, we know that

$$\sin(\angle DAE) = 2 \cdot \sin(\angle DAM) \cdot \cos(\angle DAM)$$

Now we can simply find $\sin(\angle DAM)$ and $\cos(\angle DAM)$.

$$\begin{aligned} \sin(\angle DAM) &= \frac{o}{h} = \frac{\overline{DM}}{\overline{AD}} \\ \cos(\angle DAM) &= \frac{a}{h} = \frac{\overline{AM}}{\overline{AD}} \end{aligned}$$

We can find $\overline{AM}$ easily since it's the height of the equilateral triangle which in terms of the side length s is $\frac{s\sqrt{3}}{2}$

Using this, we can find $\overline{AD}$ using the Pythagorean theorem since $\overline{AD}^2 = \overline{AM}^2 + \overline{DM}^2 = (\frac{s\sqrt{3}}{2})^2 + (\frac{s}{6})^2 = \frac{7s^2}{9}$

Now that gives us that $\overline{AD} = \frac{s\sqrt{7}}{3}$

Now since we know $\overline{AD}$, $\overline{DM}$, and $\overline{AM}$, we can use those values to compute $\sin(\angle DAM)$ and $\cos(\angle DAM)$

$$\sin(\angle DAM) = \frac{\frac{s}{6}}{\frac{s\sqrt{7}}{3}}$$

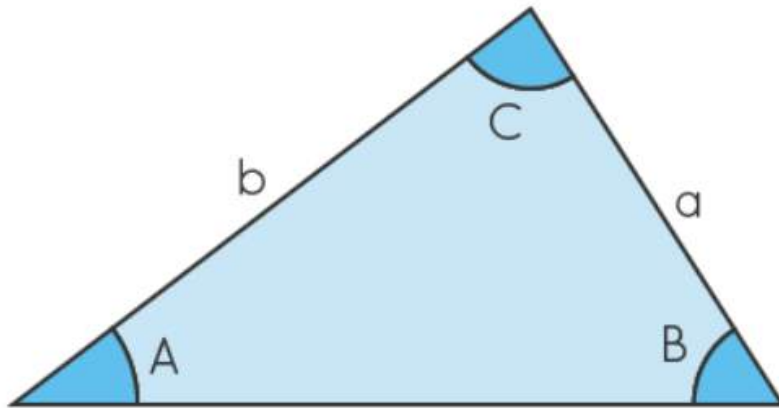
$$\cos(\angle DAM) = \frac{\frac{s\sqrt{3}}{2}}{\frac{s\sqrt{7}}{3}}$$

We multiply those two values up and then multiply it to 2 to find $\sin(\angle DAE)$ which is $\frac{3\sqrt{3}}{14}$. We add up 3 and 3 and 14 to get **20** as the answer.

§6.2 Law of Cosines and Law of Sines

Theorem 6.2.1

Law of Cosines



For any triangle with side lengths a , b , c and angles of measure A , B , and C , the following theorems apply

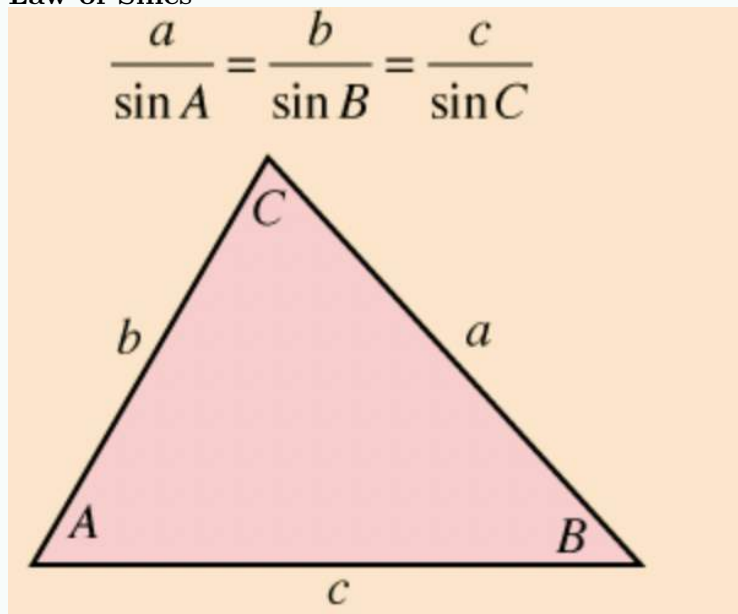
$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Theorem 6.2.2

Law of Sines

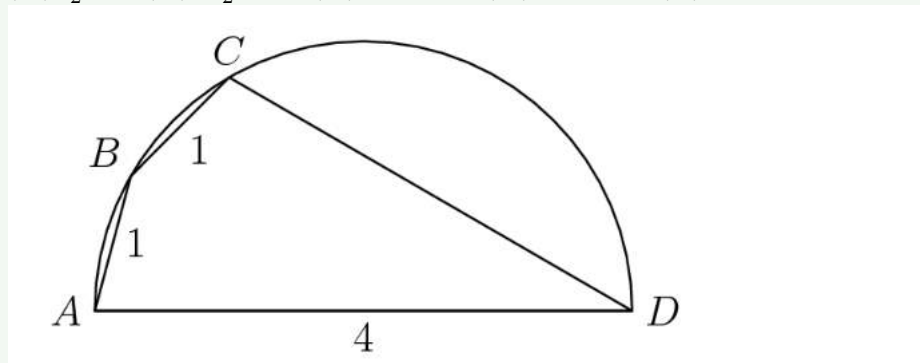


For a triangle with side lengths a, b, c and angles opposite to it A, B, C and circum-radius R , our formula is

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Problem 6.2.3 — Quadrilateral $ABCD$ is inscribed in a circle with side AD , a diameter of length 4. If sides AB and BC each have length 1, then side CD has length

- (A) $\frac{7}{2}$ (B) $\frac{5\sqrt{2}}{2}$ (C) $\sqrt{11}$ (D) $\sqrt{13}$ (E) $2\sqrt{3}$



Source: 1971 AHSME

Solution: In this problem, we can draw the line from B to O (the midpoint of AD). Then, that means $\angle AOB = \angle BOC = a$

From here, we can apply the law of cosines to $\triangle AOB$. Since we know that $c^2 = a^2 + b^2 - 2ab \cos C$ is the formula for LOC, we can apply it to our chosen triangle. Let's assume that AB is c and AO and BO are a and b . We know that both AO and BO are equivalent to the radius which is 2.

Applying LOC to it gives $\overline{AB}^2 = \overline{AO}^2 + \overline{BO}^2 - 2 \cdot \overline{AO} \cdot \overline{BO} \cdot \cos a$. Since $\overline{AB}$ is 1 and $\overline{AO}$ and $\overline{BO}$ are both 2, we plug this in to get $1 = 8 - 8 \cdot \cos a$. This gives us that $\cos a = \frac{7}{8}$.

We know that $\angle COD$ equals to $180 - 2a$. Since we can use the Law of Cosines on $\triangle COD$, we need to find the value of $\cos(180 - 2a)$. We can do this using some trigonometry identities.

Since $\cos(180 - x) = -\cos x$, we know that $\cos(180 - 2a) = -\cos(2a)$. From here, we can use our double cosine identity which states that $\cos(2x) = 2\cos^2 x - 1$. We can combine this with $-\cos(2a)$ to get that our value of $\cos 180 - 2a$ is simply equivalent to $-2\cos^2 a + 1$.

Since we know our value of $\cos a$ to be $\frac{7}{8}$, we can plug that in to get that $\cos(180 - 2a) = -2 \cdot \frac{7}{8}^2 + 1 = \frac{-17}{32}$. Now, we can use this value to apply the Law of Cosines to $\triangle COD$ to get

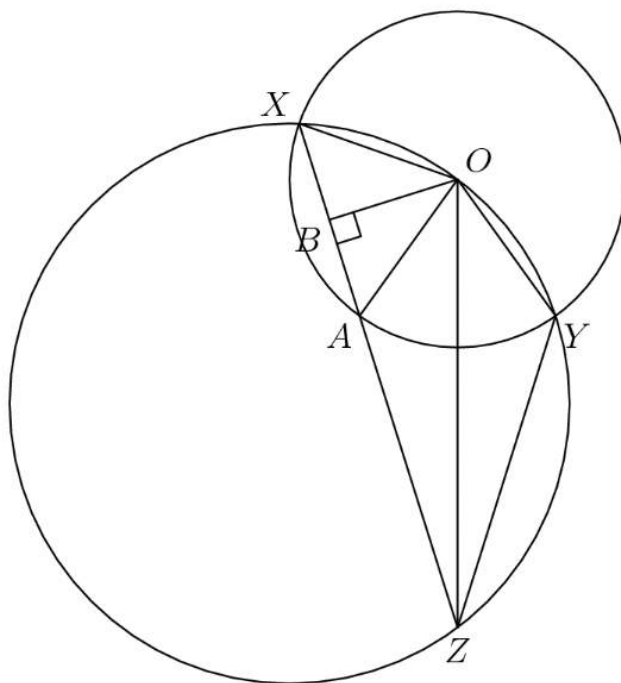
$\overline{CD}^2 = \overline{CO}^2 + \overline{DO}^2 - 2 \cdot \overline{CO} \cdot \overline{DO} \cdot \cos(180 - 2a)$. Plugging in our known values simplifies the expression to

$$\overline{CD}^2 = 2^2 + 2^2 - 2 \cdot 2 \cdot 2 \cdot \frac{-17}{32} = \frac{49}{4}.$$

This simplifies to $\overline{CD} = \frac{7}{2}$ (A).

Problem 6.2.4 — Circle C_1 has its center O lying on circle C_2 . The two circles meet at X and Y . Point Z in the exterior of C_1 lies on circle C_2 and $XZ = 13$, $OZ = 11$, and $YZ = 7$. What is the radius of circle C_1 ?

(A) 5 (B) $\sqrt{26}$ (C) $3\sqrt{3}$ (D) $2\sqrt{7}$ (E) $\sqrt{30}$



Source: 2012 AMC

Solution: In this problem, we will first apply the Law of Sines to find the relationship

between two angles.

Since we know that XO and OY have the same length, we will try to apply the extended law of sines on $\triangle XZO$ and $\triangle YZO$. Since they both have the same circumradius, we will assume the length of the circumradius is R .

For $\triangle XZO$, we get $\frac{XO}{\sin(\angle XZO)} = 2R$

For $\triangle YZO$, we get $\frac{YO}{\sin(\angle YZO)} = 2R$

Since $XO = YO$, we can equate the two equations to get that $\angle XZO = \angle YZO$.

Now we can apply the Law of Cosines to $\triangle XZO$ and $\triangle YZO$.

Applying it to triangle XZO gives $XO^2 = XZ^2 + OZ^2 - 2 \cdot XZ \cdot OZ \cdot \cos(\angle XZO)$

Applying it to triangle YZO gives $YO^2 = YZ^2 + OZ^2 - 2 \cdot YZ \cdot OZ \cdot \cos(\angle YZO)$

We plug in our side lengths to get $r^2 = 290 - 286 \cos(\angle XZO)$ and $r^2 = 170 - 154 \cos(\angle YZO)$

We can equate both equations to get $290 - 286 \cos(\angle XZO) = 170 - 154 \cos(\angle YZO)$.

Since $\angle XZO = \angle YZO$, we can simplify the equation to $290 - 286 \cos(\angle XZO) = 170 - 154 \cos(\angle XZO)$

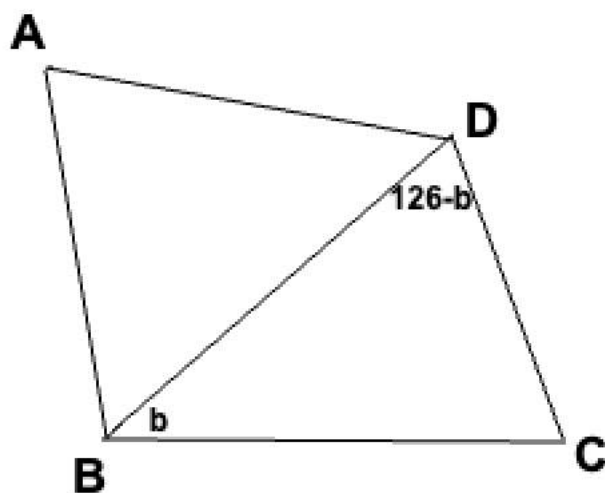
Solving the equation gives that $\cos(\angle XZO) = \frac{120}{132} = \frac{10}{11}$. We can plug this value into the equation $r^2 = 290 - 286 \cos(\angle XZO)$.

Plugging it in gives that $r^2 = 30$ which means the value of our radius is $\sqrt{30}$ (E).

Problem 6.2.5 — In quadrilateral $ABCD$, $AB = DB$ and $AD = BC$. If $\angle ABD = 36$ and $\angle BCD = 54$, find $\angle ADC$ in degrees.

Source: 2016 CHMMC

Solution: Always draw a figure for such problems. That should be your first step.



We can apply the Law of Sines to this problem. Usually in a problem in which we have multiple triangles in which some side lengths are equal, the Law of Sines can often be a helpful formula to use.

First, since we know that $\triangle ABD$ is isosceles, this gives us that $\angle BAD = \angle BDA = 72$ since both angles must be equal because we have an isosceles triangle.

The Law of Sines on $\triangle ABD$ gives

$$\frac{\overline{AD}}{\sin 36} = \frac{\overline{BD}}{\sin 72}$$

Now we can apply the Law of Sines again to $\triangle BDC$ to get

$$\frac{\overline{BD}}{\sin 54} = \frac{\overline{BC}}{\sin a}$$

Since $\overline{BC} = \overline{AD}$, we can plug in that into $\frac{\overline{BD}}{\sin 54} = \frac{\overline{BC}}{\sin a}$ to get $\frac{\overline{BD}}{\sin 54} = \frac{\overline{AD}}{\sin a}$

The equation $\frac{\overline{AD}}{\sin 36} = \frac{\overline{BD}}{\sin 72}$ can be written as $\frac{\overline{AD}}{\overline{BD}} = \frac{\sin 36}{\sin 72}$.

We can also rewrite $\frac{\overline{BD}}{\sin 54} = \frac{\overline{AD}}{\sin a}$ in the same way to get

$$\frac{\overline{AD}}{\overline{BD}} = \frac{\sin a}{\sin 54}$$

since we found two expressions for the value of $\frac{\overline{AD}}{\overline{BD}}$, we can equate both of them to get

$$\frac{\sin 36}{\sin 72} = \frac{\sin a}{\sin 54}$$

Using the double sin identity which states that $\sin 2a = 2 \cdot \sin a \cdot \cos a$ gives us that $\sin 72 = 2 \cdot \sin 36 \cdot \cos 36$. We can plug this into our expression to get

$$\frac{\sin 36}{2 \cdot \sin 36 \cdot \cos 36} = \frac{\sin a}{\sin 54}$$

We can cancel out $\sin 36$ to get $\frac{1}{2 \cdot \cos 36} = \frac{\sin a}{\sin 54}$.

We know that for any angle x , $\sin x = \cos(90 - x)$. This gives us that $\sin 54 = \cos 36$. We can apply this to our equation to cancel like terms out. Our equation becomes

$\sin a = \frac{1}{2}$. From here, we know that $\angle a$ must be 30 degrees. Since we want to find the value of $\angle ADC$, we can simply compute $\angle ADB + \angle BDC$. This gives us $72 + 30$ which is **102** degrees.

§6.3 Trigonometry Identities

There are many trigonometry identities. Take your time to practice with them since it will help you slowly memorize all. They are crucial to be able to solve many of the harder trigonometry problems that show up on the AMC and AIME. Many will rely on lots of manipulation which is where the identities come in.

Theorem 6.3.1

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

These are our Pythagorean identities. These can be helpful in many scenarios. For example, if you're given the value of $\sin \theta$ but not $\cos \theta$, then you can use the identity $\sin^2 \theta + \cos^2 \theta = 1$ and rewrite it to get that $\cos \theta = \sqrt{1 - \sin^2 \theta}$

Theorem 6.3.2

Trigonometric Functions	
Even Functions $f(-x) = f(x)$	Odd Functions $f(-x) = -f(x)$
$\cos(-x) = \cos x$ $\sec(-x) = \sec x$	$\sin(-x) = -\sin x$ $\csc(-x) = -\csc x$ $\tan(-x) = -\tan x$ $\cot(-x) = -\cot x$

These identities are extremely important when working with something like $\sin(-\theta)$ or $\cos(-\theta)$. You can use it to find the trigonometric values of negative angles.

Theorem 6.3.3

$$\sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$\sin(a-b) = \sin a \cos b - \cos a \sin b$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\cos(a-b) = \cos a \cos b + \sin a \sin b$$

$$\tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

$$\tan(a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$$

These are our sum and difference identities. It's really helpful in finding the trigonometric values for angles like 75° since you can split it up to get 30° and 45° .

For example, if you want to find $\sin(15^\circ)$, you can simply rewrite that to get $\sin(45^\circ - 30^\circ)$. Then, you can use the difference identity to find its value.

Theorem 6.3.4

$\cos x = \sin\left(\frac{\pi}{2} - x\right)$	$\sin x = \cos\left(\frac{\pi}{2} - x\right)$
$\tan x = \cot\left(\frac{\pi}{2} - x\right)$	$\cot x = \tan\left(\frac{\pi}{2} - x\right)$
$\sec x = \csc\left(\frac{\pi}{2} - x\right)$	$\csc x = \sec\left(\frac{\pi}{2} - x\right)$

These are our co-function identities. These can be helpful in situations such as when you're given the value of $\cos(\theta)$ but want to find the value of $\sin(90^\circ - \theta)$. Then, by using these identities it's easy to find out that both of the trigonometric values are equal.

Theorem 6.3.5

$$\sin(2A) = 2 \sin(A) \cos(A)$$

$$\cos(2A) = \cos^2(A) - \sin^2(A)$$

$$= 1 - 2\sin^2(A)$$

$$= 2\cos^2(A) - 1$$

$$\tan(2A) = \frac{2 \tan(A)}{1 - \tan^2(A)}$$

These are the double angle identities. They show up a lot, and it's important to know these to rapidly simplify trigonometric expressions into something simple.

Theorem 6.3.6

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

These are our half-angle identities. This can be helpful when you want to find the trigonometric value of half of a known angle in some expression.

Problem 6.3.7 — If $\sum_{n=0}^{\infty} \cos^{2n} \theta = 5$, what is the value of $\cos 2\theta$?

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{\sqrt{5}}{5}$ (D) $\frac{3}{5}$ (E) $\frac{4}{5}$

Source: 2004 AMC 12

Solution: In this problem, we can apply the sum of the infinite geometric series formula to $\sum_{n=0}^{\infty} \cos^{2n} \theta = 5$. Since we know that the first term is $\cos^0 \theta$ and the ratio is $\cos^2 \theta$, the sum is $\frac{1}{1 - \cos^2 \theta}$.

Since we know that it equates to 5, we get $1 = 5 - 5 \cos^2 \theta$. This gives us that $\cos^2 \theta = \frac{4}{5}$.

Since we want to find $\cos 2\theta$, we can simplify this by using our trig identities. We know that $\cos 2\theta = 2 \cos^2 \theta - 1$. Since we know our value of $\cos^2 \theta$, we can plug it in to get that $\cos 2\theta$ is $\frac{3}{5}$ (D).

Problem 6.3.8 — Let x and y be real numbers such that $\frac{\sin x}{\sin y} = 3$ and $\frac{\cos x}{\cos y} = \frac{1}{2}$. The value of $\frac{\sin 2x}{\sin 2y} + \frac{\cos 2x}{\cos 2y}$ can be expressed in the form $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.

Source: 2012 AIME

Solution: In this problem, we can find the terms $\frac{\sin 2x}{\sin 2y}$ and $\frac{\cos 2x}{\cos 2y}$ separately, and then we can add them.

We can simplify $\frac{\sin 2x}{\sin 2y}$ using our trigonometry identities to get

$$\frac{\sin 2x}{\sin 2y} = \frac{2 \sin x \cos x}{2 \sin y \cos y} = \frac{\sin x \cos x}{\sin y \cos y} = \frac{\sin x}{\sin y} \cdot \frac{\cos x}{\cos y}$$

Now in the expression above, we can plug in our given values to get $3 \cdot \frac{1}{2}$ which is $\frac{3}{2}$.

Now we need to simplify and use our trigonometric identities to find $\frac{\cos 2x}{\cos 2y}$.

Now we can use our identities to get that $\frac{\cos 2x}{\cos 2y} = \frac{2 \cos^2 x - 1}{2 \cos^2 y - 1}$

Now to find these terms, we can rewrite the 2 given equations as

$$\sin x = 3 \sin y$$

$$2 \cos x = \cos y$$

Now we can square both sides of both equations to get

$$\sin^2 x = 9 \sin^2 y$$

$$\cos^2 x = \frac{\cos^2 y}{4}$$

We can add both equations to get $\sin^2 x + \cos^2 x = 9 \sin^2 y + \frac{\cos^2 y}{4}$. Since we know that $\sin^2 \theta + \cos^2 \theta = 1$, we can use this identity to simplify our equation to

$$1 = 9 \sin^2 y + \frac{\cos^2 y}{4}$$

Now we can multiply all sides by 4 in the above equation to get

$$4 = 36 \sin^2 y + \cos^2 y$$

Subtracting $1 = \sin^2 y + \cos^2 y$ from the equation gives

$$3 = 35 \sin^2 y$$

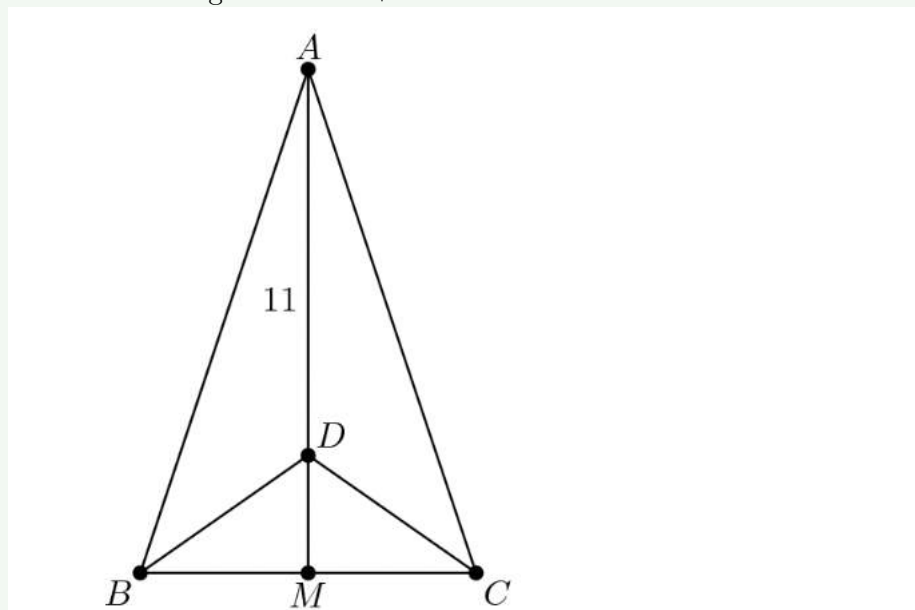
This gives us that $\sin^2 y = \frac{3}{35}$. Since we want to find the value of $\cos^2 y$ and $\cos^2 x$, we can use the equation $\sin^2 x + \cos^2 x = 1$ to get that $\cos^2 y$ is $\frac{32}{35}$.

Using the equation we found early on ($\cos^2 x = \frac{\cos^2 y}{4}$), we know that $\cos^2 x$ is $\frac{8}{35}$.

We can plug these values into our expression $\frac{2 \cos^2 x - 1}{2 \cos^2 y - 1}$. It gives us a value of $\frac{-19}{29}$.

Now since this number is our value for $\frac{\cos 2x}{\cos 2y}$, we add this to our value for $\frac{\sin 2x}{\sin 2y}$ which is $\frac{3}{2}$. We get $\frac{-19}{29} + \frac{3}{2}$ which is $\frac{49}{58}$. We sum up our numerator and denominator to get **107**.

Problem 6.3.9 — Triangle ABC is isosceles triangle, with $AB = AC$ and altitude $AM = 11$. Suppose that there is a point D on $\overline{AM}$ with $AD = 10$ and $\angle BDC = 3\angle BAC$. Then the perimeter of $\triangle ABC$ may be written in the form $a + \sqrt{b}$, where a and b are integers. Find $a + b$.



Source: 1995 AIME

Source: In this problem, we will first label $\angle BAM$ as a . This gives us that $\angle BDM$ is $3a$.

From $\triangle ABM$, we get that $\tan a = \frac{\overline{BM}}{\overline{AM}} = \frac{\overline{BM}}{11}$.

Now from $\triangle BDM$, we get that $\tan 3a = \frac{\overline{BM}}{\overline{DM}} = \frac{\overline{BM}}{1}$.

Combining the two equations gives us that $\frac{\tan 3a}{\tan a} = 11$

Now we will apply the tan addition formula which states $\tan a + b = \frac{\tan a + \tan b}{1 - \tan a \cdot \tan b}$. We apply this to our angle a to first find $\tan 2a$, and then we'll use angles a and $2a$ to find $\tan 3a$.

This identity gives that $\tan a + a = \tan 2a = \frac{2 \tan a}{1 - \tan^2 a}$

Now we use this tan addition formula again to get

$\tan 2a + a = \tan 3a = \frac{\tan a + \tan 2a}{1 - \tan a \cdot \tan 2a}$. We plug in our value of $\tan 2a$ which is $\frac{2 \tan a}{1 - \tan^2 a}$ to get that

$\tan 3a = \frac{3 \tan a - \tan^3 a}{1 - 3 \tan^2 a}$. Since we know that $\frac{\tan 3a}{\tan a} = 11$, we can plug in our value of $\tan 3a$ into this and divide it by $\tan a$ to get that $\frac{3 - \tan^2 a}{1 - 3 \tan^2 a} = 11$

Now multiplying both sides of this equation by $1 - 3 \tan^2 a$ to get that $\tan^2 a = \frac{1}{4}$. We can take the square root of both sides to get that $\tan a = \frac{1}{2}$.

We can again use $\triangle ABM$ to get that $\tan a = \frac{\overline{BM}}{11}$.

We can plug in the value we found for $\tan a$ to get that $\overline{BM} = \frac{11}{2}$.

Now the perimeter of this triangle (the value we want to find) is simply $AB + AC + BC$. Since we found $\overline{BM}$ to be $\frac{11}{2}$, we multiply this by 2 to find $\overline{BC}$ to be 11.

Now to find AB and AC (which are the same value since this is an isosceles triangle), we can apply the pythagorean theorem to $\triangle ABM$ to get that $\overline{AB}^2 = \overline{BM}^2 + \overline{AM}^2 = (\frac{11}{2})^2 + 11^2$

This gives that $\overline{AB} = \frac{11\sqrt{5}}{2}$. Now this gives that the total perimeter is $11 + 2 \cdot \frac{11\sqrt{5}}{2}$. Since we want this answer to be in the form $a + \sqrt{b}$, we rewrite the answer as $11 + \sqrt{605}$. We sum up 605 and 11 to get our answer of **616**.

Problem 6.3.10 — Given that $\log_{10} \sin x + \log_{10} \cos x = -1$ and that $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, find n .

Source: 2003 AIME

Solution: In this problem, we will first simplify the first equation by using our logarithm properties. Since we know that $\log_{10} a + \log_{10} b = \log_{10} ab$, we can apply that to the first equation to get

$$\log_{10}(\sin x \cdot \cos x) = -1$$

We can convert this to exponential form to get $\sin x \cdot \cos x = \frac{1}{10}$.

Now we can equate $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$ to a . Then this gives us two separate equations that

$$\begin{aligned}\log_{10}(\sin x + \cos x) &= a \\ \frac{1}{2}(\log_{10} n - 1) &= a\end{aligned}$$

Now we can convert the two equations above to exponential form to get

$$10^a = \sin x + \cos x$$

For the second equation which is $\frac{1}{2}(\log_{10} n - 1) = a$, we can first multiply both sides by 2 to cancel out the fraction which gives us $\log_{10} n - 1 = 2a$

We add 1 to both sides to get $\log_{10} n = 2a + 1$. In exponential form, we get $10^{2a+1} = n$.

We can rewrite $10^a = \sin x + \cos x$ so that on the lefthand side we get 10^{2a+1} . We can do that by first squaring the equation, and then multiplying both sides by 10.

$$10^{2a+1} = 10 \cdot (\sin x + \cos x)^2$$

Since we already know that our value of n equals to 10^{2a+1} , we can simplify our equation to

$$n = 10 \cdot (\sin x + \cos x)^2$$

This equation becomes $n = 10(\sin^2 x + \cos^2 x + 2 \sin x \cos x)$

We can use the identity $\sin^2 x + \cos^2 x = 1$ to get that the equation is $n = 10(1 + 2 \sin x \cos x)$

Now in the beginning of our solution, we already found that $\sin x \cdot \cos x = \frac{1}{10}$, we can plug that into our answer for n to get $10(1 + 2 \cdot \frac{1}{10}) = \mathbf{12}$.

§6.4 Complex Numbers

Definition 6.4.1

Complex numbers are in the form $a + bi$. a represents a real number while bi represents an imaginary number. i represents $\sqrt{-1}$.

Theorem 6.4.2

Note the powers for i , and notice the pattern and how the values repeat.

$$\begin{array}{llll} i^1 = i & i^2 = -1 & i^3 = -i & i^4 = 1 \\ i^5 = i & i^6 = -1 & i^7 = -i & i^8 = 1 \\ i^9 = i & i^{10} = -1 & i^{11} = -i & i^{12} = 1 \end{array}$$

Theorem 6.4.3

If $z = a + bi$ and $w = c + di$, then the following theorems apply

$\bar{z} = a - bi$ ($\bar{z}$ represents the conjugate)

$$|z| = \sqrt{a^2 + b^2}$$

$$\bar{z} \cdot \bar{w} = \overline{zw}$$

$$\bar{z} + \bar{w} = \overline{z + w}$$

$$|z| \cdot |w| = |zw|$$

$$|z|^n = |z^n|$$

$$z|z| = |z|^2$$

$|w - z|$ represents the distance between w and z in the complex plane.

Geometrically, $|z|$ represents the distance between the point and 0 in the complex plane.

Key Strategy: In many problems that involve a complex number, we can plug in $a + bi$ instead of a variable such as z for that complex number.

Problem 6.4.4 — There is a complex number z with imaginary part 164 and a positive integer n such that

$$\frac{z}{z + n} = 4i.$$

Find n .

Source: 2009 AIME

Solution: We can assume that the complex number $z = a + bi$. Since we know that 164 is the imaginary part, our value of b for z is 164. This means that $z = a + 164i$

We can plug that into the given equation to get

$$\frac{a+164i}{a+n+164i} = 4i$$

Multiplying both sides by the denominator gives

$$a + 164i = 4i(a + n) + 656i^2$$

We already know that $i^2 = -1$, so we can use that to simplify the equation to

$$a + 164i = 4i(a + n) - 656$$

We can pair up all the imaginary parts and all the real parts separately to get

$$a + 656 + (164 - 4a - 4n)i = 0.$$

$a + 656$ is our real part and $164 - 4a - 4n$ is our imaginary part. Since we know that the complex number is equivalent to 0, it means that the real part and the imaginary parts are also equal to 0. That gives us 2 equations.

$$a + 656 = 0$$

$$164 - 4a - 4n = 0$$

Using the first one gives us that $a = -656$. Plugging that into the second one gives us that $n = \mathbf{697}$.

Problem 6.4.5 — Find c if a , b , and c are positive integers which satisfy $c = (a + bi)^3 - 107i$, where $i^2 = -1$.

Source: 1985 AIME

Solution: In this problem, we know that $(a + bi)^3 - 107i$ has an imaginary part of 0 because that expression equates to c which is a positive integer. Thus, we can separate the real and imaginary parts after expanding our expression.

$$(a + bi)^3 - 107i = a^3 - 3ab^2 + (3a^2b - b^3 - 107)i = c$$

Clearly the real part is equivalent to c but the imaginary part is equivalent to 0. This gives us two equations:

$$a^3 - 3ab^2 = c$$

$$3a^2b - b^3 - 107 = 0$$

Factoring the second equation gives $b(3a^2 - b^2) = 107$

Since we know that 107 is a prime number and b must divide it, b must be equivalent to 1. This means that we're left with $3a^2 - b^2 = 107$. Plugging in 1 for b gives that $a^2 = 36$. This means $a = 6$ because a must be a positive integer.

Since we know that $a = 6$ and $b = 1$, we can plug those numbers into the equation $a^3 - 3ab^2 = c$ to get that $c = \mathbf{198}$.

Problem 6.4.6 — The complex number z is equal to $9 + bi$, where b is a positive real number and $i^2 = -1$. Given that the imaginary parts of z^2 and z^3 are the same, what is b equal to?

Source: 2007 AIME

Solution: Since we know that the imaginary parts of z^2 and z^3 are equal, we can find the value of their imaginary parts and equate it. We simply have to expand out z^2 and z^3 and take the complex part of both.

$$z^2 = (9 + bi)^2 = 81 + b^2i^2 + 18bi = 81 - b^2 + 18bi$$

Clearly the

$$\Im(z^2) = 18b$$

$$z^3 = (9 + bi)^3 = 729 - 9b^2 - 18b^2 + 162bi + 81bi - b^3i = 729 - 27b^2 + i(243b - b^3)$$

Clearly the

$$\Im(z^3) = 243b - b^3$$

Since we know that $\Im(z^2) = \Im(z^3)$, this means that $18b = 243b - b^3$.

Simplifying the equation gives $b^2 = 225$ which means $\mathbf{b = 15}$.

Problem 6.4.7 — The complex numbers z and w satisfy the system

$$z + \frac{20i}{w} = 5 + i$$

$$w + \frac{12i}{z} = -4 + 10i$$

Find the smallest possible value of $|zw|^2$.

Source: 2012 AIME

Solution: In this problem, we will first try to add both of the equations. However, that doesn't get us anywhere. Now we multiply it and see that it brings us to a good result.

$$zw - \frac{240}{zw} + 32i = 46i - 30$$

$$\text{Simplifying this gives } zw - \frac{240}{zw} = 14i - 30$$

We can multiply both sides by zw to get $(zw)^2 + (30 - 14i)zw - 240 = 0$

We can use the quadratic formula assuming zw is our ' x '. It gives us

$zw = 6 + 2i$ or $12i - 36$ for the possible solutions.

Since we want the magnitude of zw to be minimum, our value of zw will be $6 + 2i$ since its magnitude is $\sqrt{6^2 + 2^2}$ which is $\sqrt{40}$. Since we want to find the square of the magnitude, our answer is **40**.

§6.5 De Moivre's and Euler's Theorem

The previous section that covered complex numbers was on the fundamentals side. Now we'll go into more complex techniques.

Definition 6.5.1

Polar Form of Complex Numbers

The polar form is about writing a complex number such as $a + bi$ in the form $r(\cos \theta + i \sin \theta)$

Remember that $\cos \theta + i \sin \theta$ has a shortform that is $\text{cis} \theta$

The value of r is $\sqrt{a^2 + b^2}$ while θ is $\tan^{-1} \frac{b}{a}$

Theorem 6.5.2**Euler's Formula**

This is a super important theorem if you want to be able to solve many complex number problems quickly.

It's first important to know that $e^{i\pi} = -1$.

We can represent our complex number that's in polar form in such a form as shown above.

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Theorem 6.5.3**De Moivre's Theorem**

De moivre's theorem is crucial if you want to square complex numbers to a large power or simply multiply complex numbers. First of all, your complex number should be in polar form for this to work.

For two complex numbers $z = r_1 \operatorname{cis} \theta_1$ and $w = r_2 \operatorname{cis} \theta_2$, then $zw = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$

This theorem also states that if z is a complex number and $z = r(\cos \theta + i \sin \theta)$, Then, $z^n = r^n(\cos \theta \cdot n + i \sin \theta \cdot n)$

Tip: From all the theorems we just learned, it should be obvious that converting a complex number from it's rectangular form ($a + bi$) to polar form is a great idea in many problems. It can make it easy to solve complicated problems due to the simplifications that can be used in polar form.

Theorem 6.5.4**Roots of Unity**

The solutions to $z^n = 1$ where z is a complex number are $\cos(\frac{2\pi k}{n}) + i \sin(\frac{2\pi k}{n})$ which is equivalent to $\operatorname{cis}(\frac{2\pi k}{n})$ which is $e^{2\pi k/n}$. The value of k ranges from 0 to $n - 1$.

Problem 6.5.5 — Of the following complex numbers z , which one has the property that z^5 has the greatest real part?

(A) -2 (B) $-\sqrt{3} + i$ (C) $-\sqrt{2} + \sqrt{2}i$ (D) $-1 + \sqrt{3}i$ (E) $2i$

Source: 2021 AMC 12

Solution: In this problem, we will convert each of our answer choices to polar form. We don't have to do this for A and E because they are very easy to work with anyways.

For answer choice B, we can convert $-\sqrt{3} + i$ using the method described before. We first find r to be $\sqrt{(\sqrt{3})^2 + 1^2}$ which is 2.

Then, we find that $\tan^{-1} \theta = -\frac{1}{\sqrt{3}}$. We get that $\theta = 150^\circ$. This means that we can write our complex number as $2(\cos 150 + i \sin 150)$.

Sine we only want the real part of it, we can take out $2 \cos 150$. We want to find the real part raised to the power of 5. Using de Moivre's theorem, we know that it is $32 \cos(150 \cdot 5)$ which is $32 \cos(750)$ which is also equal to $32 \cos(30)$ because we can subtract out 720 (a multiple of 360) from 750. $32 \cos 30$ evaluates to $16\sqrt{3}$.

We can do the exact steps as shown above for the remaining answer choices to get that the greatest real part occurs for the complex number $-\sqrt{3} + i$ **(B)**.

Problem 6.5.6 — Let

$$z = \frac{1+i}{\sqrt{2}}.$$

What is

$$\left(z^{1^2} + z^{2^2} + z^{3^2} + \cdots + z^{12^2}\right) \cdot \left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} + \cdots + \frac{1}{z^{12^2}}\right)?$$

(A) 18 **(B)** $72 - 36\sqrt{2}$ **(C)** 36 **(D)** 72 **(E)** $72 + 36\sqrt{2}$

Source: 2021 AMC 12

Solution: Although this problem may look hard at first, it's simple if we use the tip mentioned before. Convert your complex number that is in rectangular form to polar form!

$$\frac{1+i}{\sqrt{2}} = \cos(45) + i \sin(45) = \text{cis}(45)$$

We know from De Moivre's that $(\text{cis}(45))^n = \text{cis}(45 \cdot n)$

We know that the values will repeat because if two angles mod 360 are the same, then the cosine and sine of those two angles will also be,

In this problem, if $(\text{cis}(45))^n$ is a certain value, then $(\text{cis}(45))^{n+8}$ will also be that same value or any multiple of 8 added on.

That means we can check the value of the exponents mod 8 in the expression we want to find. If any of the exponents are equivalent mod 8, then their values will be the same.

$1^2, 5^2$, and 9^2 are all 1 (mod) 8

$2^2, 6^2$, and 10^2 are all 4 (mod) 8

$3^2, 7^2$, and 11^2 are all 1 (mod) 8

$4^2, 8^2$, and 12^2 are 0 (mod) 8

This means that:

$$z^{1^2} = z^{5^2} = z^{9^2} = z^1 = \text{cis}(45) = \cos(45) + i \sin(45)$$

$$z^{2^2} = z^{6^2} = z^{10^2} = z^4 = \text{cis}(180) = \cos(180) + i \sin(180) = -1$$

$$z^{3^2} = z^{7^2} = z^{11^2} = z^1 = \text{cis}(45) = \cos(45) + i \sin(45)$$

$$z^{4^2} = z^{8^2} = z^{12^2} = z^0 = 1$$

We solve for the left side of the product to get that $z^{1^2} + z^{2^2} + z^{3^2} + \cdots + z^{12^2}$ is

$$6(\cos(45) + 6i \sin(45))$$

We solve for the right side of the product to get that $\frac{1}{z^{12}} + \frac{1}{z^{22}} + \frac{1}{z^{32}} + \cdots + \frac{1}{z^{122}}$ is $\frac{6}{\cos(45) + i \sin(45)}$

Multiplying both values cancels the $\cos(45) + i \sin(45)$ leaving us with an answer of **36**.

Problem 6.5.7 — The complex numbers z and w satisfy $z^{13} = w$, $w^{11} = z$, and the imaginary part of z is $\sin \frac{m\pi}{n}$, for relatively prime positive integers m and n with $m < n$. Find n .

Source: 2012 AIME

Solution: In this problem, we will plug in $w = z^{13}$ to our other equation to get an equation that only contains z since that's what we want to find the value for.

Doing so gives $z^{143} = z$. Then, we divide z from both sides to get $z^{142} = 1$.

Now we can apply the roots of unity theorem to get our solutions. We know that the solutions will be in the form $\cos(\frac{\pi k}{141}) + i \sin(\frac{\pi k}{141})$. Our value of k can range from 0 to 141. Clearly our value of n is **71** since it's in the denominator.

Problem 6.5.8 — There are 24 different complex numbers z such that $z^{24} = 1$. For how many of these is z^6 a real number?

(A) 0 (B) 4 (C) 6 (D) 12 (E) 24

Solution: 2017 AMC

Solution: We will use roots of unity since our equation is in the form $z^n = 1$.

We know that the solutions to this are $z = \cos(\frac{\pi k}{12}) + i \sin(\frac{\pi k}{12})$ for all values of k ranging from 0 to 23.

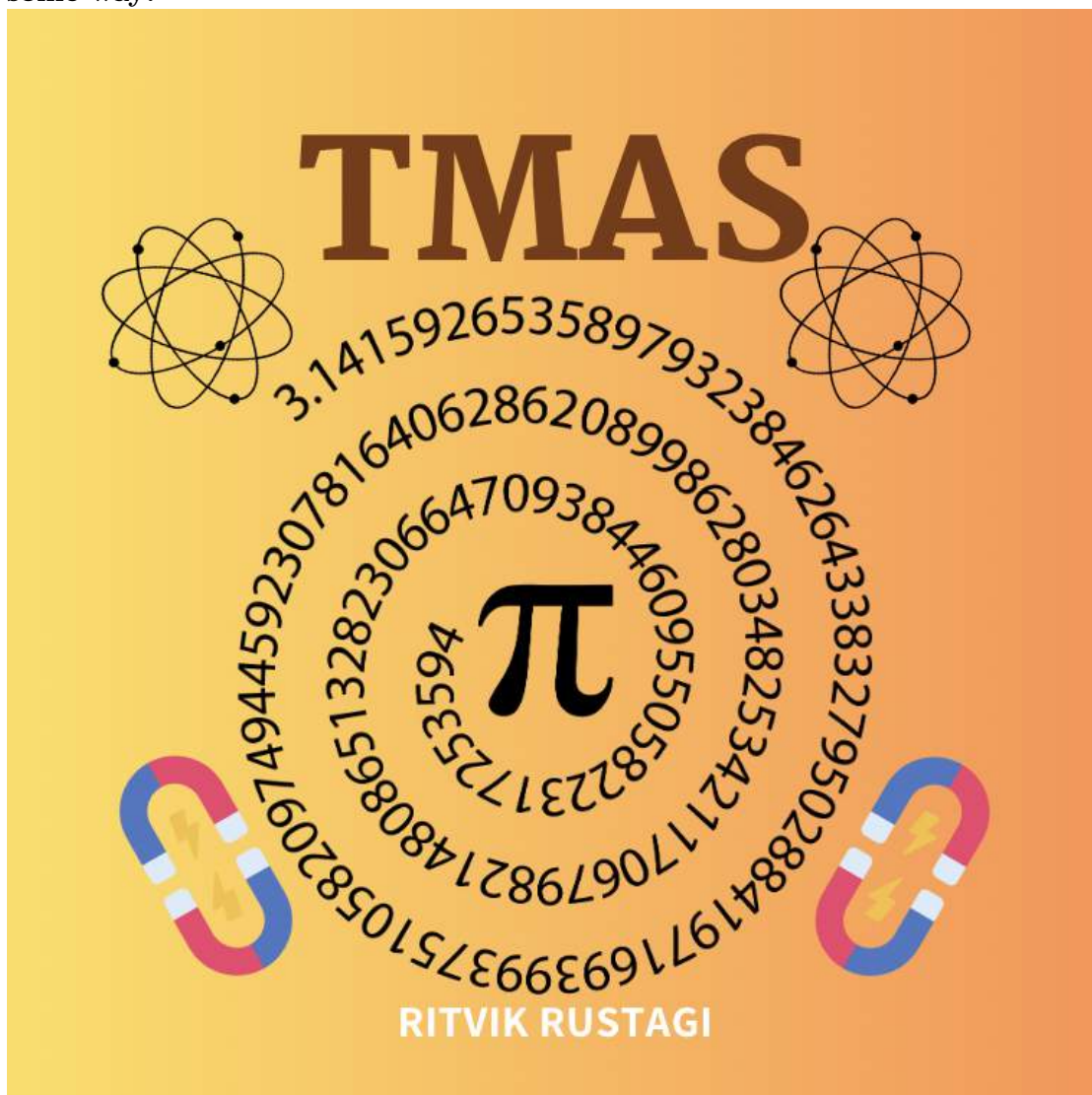
Now we can raise our expression for the value of z to the power of 6. $(\cos(\frac{\pi k}{12}) + i \sin(\frac{\pi k}{12}))^6$

From De Moivre's Theorem, we know that $z^n = r^n(\cos(\theta \cdot n) + i \sin(\theta \cdot n))$. We can use this on our expression to get that z^6 is $\cos(\frac{\pi k}{2}) + i \sin(\frac{\pi k}{2})$ for k between 0 and 23.

Since we want this expression to be real, it means that the imaginary part must be 0. The imaginary part is $i \sin(\frac{\pi k}{2})$. We want $\sin(\frac{\pi k}{2})$ to be 0. We know that $\sin$ is 0 for any multiple of π (in radians). This means that our value of k must be any even number. Since there are 12 even numbers between 0 and 23, our answer is **12** (D).

Thank you for going through this book!

It is an honor for me to have contributed to your contest math journey in some way!



Thanks,

Ritvik Rustagi

